

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**

Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the quarterly period ended September 30, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission file number: 001-36481

ASPEN AEROGELS, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

30 Forbes Road, Building B
Northborough, Massachusetts
(Address of principal executive offices)

04-3559972

(I.R.S. Employer
Identification No.)

01532

(Zip Code)

Registrant's telephone number, including area code: (508) 691-1111

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common Stock, par value \$0.00001 per share	ASPN	The New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.
Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☐	Accelerated filer	☒
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of October 31, 2023, the registrant had 70,256,926 shares of common stock outstanding.

ASPEN AEROGELS, INC.

INDEX TO FORM 10-Q

	<u>Page</u>
<u>PART I FINANCIAL INFORMATION</u>	
Item 1.	<u>Financial Statements</u>
	<u>Consolidated Balance Sheets (unaudited) as of September 30, 2023 and December 31, 2022</u>
	1
	<u>Consolidated Statements of Operations (unaudited) for the three and nine months ended September 30, 2023 and 2022</u>
	2
	<u>Consolidated Statements of Stockholders' Equity (unaudited) for the three and nine months ended September 30, 2023 and 2022</u>
	3
	<u>Consolidated Statements of Cash Flows (unaudited) for the nine months ended September 30, 2023 and 2022</u>
	4
	<u>Notes to Consolidated Financial Statements (unaudited)</u>
	5
Item 2.	<u>Management's Discussion and Analysis of Financial Condition and Results of Operations</u>
	19
Item 3.	<u>Quantitative and Qualitative Disclosures About Market Risk</u>
	35
Item 4.	<u>Controls and Procedures</u>
	35
<u>PART II OTHER INFORMATION</u>	
Item 1.	<u>Legal Proceedings</u>
	37
Item 1A.	<u>Risk Factors</u>
	37
Item 2.	<u>Unregistered Sales of Equity Securities and Use of Proceeds</u>
	39
Item 3.	<u>Defaults Upon Senior Securities</u>
	39
Item 4.	<u>Mine Safety Disclosures</u>
	39
Item 5.	<u>Other Information</u>
	39
Item 6.	<u>Exhibits</u>
	40
<u>SIGNATURES</u>	41

Trademarks, Trade Names and Service Marks

We own or have rights to use "Aspen Aerogels," "Cryogel," "Pyrogel," "Spaceloft," "PyroThin," the Aspen Aerogels logo and other trademarks, service marks and trade names of Aspen Aerogels, Inc. appearing in this Quarterly Report on Form 10-Q. Solely for convenience, the trademarks, service marks and trade names referred to in this report are presented without the ® and TM symbols, but such references are not intended to indicate, in any way, that the owner thereof will not assert, to the fullest extent under applicable law, such owner's rights to these trademarks, service marks and trade names. This report contains additional trademarks, service marks and trade names of other companies, which, to our knowledge, are the property of their respective owners.

PART I — FINANCIAL INFORMATION

Item 1. Financial Statements.

ASPEN AEROGELS, INC.
Consolidated Balance Sheets
(Unaudited)

	September 30, 2023	December 31, 2022
	(In thousands, except share and per share data)	
Assets		
Current assets:		
Cash and cash equivalents	\$ 94,643	\$ 281,335
Restricted cash	320	1,226
Accounts receivable, net of allowances of \$184 and \$255	54,413	57,350
Inventories	34,421	22,538
Prepaid expenses and other current assets	16,568	7,236
Total current assets	200,365	369,685
Property, plant and equipment, net	385,026	259,223
Operating lease right-of-use assets	17,400	11,990
Other long-term assets	2,355	2,518
Total assets	\$ 605,146	\$ 643,416
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	\$ 37,115	\$ 54,728
Accrued expenses	16,694	16,003
Deferred revenue	5,463	5,846
Operating lease liabilities	1,986	2,368
Total current liabilities	61,258	78,945
Convertible note - related party	112,088	103,580
Operating lease liabilities long-term	21,987	13,456
Total liabilities	195,333	195,981
Commitments and contingencies (Note 9)		
Stockholders' equity:		
Preferred stock, \$0.00001 par value; 5,000,000 shares authorized, no shares issued and outstanding at September 30, 2023 and December 31, 2022	—	—
Common stock, \$0.00001 par value; 250,000,000 shares authorized, 70,226,633 and 69,994,963 shares issued and outstanding at September 30, 2023 and December 31, 2022, respectively	—	—
Additional paid-in capital	1,082,896	1,075,226
Accumulated deficit	(673,083)	(627,791)
Total stockholders' equity	409,813	447,435
Total liabilities and stockholders' equity	\$ 605,146	\$ 643,416

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Operations
(Unaudited)

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2023	2022	2023	2022
	(In thousands, except share and per share data)			
Revenue	\$ 60,755	\$ 36,706	\$ 154,499	\$ 120,753
Cost of revenue	46,945	43,065	127,196	130,111
Gross profit (loss)	13,810	(6,359)	27,303	(9,358)
Operating expenses:				
Research and development	4,218	4,694	12,281	12,733
Sales and marketing	8,386	7,293	24,226	20,944
General and administrative	15,840	9,963	41,382	26,544
Total operating expenses	28,444	21,950	77,889	60,221
Loss from operations	(14,634)	(28,309)	(50,586)	(69,579)
Other income (expense)				
Interest expense, convertible note - related party	(1,938)	(1,734)	(2,424)	(4,103)
Interest income, net	1,313	448	5,532	553
Income from Employee Retention Credits	2,186	—	2,186	—
Total other income (expense), net	1,561	(1,286)	5,294	(3,550)
Net loss	\$ (13,073)	\$ (29,595)	\$ (45,292)	\$ (73,129)
Net loss per share:				
Basic and diluted	\$ (0.19)	\$ (0.75)	\$ (0.65)	\$ (2.03)
Weighted-average common shares outstanding:				
Basic and diluted	69,317,805	39,533,695	69,243,843	36,047,879

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Stockholders' Equity
(Unaudited)
(In thousands, except share data)

	Preferred Stock		Common Stock		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2022	—	\$ —	69,994,963	\$ —	\$ 1,075,226	\$ (627,791)	\$ 447,435
Net loss	—	—	—	—	—	(16,796)	(16,796)
Stock compensation expense	—	—	—	—	2,267	—	2,267
Vesting of restricted stock units	—	—	71,643	—	(385)	—	(385)
Proceeds from employee stock option exercises	—	—	2,554	—	21	—	21
Balance at March 31, 2023	—	\$ —	70,069,160	\$ —	\$ 1,077,129	\$ (644,587)	\$ 432,542
Net loss	—	—	—	—	—	(15,423)	(15,423)
Stock compensation expense	—	—	—	—	2,710	—	2,710
Issuance of restricted stock	—	—	44,928	—	—	—	—
Vesting of restricted stock units	—	—	2,464	—	(8)	—	(8)
Proceeds from employee stock option exercises	—	—	41,591	—	150	—	150
Balance at June 30, 2023	—	\$ —	70,158,143	\$ —	\$ 1,079,981	\$ (660,010)	\$ 419,971
Net loss	—	—	—	—	—	(13,073)	(13,073)
Stock compensation expense	—	—	—	—	2,789	—	2,789
Vesting of restricted stock units	—	—	2,662	—	(9)	—	(9)
Proceeds from employee stock option exercises	—	—	65,828	—	274	—	274
Issuance costs from underwritten public offering	—	—	—	—	(139)	—	(139)
Balance at September 30, 2023	—	\$ —	70,226,633	\$ —	\$ 1,082,896	\$ (673,083)	\$ 409,813

	Preferred Stock		Common Stock		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2021	—	\$ —	33,218,115	\$ —	\$ 673,461	\$ (545,053)	\$ 128,408
Net loss	—	—	—	—	—	(19,484)	(19,484)
Stock compensation expense	—	—	—	—	1,828	—	1,828
Vesting of restricted stock units	—	—	166,211	—	(2,315)	—	(2,315)
Proceeds from employee stock option exercises	—	—	4,681	—	38	—	38
Proceeds from at-the-market offering, net of commissions of \$729 and issuance costs of \$318	—	—	737,288	—	23,272	—	23,272
Proceeds from private placement of common stock, net of fees and issuance costs of \$136	—	—	1,791,986	—	49,864	—	49,864
Balance at March 31, 2022	—	\$ —	35,918,281	\$ —	\$ 746,148	\$ (564,537)	\$ 181,611
Net loss	—	—	—	—	—	(24,050)	(24,050)
Stock compensation expense	—	—	—	—	2,295	—	2,295
Issuance of restricted stock	—	—	391,324	—	—	—	—
Vesting of restricted stock units	—	—	2,569	—	(24)	—	(24)
Proceeds from employee stock option exercises	—	—	21,110	—	136	—	136
Proceeds from at-the-market offering, net of commissions of \$149 and issuance costs of \$28	—	—	145,000	—	4,786	—	4,786
Balance at June 30, 2022	—	\$ —	36,478,284	\$ —	\$ 753,341	\$ (588,587)	\$ 164,754
Net loss	—	—	—	—	—	(29,595)	(29,595)
Stock compensation expense	—	—	—	—	2,590	—	2,590
Issuance of restricted stock	—	—	3,495	—	—	—	—
Vesting of restricted stock units	—	—	236	—	—	—	—
Proceeds from at-the-market offering, net of commissions of \$1,391 and issuance costs of \$285	—	—	4,359,112	—	44,657	—	44,657
Balance at September 30, 2022	—	\$ —	40,841,127	\$ —	\$ 800,588	\$ (618,182)	\$ 182,406

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Cash Flows
(Unaudited)

	Nine Months Ended September 30,	
	2023	2022
	(In thousands)	
Cash flows from operating activities:		
Net loss	\$ (45,292)	\$ (73,129)
Adjustments to reconcile net loss to net cash used in operating activities:		
Depreciation	10,757	6,692
Accretion of interest on convertible note - related party	1,721	4,103
Amortization of convertible note issuance costs	28	23
Amortization of debt discount due to modification of convertible note – related party	675	—
Provision for bad debt	(89)	42
Stock-compensation expense	7,766	6,713
Reduction in the carrying amount of operating lease right-of-use assets	2,186	1,925
Changes in operating assets and liabilities:		
Accounts receivable	3,026	(6,733)
Inventories	(11,883)	(8,613)
Prepaid expenses and other assets	(6,771)	(3,055)
Accounts payable	(420)	481
Accrued expenses	691	1,918
Deferred revenue	(383)	1,035
Operating lease liabilities	(1,845)	(1,721)
Net cash used in operating activities	(39,833)	(70,319)
Cash flows from investing activities:		
Capital expenditures	(147,669)	(119,348)
Net cash used in investing activities	(147,669)	(119,348)
Cash flows from financing activities:		
Proceeds from issuance of convertible note - related party	—	100,000
Issuance costs from convertible note	—	(185)
Proceeds from employee stock option exercises	445	174
Payments made for employee restricted stock tax withholdings	(402)	(2,340)
Proceeds from at-the-market offering, net of commissions of \$2,269	—	73,348
Fees and issuance costs from at-the-market offering	—	(631)
Proceeds from private placement of common stock	—	50,000
Fees and issuance costs from private placement of common stock	(139)	(137)
Repayment of prepayment liability	—	(4,728)
Net cash provided by (used in) financing activities	(96)	215,501
Net (decrease) increase in cash, cash equivalents and restricted cash	(187,598)	25,834
Cash, cash equivalents and restricted cash at beginning of period	282,561	76,564
Cash, cash equivalents and restricted cash at end of period	\$ 94,963	\$ 102,398
Supplemental disclosures of cash flow information:		
Interest paid	\$ 1	\$ 130
Income taxes paid	\$ —	\$ —
Supplemental disclosures of non-cash activities:		
Right-of-use assets obtained in exchange for new operating lease liabilities	\$ 9,994	\$ 5,857
Capitalized interest	\$ 6,084	\$ 1,277
Changes in accrued capital expenditures	\$ (17,193)	\$ 40,402

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(UNAUDITED)

(1) Description of Business and Basis of Presentation

Nature of Business

Aspen Aerogels, Inc. (the Company) is an aerogel technology company that designs, develops and manufactures innovative, high-performance aerogel insulation used primarily in the energy industrial and sustainable insulation materials markets. In addition, the Company has introduced a line of aerogel thermal barriers for use in battery packs in the electric vehicle market. The Company is also developing applications for its aerogel technology in the battery materials and a number of other high-potential markets.

The Company maintains its corporate offices in Northborough, Massachusetts. The Company has three wholly owned subsidiaries: Aspen Aerogels Rhode Island, LLC, Aspen Aerogels Germany, GmbH and Aspen Aerogels Georgia, LLC.

Liquidity

During the nine months ended September 30, 2023, the Company incurred a net loss of \$45.3 million, used \$39.8 million of cash in operations and used \$147.7 million of cash for capital expenditures. The Company had unrestricted cash and cash equivalents of \$94.6 million as of September 30, 2023.

On November 28, 2022, the Company entered into a loan agreement with (the GM Loan Agreement) General Motors Holdings LLC (GM), an entity affiliated with General Motors LLC, which provides for a multi-draw senior secured term loan (the GM Loan) in an aggregate principal amount of up to \$100.0 million, available to the Company on a delayed draw basis beginning January 1, 2023 to September 30, 2023, subject to certain conditions precedent to funding. On September 28, 2023, the Company amended the GM Loan Agreement to extend the draw period for the delayed GM Loan to a period beginning on the date that is twelve months prior to the date agreed upon by the Company and GM for the start of production at an aerogel manufacturing facility in Bulloch County, Georgia (the Plant) and ending on March 31, 2024 (or any later date approved in writing by GM at its sole discretion); extend the maturity date of the GM Loan from March 31, 2025 to September 30, 2025; and add financial covenants measured starting from the fiscal quarter ending December 31, 2024 and at the end of each fiscal quarter thereafter.

The revised GM Loan Agreement also amended the conditions precedent to funding to require the Company to provide evidence that cash proceeds of one or more equity and/or debt financing arrangements of not less than \$500.0 million in the aggregate have been contributed to and disbursed by the Company in the manner prescribed in a pre-determined project budget (subject to a permitted variance) to fund the construction of and equipment for the first phase of the Plant and require that 70% of the total cost in connection with the construction and operation of the Plant has been fully funded prior to such applicable borrowing under the GM Loan Agreement.

The Company is increasing investment in the research and development of next-generation aerogel products and manufacturing process technologies. In addition, the Company has developed a number of promising aerogel products and technologies for the electric vehicle market. The Company believes that the commercial potential for the Company's products and technology in the electric vehicle market is significant. Accordingly, the Company is hiring additional personnel, incurring additional operating expenses, and incurring significant capital expenditures to expand silica aerogel manufacturing capacity, build an automated thermal barrier fabrication operation, enhance research and development laboratory facilities and equipment, and construct a battery materials facility, among other efforts.

The Company expects its existing cash balance will be sufficient to support current operating requirements, current research and development activities and the capital expenditures required to support the evolving commercial opportunity in the electric vehicle market and other strategic business initiatives. However, the Company plans to supplement its cash balance with equity financings, debt financings, equipment leasing, sale-leaseback transactions, customer prepayments, or government grant and loan programs to provide the additional capital necessary to purchase the capital equipment, construct the new facilities, establish the operations and complete the aerogel capacity expansions required to support these evolving commercial opportunities and strategic business initiatives.

Unaudited Interim Financial Information

The accompanying unaudited interim consolidated financial statements include the accounts of the Company and have been prepared in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP) and with the instructions to Form 10-Q and Article 10 of Regulation S-X. Certain information and disclosures normally included in the consolidated financial statements prepared in accordance with U.S. GAAP have been condensed or omitted pursuant to such rules and regulations. As such, the information included in this Quarterly Report on Form 10-Q should be read in conjunction with the audited consolidated financial statements and accompanying notes in our Annual Report on Form 10-K for the year ended December 31, 2022 (the Annual Report), filed with the U.S. Securities and Exchange Commission on March 16, 2023.

In the opinion of the Company's management, the unaudited interim consolidated financial statements have been prepared on the same basis as the audited consolidated financial statements and include all adjustments that are of a normal recurring nature and necessary for the fair statement of the Company's financial position as of September 30, 2023 and the results of its operations and stockholders' equity for the three and nine months ended September 30, 2023 and 2022 and the cash flows for the nine-month periods then ended. The Company has evaluated subsequent events through the date of this filing.

The Company's results of operations for the three and nine months ended September 30, 2023 are not necessarily indicative of the results to be expected for the year ending December 31, 2023 or any other period.

(2) Significant Accounting Policies

Principles of Consolidation

The accompanying condensed consolidated financial statements of the Company include the accounts of all its subsidiaries which are majority-owned, controlled by the Company or a variable interest entity (VIE) where the Company is the primary beneficiary. All significant intercompany accounts and transactions have been eliminated in consolidation.

A VIE is an entity that lacks one or more of the characteristics of a voting interest entity. A VIE is defined as an entity in which equity investors do not have the characteristics of a controlling financial interest or do not have sufficient equity at risk for the entity to finance its activities without additional subordinated financial support from other parties. The Company consolidates a VIE in accordance with ASC 810, Consolidation (ASC 810) when it is the primary beneficiary of such VIE. As primary beneficiary, the Company has both the power to direct the activities that most significantly impact the economic performance of the VIE and a right to receive benefits or absorb losses of the entity that could be potentially significant to the VIE. The Company is required to reconsider its evaluation of whether to consolidate a VIE each reporting period, based upon changes in the facts and circumstances pertaining to the VIE.

The Company evaluates the initial consolidation of each Consolidated VIE, which includes a determination of whether the VIE constitutes the definition of a business in accordance with ASC 805, Business Combinations (ASC 805), by considering if substantially all of the fair value of the gross assets within the VIE are concentrated in either a single identifiable asset or group of single identifiable assets. Upon consolidation, the Company recognizes the assets acquired, the liabilities assumed, and any third-party ownership of membership interests as non-controlling interest as of the consolidation or acquisition date, measured at their relative fair values.

In April 2022, the Company engaged Prodensa Servicios de Consultora to establish OPE Manufacturer Mexico S de RL de CV, a maquiladora located in Mexico with the express purposes of manufacturing thermal barrier PyroThin products and ultimately constructing an automated fabrication facility for PyroThin. OPE is currently owned by Prodensa, which charges a management fee though there is an option for OPE to be purchased by the Company after a period of 18 months. During the period between inception and the purchase option, OPE operations are consolidated within the Company's financial statements as of and for the nine months ended September 30, 2023.

Use of Estimates

The preparation of the consolidated financial statements requires the Company to make a number of estimates and assumptions that affect the reported amounts of assets and liabilities at the date of the consolidated financial statements and the reported amounts of revenues and expenses during the reporting period. Significant items subject to such estimates and assumptions include allowances for doubtful accounts, sales returns and allowances, product warranty costs, inventory valuation, the carrying amount of property and equipment, right-of-use assets, lease liabilities, stock-based compensation, and deferred income taxes. The Company evaluates its

estimates and assumptions on an on-going basis using historical experience and other factors, including current economic conditions, which are believed to be reasonable under the circumstances. Management adjusts such estimates and assumptions when facts and circumstances warrant. Illiquid credit markets, volatile equity markets and declines in business investment can increase the uncertainty inherent in such estimates and assumptions. As future events and their effects cannot be determined with precision, actual results could differ significantly from these estimates. Changes in these estimates resulting from continuing changes in the economic environment will be reflected in the financial statements in future periods.

Reclassifications

Certain prior year amounts have been reclassified to conform to the current year presentation.

Cash and Cash Equivalents

Cash equivalents include short-term, highly liquid instruments, which consist of money market accounts and high-quality debt securities issued by the U.S. government via cash sweep and investment accounts. Cash and cash equivalents are maintained primarily with major financial institutions in North America. Deposits with these financial institutions may exceed the amount of insurance provided on such deposits; however, these deposits typically may be redeemed upon demand and, therefore, bear minimal risk.

In light of the failure of Silicon Valley Bank and other regional banks, the Company continues to establish commercial banking relationships with additional large financial institutions.

Restricted Cash

As of September 30, 2023, the Company had \$0.3 million of restricted cash to support its outstanding letters of credit to secure obligations under certain commercial contracts and other obligations.

Concentration of Credit Risk

Financial instruments, which potentially expose the Company to concentrations of credit risk, consist principally of accounts receivable. The Company's customers are primarily insulation distributors, insulation contractors, insulation fabricators and select energy and automotive end-users located throughout the world. The Company performs ongoing credit evaluations of its customers' financial condition and generally requires no collateral to secure accounts receivable. The Company maintains an allowance for doubtful accounts based on its assessment of the collectability of accounts receivable. The Company reviews the allowance for doubtful accounts quarterly. During the nine months ended September 30, 2023, the Company recorded a reduction for estimated customer uncollectible accounts receivable of less than \$0.1 million. During the nine months ended September 30, 2022, the Company recorded a reduction for estimated customer uncollectible accounts receivable of less than \$0.1 million.

For the nine months ended September 30, 2023 and 2022, two customers represented 50% and 44% of total revenue, respectively.

At September 30, 2023, the Company had one customer which accounted for 61% of accounts receivable. At December 31, 2022, the Company had two customers which accounted for 44% and 10% of accounts receivable, respectively.

Revenue Recognition

The Company recognizes revenue in accordance with Accounting Standards Codification 606, Revenue from Contracts with Customers (ASC 606). See note 3 for further details.

Leases

The Company accounts for its leases in accordance with Accounting Standards Update (ASU) 2016-02 (Topic 842). See note 10 for further details.

Stock-based Compensation

The Company grants share-based awards to its employees and non-employee directors under its equity incentive plans: the 2023 Equity Incentive Plan (the 2023 Equity Plan) and its predecessor, the 2014 Employee, Director, and Consultant Equity Incentive Plan (the 2014 Equity Plan). All share-based awards granted, including grants of stock options, restricted stock and restricted stock units (RSUs), are recognized in the statement of operations based on their fair value as of the date of grant. Expense is recognized on a straight-line basis over the requisite service period for all awards with service conditions. For performance-based awards, the grant date fair value is recognized as expense when the condition is probable of being achieved, and then on a graded basis over the requisite service period. The Company uses the Black-Scholes option-pricing model to determine the fair value of service-based option awards. The Black-Scholes model requires the use of a number of complex and subjective assumptions including fair value of the underlying security, the expected volatility of the underlying security, a risk-free interest rate and the expected term of the option.

The fair value of restricted stock and RSUs is determined using the closing price of the Company's common stock on the date of grant. All shares of restricted stock are not transferable until vested. Restricted stock is typically issued to non-employee directors and typically vests over a one-year period from the date of issuance. RSUs are issued to employees and typically vest over a three-year period from the date of issuance. The fair value of restricted stock and RSUs upon which vesting is solely service-based is expensed ratably over the vesting period. If the service condition for shares of restricted stock is not met for any reason, the shares of unvested restricted stock will be forfeited and returned to the Company.

For stock options that contain a market condition, the Company uses the Monte-Carlo simulation option-pricing model to determine the fair value of the awards. In addition to the input assumptions used in the Black-Scholes model, the Monte-Carlo simulation option-pricing model factors the probability that the specific market condition may or may not be satisfied into the valuation. Stock-based compensation expense for awards with a market condition is recognized on a straight-line basis over the requisite service period for each such award.

During the nine months ended September 30, 2023, the Company granted 511,241 restricted common stock units (RSUs) with an aggregate grant date fair value of \$4.5 million and non-qualified stock options (NSOs) to purchase 1,956,464 shares of common stock with an aggregate grant date fair value of \$9.6 million to employees under its equity incentive plans. The RSUs and NSOs granted to employees will typically vest over a three-year period. During the nine months ended September 30, 2023, the Company also granted 44,928 shares of restricted common stock with a grant date fair value of \$0.3 million and NSOs to purchase 46,272 shares of common stock with a grant date fair value of \$0.2 million to its non-employee directors under the 2023 Equity Plan. The restricted common stock and NSOs granted to non-employee directors vest upon the earlier of the date that is the one-year anniversary of the grant date or the day prior to the Company's annual meeting of stockholders to be held in 2024.

Stock-based compensation is included in cost of revenue or operating expenses, as applicable, and consists of the following:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2023	2022	2023	2022
	(In thousands)		(In thousands)	
Cost of product revenue	\$ 93	\$ 270	\$ 420	\$ 656
Research and development expenses	255	306	510	843
Sales and marketing expenses	378	548	1,110	1,328
General and administrative expenses	2,063	1,466	5,726	3,886
Total stock-based compensation	<u>\$ 2,789</u>	<u>\$ 2,590</u>	<u>\$ 7,766</u>	<u>\$ 6,713</u>

The 2023 Equity Plan was approved by stockholders at the Company's annual meeting of stockholders on June 1, 2023 as the successor to the 2014 Equity Plan, and no further awards may be made under the 2014 Equity Plan after that date. As of September 30, 2023, 6,030,483 shares of common stock were reserved for issuance upon the exercise or vesting of outstanding stock-based awards granted under the Company's equity incentive plans. Any cancellations or forfeitures of awards outstanding under the 2023 Equity Plan, the 2014 Equity Plan or the 2001 Equity Incentive Plan, as amended (the 2001 Equity Plan) will result in the shares reserved for issuance pursuant to such awards becoming available for grant under the 2023 Equity Plan. As of September 30, 2023, the Company has either reserved in connection with statutory tax withholdings or issued a total of 4,956,866 shares under the Company's equity incentive plans. As of September 30, 2023, there were 2,134,645 shares of common stock available for future grant under the 2023 Equity Plan.

Net Loss per Share

The Company calculates net loss per share of common stock based on the weighted-average number of shares of common stock outstanding during each period. Potential common stock equivalents are determined using the treasury stock method. The weighted-average number of shares of common stock included in the computation of diluted net loss gives effect to all potentially dilutive common equivalent shares, including outstanding stock options and RSUs. Common equivalent shares are excluded from the computation of diluted net loss per share if their effect is antidilutive.

Warranty

The Company provides warranties for its products and records the estimated cost within cost of revenue in the period that the related revenue is recorded.

The Company's standard warranty period for energy industrial products extends to one year from the date of shipment. This standard warranty provides that the Company's products will be free from defects in material and workmanship, and will, under normal use, conform to the specifications for the product.

The Company's thermal barrier products provide quality and warranty provisions customary in the automotive industry.

The Company recorded warranty expense related to its thermal barrier products of \$0.2 million during each of nine months ended September 30, 2023 and 2022.

Employee Retention Credits

In March 2020, the Coronavirus Aid, Relief, and Economic Security Act (CARES Act) was signed into law, providing numerous tax provisions and other stimulus measures, including the Employee Retention Credit: a refundable tax credit against certain employment taxes. The Taxpayer Certainty and Disaster Tax Relief Act of 2020 and the American Rescue Plan Act of 2021 extended and expanded the availability of the Employee Retention Credits. We qualified for the Employee Retention Credits in the third and fourth quarters of 2020 and the first quarter of 2021. In September 2023, we submitted filings for CARES Employee Retention Credits totaling \$2.2 million that are reported in the accompanying condensed consolidated balance sheet within prepaid expenses and other current assets as of September 30, 2023, and in the accompanying statement of operations for the three and nine months ended September 30, 2023.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the Financial Accounting Standards Board or other standard setting bodies. Recently issued standards typically do not require adoption until a future effective date. Prior to their effective date, the Company evaluates the pronouncements to determine the potential effects of adoption to its consolidated financial statements.

Standards Implemented Since December 31, 2022

The Company has not implemented any accounting standards that had a material impact on its consolidated financial statements during the nine months ended September 30, 2023.

Standards to be Implemented

The Company believes that the impact of recently issued accounting standards that are not yet effective will not have a material impact on its consolidated financial statements.

(3) Revenue from Contracts with Customers

Revenue Recognition

Revenue is recognized when a customer obtains control of promised goods or services in an amount that reflects the consideration which the entity expects to receive in exchange for those goods or services. To determine revenue recognition for

arrangements within the scope of ASC 606, the Company performs the following five steps: (i) identification of the contract with a customer; (ii) identification of the performance obligations in the contract; (iii) determination of the transaction price; (iv) allocation of the transaction price to the separate performance obligations in the contract; and (v) recognition of the revenue associated with performance obligations as they are satisfied. The Company applies the five-step model to contracts when it is probable that the Company will collect the consideration it is entitled to in exchange for the goods or services it transfers to the customer. At contract inception, once the contract is determined to be within the scope of ASC 606, the Company assesses the goods or services promised within each contract and determines those that are performance obligations and assesses whether each promised good or service is distinct. If the contract contains a single performance obligation, the entire transaction price is allocated to the single performance obligation. Contracts that contain multiple performance obligations require an allocation of the transaction price based on the estimated relative standalone-selling prices of the promised products or services underlying each performance obligation. The Company determines standalone-selling prices based on the price at which the performance obligation is sold separately. If the standalone-selling price is not observable through past transactions, the Company estimates the standalone-selling price considering available information such as market conditions and internally approved pricing guidelines related to the performance obligations. The Company then recognizes as revenue the amount of the transaction price that is allocated to the respective performance obligation when (or as) the performance obligation is satisfied.

When determining the transaction price of a contract, an adjustment is made if payment from a customer occurs either significantly before or significantly after performance, resulting in a significant financing component. Applying the practical expedient in paragraph ASC 606-10-32-18, the Company does not assess whether a significant financing component exists if the period between when the Company performs its obligations under the contract and when the customer pays is one year or less. The Company did not have any contracts outstanding at December 31, 2022 and did not enter into any contracts during the nine months ended September 30, 2023 that contained a significant financing component.

The Company records deferred revenue for product sales when (i) the Company has delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

Energy Industrial

The Company generally enters into contracts containing one type of performance obligation. For a majority of the contracts, the Company recognizes revenue at a point in time when transfer of control of the products is passed to the customer, which is generally upon delivery according to contractual shipping terms within customer purchase orders. For a limited number of customer arrangements for customized products with no alternative use to the Company and an enforceable right to payment for progress completed to date, the Company recognizes revenue over time using units of production to measure progress toward satisfying the performance obligations. Units of production represent work performed as we do not generate significant work in process and thereby best depicts the transfer of control to the customer. Customer invoicing terms for contracts for which revenue is recognized under the over time methodology are typically based on certain milestones within the production and delivery schedule. The timing of revenue recognition is assessed on a contract-by-contract basis.

The Company also enters into rebate agreements with certain customers. These agreements may be considered an additional performance obligation of the Company or variable consideration within a contract. Rebates are recorded as a reduction of revenue in the period the related revenue is recognized. A corresponding liability is recorded as a component of deferred revenue on the consolidated balance sheets. These arrangements are primarily based on the customer attaining contractually specified sales volumes.

The Company estimates the amount of its sales that may be returned by its customers and records this estimate as a reduction of revenue in the period the related revenue is recognized. The Company currently estimates return liabilities using historical rates of return, current quarter credit sales, and specific items of exposure on a contract-by-contract basis. Sales return reserves were approximately \$0.1 million at both September 30, 2023 and December 31, 2022.

Thermal Barriers

The Company supplies fabricated, multi-part thermal barriers for use in battery packs in the electric vehicle market. These thermal barriers are customized to meet customer specifications. Although thermal barrier products are customized with no alternative use to the Company, the Company does not always have an enforceable right to payment. Under the provisions of ASC 606, the Company recognizes revenue at a point in time when transfer of the control of the products is passed to the customer according to the terms of the contract. The timing of revenue recognition is assessed on a contract-by-contract basis.

Shipping and Handling Costs

Shipping and handling costs associated with outbound freight after control over a product has transferred to a customer are accounted for as fulfillment costs and are included in the cost of product revenue. The associated amount of revenue recognized includes the consideration to which the Company expects to be entitled to receive in exchange for incurring these shipping and handling costs.

Disaggregation of Revenue

In the following tables, revenue is disaggregated by primary geographical region and source of revenue:

	Three Months Ended September 30,					
	2023			2022		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 5,801	\$ 5,801	\$ —	\$ 7,296	\$ 7,296
Canada	—	633	633	—	928	928
Europe	—	11,361	11,361	—	4,281	4,281
Latin America	—	1,176	1,176	—	84	84
U.S.	41,784	—	41,784	24,117	—	24,117
Total revenue	<u>\$ 41,784</u>	<u>\$ 18,971</u>	<u>\$ 60,755</u>	<u>\$ 24,117</u>	<u>\$ 12,589</u>	<u>\$ 36,706</u>
Source of revenue						
Energy industrial	\$ 12,249	\$ 15,663	\$ 27,912	\$ 13,510	\$ 11,242	\$ 24,752
Thermal barrier	29,535	3,308	32,843	10,607	1,347	11,954
Total revenue	<u>\$ 41,784</u>	<u>\$ 18,971</u>	<u>\$ 60,755</u>	<u>\$ 24,117</u>	<u>\$ 12,589</u>	<u>\$ 36,706</u>

	Nine Months Ended September 30,					
	2023			2022		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 27,522	\$ 27,522	\$ —	\$ 26,271	\$ 26,271
Canada	—	1,519	1,519	—	3,240	3,240
Europe	—	26,735	26,735	—	14,352	14,352
Latin America	—	5,065	5,065	—	2,985	2,985
U.S.	93,658	—	93,658	73,905	—	73,905
Total revenue	<u>\$ 93,658</u>	<u>\$ 60,841</u>	<u>\$ 154,499</u>	<u>\$ 73,905</u>	<u>\$ 46,848</u>	<u>\$ 120,753</u>
Source of revenue						
Energy industrial	\$ 43,994	\$ 53,317	\$ 97,311	\$ 47,989	\$ 42,415	\$ 90,404
Thermal barrier	49,664	7,524	57,188	25,916	4,433	30,349
Total revenue	<u>\$ 93,658</u>	<u>\$ 60,841</u>	<u>\$ 154,499</u>	<u>\$ 73,905</u>	<u>\$ 46,848</u>	<u>\$ 120,753</u>

Contract Balances

The following table presents changes in the Company's contract assets and contract liabilities during the nine months ended September 30, 2023:

	Balance at December 31, 2022	Additions	Deductions	Balance at September 30, 2023
	(In thousands)			
Contract assets				
Thermal barrier	\$ 143	\$ —	\$ (143)	\$ —
Total contract assets	<u>\$ 143</u>	<u>\$ —</u>	<u>\$ (143)</u>	<u>\$ —</u>
Contract liabilities				
Deferred revenue				
Energy industrial	\$ 5,846	\$ 14,308	\$ (14,691)	\$ 5,463
Total contract liabilities	<u>\$ 5,846</u>	<u>\$ 14,308</u>	<u>\$ (14,691)</u>	<u>\$ 5,463</u>

During the nine months ended September 30, 2023, the Company recognized \$5.7 million of revenue that was included in deferred revenue as of December 31, 2022.

A contract asset is recorded when the Company satisfies a performance obligation by transferring a promised good or service and has earned the right to consideration from its customer. These assets may represent a conditional right to consideration and are included within accounts receivable and other current assets on the consolidated balance sheets.

A contract liability is recorded when consideration is received, or such consideration is unconditionally due, from a customer prior to transferring goods or services under the terms of the contract. Contract liabilities are recognized as revenue after control of the products or services is transferred to the customer and all revenue recognition criteria have been met.

(4) Inventories

Inventories consist of the following:

	September 30, 2023	December 31, 2022
	(In thousands)	
Raw materials	\$ 22,361	\$ 19,876
Work in process	11,015	2,204
Finished goods	1,045	458
Total	<u>\$ 34,421</u>	<u>\$ 22,538</u>

(5) Property, Plant and Equipment, Net

Property, plant and equipment consist of the following:

	September 30, 2023	December 31, 2022	Useful life
	(In thousands)		
Construction in progress	\$ 299,942	\$ 209,056	—
Buildings	24,016	24,016	30 years
Machinery and equipment	166,060	136,607	3-10 years
Computer equipment and software	10,890	10,239	3 years
Leasehold improvements	23,171	9,226	Shorter of useful life or lease term
Total	524,079	389,144	
Accumulated depreciation	(139,053)	(129,921)	
Property, plant and equipment, net	<u>\$ 385,026</u>	<u>\$ 259,223</u>	

Depreciation expense was \$10.8 million and \$6.7 million for the nine months ended September 30, 2023 and 2022, respectively.

Construction in progress totaled \$299.9 million and \$209.1 million at September 30, 2023 and December 31, 2022, respectively. The balance at September 30, 2023 and December 31, 2022 included engineering designs and construction costs totaling \$255.2 million and \$164.5 million, respectively, for a planned aerogel manufacturing facility in Bulloch County, Georgia. Capitalized interest totaled \$8.8 million and \$2.7 million at September 30, 2023 and December 31, 2022, respectively.

(6) Accrued Expenses

Accrued expenses consist of the following:

	September 30, 2023	December 31, 2022
	(In thousands)	
Employee compensation	\$ 14,001	\$ 12,467
Other accrued expenses	2,693	3,536
Total	<u>\$ 16,694</u>	<u>\$ 16,003</u>

(7) Related Party Transactions

Convertible Note

During the year ended December 31, 2022, the Company issued a \$100.0 million aggregate principal amount convertible note to Wood River Capital, LLC, an entity affiliated with Koch Disruptive Technologies, LLC (the 2022 Convertible Note). Refer to note 8 for more information.

During the nine months ended September 30, 2023, the Company incurred \$7.8 million of interest from the 2022 Convertible Note, and capitalized \$6.1 million as part of the construction in progress for the planned manufacturing facility in Bulloch County, Georgia.

Other

During the nine months ended September 30, 2023, the Company recorded costs of \$8.6 million as a component of construction in progress, in connection with the planned aerogel manufacturing facility in Bulloch County, Georgia in fees from an entity affiliated with Koch Disruptive Technologies, LLC for project management service. The Company had \$2.8 million in accounts payable as of September 30, 2023 due to the entity affiliated with Koch Disruptive Technologies, LLC.

(8) Convertible Note – Related Party

2022 Convertible Note

On February 15, 2022, the Company entered into a note purchase agreement (the Note Purchase Agreement) with Wood River Capital LLC, an entity affiliated with Koch Disruptive Technologies, LLC (Koch), relating to the issuance and sale to Koch of the 2022 Convertible Note in the aggregate principal amount of \$100.0 million. The transactions contemplated by the Note Purchase Agreement closed on February 18, 2022 (the Issue Date). The maturity date of the 2022 Convertible Note is February 18, 2027, subject to earlier conversion, redemption, or repurchase.

The 2022 Convertible Note is a senior unsecured obligation of the Company and ranks equal in right of payment to all senior unsecured indebtedness of the Company, and will rank senior in right of payment to any indebtedness that is contractually subordinated to the 2022 Convertible Note.

In accordance with ASU 2020-06, the 2022 Convertible Note is accounted for as a single unit of account and consists of the following:

	September 30, 2023
	(In thousands)
Convertible note, principal	\$ 100,000
Payment in-kind	12,952
Accrued interest	2,683
Discount on convertible note, net of accumulated amortization	(3,422)
Debt issuance costs, net of accumulated amortization	(125)
Convertible note	\$ 112,088

In general, fair values determined by Level 1 inputs utilize observable inputs such as quoted prices in active markets for identical assets or liabilities. Fair values determined by Level 2 inputs utilize data points that are either directly or indirectly observable, such as quoted prices for similar instruments in active markets, interest rates and yield curves. Fair values determined by Level 3 inputs utilize unobservable data points in which there is little or no market data, which require the Company to develop its own assumptions for the asset or liability. The 2022 Convertible Note does not have current observable inputs such as recent trading prices (Level 3) and is measured at fair value using a combination of option pricing and discounted cash flow models and incorporate management's assumptions for stock price, volatility and risk rate.

The Company estimated the fair value of the 2022 Convertible Notes is approximately \$97.5 million as of September 30, 2023. However, as the Company has not elected to utilize the fair value option, it is carried at amortized cost of \$112.1 million.

Contractual Interest Rates

The 2022 Convertible Note was issued at par and bears interest at the Secured Overnight Financing Rate (SOFR) plus 5.50% per annum if interest is paid in cash, or, if interest is paid in-kind as an increase in the principal amount of the outstanding note, at the SOFR plus 6.50% per annum. Under the terms of the 2022 Convertible Note, SOFR has a floor of 1% and a cap of 3%. Interest on the 2022 Convertible Note is payable semi-annually in arrears on June 30 and December 30. The Company, at its option, is permitted to settle each semi-annual interest payment in cash, in-kind, or any combination thereof. It is expected that the Notes will mature on February 18, 2027, subject to earlier conversion, redemption or repurchase.

The Company elected to repay the contractual interest due on June 30, 2022, December 30, 2022 and June 30, 2023 in-kind as an increase to the principal amount of \$2.9 million, \$4.9 million, and \$5.1 million, respectively. The contractual interest attributable to the 2022 Convertible Note was recorded as an addition to the convertible note – related party balance on the condensed consolidated balance sheets.

Debt issuance costs, net of accumulated amortization is \$0.1 million as of September 30, 2023. The effective interest rate approximated the contract interest rate for the nine months ended September 30, 2023. The Company amortized \$0.7 million of the \$4.1 million discount on the convertible note as of September 30, 2023 utilizing an effective interest rate of 10.7%.

Conversion Rights

On November 28, 2022, the Company entered into an amendment to the 2022 Convertible Note to reduce the initial Conversion Price by \$5.00 per share from \$34.936625 per share to \$29.936625 per share, by increasing the initial Conversion Rate from 28.623257 shares per \$1,000 of Capitalized Principal Amount to 33.400100 shares per \$1,000 of Capitalized Principal Amount under the Convertible Note. Accordingly, the 2022 Convertible Note is convertible at the option of the holder at any time prior to the business day immediately preceding the maturity date at an initial conversion rate of 33.400100 shares of the Company's common stock per \$1,000 of capitalized principal. The effective conversion price is approximately \$29.936625 per share (the Conversion Price). The Conversion Price is subject to adjustment upon the occurrence of certain dilutive events such as stock splits and combinations, stock dividends, mergers and spin-off. As of September 30, 2023, 3,862,221 shares of the Company's common stock were issuable upon conversion of the 2022 Convertible Note. The Company has the right to settle conversions in shares of common stock, cash, or any combination thereof. If the closing price per share of the Company's common stock on the New York Stock Exchange is at least 130% of the Conversion Price for 20 consecutive trading days, the Company may elect to convert the principal and accrued interest owing under the Notes, plus a make-whole amount equal to the sum of the present values of the remaining interest payments that would have otherwise been payable from the date of such conversion, redemption or repurchase, as applicable, through maturity (the Make-Whole Amount), into the Company's common stock at the Conversion Price.

Optional Redemption

The 2022 Convertible Note is redeemable at the Company's option at any time and in the event that the volume weighted average price of the Company's common stock for the 10 trading days immediately preceding the date on which the Company provides the redemption notice has been at least 130% of the Conversion Price then in effect at a redemption price of 100% of the principal amount, plus accrued and unpaid interest (excluding the redemption date), plus the Make-Whole Amount.

Contingent Redemption

Upon the occurrence of certain fundamental changes described in the Indenture (each, a Fundamental Change), the Holder of the Note may require that the Company repurchase all or part of the principal amount of the Note at a purchase price of 100% of the principal amount of such Note, plus accrued and unpaid interest to, but excluding, the Fundamental Change repurchase date, plus the Make-Whole Amount. The Indenture includes customary "events of default," which may result in the acceleration of the maturity of the Note.

Embedded Derivatives

The Company determined that the Make-Whole feature of the 2022 Convertible Note requires bifurcation in accordance with Accounting Standards Codification 815, Derivatives and Hedging (ASC 815). Accordingly, the Company must separately account for the feature at fair value with changes in fair value reported in current period earnings. The fair value of the Make-Whole was determined to be immaterial as of February 18, 2022 and September 30, 2023.

(9) Commitments and Contingencies

Cloud Computing Agreement

The Company is party to a cloud computing agreement that is a service contract for enterprise resource planning software. During the year ended December 31, 2022, the Company amended the agreement to a new five-year term. As of September 30, 2023, the Company had \$1.5 million of amortized costs related to implementation of the agreement that began to amortize during 2022. The capitalized implementation costs are classified on the consolidated balance sheets as follows:

	September 30, 2023	December 31, 2022
	(In thousands)	
Cloud computing costs included in other current assets	\$ 420	\$ 420
Cloud computing costs included in other assets	1,590	1,590
Amortization of cloud computing costs	(556)	(242)
Total capitalized cloud computing costs	<u>\$ 1,454</u>	<u>\$ 1,768</u>

Thermal Barrier Contracts

The Company is party to production contracts with General Motors to supply fabricated, multi-part thermal barriers (Barriers) for use in the battery system of its next-generation electric vehicles (Contracts). Pursuant to the Contracts, the Company is obligated to supply Barriers at fixed annual prices and at volumes to be specified by General Motors up to a daily maximum quantity through the respective terms of the agreements, which expire at various times from 2026 through 2034. While General Motors has agreed to purchase its requirement for Barriers from the Company for locations to be designated from time to time by General Motors, it has no obligation to purchase any minimum quantity of Barriers under the Contracts. In addition, General Motors may terminate the Contracts at any time and for any or no reason. All other terms of the Contracts are generally consistent with General Motors' standard purchase terms, including quality and warranty provisions customary in automotive industry.

Federal, State and Local Environmental Regulations

The Company is subject to federal, state and local environmental laws and regulations. These laws generally provide for control of pollutants released into the environment and require responsible parties to undertake remediation. Penalties may be imposed for noncompliance.

Litigation

The Company is, from time to time, a party to litigation that arises in the normal course of its business operations. See Part II, Item 1 “Legal Proceedings” of this Quarterly Report on Form 10-Q for a description of certain of the Company’s current legal proceedings. The Company is not presently a party to any litigation for which it believes a loss is probable requiring an amount to be accrued or a possible loss contingency requiring disclosure.

(10) Leases

The Company leases office, laboratory, warehouse and fabrication space in Massachusetts, Rhode Island and Monterrey, Mexico under operating leases. Under these agreements, the Company is obligated to pay annual rent, real estate taxes, and certain other operating expenses. The Company also leases equipment under operating leases. The Company’s operating leases expire at various dates through 2034.

The Company determines if an arrangement is a lease at inception. Right-of-use (ROU) assets represent the Company’s right to use an underlying asset for the lease term and lease liabilities represent the Company’s payment obligations under the lease. Operating lease ROU assets and liabilities are recognized based on the present value of lease payments over the lease term. To measure its lease liabilities, the Company uses its incremental borrowing rate or the rate implicit in the lease, if available. The Company calculates its incremental borrowing rate using a synthetic credit rating analysis based on Moody’s Building Materials Industry Rating Methodology. ROU assets also include any direct costs and prepaid lease payments but exclude any lease incentives received. The Company’s lease terms may include options to extend or terminate the lease when it is reasonably certain that the Company will exercise that option. Lease expense for lease payments is recognized on a straight-line basis over the lease term.

The Company elected the short-term lease recognition exemption for all leases that qualify. For leases that qualify for this exemption, the Company does not recognize ROU assets or lease liabilities. For lease agreements with lease and non-lease components, the Company accounts for each component separately. However, in the case of equipment leases, the Company accounts for lease and non-lease components as a single component.

Maturities of operating lease liabilities as of September 30, 2023 are as follows:

Year	Operating Leases (In thousands)
2023 (excluding the nine months ended September 30, 2023)	\$ 1,254
2024	4,431
2025	4,271
2026	3,958
2027	3,683
Thereafter	23,018
Total lease payments	40,615
Less imputed interest	(16,642)
Total lease liabilities	<u>\$ 23,973</u>

The Company incurred operating lease costs of \$4.2 million and \$2.9 million during the nine months ended September 30, 2023 and 2022, respectively. Cash payments related to operating lease liabilities were \$3.6 million and \$2.6 million during the nine months ended September 30, 2023 and 2022, respectively.

As of September 30, 2023, the weighted average remaining lease term for operating leases was 9.3 years. As of September 30, 2023, the weighted average discount rate for operating leases was 11.9%.

As of September 30, 2023, the Company had no additional operating real estate or equipment leases that would commence during 2023.

(11) Net Loss Per Share

The computation of basic and diluted net loss per share consists of the following:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2023	2022	2023	2022
(In thousands, except share and per share data)				
Numerator:				
Net loss	\$ (13,073)	\$ (29,595)	\$ (45,292)	\$ (73,129)
Denominator:				
Weighted average shares outstanding, basic and diluted	69,317,805	39,533,695	69,243,843	36,047,879
Net loss per share, basic and diluted	\$ (0.19)	\$ (0.75)	\$ (0.65)	\$ (2.03)

Potentially dilutive common shares that were excluded from the computation of diluted net loss per share because they were anti-dilutive consist of the following:

	Three and Nine Months Ended September 30,	
	2023	2022
Common stock options	5,462,015	3,989,342
Restricted common stock units	568,469	259,766
Restricted common stock awards	889,366	856,435
Convertible note, if converted	3,862,221	3,016,319
Total	10,782,071	8,121,862

As the Company incurred a net loss for the three and nine months ended September 30, 2023 and 2022, the potential dilutive shares from common stock options, restricted common stock units, restricted common stock awards, and the convertible note were excluded from the calculation of diluted net loss per share because their effect would have been anti-dilutive for the periods presented. The Company excludes the shares issued in connection with restricted stock awards from the calculation of basic weighted average common shares outstanding until the restrictions lapse.

(12) Income Taxes

The Company incurred net operating losses and recorded a full valuation allowance against net deferred tax assets for all periods presented. Accordingly, the Company has not recorded a provision for federal or state income taxes.

(13) Segment Information

Operating segments are identified as components of an enterprise about which separate, discrete financial information is available for evaluation by the chief operating decision maker in making decisions on how to allocate resources and assess performance. The Company's chief operating decision maker is the Chief Executive Officer. The Company's chief operating decision maker reviews consolidated operating results to make decisions about allocating resources and assessing performance for the entire Company. The Company reports two segments: Energy Industrial and Thermal Barrier. We evaluate segment performance based on the segment profit (loss) before corporate expenses.

Summarized below are the Revenue and Segment Operating Profit for each reporting segment:

	Revenue		Segment Operating Profit (Loss)		Revenue		Segment Operating Profit (Loss)	
	Three Months Ended		Three Months Ended		Nine Months Ended		Nine Months Ended	
	September 30,		September 30,		September 30,		September 30,	
	2023	2022	2023	2022	2023	2022	2023	2022
	(In thousands)		(In thousands)		(In thousands)		(In thousands)	
Energy industrial	\$ 27,912	\$ 24,752	\$ 5,825	\$ 2,063	\$ 97,311	\$ 90,404	\$ 24,246	\$ 11,044
Thermal barrier	32,843	11,954	7,985	(8,422)	57,188	30,349	3,057	(20,402)
Total	<u>\$ 60,755</u>	<u>\$ 36,706</u>	<u>\$ 13,810</u>	<u>\$ (6,359)</u>	<u>\$ 154,499</u>	<u>\$ 120,753</u>	<u>\$ 27,303</u>	<u>\$ (9,358)</u>
Corporate expenses			28,444	21,950			77,889	60,221
Operating loss			(14,634)	(28,309)			(50,586)	(69,579)
Other income (expense), net			1,561	(1,286)			5,294	(3,550)
Net loss			<u>\$ (13,073)</u>	<u>\$ (29,595)</u>			<u>\$ (45,292)</u>	<u>\$ (73,129)</u>

	Total Assets	
	September 30,	December 31,
	2023	2022
	(In thousands)	
Energy industrial	\$ 88,940	\$ 94,415
Thermal barrier	91,860	39,320
Total assets of reportable segments	180,800	133,735
Construction in progress	299,943	209,050
All other corporate assets	124,403	300,631
	<u>\$ 605,146</u>	<u>\$ 643,416</u>

(14) Other Current Assets

The CARES Act provides an employee retention credit (CARES Employee Retention Credit), which is a refundable tax credit against certain employment taxes of up to \$5,000 per employee for eligible employers. The tax credit is equal to 50% of qualified wages paid to employees during a quarter, capped at \$10,000 of qualified wages per employee through December 31, 2020. Additional relief provisions were passed by the United States government, which extend and slightly expand the qualified wage caps on these credits through December 31, 2021. Based on these additional provisions, the tax credit is now equal to 70% of qualified wages paid to employees during a quarter, and the limit on qualified wages per employee has been increased to \$10,000 of qualified wages per quarter. The Company qualified for the tax credit under the CARES Act for qualified wages for the years ended December 31, 2020 and 2021. In September 2023, the Company submitted filings for CARES Employee Retention Credits totaling \$2.2 million that are reported in the accompanying condensed consolidated balance sheet within prepaid expenses and other current assets as of September 30, 2023, and in the accompanying statement of operations for the three and nine months ended September 30, 2023.

(15) Subsequent Events

The Company has evaluated subsequent events through November 2, 2023, the date of issuance of the consolidated financial statements for the three and nine months ended September 30, 2023.

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations.

The following information should be read in conjunction with the unaudited financial information and the notes thereto included in this Quarterly Report on Form 10-Q and the audited financial information and the notes thereto included in the Annual Report on Form 10-K for the year ended December 31, 2022, filed with the U.S. Securities and Exchange Commission (SEC) on March 16, 2023, which we refer to as the Annual Report.

Certain matters discussed in this Quarterly Report on Form 10-Q may be deemed to be forward-looking statements that involve risks and uncertainties. We make such forward-looking statements pursuant to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995 and other federal securities laws. In this Quarterly Report on Form 10-Q, words such as "may," "will," "anticipate," "estimate," "expects," "projects," "intends," "plans," "believes" and similar expressions (as well as other words or expressions referencing future events, conditions or circumstances) are intended to identify forward-looking statements.

Our actual results and the timing of certain events may differ materially from the results discussed, projected, anticipated, or indicated in any forward-looking statements. We caution you that forward-looking statements are not guarantees of future performance and that our actual results of operations, financial condition and liquidity, and the development of the industry in which we operate may differ materially from the forward-looking statements contained in this Quarterly Report on Form 10-Q. In addition, even if our results of operations, financial condition and liquidity, and the development of the industry in which we operate are consistent with the forward-looking statements contained in this Quarterly Report on Form 10-Q, they may not be predictive of results or developments in future periods.

The following information and any forward-looking statements should be considered in light of factors discussed elsewhere in this Quarterly Report on Form 10-Q and under "Risk Factors" in Item 1A of the Annual Report and the Quarterly Reports on Form 10-Q for the quarters ended March 31, 2023 and June 30, 2023.

We caution readers not to place undue reliance on any forward-looking statements made by us, which speak only as of the date they are made. We disclaim any obligation, except as specifically required by law and the rules of the SEC, to publicly update or revise any such statements to reflect any change in our expectations or in events, conditions or circumstances on which any such statements may be based, or that may affect the likelihood that actual results will differ from those set forth in the forward-looking statements.

You should read the following discussion and analysis of financial condition and results of operations together with Part I Item 1 "Financial Statements," which includes our financial statements and related notes, elsewhere in this Quarterly Report on Form 10-Q.

Investors and others should note that we routinely use the Investors section of our website to announce material information to investors and the marketplace. While not all of the information that we post on the Investors section of our website is of a material nature, some information could be deemed to be material. Accordingly, we encourage investors, the media, and others interested in us to review the information that we share on the Investors section of our website, <https://www.aerogel.com>.

Products

Our core businesses are organized into two reportable segments: Energy Industrial and Thermal Barrier. The following describes our key product offerings and new product innovations by reportable segment.

Energy Industrial

We design, develop and manufacture innovative, high-performance aerogel insulation used primarily in the energy industrial and sustainable insulation materials markets. We believe our aerogel blankets deliver the best thermal performance of any widely used insulation product available on the market today and provide a combination of performance attributes unmatched by traditional insulation materials. Our end-user customers select our products where thermal performance is critical and to save money, improve resource efficiency, enhance sustainability, preserve operating assets and protect workers. Our insulation is used by oil producers and the owners and operators of refineries, petrochemical plants, liquefied natural gas facilities, power generating assets and other energy industrial. Our Pyrogel and Cryogel product lines have undergone rigorous technical validation by industry leading end-users and achieved significant market adoption. Our Spaceloft sustainable insulation materials are increasingly used by building owners to improve the energy efficiency and to enhance fire protection in buildings ranging from historic brownstones to modern high rises.

We also derive revenue from a number of other end markets. Customers in these markets use our products for applications as diverse as military and commercial aircraft, trains, buses, appliances, apparel, footwear and outdoor gear. As we continue to enhance

our Aerogel Technology Platform, we believe we will have additional opportunities to address high-value applications in the global insulation market, the electric vehicle market and in a number of new, high-value markets, including hydrogen energy, filtration, water purification, and gas sorption.

We market and sell our products primarily through a sales force based in North America, Europe and Asia. The efforts of our sales force are supported by a small number of sales consultants with extensive knowledge of a particular market or region. Our sales force is responsible for establishing and maintaining customer and partner relationships, delivering highly technical information and ensuring high-quality customer service.

Our salespeople work directly with end-user customers and engineering firms to promote the qualification, specification and acceptance of our aerogel and thermal barrier products. We also rely on an existing and well-established channel of qualified insulation distributors and contractors in more than 50 countries around the world to ensure rapid delivery of our aerogel products and strong end-user support.

Thermal Barrier

We are also actively developing a number of promising aerogel products and technologies for the electric vehicle market. We have developed and are commercializing our proprietary line of PyroThin aerogel thermal barriers for use in battery packs in electric vehicles. Our PyroThin product is an ultra-thin, lightweight and flexible thermal barrier designed with other functional layers to impede the propagation of thermal runaway across multiple lithium-ion battery system architectures. Our thermal barrier technology is designed to offer a unique combination of thermal management, mechanical performance and fire protection properties. These properties enable electric vehicle manufacturers to achieve critical battery performance and safety goals. In addition, we are seeking to leverage our patented carbon aerogel technology to develop industry-leading battery materials for use in lithium-ion battery cells. These battery materials have the potential to increase the energy density of the battery cells, thus enabling an increase in the driving range of electric vehicles.

The commercial potential for our PyroThin thermal barriers and our carbon aerogel battery materials in the electric vehicle market is significant. Accordingly, we are hiring additional personnel, incurring additional operating expenses, incurring significant capital expenditures to expand aerogel manufacturing capacity, establishing an automated thermal barrier fabrication operation, enhancing research and development resources and expanding our battery material research facilities, among other items.

We have entered into production contracts with certain major OEMs, including General Motors LLC ("General Motors"), to supply fabricated, multi-part thermal barriers for use in the battery system of its next-generation electric vehicles. Pursuant to the contracts with General Motors, we are obligated to supply the barriers at fixed annual prices and at volumes to be specified by the customer up to a daily maximum quantity through the term of the agreements, which expire at various times from 2026 through 2034. While General Motors has agreed to purchase its requirement for the barriers from us at locations to be designated from time to time, it has no obligation to purchase any minimum quantity of barriers under the contracts. In addition, General Motors may terminate the contracts any time and for any or no reason. All other terms of the contracts are generally consistent with General Motors's standard purchase terms, including quality and warranty provisions customary in the automotive industry.

Manufacturing Operations

We manufacture our products using our proprietary technology at our facility in East Providence, Rhode Island. We have operated the East Providence facility since 2008 and have increased our capacity in phases to approximately \$250.0 million in annual revenue. To meet expected growth in demand for our aerogel products in the electric vehicle market, we have been in the process of expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations as to demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we ramp up construction, we expect to be able to substantially reduce our planned capital expenditures for 2023 and 2024. At the same time, we believe that productivity improvements in our existing Rhode Island facility combined with supply of our energy industrial products from one or more contract manufacturers in China beginning in 2024 will permit us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of the second plant. Nonetheless, there can be no assurance as to when we will ramp up construction on the second plant. There can also be no assurance that our contract manufacturing strategy of meeting the demand of our energy industrial customers with supply from one or more contract manufacturers in China will provide us with adequate manufacturing capacity or supply for that expected demand. Furthermore, when we ramp up construction on the second plant, further cost inflation and/or supply chain disruptions, as well as potential changes in the scope of the facilities, could lead to increases to our prior estimates for completion of the second plant.

Recent Developments

On September 28, 2023 (the “First Amendment Effective Date”), Aspen Aerogels Georgia, LLC, a Georgia limited liability company (the “Borrower”), our wholly-owned subsidiary, entered into a first amendment (the “First Amendment”) to amend that certain loan agreement, dated as of November 28, 2022 (the “Loan Agreement”), by and among (i) the Borrower, (ii) us, as a guarantor and (iii) Aspen Aerogels Rhode Island, LLC, a Rhode Island limited liability company (“Aspen RI” and, together with the Borrower and us, each, a “Loan Party” and collectively, the “Loan Parties”), as a guarantor, and (iv) General Motors Holdings LLC (“GM”), as lender.

Pursuant to the First Amendment, the Loan Parties and GM agreed to, among other things, (i) extend the draw period for the delayed draw senior secured term loans (the “Loan”), from beginning on January 1, 2023 and ending on September 30, 2023, to instead a period beginning on the date that is twelve (12) months prior to the date agreed upon by Borrower and GM for the start of production at an aerogel manufacturing facility in Bulloch County, Georgia (the “Plant”) under the applicable GM Purchase Contracts (as defined therein) and ending on March 31, 2024 (or any later date approved in writing by GM at its sole discretion); (ii) extend the maturity date of the Loan Agreement from March 31, 2025 to September 30, 2025; and (iii) add the following financial covenants measured starting from the fiscal quarter ending December 31, 2024 and at the end of each fiscal quarter thereafter (each such fiscal quarter end, a “Test Date”): (A) a covenant requiring that the Total Leverage Ratio (as defined therein) shall not exceed 5.00:1.00 as of the applicable Test Date and (B) a covenant requiring that the ratio of our total indebtedness and our subsidiaries (with certain exceptions) to our consolidated equity shall not be greater than 1.20x as of the applicable Test Date, in each case, with compliance demonstrated through the submission of a certain compliance certificate (the “Financial Covenants”), subject to customary equity cure rights with respect to the Financial Covenants.

The First Amendment also amended the conditions precedent to funding to (i) require the Borrower to provide evidence that cash proceeds of one or more equity and/or debt financing arrangements of not less than \$500.0 million in the aggregate (inclusive of all equity investments contributed to the Borrower for use in connection with the Plant prior to the First Amendment Effective Date) have been contributed to and disbursed by the Borrower in the manner prescribed in a pre-determined project budget (subject to a permitted variance) to fund the construction of and equipment for the first phase of the Plant and (ii) require that 70% of the total cost in connection with the construction and operation of the Plant has been fully funded prior to such applicable borrowing under the Loan Agreement.

Financial Summary

Our revenue for the nine months ended September 30, 2023 was \$154.5 million, which represented an increase of \$33.7 million, or 28%, from \$120.8 million for the nine months ended September 30, 2022. Net loss for the nine months ended September 30, 2023 was \$45.3 million and net loss per share was \$0.65. Net loss for the nine months ended September 30, 2022 was \$73.1 million and net loss per share was \$2.03.

Key Metrics and Non-GAAP Financial Measures

We regularly review a number of metrics, including the following key metrics, to evaluate our business, measure our performance, identify trends affecting our business, formulate financial projections and make strategic decisions.

Square Foot Operating Metric

We price our energy industrial product and measure our shipments in square feet. We believe the square foot operating metric allows us and our investors to measure our manufacturing capacity and energy industrial product shipments on a uniform and consistent basis. The following chart sets forth energy industrial product shipments in square feet associated with recognized revenue, including revenue recognized over time, for the periods presented:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2023	2022	2023	2022
	(In thousands)			
Product shipments in square feet	5,909	6,711	22,337	24,074

Adjusted EBITDA

We use Adjusted EBITDA, a non-GAAP financial measure, as a means to assess our operating performance. We define Adjusted EBITDA as net income (loss) before interest expense, taxes, depreciation, amortization, stock-based compensation expense and other items, from time to time, which we do not believe are indicative of our core operating performance. Adjusted EBITDA is a supplemental measure of our performance that is not presented in accordance with U.S. GAAP. Adjusted EBITDA should not be considered as an alternative to net income (loss) or any other measure of financial performance calculated and presented in accordance with U.S. GAAP. In addition, our definition and presentation of Adjusted EBITDA may not be comparable to similarly titled measures presented by other companies.

We use Adjusted EBITDA:

- as a measure of operating performance because it does not include the impact of items that we do not consider indicative of our core operating performance;
- for planning purposes, including the preparation of our annual operating budget;
- to allocate resources to enhance the financial performance of our business; and
- as a performance measure used under our bonus plan.

We also believe that the presentation of Adjusted EBITDA provides useful information to investors with respect to our results of operations and in assessing the performance and value of our business. Various measures of EBITDA are widely used by investors to measure a company's operating performance without regard to items that can vary substantially from company to company depending upon financing and accounting methods, book values of assets, capital structures and the methods by which assets were acquired.

Although measures similar to Adjusted EBITDA are frequently used by investors and securities analysts in their evaluation of companies, we understand that Adjusted EBITDA has limitations as an analytical tool, and you should not consider it in isolation or as a substitute for net income (loss), income (loss) from operations, net cash provided by (used in) operating activities or an analysis of our results of operations as reported under U.S. GAAP. Some of these limitations are:

- Adjusted EBITDA does not reflect our historical cash expenditures or future requirements for capital expenditures or other contractual commitments;
- Adjusted EBITDA does not reflect changes in, or cash requirements for, our working capital needs;
- Adjusted EBITDA does not reflect stock-based compensation expense;
- Adjusted EBITDA does not reflect our income tax expense or cash requirements to pay our income taxes;
- Adjusted EBITDA does not reflect our interest expense, or the cash requirements necessary to service interest or principal payments on our debt;
- although depreciation, amortization and impairment charges are non-cash charges, the assets being depreciated, amortized or impaired will often have to be replaced in the future, and Adjusted EBITDA does not reflect any cash requirements for these replacements; and
- other companies in our industry may calculate EBITDA or Adjusted EBITDA differently than we do, limiting their usefulness as a comparative measure.

Because of these limitations, our Adjusted EBITDA should not be considered as a measure of discretionary cash available to us to reinvest in the growth of our business or as a measure of cash available for us to meet our obligations.

To properly and prudently evaluate our business, we encourage you to review the U.S. GAAP financial statements included elsewhere in this Quarterly Report on Form 10-Q, and not to rely on any single financial measure to evaluate our business.

The following table presents a reconciliation of net loss, the most directly comparable U.S. GAAP measure, to Adjusted EBITDA for the periods presented:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2023	2022	2023	2022
	(In thousands)			
Net loss	\$ (13,073)	\$ (29,595)	\$ (45,292)	\$ (73,129)
Depreciation and amortization	4,550	2,531	10,757	6,692
Stock-based compensation ⁽¹⁾	2,789	2,590	7,766	6,713
Other (income) expense	(1,561)	1,286	(5,294)	3,550
Adjusted EBITDA	<u>\$ (7,295)</u>	<u>\$ (23,188)</u>	<u>\$ (32,063)</u>	<u>\$ (56,174)</u>

⁽¹⁾ Represents non-cash stock-based compensation related to vesting and modifications of stock option grants, vesting of restricted stock units and vesting of restricted common stock.

Our financial performance, including such measures as net income (loss), earnings per share and Adjusted EBITDA, are affected by a number of factors including volume and mix of aerogel products sold, average selling prices, our material costs and manufacturing expenses, the costs associated with capacity expansions and start-up of additional production capacity, and the amount and timing of operating expenses. Accordingly, we expect that our net income (loss), earnings per share and Adjusted EBITDA will vary from period to period.

We expect to maintain strong revenue growth during 2023 driven by a continued post-COVID recovery in the energy industrial market, accelerating demand in the electric vehicle market and continued market share gains in the sustainable insulation materials market. Our expectation to maintain strong revenue growth is based, in part, on our OEM customers' production volume forecasts and targets as well as our expectation to successfully scale our manufacturing capabilities and address any potential supply chain issues to meet this expected demand. As a result, we expect to experience a decrease in both net loss and Adjusted EBITDA during 2023.

Components of Our Results of Operations

Revenue

We recognize revenue from the sale of our energy industrial aerogel products and thermal barriers. Revenue is recognized upon the satisfaction of contractual performance obligations.

We record deferred revenue for sales when (i) we have delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

We project revenue growth during 2023 due to accelerating demand in the electric vehicle market and continued market share gains in the sustainable insulation materials market.

Cost of Revenue

Cost of product revenue consists primarily of materials and manufacturing expense. Cost of product revenue is recorded when the related product revenue is recognized.

Material is our most significant component of cost of product revenue and includes fibrous batting, silica materials and additives. Material costs as a percentage of product revenue vary from product to product due to differences in average selling prices, material requirements, product thicknesses, and manufacturing yields. In addition, we provide warranties for our products and record the estimated cost within cost of revenue in the period that the related revenue is recorded or when we become aware that a potential warranty claim is probable and can be reasonably estimated. As a result of these factors, material costs as a percentage of product revenue will vary from period to period due to changes in the mix of aerogel products sold, the costs of our raw materials or the estimated cost of warranties. In addition, global supply chain disturbances, increased reliance on foreign materials procurement, industrial gas supply constraints, increases in the cost of our raw materials, and other factors may significantly impact our material costs and have a material impact on our operations. We expect that material costs will increase in absolute dollars during 2023 due to projected growth in product shipments, but decrease as a percentage of revenue due to projected increases in average selling prices, improved manufacturing, and fabrication yields and a favorable mix of products sold.

Manufacturing expense is also a significant component of cost of revenue. Manufacturing expense includes labor, utilities, maintenance expense, and depreciation on manufacturing assets. Manufacturing expense also includes stock-based compensation of manufacturing employees and shipping costs. We expect that manufacturing expense will increase in absolute dollars and increase as a percentage of revenue during 2023 due to increased staffing and spending levels in support of our thermal barrier business, including the start-up and operation of an automated fabrication facility in Monterrey, Mexico. We are also continuing to monitor the impact of engaging one or more contract manufacturers in China to supply our aerogel products for the energy industrial market beginning in 2024 on our manufacturing expense and cost of product revenue.

In total, we expect that cost of product revenue will increase in absolute dollars during 2023 versus 2022 and decrease as a percentage of revenue versus 2022 driven by the costs to support our expected higher run-rate revenue in future periods.

Gross Profit

Our gross profit as a percentage of revenue is affected by a number of factors, including the volume of products produced and sold, the mix of products sold, average selling prices, our material and manufacturing costs, realized capacity utilization and the costs associated with expansions and start-up of production capacity. Accordingly, we expect our gross profit to vary significantly in absolute dollars and as a percentage of revenue from period to period.

During 2023, we expect gross profit to increase in both absolute dollars and as a percentage of total revenue due to the combination of a projected increase in total revenue combined with projected reduction in material costs as a percentage of total revenue, offset, in part, by a projected increase in manufacturing expense as a percentage of revenue.

In the longer term, we expect gross profit to improve in absolute dollars and as a percentage of revenue due to expected increases in total revenue, production volumes and manufacturing productivity. In addition, we expect the gross profit improvement derived from the increases in revenue, volume and productivity will be supported by the continued implementation of lower cost product formulations and realization of material purchasing efficiencies.

Operating Expenses

Operating expenses consist of research and development, sales and marketing, and general and administrative expenses. Operating expenses include personnel costs, legal fees, professional fees, service fees, insurance premiums, travel expense, facilities related costs and other costs, expenses and fees. The largest component of our operating expenses is personnel costs, consisting of salaries, benefits, incentive compensation and stock-based compensation. In any particular period, the timing and extent of personnel additions or reductions, legal activities, including patent enforcement actions, marketing programs, research efforts and a range of similar activities or actions could materially affect our operating expenses, both in absolute dollars and as a percentage of revenue.

During 2023, we expect to continue to hire additional personnel and incur additional operating expenses to support the anticipated multi-year growth in our PyroThin thermal barrier business. As a result, we expect that operating expenses will increase in absolute dollars, and remain consistent as a percentage of revenue during the year. In the longer term, we expect that operating expenses will increase in absolute dollars, but decrease as a percentage of revenue.

Research and Development Expenses

Research and development expenses consist primarily of expenses for personnel engaged in the development of next generation aerogel compositions, form factors and manufacturing technologies. These expenses also include testing services, prototype expenses, consulting services, trial formulations for new products, equipment depreciation, facilities costs and related overhead. We expense research and development costs as incurred. We expect to continue to devote substantial resources to the development of new aerogel technologies, including our carbon aerogel battery materials. We believe that these investments are necessary to maintain and improve our competitive position. We also expect to continue to invest in research and engineering personnel and the infrastructure required in support of their efforts. We expect our research and development expenses will increase in absolute dollars, while decreasing as a percentage of revenue in 2023 and in the longer term.

Sales and Marketing Expenses

Sales and marketing expenses consist primarily of personnel costs, incentive compensation, marketing programs, travel and related costs, consulting expenses and facilities related costs. We expect our sales and marketing expenses will increase in absolute dollars, while decreasing as a percentage of revenue in 2023 and in the longer term.

General and Administrative Expenses

General and administrative expenses consist primarily of personnel costs, legal expenses, consulting and professional services, audit fees, compliance with securities, corporate governance and related laws and regulations, investor relations expenses and insurance premiums, including director and officer insurance.

We expect our general and administrative expenses to increase as we add general and administrative personnel to support the anticipated growth of our business. We also expect that the patent enforcement actions, described in more detail under “Legal Proceedings” in Part I, Item 3 of our Annual Report on Form 10-K for the year ended December 31, 2022 and “Legal Proceedings” in Part II, Item 1 of this Quarterly Report on Form 10-Q, if protracted, could result in significant legal expense over the medium to long-term. We expect that our general and administrative expenses will increase in absolute dollars but decrease as a percentage of revenue in the longer term. In 2023, we expect such expenses will increase in both absolute dollars and as a percentage of revenue.

Interest Expense, Convertible Note - Related Party

Interest expense, convertible note - related party is net of the capitalized interest related to the \$100.0 million in aggregate initial principal amount of our 2022 Convertible Note and \$13.0 million relating to payment in-kind.

Interest Income (Expense), Net

Interest expense, net consists of interest expense related to our revolving credit facility and interest earned on the cash balances invested in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government.

Income from Employee Retention Credit

Employee retention credit consists of other income related to our submitted filings for CARES Employee Retention Credits.

Provision for Income Taxes

We have incurred net losses since inception and have not recorded benefit provisions for U.S. federal income taxes or state income taxes since the tax benefits of our net losses have been offset by valuation allowances due to the uncertainty associated with the utilization of net operating loss carryforwards.

Results of Operations

Three months ended September 30, 2023 compared to the three months ended September 30, 2022

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Revenue:						
Energy industrial	\$ 27,912	46%	\$ 24,752	67%	\$ 3,160	13%
Thermal barrier	32,843	54%	11,954	33%	20,889	175%
Total revenue	\$ 60,755	100%	\$ 36,706	100%	\$ 24,049	66%

Total revenue increased \$24.0 million, or 66%, to \$60.7 million for the three months ended September 30, 2023 from \$36.7 million in the comparable period in 2022. The increase in total revenue was the result of an increase in both thermal barrier and energy industrial revenue.

The following chart sets forth energy industrial product shipments in square feet associated with recognized revenue, including revenue recognized over time, for the periods presented:

	Three Months Ended September 30,		Change	
	2023	2022	Amount	Percentage
Product shipments in square feet (in thousands)	5,909	6,711	(802)	(12)%

Energy industrial revenue increased by \$3.1 million, or 13%, to \$27.9 million for the three months ended September 30, 2023 from \$24.8 million in the comparable period in 2022. This increase was driven by project-based demand in the subsea market and a more favorable mix of product shipments in the global petrochemical and refinery markets in Latin America, offset, in part, by a decrease in the volume of shipments in the global petrochemical and refinery markets of Asia, North America and Europe.

Energy industrial revenue for the three months ended September 30, 2023 included \$8.0 million to a North American distributor, in comparison to \$10.5 million for the comparable period of 2022.

The average selling price per square foot of our energy industrial products increased by \$1.03, or 28%, to \$4.72 per square foot for the three months ended September 30, 2023 from \$3.69 per square foot for the three months ended September 30, 2022. The increase in average selling price reflected the impact of price increases enacted in 2023 and a change in the mix of products sold, as we strive to maximize capacity in our aerogel manufacturing facility. This increase in average selling price had the effect of increasing product revenue by \$6.1 million for the three months ended September 30, 2023 from the comparable period in 2022.

In volume terms, energy industrial product shipments decreased by 0.8 million square feet, or 12%, to 5.9 million square feet for the three months ended September 30, 2023, as compared to 6.7 million square feet for the three months ended September 30, 2022. The decrease in volume had the effect of decreasing product revenue by \$3.0 million for the three months ended September 30, 2023 from the comparable period in 2022.

Thermal barrier revenue was \$32.8 million for the three months ended September 30, 2023 as compared to \$11.9 million for the three months ended September 30, 2022. During the three months ended September 30, 2023 and 2022, thermal barrier revenue included \$29.5 million and \$9.2 million, respectively, to a major U.S. automotive OEM.

Cost of Revenue

	Three Months Ended September 30,				Change	
	2023		2022		Amount	Percentage
	Amount	Percentage of Related Revenue	Amount	Percentage of Related Revenue	Amount	Percentage
	(\$ in thousands)					
Cost of revenue:						
Energy industrial	\$ 22,087	79%	\$ 22,689	92%	\$ (602)	(3)%
Thermal barrier	24,858	76%	20,376	170%	4,482	22%
Total cost of revenue	\$ 46,945	77%	\$ 43,065	117%	\$ 3,880	9%

Total cost of revenue increased \$3.9 million, or 9%, to \$47.0 million for the three months ended September 30, 2023 from \$43.1 in the comparable period in 2022. The increase in total cost of revenue was the result of an increase in thermal barrier cost of revenue, offset by a decrease in energy industrial cost of revenue.

Energy industrial cost of revenue decreased \$0.6 million, or 3%, to \$22.1 million for the three months ended September 30, 2023 from \$22.7 million in the comparable period in 2022. The \$0.6 million decrease was the result of a \$0.2 million decrease in material costs due to change in the product mix and a \$0.4 million decrease in manufacturing and other operating costs from the comparable period in 2022. The decrease in manufacturing costs was driven by decrease in compensation and related costs of \$0.7 million, depreciation and facility related expenses of \$0.4 million and other manufacturing and operating costs of \$0.2 million, offset by an increase in utilities expenses of \$0.9 million.

Thermal barrier cost of revenue increased \$4.5 million to \$24.9 million for the three months ended September 30, 2023 as compared to \$20.4 million for the three months ended September 30, 2022. The \$4.5 million increase was the result of a \$1.3 million increase in material costs and a \$3.2 million increase in manufacturing costs. The increase in manufacturing costs was driven by a \$1.9 million increase in depreciation and facility costs, \$1.0 million increase in utilities expenses and other manufacturing and operating costs of \$0.6 million, offset by a decrease in compensation and related costs of \$0.3 million.

Gross Profit

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Gross profit:						
Energy industrial	\$ 5,825	21%	\$ 2,063	8%	\$ 3,762	182%
Thermal barrier	7,985	24%	(8,422)	(70)%	16,407	195%
Total gross profit (loss)	\$ 13,810	23%	\$ (6,359)	(17)%	\$ 20,169	317%

Gross profit increased by \$20.1 million, or 317%, to \$13.8 million for the three months ended September 30, 2023 from \$6.3 million of gross loss in the comparable period in 2022. The increase in gross profit was the result of the \$24.0 million increase in total revenue, offset by the \$3.9 million increase in total cost of revenue. The increase in gross profit reflects the increase in revenue, offset, in part by, additional resources to support our expected higher run-rate revenue in future periods for both our energy industrial and thermal barrier products from the comparable period in 2022.

Research and Development Expenses

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Research and development expenses	\$ 4,218	7%	\$ 4,694	13%	\$ (476)	(10)%

Research and development expenses decreased by \$0.5 million, or 10%, to \$4.2 million for the three months ended September 30, 2023 from \$4.7 million in the comparable period in 2022. The \$0.5 million decrease reflects decreases in professional fees of \$0.6 million, compensation and related expenses of \$0.1 million and other research and development expenses of \$0.1 million, offset by an increase in facility related expenditures of \$0.3 million.

Research and development expenses as a percentage of total revenue decreased to 7% of total revenue for the three months ended September 30, 2023 from 13% in the comparable period in 2022.

Sales and Marketing Expenses

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Sales and marketing expenses	\$ 8,386	14%	\$ 7,293	20%	\$ 1,093	15%

Sales and marketing expenses increased by \$1.1 million, or 15%, to \$8.4 million for the three months ended September 30, 2023 from \$7.3 million in the comparable period in 2022. The \$1.1 million increase was principally the result of increases in depreciation and facility related expenses of \$0.9 million and an increase in compensation and related expenses of \$0.4 million, offset by a decrease in professional fees of \$0.2 million.

Sales and marketing expenses as a percentage of total revenue decreased to 14% of total revenue for the three months ended September 30, 2023 from 20% in the comparable period in 2022.

General and Administrative Expenses

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
General and administrative expenses	\$ 15,840	26%	\$ 9,963	27%	\$ 5,877	59%

General and administrative expenses increased by \$5.8 million, or 59%, to \$15.8 million for the three months ended September 30, 2023 from \$10.0 million in the comparable period in 2022. The \$5.8 million increase was the result of additional staffing combined with increases in compensation and related costs of \$5.2 million and professional services expenses of \$1.0 million, and other general and administrative expenses of \$0.1 million, partially offset by a decrease in facility related expenditures of \$0.5 million.

General and administrative expenses as a percentage of total revenue decreased to 26% for the three months ended September 30, 2023 from 27% in the comparable period in 2022.

Other Income (Expense), net

	Three Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Other income (expense):						
Interest (expense), related party	\$ (1,938)	(3)%	\$ (1,734)	(5)%	\$ (204)	12%
Interest income, net	1,313	2%	448	1%	865	193%
Income from Employee Retention Credits	2,186	4%	—	—	2,186	NM
Total other income (expense), net	\$ 1,561	3%	\$ (1,286)	(4)%	\$ 2,847	(221)%

Other income (expense), net increased by \$2.8 million to \$1.5 million of other income for the three months ended September 30, 2023 from \$1.3 million of other expense in the comparable period in 2022. The \$2.8 million increase was the result of \$2.2 million of Employee Retention Credits, \$0.8 million of interest income and a \$0.2 million net impact of capitalized interest relating to our Convertible Note in the comparable period in 2022.

Results of Operations

Nine months ended September 30, 2023 compared to the nine months ended September 30, 2022

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Revenue:						
Energy industrial	\$ 97,311	63%	\$ 90,404	75%	\$ 6,907	8%
Thermal barrier	57,188	37%	30,349	25%	26,839	88%
Total revenue	\$ 154,499	100%	\$ 120,753	100%	\$ 33,746	28%

Total revenue increased \$33.7 million, or 28%, to \$154.5 million for the nine months ended September 30, 2023 from \$120.8 million in the comparable period in 2022. The increase in total revenue was the result of an increase in both thermal barrier and energy industrial revenue.

The following chart sets forth energy industrial product shipments in square feet associated with recognized revenue, including revenue recognized over time, for the periods presented:

	Nine Months Ended September 30,		Change	
	2023	2022	Amount	Percentage
Product shipments in square feet (in thousands)	22,337	24,074	(1,737)	(7)%

Energy industrial revenue increased by \$6.9 million, or 8%, to \$97.3 million for the nine months ended September 30, 2023 from \$90.4 million in the comparable period in 2022. This increase was driven by project-based demand in the subsea market and a more favorable mix of product shipments in the global petrochemical and refinery markets in Latin America and Asia, offset, in part, by a decrease in the volume of shipments in the global petrochemical and refinery markets of North America and Europe.

Energy industrial revenue for the nine months ended September 30, 2023 included \$27.6 million to a North American distributor, in comparison to \$30.7 million for the comparable period of 2022.

The average selling price per square foot of our energy industrial products increased by \$0.60, or 16%, to \$4.36 per square foot for the nine months ended September 30, 2023 from \$3.76 per square foot for the nine months ended September 30, 2022. The increase in average selling price reflected the impact of price increases enacted in 2023 and a change in the mix of products sold, as we strive to maximize capacity in our aerogel manufacturing facility. This increase in average selling price had the effect of increasing product revenue by \$13.4 million for the nine months ended September 30, 2023 from the comparable period in 2022.

In volume terms, energy industrial product shipments decreased by 1.7 million square feet, or 7%, to 22.3 million square feet for the nine months ended September 30, 2023, as compared to 24.0 million square feet for the nine months ended September 30, 2022. The decrease in volume had the effect of decreasing product revenue by \$6.5 million the nine months ended September 30, 2023 from the comparable period in 2022.

Thermal barrier revenue was \$57.2 million for the nine months ended September 30, 2023 as compared to \$30.3 million for the nine months ended September 30, 2022. During the nine months ended September 30, 2023 and 2022, thermal barrier revenue included \$49.0 million and \$22.8 million to a major U.S. automotive OEM, respectively.

Cost of Revenue

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Related Revenue	Amount	Percentage of Related Revenue	Amount	Percentage
Cost of revenue:						
Energy industrial	\$ 73,065	75%	\$ 79,360	88%	\$ (6,295)	(8)%
Thermal barrier	54,131	95%	50,751	167%	3,380	7%
Total cost of revenue	\$ 127,196	82%	\$ 130,111	108%	\$ (2,915)	(2)%

Total cost of revenue decreased \$2.9 million, or 2%, to \$127.2 million for the nine months ended September 30, 2023 from \$130.1 in the comparable period in 2022. The decrease in total cost of revenue was the result of a decrease in energy industrial cost of revenue, offset by an increase in thermal barrier cost of revenue.

Energy industrial cost of revenue decreased \$6.3 million, or 8%, to \$73.1 million for the nine months ended September 30, 2023 from \$79.4 million in the comparable period in 2022. The \$6.3 million decrease was the result of a \$6.9 million decrease in material costs due to change in the product mix, offset by a \$0.6 million increase in manufacturing and other operating costs from the comparable period in 2022. The increase in manufacturing costs was driven by an increase in utilities expenses of \$0.7 million, offset by a decrease in compensation and related costs of \$0.1 million.

Thermal barrier cost of revenue increased \$3.4 million to \$54.1 million for the nine months ended September 30, 2023 as compared to \$50.7 million for the nine months ended September 30, 2022. The \$3.4 million increase was the result of a \$6.3 million increase in manufacturing costs, offset by a \$2.9 million decrease in material costs. The increase in manufacturing costs was driven by increases in depreciation and facility costs of \$5.3 million and utilities expenses of \$2.6 million and other manufacturing and operating costs of \$1.0 million, offset by a decrease in compensation and related costs of \$2.6 million.

Gross Profit

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Gross profit:						
Energy industrial	\$ 24,246	25%	\$ 11,044	12%	\$ 13,202	120%
Thermal barrier	3,057	5%	(20,402)	(67)%	23,459	115%
Total gross profit (loss)	\$ 27,303	18%	\$ (9,358)	(8)%	\$ 36,661	392%

Gross profit increased by \$36.6 million, or 392%, to \$27.3 million for the nine months ended September 30, 2023 from \$9.3 million of gross loss in the comparable period in 2022. The increase in gross profit was primarily the result of the \$33.7 million

increase in total revenue and the \$2.9 million decrease in total cost of revenue. The increase in gross profit reflects the decrease in costs, offset, in part by, additional resources to support our expected higher run-rate revenue in future periods for both our energy industrial and thermal barrier products from the comparable period in 2022.

Research and Development Expenses

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Research and development expenses	\$ 12,281	8%	\$ 12,733	11%	\$ (452)	(4)%

Research and development expenses decreased by \$0.4 million, or 4%, to \$12.3 million for the nine months ended September 30, 2023 from \$12.7 million in the comparable period in 2022. The \$0.4 million decrease reflects a decrease in professional fees of \$1.4 million, compensation and related costs of \$0.2 million and other research and development expenses of \$0.4 million, offset by an increase in depreciation and facility related expenses of \$1.6 million.

Research and development expenses as a percentage of total revenue decreased to 8% of total revenue for the nine months ended September 30, 2023 from 11% in the comparable period in 2022.

Sales and Marketing Expenses

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Sales and marketing expenses	\$ 24,226	16%	\$ 20,944	17%	\$ 3,282	16%

Sales and marketing expenses increased by \$3.3 million, or 16%, to \$24.2 million for the nine months ended September 30, 2023 from \$20.9 million in the comparable period in 2022. The \$3.3 million increase was principally the result of increases in compensation and related costs of \$1.9 million, facility related expenditures of \$1.2 million and other sales and marketing expenses of \$0.2 million.

Sales and marketing expenses as a percentage of total revenue decreased to 16% for the nine months ended September 30, 2023 from 17% in the comparable period in 2022.

General and Administrative Expenses

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
General and administrative expenses	\$ 41,382	27%	\$ 26,544	22%	\$ 14,838	56%

General and administrative expenses increased by \$14.8 million, or 56%, to \$41.3 million for the nine months ended September 30, 2023 from \$26.5 million in the comparable period in 2022. The \$14.8 million increase was the result of additional staffing combined with increases in compensation and related costs of \$12.6 million and professional services expenses of \$2.7 million, offset by a \$0.5 million decrease in other general and administrative expenses.

General and administrative expenses as a percentage of total revenue increased to 27% for the nine months ended September 30, 2023 from 22% in the comparable period in 2022.

Other Income (Expense), net

	Nine Months Ended September 30,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Other income (expense):						
Interest (expense), related party	\$ (2,424)	(2)%	\$ (4,103)	(3)%	\$ 1,679	(41)%
Interest income, net	5,532	4%	553	—	4,979	NM
Income from Employee Retention Credits	2,186	1%	—	—	2,186	NM
Total other income (expense), net	\$ 5,294	3%	\$ (3,550)	(3)%	\$ 8,844	NM

Other income (expense), net increased by \$8.8 million to \$5.3 million of other income for the nine months ended September 30, 2023 from \$3.5 million of other expense in the comparable period in 2022. The \$8.8 million increase was the result of \$5.0 million of interest income, \$2.2 million of Employee Retention Credits and a \$1.6 million net impact of capitalized interest relating to our Convertible Note in the comparable period in 2022.

Liquidity and Capital Resources

Overview

We have experienced significant losses and invested substantial resources since our inception to develop, commercialize and protect our aerogel technology and to build a manufacturing infrastructure capable of supplying aerogel products at the volumes and costs required by our customers. These investments have included research and development and other operating expenses, capital expenditures and investment in working capital balances.

Our long-term financial projections anticipate revenue growth, increasing levels of gross profit, and improved cash flows from operations. To meet expected growth in demand for our aerogel products in the electric vehicle market, we have been in the process of expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations as to demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we ramp up construction, we expect to be able to substantially reduce our planned capital expenditures for 2023 and 2024. At the same time, we believe that productivity improvements in our existing Rhode Island facility combined with supply of our energy industrial products from one or more contract manufacturers in China beginning in 2024 will permit us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of the second plant. Nonetheless, there can be no assurance as to when we will ramp up construction on the second plant. There can also be no assurance that our contract manufacturing strategy of meeting the demand of our energy industrial customers with supply from one or more contract manufacturers in China will provide us with adequate manufacturing capacity or supply for that expected demand. Furthermore, when we ramp up construction on the second plant, further cost inflation and/or supply chain disruptions, as well as potential changes in the scope of the facilities, could lead to increases to our prior estimates for completion of the second plant.

We are also increasing our investment in the research and development of next-generation aerogel products and technologies. During 2023, we will continue to develop aerogel products and technologies for the electric vehicle market. We believe the commercial potential for our technology in the electric vehicle market is significant. To meet the anticipated revenue growth and take advantage of this market opportunity, we are adding personnel, incurring additional operating expenses, and planning to construct a carbon aerogel battery materials facility, among other items.

In February 2022, we sold and issued to an affiliate of Koch \$100.0 million in aggregate principal amount of our 2022 Convertible Note. In addition, in March 2022, pursuant to a securities purchase agreement dated February 15, 2022, we sold to an affiliate of Koch 1,791,986 shares of our common stock, at a price of \$27.902 per share, for net proceeds of \$49.9 million after deducting fees and offering expenses of \$0.1 million.

We believe that our September 30, 2023 cash and cash equivalents balance of \$94.6 million will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunities in the electric vehicle market and other strategic business opportunities.

However, we plan to supplement our cash balance with equity financings, debt financings, equipment leasing, sale-leaseback transactions, customer prepayments, or government grant and loan programs to provide the additional capital necessary to purchase the capital equipment, construct the new facilities, establish the operations and complete the aerogel capacity expansions required to support our evolving commercial opportunities and strategic business initiatives. We also intend to enter into a new revolving credit facility. We believe that the consummation of equity financings could potentially result in an ownership change under Section 382 of the Internal Revenue Code. Such an ownership change would lead to the use of our net operating loss carryforwards being restricted. Our inability to use a substantial portion of our net operating loss carryforwards would result in a higher effective tax rate and adversely affect our financial condition and results of operations.

Primary Sources of Liquidity

Our principal sources of liquidity are currently our cash and cash equivalents. Cash and cash equivalents consist primarily of cash, money market accounts, and sweep accounts on deposit with banks. As of September 30, 2023, we had \$94.6 million of unrestricted cash and cash equivalents.

In February 2022, we sold and issued to an affiliate of Koch \$100.0 million in aggregate principal amount of our 2022 Convertible Note. In addition, in March 2022, pursuant to a securities purchase agreement dated February 15, 2022, we sold to an affiliate of Koch 1,791,986 shares of our common stock, at a price of \$27.902 per share, for net proceeds of \$49.9 million after deducting fees and offering expenses of \$0.1 million.

Analysis of Cash Flow

Net Cash Used in Operating Activities

During the nine months ended September 30, 2023, we used \$39.8 million in net cash in operating activities, as compared to the use of \$70.3 million in net cash during the comparable period in 2022, a decrease in the use of cash of \$30.5 million. This decrease in use of cash was the result of lower net loss adjusted for non-cash items of \$31.4 million offset by net cash used by changes in operating assets and liabilities of \$0.9 million.

During the nine months ended September 30, 2022, we used \$70.3 million in net cash in operating activities, as compared to the use of \$6.6 million in net cash during the comparable period in 2021, an increase in the use of cash of \$63.7 million. This increase in use of cash was the result of increases in net loss adjusted for non-cash items of \$40.6 million and in net cash used by changes in operating assets and liabilities of \$23.1 million.

Net Cash Used in Investing Activities

Net cash used in investing activities is for capital expenditures for machinery and equipment principally to improve the throughput, efficiency and capacity of our East Providence facility and engineering designs and construction costs for the planned aerogel manufacturing facility in Bulloch County, Georgia. Net cash used in investing activities for the nine months ended September 30, 2023 and 2022 was \$147.7 million and \$119.3 million, respectively.

Net Cash Provided by Financing Activities

Net cash used in financing activities for the nine months ended September 30, 2023 totaled \$0.1 million and consisted of \$0.4 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units and \$0.1 million issuance costs from private placement of common stock, offset, in part, by \$0.4 million in proceeds from employee stock option exercises.

Net cash provided by financing activities for the nine months ended September 30, 2022 totaled \$215.5 million and consisted of \$99.8 million in net proceeds from the issuance of convertible debt, \$49.9 million in net proceeds from the private placement of our common stock, \$72.7 million in net proceeds from the ATM offering program, and less than \$0.2 million in proceeds from employee stock option exercises, offset, in part, by \$4.7 million in cash used for repayments of a prepayment liability and \$2.4 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units.

Contractual Obligations and Commitments

There have been no material changes to our contractual obligations and commitments as reported in our Annual Report.

Recent Accounting Pronouncements

Information regarding new accounting pronouncements is included in note 2 to our unaudited consolidated financial statements contained in Item 1 of this Quarterly Report on Form 10-Q.

Critical Accounting Policies and Estimates

Our financial statements are prepared in accordance with U.S. GAAP. The preparation of our financial statements and related disclosures requires us to make estimates, assumptions and judgments that affect the reported amount of assets, liabilities, revenue, costs and expenses and related disclosures. We believe that the estimates, assumptions and judgments involved in these accounting policies have the greatest potential impact on our financial statements and, therefore, we consider these to be our critical accounting policies. Accordingly, we evaluate our estimates and assumptions on an ongoing basis. Our actual results may differ from these estimates under different assumptions and conditions. See our Annual Report and note 2 to our consolidated financial statements included elsewhere in this Quarterly Report on Form 10-Q for information about these critical accounting policies, as well as a description of our other significant accounting policies.

Certain Factors That May Affect Future Results of Operations

The SEC encourages companies to disclose forward-looking information so that investors can better understand a company's future prospects and make informed investment decisions. This Quarterly Report on Form 10-Q contains such "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995. These statements involve known and unknown risks, uncertainties and other important factors, which may cause our actual results, performance or achievements to be materially different from any future results, performances or achievements expressed or implied by the forward-looking statements. Forward-looking statements include, but are not limited to, statements about: the expected future growth of the market for our aerogel products and our continued gain in market share, in particular in the electric vehicle market, the energy infrastructure insulation market, the lithium-ion battery thermal barrier markets, and other markets we target; our beliefs in the appropriateness of our assumptions, the accuracy of our estimates regarding expenses, loss contingencies, future revenues, revenue capacity, future profits, uses of cash, available credit, capital requirements, and the need for additional financing to operate our business and for capital expenditures and to fund our planned strategic business initiatives; the performance of our aerogel blankets; our expectation that we will be successful in obtaining, enforcing and defending our patents against competitors and that such patents are valid and enforceable; our expectations regarding the investment to open a second manufacturing facility in Georgia, the extended construction and commissioning timeframe for the planned second manufacturing facility, our efforts to manage the construction of the second plant to align with our expectations of demand from EV customers; our estimates of annual production capacity; beliefs about the commercial potential for our technology in the electric vehicle market; beliefs about our ability to produce and deliver products to electric vehicle customers; beliefs about Aspen's contracts with the major automotive manufacturers; our expectations about the size and timing of awarded business in the electric vehicle market, future revenues and profit margins, arising from our supply relationship and contract with automotive OEMs and our ability to win more business and increase revenue in the electric vehicle market; beliefs about the performance of our thermal barrier products in the battery systems of electric vehicles; the current or future trends in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, electric vehicle thermal barrier, electric vehicle battery materials or other markets and the impact of these trends on our business; our investments in the electric vehicle market and aerogel technology platform; our beliefs about the financial metrics that are indicative of our core performance; our expectations about the effect of manufacturing capacity on financial metrics such as Adjusted EBITDA; our expectations about future revenues, expenses, gross profit, net loss, loss per share and Adjusted EBITDA, sources and uses of cash, capital requirements and the sufficiency of our existing cash balance and available credit; our beliefs about the outcome, effects or estimated costs of current or potential litigation or their respective timing, including expected legal expense in connection with our patent enforcement actions; our expectations about future material costs and manufacturing expenses as a percentage of revenue, including the impact of engaging one or more contract manufacturers in China for supply of our energy industrial products; our expectation about the ability of the Chinese contract manufacturers that we engage to consistently supply the aerogel product that we order in a timely manner; our expectations of future gross profit and the effect of manufacturing expenses, manufacturing capacity and productivity on gross profit; our expectations about our resources and other investments in new technology and related research and development activities and associated expenses; our expectations about short and long term (a) research and development (b) general and administrative and (c) sales and marketing expenses; our expectations of revenue growth, increased gross profit, and improving cash flows over the long term; our intentions about managing capital expenditures and working capital balances; and our expectations about potential sources of future financing.

Words such as “may,” “will,” “anticipate,” “estimate,” “expects,” “projects,” “intends,” “plans,” “believes” and words and terms of similar substance used in connection with any discussion of future operating or financial performance, identify forward-looking statements. All forward-looking statements are management’s present expectations of future events and are subject to a number of risks and uncertainties that could cause actual results to differ materially and adversely from those described in the forward-looking statements. These risks include, but are not limited to, those set forth in this Quarterly Report on Form 10-Q and under the heading “Risk Factors” contained in Item 1A of our Annual Report.

In light of these assumptions, risks and uncertainties, the results and events discussed in the forward-looking statements contained in this Quarterly Report on Form 10-Q might not occur. Stockholders and other readers are cautioned not to place undue reliance on the forward-looking statements, which speak only as of the date of this Quarterly Report on Form 10-Q. We are not under any obligation, and we expressly disclaim any obligation, to update or alter any forward-looking statements, whether as a result of new information, future events or otherwise. All subsequent forward-looking statements attributable to Aspen Aerogels, Inc. or to any person acting on its behalf are expressly qualified in their entirety by the cautionary statements contained or referred to in this section.

Item 3. Quantitative and Qualitative Disclosures About Market Risk.

Market risk represents the risk of loss that may impact our financial position due to adverse changes in financial market prices and rates. Our market risk exposure results primarily from fluctuations in interest rates, as well as from inflation. In the normal course of business, we are exposed to market risks, including changes in interest rates which affect our cash flows. We may also face additional exchange rate risk in the future as we expand our business internationally.

Interest Rate Risk

We are exposed to changes in interest rates in the normal course of our business. As of September 30, 2023, we had unrestricted cash and cash equivalents of \$94.6 million. These amounts were held for working capital and capital expansion purposes and were invested primarily in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government via cash sweep accounts primarily at major financial institutions in North America. Due to the short-term nature of these investments, we believe that our exposure to changes in the fair value of our cash as a result of changes in interest rates is not material.

As of September 30, 2023, we had a convertible note outstanding with principal balance of \$113.0 million. Our convertible note bears interest at the Secured Overnight Financing Rate (SOFR) plus 5.50% per annum if interest is paid in cash, or, if interest is paid in-kind as an increase in the principal amount of the outstanding note, at the SOFR plus 6.50% per annum. Under the terms of the investment, SOFR has a floor of 1% and a cap of 3%. Interest is paid semi-annually in arrears on June 30 and December 30. We, at our option, are permitted to settle each semi-annual interest payment in cash, in-kind, or any combination thereof.

As of September 30, 2023, we had \$0.3 million of restricted cash to support our outstanding letters of credit to secure obligations under certain commercial contracts and other obligations. We terminated our revolving credit facility agreement on November 28, 2022.

Inflation Risk

Although we expect that our operating results will be influenced by general economic conditions, we do not believe that inflation has had a material effect on our results of operations during the periods presented in this report. However, our business may be affected by inflation in the future.

Foreign Currency Exchange Risk

We are subject to inherent risks attributed to operating in a global economy. Principally all our revenue, receivables, purchases and debts are denominated in U.S. dollars.

Item 4. Controls and Procedures.

(a) Evaluation of Disclosure Controls and Procedures.

We maintain disclosure controls and procedures that are designed to ensure that information required to be disclosed in the reports that we file or submit under the Securities Exchange Act of 1934, as amended (the Exchange Act), is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

As of September 30, 2023, our management, with the participation of our principal executive officer and principal financial officer, evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act). Our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives, and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on such evaluation, our principal executive officer and principal financial officer have concluded that, as of September 30, 2023, our disclosure controls and procedures were effective to ensure that information required to be disclosed by us in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

(b) *Changes in Internal Controls.*

During the nine months ended September 30, 2023, there were no changes in our internal control over financial reporting, as such term is defined in Rules 13a-15(f) and 15(d)-15(f) promulgated under the Exchange Act, that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

PART II — OTHER INFORMATION

Item 1. Legal Proceedings.

We are involved in various legal claims and proceedings in the normal course of operations. We believe the outcome of these matters will not have a material adverse effect on our consolidated financial position, results of operations or liquidity, except as described in Part 1, Item 3. “Legal Proceedings” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our legal proceedings from those disclosed therein, other than as noted below.

Our patent infringement proceedings in Italy against of AMA S.p.A. and AMA Composites S.r.l. (collectively, AMA) are ongoing. In July 2023, the technical experts appointed by the judge issued a report finding key claims of our process patents valid and infringed by the aerogel products manufactured by Nano Tech Co., Ltd. and sold by AMA.

Our patent infringement proceedings in Korea against Beerenberg Services AS, Beerenberg Korea Ltd., and Bronx (China) Co., Ltd. (collectively, “Beerenberg”) at the Seoul District Court and Korea Trade Commission (KTC) are ongoing. Beerenberg Korea Ltd. and Bronx (China) Co., Ltd. have confirmed in their answers to the KTC investigation that the accused infringing products are manufactured in China by Bronx (China) Co., Ltd. and imported into Korea by Beerenberg Korea Ltd.

In August and September 2023, LG Chem Ltd. filed oppositions at the Korean Intellectual Property Trial and Appeal Board and at the Japanese Patent Office against one of the Korean patents we are asserting against Beerenberg in Korea and a Japanese counterpart of the Korean patents. We intend to vigorously defend the validity of these patents.

In October 2022, we were served with a summons from Aerogels Poland Nanotechnology LLC (“APN”), a former distributor of our products in Poland with whom we previously terminated our distribution agreements because of APN’s failure to pay amounts due to us. The summons asserts causes of action for declaratory judgment, breach of contract, breach of implied contract, equitable estoppel and fraud, and states that plaintiffs will seek declaratory judgment, actual and liquidated damages in the sum of \$20 million, in addition to attorneys’ fees. We were not served with any complaint at the time the summons was served. In December 2022, we filed a notice of appearance in New York County Supreme Court and a demand upon plaintiffs to file and serve a complaint. In March 2023, plaintiffs filed a complaint asserting various causes of action consistent with those set forth in the October 2022 summons, and a demand for monetary damages and other relief in excess of \$16 million. On June 30, 2023, we filed a motion for an order to compel arbitration, seeking dismissal of certain claims, and for attorney’s fees. We intend to vigorously defend this matter.

Item 1A. Risk Factors.

The ownership of our common stock involves a number of risks and uncertainties. When evaluating the Company and our business before making an investment decision regarding our securities, potential investors should carefully consider the risk factors and uncertainties described in Part 1, Item 1A. “Risk Factors” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our risk factors from those disclosed therein, other than as provided below.

We are engaging third-party contract manufacturers in China to supply our energy industrial products beginning in 2024. If such contract manufacturers are unable to manufacture and deliver a sufficient quantity of high-quality products on a timely and cost-efficient basis, our net revenue and business operations may be harmed and our reputation may suffer.

We are engaging one or more contract manufacturers in China for supply of our energy industrial products beginning in 2024, which we believe will enable us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of our planned second manufacturing plant. If our contract manufacturers are unable to deliver the required aerogel product on a timely basis, we may experience delays in delivering our finished aerogel product to customers in the energy industrial market. In addition, because our third party contract manufacturers have manufacturing facilities in China, their ability to provide us with adequate supplies of high-quality products on a timely and cost-efficient basis is subject to a number of additional risks and uncertainties, including political, social and economic instability and other factors that could impact the shipment of supplies. If our manufacturers are unable to provide us with adequate supplies of high-quality aerogel products on a timely and cost-efficient basis, our operations could be disrupted and our revenue and business operations may suffer. Moreover, if our third-party contract manufacturers cannot consistently produce high-quality products that are free of defects, we may experience a loss of customers, which may also reduce our revenues and may harm our reputation and brand. Furthermore, our third-party contract manufacturers may become subject to various supply chain disruptions, including but not limited as a result of COVID-19, other pandemics or public health crises, and geopolitical disputes and conflicts, any of which could slow or halt the delivery of products to us and increase the price of certain materials due to resulting increases in costs of raw materials and shipping costs.

Our potential inability to adequately protect our intellectual property as a result of engaging contract manufacturers in China for the supply of our aerogel products for our customers in the energy industrial market could negatively impact our performance.

In connection with our engagement of contract manufacturers in China, we expect to implement customary manufacturer safeguards onsite, such as the use of confidentiality agreements with employees, to protect our proprietary information and technologies during the manufacturing process of our aerogel products for the energy industrial market. However, these safeguards may not effectively prevent unauthorized use of such information and technical know-how, or prevent the contract manufacturer from retaining them. Although the courts in China are increasing and broadening their protection of intellectual property rights, the legal regime governing intellectual property rights in China is relatively immature and it is often difficult to create and enforce intellectual property rights or protect trade secrets there. We face risks that our proprietary information may not be afforded the same protection in China as it is in countries with well-developed intellectual property laws, and local laws may not provide an adequate remedy in the event of unauthorized disclosure of confidential information. Costly and time-consuming litigation could be necessary to enforce and determine the scope of our proprietary rights in China, and failure to obtain or maintain trade secret protection could adversely affect our competitive business position. In the event that the third-party contract manufacturer of our proprietary aerogel product misappropriates our intellectual property, our business, prospects and financial condition could be materially and adversely affected.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds.

(a) Unregistered Sales of Equity Securities.

None.

(b) Use of Proceeds from Initial Public Offering of Common Stock.

Not applicable.

(c) Purchases of Equity Securities by the Issuer and Affiliated Purchasers.

We did not repurchase any of our equity securities during the quarter ended September 30, 2023.

Item 3. Defaults Upon Senior Securities.

None.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Other Information.

During the fiscal quarter ended September 30, 2023, none of our directors or executive officers adopted, modified or terminated any contract, instruction or written plan for the purchase or sale of our securities that was intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) or any “non-Rule 10b5-1 trading arrangement.”

Item 6. Exhibits.

(a) Exhibits

10.1+	<u>Employment Offer Letter, dated August 23, 2023, by and between the Company and Santhosh Daniel.</u>
10.2	<u>First Amendment, dated September 28, 2023, to the Loan Agreement, dated November 28, 2022, by and among the Company, Aspen Aerogels Georgia, LLC, Aspen Aerogels Rhode Island, LLC and General Motors Holdings LLC.</u>
10.3+	<u>Aspen Aerogels, Inc. Bonus Plan (Amended and Restated Effective as of January 1, 2024).</u>
10.4+	<u>Executive Employment Agreement, dated September 5, 2023, by and between the Company and Stephanie Pittman.</u>
10.5+	<u>Separation Agreement, dated September 5, 2023, by and between the Company and Kelley Conte.</u>
10.6+	<u>Consulting Agreement, dated September 5, 2023, by and between the Company and Kelley Conte.</u>
31.1	<u>Certification of principal executive officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
31.2	<u>Certification of principal financial officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
32	<u>Certifications of the principal executive officer and the principal financial officer under Section 906 of the Sarbanes-Oxley Act of 2002.</u>
101.INS	XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
101.SCH	Inline XBRL Taxonomy Extension Schema Document.
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document.
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document.
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document.
104	Cover Page Interactive Data File (embedded within the Inline XBRL document).

+ Management contract or compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

ASPEN AEROGELS, INC.

Date: November 2, 2023

By: /s/ Donald R. Young
Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Date: November 2, 2023

By: /s/ Ricardo C. Rodriguez
Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer)

8/23/2023

Dear
Santhosh,

It is my pleasure to invite you to join Aspen Aerogels, Inc. ("Aspen" or the "Company"). As we discussed, this letter confirms certain terms and conditions of our offer of employment. We hope that you choose to join the Aspen team and look forward to a mutually beneficial relationship.

1. Compensation:

- a. **Job Title; Duties:** Aspen will employ you as a Chief Accounting Officer, reporting to Ricardo Rodriguez, Chief Financial Officer, and located in our Marlborough, MA office. We expect you to perform all duties and responsibilities that are associated with this position or otherwise that may be assigned to you by the Company, including but not limited to performing services for other Company affiliates or subsidiaries at the Company's request without further compensation other than as set forth below, and not to undertake any other employment or consultancy without the prior written consent of the Company. You agree to abide by all policies instituted by the Company, as they may be amended from time to time.
- b. **Expected Start Date:** We expect that your employment will commence on or about September 11, 2023, or such other date as you and the Company shall agree to in writing (the "Start Date").
- c. **Base Salary:** Aspen will pay you an annual base salary of \$300,000. This base salary will be subject to applicable withholdings, payable on a biweekly basis, and prorated during your first calendar year of employment based on your Start Date.
- d. **Corporate Bonus Plan:** In addition to your base salary, and to the extent you are a full-time employee, you may be eligible to receive a discretionary bonus, determined in the Company's sole discretion and subject to the terms and conditions of the Company's Corporate Bonus Plan ("Bonus Plan"). Your target annual bonus will be 35% of your annual salary ("Bonus"). This Bonus is contingent upon the achievement of defined targets and will be prorated based on your date of hire for 2023. Participation in the Bonus Plan is also contingent upon necessary Company approvals.
You will receive a separate Participation Letter outlining the details of your participation in the Bonus Plan upon such approval.

Other Compensation

- e. **New Hire Stock Equity:** As a means of rewarding you for your future contributions to Aspen, subject to the Company's approval, you will receive, pursuant to Aspen Aerogels 2023 Equity Incentive Plan, an equity grant of about \$200,000 in fair market value which, at the discretion of the Company, may include grants of restricted stock units and non-qualified stock options, each of which
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typically have a three-year graded service-based vesting period. You will receive additional governing documents upon the grant.

- f. **LTIP:** Subject to the approval of the Company, you will be eligible to participate in Aspen's Long Term Incentive Plan program. You will receive additional information on this program in conjunction with the program timeline.

2. **Employee Benefits:**

- a. As an employee of the Company, you will generally be eligible to participate in the Company's benefits plans and programs in accordance with Company policies, and subject to the terms and conditions set forth in the benefit programs themselves. You will be able to participate in any other employee benefit plans that the Company offers its employees for which you meet the eligibility requirements and consistent with the Company's policies. Please refer to the benefits summary plan descriptions for further information. By signing below, you understand that nothing in this offer letter requires the Company to establish or maintain any benefit plan, policy or arrangement and that the Company's benefit plans are governed by their own documents, which may be amended by the Company or its affiliates, as the case may be, from time to time. Benefits log in information will be provided to you under separate cover.
- b. You will accrue twenty (20) days of vacation per year at the bi-weekly rate of 6.16 hours per pay period (pro-rated as applicable), subject to any Company policy then in effect. We strongly encourage all employees to take all available vacation time to rest, relax and recharge. Holidays and other time off will be in accordance with the then-current Company policy.
3. **No Restrictions.** By countersigning this offer, you certify to the Company that: (i) you are free to enter into and fully perform the duties of this position and that you are not subject to any employment, confidentiality, non-competition or other agreement that would restrict your employment by the Company; (ii) no trade secret or proprietary information belonging to any previous employer will be disclosed by you to the Company and that no such information will be retained by you or brought with you to the Company; and (iii) all facts you have presented or will present to the Company are accurate and true, and this includes, but is not limited to, all oral and written statements you have made (including those pertaining to your education, training, qualifications, licensing and prior work experience) on any job application, resume or c.v., or in any interview or discussion with the Company. Additionally, if there are any written agreement(s) containing restrictive covenants (e.g. non-competition) with a previous employer, you have provided any such agreement(s) to the Company for review.
4. **Additional Conditions to Employment:** Your offer of employment is also expressly conditioned upon the following:
- Your ability to give us, within three (3) days of starting work, proof that you are eligible to be employed in the United States consistent with Form I-9 of the United States Department of Homeland Security. You will be required to present documents proving your identity and authorization to work in the United States, in order to complete Form I-9 and E-Verify process. This is a requirement of the Department of Homeland Security. A list of acceptable documents will be included in your New Hire packet provided by HR.
-

- Satisfactory completion of a background check, including, but not limited to, verification of previous employment, education, criminal, drug and other checks. A credit check may also be required for certain positions, as permitted by applicable law. We will provide you with a disclosure and authorization of background check forms that you will need to sign before we run the background check in connection with your hire and which will also permit the Company to run background checks at other points during your employment as needed. You will be contacted by our background check vendor to coordinate a drug screen. If this test is not completed within the required timeframe, this offer may be rescinded.
- Your execution of a Restrictive Covenant Agreement, which we have enclosed as **Exhibit A** with this offer letter, and which, among other things, prohibits the unauthorized use or disclosure of the Company's confidential and proprietary information and solicitation of its customers, clients, and employees.
- Additionally, upon your acceptance and completion of hiring paperwork, you will be provided with the Company's Employee Handbook. This document will provide further information regarding the Company's policies, procedures, and benefits.
- You acknowledge that this offer letter, (along with the final form of any referenced documents), represents the entire agreement between you and the Company and that no verbal or written agreements, promises or representations that are not specifically stated in this offer, are or will be binding upon the Company.

By signing below, you expressly acknowledge and agree that your employment with the Company will be "at will." This means that either you or the Company may terminate your employment at any time and for any reason with or without notice. It also means that the Company is not promising to employ you for any definite period of time and this is not a contract of employment. While supervisors and managers have certain hiring authority, no supervisor, manager, or representative of the Company has any authority to alter the at-will relationship.

This offer is valid until August 30, 2023, after which time it will lapse. If you have any questions regarding this offer of employment, please contact Annie Buttitta in Human Resources at abuttitta@aerogel.com. To accept your employment with the Company, please sign below using our electronic signature system via DocuSign.

We look forward to having you join the Company.

Sincerely Yours,
/s/ Kelley Conte

Kelley Conte
Senior Vice President, Human Resources

/s/ Santhosh Daniel
Acknowledged and Accepted
Santhosh Daniel

August 24, 2023
Date

Exhibit
A

Restrictive Covenant Agreement

This Restrictive Covenant Agreement ("Agreement") is made and entered into as of the date set forth below by Aspen Aerogels, Inc. ("Aspen" or the "Company") and Santhosh Daniel, on behalf of themselves and their heirs, executors, administrators, successors and assigns (collectively referred to throughout this Agreement as "Employee," and, together with Aspen, the "Parties").

WHEREAS, the Company desires to employ the Employee, pay the Employee a salary and other compensation and provide other benefits, and provide the Employee with access to its confidential information and relationships, on the express condition that Employee executes this Agreement.

NOW, THEREFORE, in consideration of the mutual covenants and agreements herein contained and other good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, the parties agree as follows:

1. Employee Acknowledgments. The Employee expressly acknowledges that:

(a) the Company is in engaged in the business of research, development, manufacturing, commercialization, marketing, sales, distribution of products, services and technologies involving aerogels (the "Business");

(b) the Business in which the Company is engaged is intensely competitive and Employee's employment by the Company requires that Employee have access to and knowledge of the Confidential Information (as that term is defined below) of the Company and of the Company Group. For purposes of this Agreement, "Company Group" means, individually and collectively, Aspen Aerogels, Inc., and its direct and indirect subsidiaries and affiliates.

(c) the Company's Confidential Information is vital to the success of the Company's Business and has been or will be developed or attained by great efforts and at great expense to the Company and the direct or indirect unauthorized disclosure of any Confidential Information would place the Company at a serious competitive disadvantage and would do serious damage, financial and otherwise, to the Company's Business and its shareholders and the job security of its employees;

(d) in the course of Employee's employment with the Company, the Employee has been introduced and provided access to those with important business relationships to the Company Group, and that any and all goodwill created through such introductions and access belongs exclusively to the Company, including, but not limited to, any goodwill created as a result of direct or indirect contacts or relationships between the Employee and any Customer or Potential Customer (as defined below));

(e) by Employee's training, experience and expertise, Employee's services to the Company are special and unique;

(f) the Company would not have provided and will not agree to continue to provide Employee with access to Confidential Information and introductions and access to Customers or Potential Customers, unless Employee agrees to the restrictive covenants set forth in this Agreement and

acknowledges that the restrictive covenants set forth below are necessary to protect the legitimate business interests of the Company;

(g) the execution and full compliance with this Agreement is an express term and condition of Employee's employment with the Company, and the salary and benefits provided and access to Confidential Information and Customers or Potential Customers to Employee are good and reasonable consideration for the restrictive covenants and obligations imposed on Employee by this Agreement; and

(h) Employee has been given a reasonable opportunity to consider the terms of this Agreement, and to consult with an attorney of their choosing, prior to executing this Agreement.

2. Non-Disclosure.

(a) Confidential Information. "Confidential Information" for purposes of this Agreement means information disclosed to Employee or to which Employee had access that includes, but is not limited to, trade secrets and confidential and proprietary information of the Company Group, or any information provided to the Employee or the Company under an obligation of confidentiality to a third party, or any confidential, trade secret, or proprietary information acquired by the Company from others with whom the Company or any affiliate has a business relationship, whether in written, oral, electronic or other form, including, but not limited to (i) trade secrets, inventions, mask works, ideas, processes, algorithms, formulae, software in source or object code, data programs, other works of authorship, know-how, improvements, technology direction, product or technology development methodology, technology assessment, experimental procedures, results, process development, product plans, development plans, testing procedures, quality control and testing processes, discoveries, developments, designs and techniques, any other proprietary technology and all Inventions (as defined in Section 3); (ii) information regarding research, development, new products, marketing and selling, business plans, budgets and unpublished financial statements, licenses, prices and costs, margins, discounts, credit terms, pricing and billing policies, quoting procedures, methods of obtaining business, forecasts, future plans and potential strategies, financial projections and business strategies, operational plans, financing and capital-raising plans, activities and agreements, internal services and operational manuals, methods of conducting Company business, suppliers and supplier information, sales techniques and strategies, training methods and materials, and purchasing; (iii) information regarding customers and potential customers of the Company, including customer lists, names, representatives, their needs or desires with respect to the types of products or services offered by the Company, proposals, bids, contracts and their contents and parties, the type and quantity of products and services provided or sought to be provided to customers and potential customers of Company and other non-public information relating to customers and potential customers; (iv) information regarding any of the Company's business partners and its services, including names, representatives, proposals, bids, contracts and their contents and parties, the type and quantity of products and services received by the Company, and other non-public information relating to business partners; (v) information regarding personnel, employee lists, compensation, and employee skills; (vi) any other non- public information which a competitor of the Company could use to the competitive disadvantage of the Company; and (vii) any other scientific, technical or trade secrets of the Company or of any third party provided to the Employee or the Company under a condition of confidentiality, provided that Confidential Information shall not include information that was known to the Employee prior to joining the Company, general industry knowledge, or is in the public domain other than through any fault or act by the Employee.,.

(b) Trade Secrets. the term "trade secrets" as used in this Agreement shall be given its broadest possible interpretation under applicable law and shall mean all forms and types of financial,

business, scientific, technical, economic, or engineering information, including patterns, plans, compilations, program devices, formulas, designs, prototypes, methods, techniques, processes, procedures, programs, or codes, whether tangible or intangible, and whether or how stored, compiled, or memorialized physically, electronically, graphically, photographically, or in writing that (i) the Company has taken reasonable measures to keep secret, and that (ii) derives independent economic value, actual or potential, from not being generally known to, and not being readily ascertainable through proper means by, another person who can obtain economic value from the disclosure or use of the information.

(c) Non-Disclosure. The Employee expressly acknowledges and agrees that all Confidential Information is and shall remain the sole property of the Company or the third party to whom the Company owes an obligation of confidentiality and that the Employee shall hold it in strictest confidence. The Employee shall at all times, both during the Employee's employment with the Company and after the termination of the Employee's employment for any reason or for no reason, maintain in confidence and shall not, without the prior written consent of the Company, use (except in the course of performance of the Employee's duties for the Company or by court order), disclose, or give to others any Confidential Information. The terms of this Section 2(c) are in addition to, and not in lieu of, any statutory or other contractual or legal obligation that the Employee may have relating to the protection of the Company's Confidential Information.

(d) Defend Trade Secrets Act. By executing this Agreement, Employee has been notified by this writing, in accordance with the Defend Trade Secrets Act of 2016, that (i) an individual shall not be held criminally or civilly liable under any federal or state trade secret law for the disclosure of a trade secret that is made in confidence to a federal, state, or local government official, or to an attorney, solely for the purpose of reporting or investigating a suspected violation of law, (ii) an individual shall not be held criminally or civilly liable under any federal or state trade secret law for the disclosure of a trade secret that is made in a complaint or other document filed in a lawsuit or other proceeding, if such filing is made under seal; and (iii) an individual who files a lawsuit for retaliation by an employer for reporting a suspected violation of law may disclose the trade secret to the attorney of the individual and use the trade secret information in the court proceeding, if the individual files any document containing the trade secret under seal and does not disclose the trade secret except pursuant to court order;

(e) Notification to Company. In the event the Employee is questioned by anyone not employed by the Company or by an employee of or a consultant to the Company not authorized to receive Confidential Information, in regard to any Confidential Information or concerning any fact or circumstance relating thereto, the Employee agrees to promptly notify the Company's Chief Legal Officer and General Counsel.

(f) Return of Confidential Information and Company Property. Upon the termination of the Employee's employment with the Company for any reason or for no reason, or if the Company otherwise requests at any other time before such termination, the Employee shall: (i) return to the Company all tangible Confidential Information and copies thereof (regardless how such Confidential Information or copies are maintained); and (ii) deliver to the Company any property of the Company which may be in the Employee's possession, including, but not limited to, products, materials, memoranda, notes, records, reports, or other documents or photocopies of the same. The Employee agrees that, when the Employee leaves the employ of the Company, or at any time the Company may request, the Employee shall deliver to the Company any and all Company equipment provided to the Employee including laptops and other electronic devices, as well as drawings, notes, memoranda, specifications, devices, formulas, and any other

documents pertaining to the Company and/or the Company's business, including, but not limited to, computer files, together with all copies thereof, and any other material containing or disclosing any Confidential Information as defined in Section 2(a) above (collectively, "such Documents"). The above shall include any and all such Documents contained on, for example, a home computer system, tablet or smart phone. The Employee further agrees not to retrieve or retain in any way any such Documents, and the Employee shall, for example, first return such Documents to the Chief Legal Officer and General Counsel and then consult with the Company's Chief Legal Officer and General Counsel regarding the removal and deletion of such Documents from any home computer system, personal electronic device, or other personal property and act in accordance with their instructions. The Employee expressly authorizes the Company's designated representatives to access such equipment or devices for this limited purpose and shall provide any passwords or access codes necessary to accomplish this task. The Employee further agrees that any property situated on the Company's premises and/or owned by the Company, including disks and other storage media, filing cabinets or other work areas, is subject to inspection by Company personnel at any time with or without notice.

3. Inventions.

(a) **Property of the Company.** All ideas, discoveries, creations, manuscripts and properties, innovations, improvements, know-how, inventions, designs, developments, apparatus, techniques, methods, laboratory notebooks, formulae, data, protocols, writings, specifications, sound recordings, and pictorial and graphical representations; the exclusive legal right to reproduce, perform, display, distribute and make derivative works of a work of authorship ("Copyright"); and all paternity, integrity, disclosure, modification, withdrawal, special and any other similar rights recognized by the laws of any jurisdiction or country as "moral rights," "artist's rights," "droit moral," or the like (collectively, "Moral Rights") (collectively, the "Inventions"), which may be used in the business of the Company, whether patentable, copyrightable or not, which the Employee may conceive, reduce to practice or develop during the Employee's employment with the Company, whether alone or in conjunction with another or others, whether during or out of regular business hours, whether or not on the Company's premises or with the use of its equipment, and whether at the request or upon the suggestion of the Company or otherwise, shall be and are the sole and exclusive property of the Company, and that the Employee shall not publish any of the Inventions without the prior written consent of the Company or its designee. The Employee acknowledges and agrees that any Inventions conceived or made by the Employee, alone or with others, within two (2) years following termination of the Employee's employment are likely to have been conceived in significant part while employed by the Company; accordingly, the Employee agrees that such Inventions shall be presumed to have been conceived during the Employee's employment with the Company until the Employee has established the contrary by clear and convincing evidence, and that such Inventions are subject to the terms and conditions of this Section 3. The Employee also acknowledges that all original works of authorship which are made by the Employee (solely or jointly with others) within the scope of the Employee's employment or which relate to the business of the Company or a Company affiliate and which are protectable by copyright are "works made for hire" pursuant to the United States Copyright Act (17 U.S.C. § 101). The Employee hereby assigns to the Company or its designee all of the Employee's right, title and interest in and to all of the foregoing. During and after Employee's employment, Employee agrees promptly to inform and disclose all Inventions to the Company in writing, to assign all Inventions to the Company, to provide all assistance reasonably requested by the Company to preserve its interests in the Inventions (such as by executing assignments and other documents, testifying, etc.), such assistance to be provided at the Company's expense but without additional compensation to Employee. The Employee further represents and warrants that, to the best of the Employee's knowledge and belief, none of the

Inventions shall violate or infringe upon any right, patent, copyright, trademark or right of privacy, or constitute libel or slander against or violate any other rights of any person, firm or corporation, and that the Employee shall use the Employee's best efforts to prevent any such violation.

(b) **Cooperation; Power of Attorney.** At any time during or after the Employee's employment with the Company, the Employee shall fully cooperate with the Company and its attorneys and agents in securing and protecting the Company's rights to Inventions, including but not limited to the preparation and filing of all papers and other documents as may be required to perfect the Company's rights in and to any of such Inventions, and joining in any proceeding to obtain letters patent, copyrights, trademarks or other legal rights with respect to any such Inventions in the United States and in any and all other countries, provided that the Company shall bear the expense of such proceedings, and that any patent or other legal right so issued to the Employee personally shall be assigned by the Employee to the Company or its designee without charge by the Employee. If the Company is unable, after reasonable effort, to secure the Employee's signature on any such papers and/or other documents, the Employee hereby irrevocably designates and appoints each officer of the Company as the Employee's agent and attorney-in-fact to execute any such papers on the Employee's behalf, and to take any and all actions as the Company may deem necessary or desirable in order to protect its rights and interests in any Invention. Additionally, Employee promises to cooperate reasonably with the Company in the transition of matters that the Employee worked on or were involved in prior to their termination of employment. In addition, at the Company's request, Employee agrees to assist, consult with, and cooperate with the Company in any litigation, intellectual property procedure or administrative procedure or inquiry that involves the Employee's prior duties with the Company, subject to reimbursement for their reasonable out of pocket expenses, such as travel, meals or lodging.

(c) **Licensing and Use of Innovations.** With respect to any Inventions, and work of any similar nature (from any source), whenever created, which the Employee has not conceived, reduced to practice or developed during the Employee's employment with the Company, but which the Employee provides to the Company or incorporates in any Company product or system, the Employee hereby grants to the Company a royalty-free, fully paid-up, non-exclusive, perpetual and irrevocable license throughout the world to use, modify, create derivative works from, disclose, publish, translate, reproduce, deliver, perform, sell, license, dispose of, and to authorize others so to do, all such Inventions. The Employee shall not include in any Inventions they deliver to the Company or use on its behalf, without the prior written consent of the Company, any material which is or shall be patented, copyrighted or trademarked by the Employee or others unless the Employee provides the Company with the written permission of the holder of any patent, copyright or trademark owner for the Company to use such material in a manner consistent with then- current Company policy.

(d) **Disclosure of Inventions.** Promptly upon conception of each Invention, the Employee agrees to disclose the same to the Company and the Company shall have full power and authority to file and prosecute patent applications thereon and to procure and maintain patents thereon. The Employee agrees that such Inventions shall remain subject to all provisions of this Agreement, including but not limited to the ownership, cooperation and licensing provisions described in this Section 3. The Employee acknowledges that the Employee's obligation to disclose such information is ongoing during the Employee's employment with the Company, and that after the Employee executes this Agreement, if the Employee determines that any additional Inventions in which the Employee claims or intends to claim any right, title or interest (including but not limited to patent, copyright and trademark interest) has been or is

likely to be delivered to the Company or incorporated in any company product or system, the Employee shall make immediate written disclosure of the same to the Company.

(e) **Prior Inventions.** Employee shall provide the Company a listing which shall be appended to this Agreement of any and all Inventions, that may relate to the business of the Company or actual or demonstrably anticipated research or development and that were made by the Employee or acquired by the Employee prior to the commencement of the Employee's employment with the Company, and which are not to be assigned to the Company ("Prior Inventions"). If no such list is attached, the Employee represents and agrees that it is because the Employee has no rights in any existing Inventions that may relate to the Company's business or actual or demonstrably anticipated research or development. The Employee acknowledges and agrees that if the Employee uses or includes any Prior Inventions in the scope of the Employee's employment or in any product or service in the Company, or if the Employee's rights in any Prior Inventions may block or interfere with, or may otherwise be required for, the exercise by the Company of any rights assigned to the Company under this Agreement, the Employee shall immediately notify the Company in writing. Unless the Company and the Employee agree otherwise in writing as to particular Prior Inventions, the Employee grants the Company, in such circumstances (whether or not the Employee gives the Company notice as required above), a non-exclusive, perpetual, transferrable, fully-paid and royalty free, irrevocable and worldwide license, with rights to sublicense through multiple levels of sublicensees, to reproduce, make derivative works of, distribute, publicly perform, and publicly display in any form or medium, whether now known or later developed, make, have made, use, sell, import, offer for sale, and exercise any and all present or future rights in, such Prior Inventions. To the extent that any third parties have rights in any such Prior Inventions, the Employee hereby represents and warrants that such third party or parties have validly and irrevocably granted the Employee the right to grant the license stated above. In the event Employee becomes aware of any potential infringement regarding Prior Inventions, Employee shall immediately notify the Company. Employee agrees to indemnify and hold harmless the Company, its affiliates and its and their officers, directors, employees and representatives from any claims, losses or costs associated with a potential infringement claim related to any Prior Inventions.

(f) **Assignment/Transfer of Web Properties.** The Employee agrees to transfer and assign (both during and after employment), and does hereby assign to the Company all rights, titles, and interests in and to any domain name or social media account (collectively called "Web Properties") registered or owned by the Employee that: (1) was registered with the intent to be used by the Company; and/or (2) relates in any manner to, or is used to comment on, the actual or anticipated business of the Company; and/or (3) contains a registered or common law trademark of the Company.

(g) **Incorporation of Software Code.** The Employee agrees that they shall not incorporate into any Company software or otherwise deliver to Company any software code licensed under the GNU General Public License or Lesser General Public License or any other license that, by its terms, requires or conditions the use or distribution of such code on the disclosure, licensing, or distribution of any source code owned or licensed by Company except in strict compliance with Company policies regarding the use of such software.

4. Non-Disparagement. Employee agrees that during the Employee's employment by the Company and at any time thereafter, the Employee shall not take any action or make any statement, verbally or in writing, or via social media which has the purpose or effect of disparaging the Company, or its directors, officers, employees or its products.

5. Non-Solicitation.

(a) Employee agrees that during Employee's employment (except as necessary to perform services on the Company's behalf) and for a period of twelve (12) months after the Employee's termination date ("Non-Solicitation Period"), regardless of the reason Employee's employment terminates, Employee agrees that Employee shall not, directly or indirectly, for the Employee's own account or on behalf of any other person or entity, (i) solicit, call upon or accept business from, any Customer or Potential Customer; or (ii) interfere with the business relationship between any Customer or Potential Customer and the Company; or (iii) solicit, induce, persuade or hire, or attempt to solicit, induce, persuade or hire, or assist any third party in the solicitation, inducement, persuasion or hiring of, any employee or contractor of the Company who worked for the Company during the Employee's tenure with the Company, to leave the employ of the Company. Employee acknowledges and agrees that the Non-Solicitation Period shall be tolled and shall not run, during any period in which the Employee is in violation of the terms herein.

(b) "Customer or Potential Customer" shall mean any person or entity who or which, at any time during the one year period prior to the Employee's contact with such person or entity, if such contact occurs during the Employee's employment or, if such contact occurs following the termination of the Employee's employment, during the one year period prior to the date the Employee's employment with the Company ends: (i) contracted for, was billed for, or received from the Company any product, service, or process with which the Employee worked directly or indirectly during the Employee's employment by the Company or about which the Employee acquired Confidential Information; or (ii) was in contact with the Employee or in contact with any other employee, owner, or agent of the Company, of which contact the Employee was or should have been aware, concerning the sale or purchase of, or contract for, any product, service, or process with which the Employee worked directly or indirectly during the Employee's employment with the Company or about which the Employee acquired Confidential Information; or (iii) was solicited by the Company in an effort in which the Employee was involved or of which the Employee was aware.

(c) Interests to be Protected. The Company and the Employee acknowledge that Employee shall perform essential services for the Company, its employees, and/or its stockholders during their employment. The Employee shall be exposed to, have access to, and work with, a considerable amount of Confidential Information. The Company and the Employee also expressly recognize and acknowledge that the personnel of the Company have been trained by, and are valuable to, the Company and that the Company shall incur substantial recruiting and training expenses if the Company must hire new personnel or retrain existing personnel to fill vacancies. The Company and the Employee acknowledge this covenant is reasonable and it is necessary for the protection of the Company, its stockholders, and employees.

6. Litigation and Regulatory Cooperation. During and for a reasonable period of time after the Employee's employment, the Employee shall cooperate fully with the Company in the defense or prosecution of any claims actions, investigations or examinations now in existence or which may be brought in the future against or on behalf of the Company which relate to events or occurrences that transpired while the Employee was employed by the Company.

7. Workplace Rules. The Company may, from time to time, make, adopt, and/or modify workplace policies and rules and provide to Employee (directly or through designated electronic locations) copies of such policies and rules, as the Company determines, at its discretion, as necessary to comply with laws and regulations and to provide safe, efficient, productive and effective workplace and work environment, including but not limited to the Company's Code of Business Conduct and Ethics, as well as

its Employee Handbook. Employee hereby agrees to comply with all such policies and rules along with all applicable laws and regulations and will complete any corresponding annual or periodic attestations associated with any Company policies.

8. Protected Rights. For the avoidance of doubt, nothing in this Agreement prohibits or otherwise restricts Employee from (a) initiating, testifying, assisting, complying with a subpoena from, or participating in any manner with an investigation conducted by a governmental agency, including without limitation; (b) discussing or disclosing information about unlawful acts in or related to the workplace, including, but not limited to discrimination, harassment, sexual assault, retaliation, wage and hour violations, conduct that is against a clear mandate of public policy, or any other conduct Employee has reason to believe is unlawful; (c) making any necessary disclosures as otherwise required by law.

9. No Conflicting Obligation. Employee hereby represents and warrants to the Company that they (a) are not presently under and will not in the future become subject to any obligation to any person, entity or prior employer which is inconsistent or in conflict with this Agreement or which would prevent, limit or impair in any way their performance of their employment with the Company and (b) has not disclosed and will not disclose to the Company, nor use for the Company's benefit, any confidential information and trade secrets of any other person or entity, including any prior employer.

10. Change; Reaffirmation. Employee understands, acknowledges and agrees that: (a) Employee's covenants and promises under this Agreement, including but not limited to Sections 1, 2, 3, 4 and 5, shall continue in accordance with their express terms regardless of any material changes in the Employee's title, position, duties, salary, compensation or benefits or other terms and conditions of employment, and Employee agrees to comply with such obligations; and (b) if Employee should transfer between or among any member of the Company Group, wherever situated, or be promoted or reassigned to functions other than Employee's present functions, the terms of this Agreement shall continue to apply with full force, and Employee agrees to comply with such terms. Employee expressly consents to be bound by this Agreement, including but not limited to Sections 1, 2, 3, 4 and 5, for the benefit of the Company or member of the Company Group to whose employ Employee may be transferred, without the necessity that this Agreement be resigned at the time of such transfer.

11. Severability; Enforceability. The parties intend for this Agreement to be enforced as written. However, if any portion or provision of this Agreement shall to any extent be declared illegal or unenforceable by a duly authorized court having jurisdiction, (a) then the remainder of this Agreement, or the application of such portion or provision in circumstances other than those as to which it is so declared illegal or unenforceable, shall not be affected thereby, and each portion and provision of this Agreement shall be valid and enforceable to the fullest extent permitted by law; and/or (b) because of the scope of the restrictive covenant is found to be unreasonable, the Company and the Employee agree that the court making such determination shall have the power to "blue-pencil" the Agreement as necessary to make it reasonable in scope; and in its reduced or blue-penciled form such portion or provision shall then be enforceable and shall be enforced.

12. Injunctive Relief. Employee acknowledges that the restrictive covenants contained in this Agreement are reasonable and necessary for the protection of the Company's legitimate business interests, and any violation of these restrictive covenants would cause substantial injury and irreparable harm to the Company, and therefore, in the event of a breach or threatened breach by Employee of any of these restrictive covenants, the Company shall be entitled to seek and obtain equitable relief, in the form of specific performance, and/or temporary, preliminary or permanent injunctive relief, or any other equitable

remedy which then may be available restraining Employee from such breach or threatened breach; provided, however, that the right to apply for such relief shall not be construed as prohibiting the Company from pursuing any other available remedies for such breach or threatened breach.

13. Waiver of Jury Trial. The parties to this Agreement each hereby waives, to the fullest extent permitted by law, any right to trial by jury of any claim, demand, action, or cause of action (a) arising under this Agreement or (b) in any way connected with or related or incidental to the dealings of the parties hereto in respect of this Agreement or any of the transactions related hereto, in each case whether now existing or hereafter arising, and whether in contract, tort, equity, or otherwise. The parties to this Agreement each hereby agree and consent that any such claim, demand, action, or cause of action shall be decided by court trial without a jury and that the parties to this Agreement may file an original counterpart or a copy of this Agreement with any court as written evidence of the consent of the parties hereto to the waiver of their right to trial by jury.

14. Survival. This Agreement, and the restrictive covenants contained in this Agreement, shall survive the termination of Employee's employment for any reason.

15. Amendment. This Agreement may not be modified, altered or changed except in writing and signed by both Parties wherein specific reference is made to this Agreement.

16. Successors. The Company may require any successor to the Company to expressly assume and agree to perform this Agreement in the same manner and to the same extent that the Company would be required to perform it if no such succession had taken place. The Employee's obligations under this Agreement may not be assigned by the Employee without the prior written consent of the Company.

17. Waiver. No waiver of any provision hereof shall be effective unless made in writing and signed by the waiving party. The failure of any party to require the performance of any term or obligation of this Agreement, or the waiver by any party of any breach of this Agreement, shall not prevent any subsequent enforcement of such term or obligation or be deemed a waiver of any subsequent breach.

18. Governing Law. All issues concerning this Agreement shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts, without giving effect to any choice of law or conflict of law provision or rule (whether of the Commonwealth of Massachusetts or any other jurisdiction) that would cause the application of the law of any jurisdiction other than the Commonwealth of Massachusetts.

19. Integration. This Agreement embodies the entire agreement and understanding between the parties hereto with respect to the subject matter hereof and supersedes all prior oral or written agreements and understandings relating to the subject matter hereof. No statement, representation, warranty, covenant or agreement of any kind not expressly set forth in this Agreement shall affect, or be used to interpret, change or restrict, the express terms and provisions of this Agreement. Notwithstanding the foregoing, the obligations set forth in this Agreement will be in addition to any obligations undertaken by Employee under any other agreement containing restrictive covenants entered into between the Company and Employee (each an "Other Agreement"), and such obligations shall remain in full force and effect according to their terms.

20. Counterparts. This Agreement may be executed in any number of counterparts, each of which when so executed and delivered shall be taken to be an original; but such counterparts shall together constitute one and the same document.

The Parties knowingly and voluntarily sign this Agreement as of the date(s) set forth below:

Employee

By: /s/ Santhosh Daniel
Name: Santhosh Daniel
Date: September 9, 2023

Company
Aspen Aerogels, Inc.

By: /s/ Kelley Conte
Name: Kelley Conte
Title: Senior Vice President, Human Resources
Date: August 23, 2023

FIRST AMENDMENT TO LOAN AGREEMENT

This FIRST AMENDMENT TO LOAN AGREEMENT (this “Amendment”) is entered into as of September 28, 2023 by and among ASPEN AEROGELS, INC., a Delaware corporation (the “Parent”), ASPEN AEROGELS GEORGIA, LLC, a Georgia limited liability company (“Borrower”), ASPEN AEROGELS RHODE ISLAND, LLC, a Rhode Island limited liability company (“Aspen RI”) and GENERAL MOTORS HOLDINGS LLC (the “Lender”).

WITNESSETH:

WHEREAS, the Parent, Borrower, Aspen RI and the Lender are parties to that certain Loan Agreement, dated as of November 28, 2022 (as the same has been and may be further amended, restated, amended and restated, supplemented or otherwise modified from time to time prior to the effectiveness of this Amendment, the “Loan Agreement”); and

WHEREAS, the Parent and Borrower have requested, and the Lender has agreed to, amend certain provisions of the Loan Agreement on the terms set forth herein and subject to the satisfaction of the conditions set forth herein.

NOW, THEREFORE, in consideration of the mutual agreements, provisions and covenants contained herein, the parties hereby agree as follows:

SECTION 1. Defined Terms. Capitalized terms used but not defined herein shall have the meanings ascribed to them in the Loan Agreement.

SECTION 2. Amendments to Loan Agreement. Upon satisfaction of the conditions set forth in Section 5 hereof, the Loan Agreement is hereby amended:

- (a) to delete red or green stricken text (indicated textually in the same manner as the following examples: ~~stricken text~~ and ~~stricken text~~) and (b) to add the blue or green double-underlined text (indicated textually in the same manner as the following examples: double-underlined text and double-underlined text), in each case, as set forth in the Loan Agreement attached hereto as Annex I;
- (b) to replace Schedule 7.6 of the Loan Agreement with Schedule 7.6 attached hereto; and
- (c) to add the Form of Compliance Certificate attached hereto to the Loan Agreement as a new Exhibit H.

SECTION 3. Waiver. Any Default or Event of Default arising from any failure by Borrower or any Loan Party to comply with Milestone number 3 (“Electrical Power Distribution to be finished by August 15, 2023”) set forth in Schedule 7.6 of the Loan Agreement, prior to giving effect to this Amendment (the “Specified Milestone”), are hereby waived by the Lender; provided, that the foregoing waiver shall be effective solely with respect to the Specified Milestone and shall not, except as specifically set forth herein with respect to the Specified Milestone, (i) constitute or be construed as a waiver of any other Default or Event of Default that has occurred and is continuing under the Loan Agreement and the other Loan Documents or (ii) constitute or be construed as a consent or release of, or a limitation upon, the Lender’s rights and remedies under the Loan Agreement or any other Loan Document, whether arising as a

consequence of any other such Default or Event of Default which may now exist or otherwise.

SECTION 4. Representation and Warranties.

(a) each of the representations and warranties made by any Loan Party pursuant to the Loan Agreement, the other Loan Documents and any related agreements to which it is a party, shall be true and correct in all material respects (without duplication of any materiality provisions or qualifier contained therein) on and as of the date hereof as if made on and as of such date (except to the extent any such representation or warranty expressly relates only to any earlier and/or specified date); and

(b) on the date hereof, after giving effect to the transactions contemplated by this Amendment, no Default or Event of Default has occurred or is continuing.

SECTION 5. Conditions. This Amendment shall become effective upon the satisfaction of the following conditions precedent in a manner satisfactory to Lender:

(a) Lender shall have received an executed counterpart hereof signed by each of the Lender, the Parent, Borrower and each other Loan Party;

(b) Lender shall have received from the Borrower the payment of all fees due and payable by any Loan Party pursuant to the Loan Agreement (as amended by this Amendment), including the reasonable and documented fees, charges and disbursements of counsel to Lender, incurred in connection with the Loan Agreement prior to the date hereof and in connection with the preparation, negotiation, execution and delivery of this Amendment and the documents related hereto; and

(c) Lender shall have received an updated Project Budget as of the date hereof in form and substance reasonably acceptable to the Lender (to be substantially consistent with the form of the Project Budget delivered on the Closing Date).

SECTION 6. No Modification. This Amendment is a Loan Document. Except as expressly set forth herein (including under Section 3 above), nothing contained herein shall be deemed to constitute a waiver of compliance with any term or condition contained in the Loan Agreement or any of the other Loan Documents or constitute a course of conduct or dealing among the parties. Except as expressly stated herein, the Lender reserves all rights, privileges and remedies under the Loan Documents. Except as amended or consented to hereby, the Loan Agreement and other Loan Documents remain unmodified and in full force and effect. Upon satisfaction of the conditions set forth in Section 5 hereof, all references in the Loan Documents to the Loan Agreement shall be deemed to reference the Loan Agreement as amended hereby.

SECTION 7. Counterparts; Electronic Execution. This Amendment may be executed in any number of counterparts and by different parties on separate counterparts, each of which, when executed and delivered, shall be deemed to be an original, and all of which, when taken together, shall constitute but one and the same Amendment. Delivery of an executed counterpart of this Amendment by facsimile or other electronic method of transmission shall be equally as effective as delivery of an original executed counterpart of this Amendment. Any party delivering an executed counterpart of this Amendment by facsimile or other electronic method of transmission also shall deliver an original executed counterpart of this Amendment but the failure to

deliver an original executed counterpart shall not affect the validity, enforceability, and binding effect of this Amendment.

SECTION 8, Severability of Provisions. Any provision of this Amendment that is prohibited or unenforceable in any jurisdiction shall, as to such jurisdiction, be ineffective to the extent of such prohibition or unenforceability without invalidating the remaining provisions hereof, and any such prohibition or unenforceability in any jurisdiction shall not invalidate or render unenforceable such provision in any other jurisdiction.

SECTION 9, Choice of Law and Venue; Jury Trial Waiver; Judicial Reference Provision.

(a) THE VALIDITY OF THIS AMENDMENT, THE CONSTRUCTION, INTERPRETATION, AND ENFORCEMENT HEREOF, THE RIGHTS OF THE PARTIES HERETO WITH RESPECT TO ALL MATTERS ARISING HEREUNDER OR RELATED HERETO, AND ANY CLAIMS, CONTROVERSIES OR DISPUTES ARISING HEREUNDER OR RELATED HERETO SHALL BE DETERMINED UNDER, GOVERNED BY, AND CONSTRUED IN ACCORDANCE WITH THE LAWS OF THE STATE OF MICHIGAN.

(b) THE PARTIES AGREE THAT ALL ACTIONS OR PROCEEDINGS ARISING IN CONNECTION WITH THIS AMENDMENT SHALL BE TRIED AND LITIGATED ONLY IN THE STATE OF MICHIGAN AND, TO THE EXTENT PERMITTED BY APPLICABLE LAW, THE STATE COURTS LOCATED IN WAYNE COUNTY, MICHIGAN OR THE FEDERAL COURTS LOCATED IN THE EASTERN DISTRICT OF MICHIGAN. EACH LOAN PARTY AND THE LENDER WAIVES, TO THE EXTENT PERMITTED UNDER APPLICABLE LAW, ANY RIGHT IT MAY HAVE TO ASSERT THE DOCTRINE OF FORUM *NON CONVENIENS* OR TO OBJECT TO VENUE TO THE EXTENT ANY PROCEEDING IS BROUGHT IN ACCORDANCE WITH THIS SECTION 9(b).

(c) TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, EACH LOAN PARTY AND THE LENDER HEREBY WAIVES ITS RIGHTS, IF ANY, TO A JURY TRIAL OF ANY CLAIM, CONTROVERSY, DISPUTE OR CAUSE OF ACTION DIRECTLY OR INDIRECTLY BASED UPON OR ARISING OUT OF ANY OF THIS AMENDMENT OR ANY OF THE TRANSACTIONS CONTEMPLATED THEREIN, INCLUDING CONTRACT CLAIMS, TORT CLAIMS, BREACH OF DUTY CLAIMS, AND ALL OTHER COMMON LAW OR STATUTORY CLAIMS (EACH A "CLAIM"). EACH LOAN PARTY AND THE LENDER REPRESENTS THAT EACH HAS REVIEWED THIS WAIVER AND EACH KNOWINGLY AND VOLUNTARILY WAIVES ITS JURY TRIAL RIGHTS FOLLOWING CONSULTATION WITH LEGAL COUNSEL. IN THE EVENT OF LITIGATION, A COPY OF THIS AMENDMENT MAY BE FILED AS A WRITTEN CONSENT TO A TRIAL BY THE COURT.

(d) EACH OF THE LENDER AND THE LOAN PARTIES HEREBY IRREVOCABLY AND UNCONDITIONALLY SUBMITS TO THE EXCLUSIVE JURISDICTION OF THE STATE AND FEDERAL COURTS LOCATED IN THE COUNTY OF MICHIGAN AND THE STATE OF MICHIGAN, IN ANY ACTION OR PROCEEDING ARISING OUT OF OR RELATING TO THIS AMENDMENT, OR FOR RECOGNITION OR ENFORCEMENT OF ANY JUDGMENT. EACH OF THE PARTIES HERETO AGREES THAT A FINAL JUDGMENT IN ANY SUCH ACTION OR PROCEEDING SHALL BE CONCLUSIVE AND MAY BE ENFORCED IN OTHER JURISDICTIONS BY SUIT ON THE JUDGMENT OR IN ANY OTHER MANNER PROVIDED BY LAW. NOTHING IN THIS AMENDMENT SHALL AFFECT ANY RIGHT THAT THE LENDER MAY OTHERWISE HAVE TO BRING ANY ACTION OR PROCEEDING RELATING TO THIS AMENDMENT AGAINST ANY LOAN PARTY OR ITS PROPERTIES IN THE COURTS OF ANY JURISDICTION.

[Remainder of Page Intentionally Left Blank; Signature Pages Follow]

IN WITNESS WHEREOF, each of the undersigned has executed this Amendment as of the date set forth above.

BORROWER:

ASPEN AEROGELS GEORGIA, LLC

By: /s/ Ricardo C. Rodriguez

Name: Ricardo C. Rodriguez

Title: Chief Financial Officer and Treasurer

GUARANTORS :

ASPEN AEROGELS, INC.

By: /s/ Ricardo C. Rodriguez

Name: Ricardo C. Rodriguez

Title: Chief Financial Officer and Treasurer

ASPEN AEROGELS RHODE ISLAND, LLC

By: /s/ Ricardo C. Rodriguez

Name: Ricardo C. Rodriguez

Title: Chief Financial Officer and Treasurer

[Signature Page to First Amendment to Loan Agreement]

LENDER:

GENERAL MOTORS HOLDINGS LLC,

By: /s/ Gustavo Vello

Name: Gustavo Vello

Title: Assistant Treasurer, GM

[Signature Page to First Amendment to Loan Agreement]

Exhibit H

Form of Compliance Certificate

(see attached)

Schedule I – Financial Covenant Calculations¹

Schedule I(a): Total Leverage Ratio § 6.13(a)

A. Required Covenant:

<u>Test Date</u>	<u>Ratio</u>
As of each Test Date, commencing with the Fiscal Quarter ending December 31, 2024	Less than or equal to 5.00 to 1.00

B. Actual Computation: _____ : **1.00**

1.	Total Indebtedness (see <u>Section D</u> hereof) of Parent and its Subsidiaries	\$ _____
2.	<u>MINUS</u> Indebtedness outstanding under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000 and undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness	\$ (_____)
3.	Subtotal (1-2) =	
	<u>Divided by:</u>	
4.	Consolidated EBITDA of Parent and its Subsidiaries for the most recently completed Measurement Period	\$ _____
5	Ratio (3 ÷ 4) =	_____

C. Consolidated EBITDA

1.	Consolidated Net Income of Parent and its Subsidiaries for the most recently completed Measurement Period	\$ _____
	<u>PLUS</u> (to the extent included in calculating Consolidated Net Income for the most recently completed Measurement Period, without duplication)	
2.	the sum of all interest, premium payments, debt discount, fees, charges and related expenses in connection with Indebtedness for borrowed money to the extent treated as interest in accordance with GAAP	\$ _____

¹ To be delivered also following the delivery of the audited F/S if such statements would result in a different calculation than as set forth in the certificate delivered concurrently with the unaudited financial statements for the applicable period.

3. provision for federal, state, local and foreign income Taxes, taxes on profit or capital, including, without limitation, state franchise and similar Taxes, and foreign withholding taxes	\$ _____
4. depreciation and amortization expenses	\$ _____
5. noncash (A) charges, expenses, costs or losses solely to the extent constituting extraordinary, unusual or non-recurring items, (B) compensation charges, (C) charges for the excess of GAAP rent expense over actual rent paid during such period due to the use of straight line rent for GAAP purposes or write-off and/or write-downs of inventory or other tangible assets and (D) asset write-offs and/or write-downs or any impairment charges	\$ _____
6. charges or expenses incurred in connection with the issuance or incurrence of permitted Indebtedness or the issuance of Equity Interests (that do not result in a Change of Control) in an aggregate amount during the term of this Agreement not to exceed \$500,000	
7. any other adjustments as shall be mutually agreed	\$ _____
8. Subtotal (1+2+3+4+5+6+7) =	\$ _____
<u>MINUS</u> (to the extent included in calculating such Consolidated Net Income for the most recently completed Measurement Period, without duplication)	
9. federal, state, local and foreign income Tax credits, Tax credits on profit or capital, including, without limitation, state franchise and similar Tax credits, and foreign withholding Tax credits	\$ _____
10 any noncash gain from extraordinary, unusual or non-recurring items .	\$ _____
11 Subtotal (9+10) = .	\$ _____
Total (8-11) =	\$ _____

D. Indebtedness

(i) The amount of any Indebtedness represented by a guaranty or other similar instrument shall be the lesser of the principal amount of the obligations guaranteed and still outstanding and the maximum amount for which the guaranteeing Person may be liable pursuant to the terms of the instrument embodying such Indebtedness, and (ii) the amount of any Indebtedness which is limited or is non-recourse to a Person or for which recourse is limited to an identified asset shall be valued at the lesser of (A) if applicable, the limited amount of such obligations, and (B) if applicable, the fair market value of such assets securing such obligation	
1. All obligations of such Person for borrowed money	\$ _____
2. all obligations of such Person evidenced by bonds, debentures, notes, or other similar instruments and all reimbursement or other obligations in respect of letters of credit, bankers acceptances, or other financial products	\$ _____
3. all obligations or liabilities of others secured by a Lien on any asset of such Person, irrespective of whether such obligation or liability is assumed	\$ _____
4. all obligations of such Person to pay the deferred purchase price of assets (other than trade payables incurred in the ordinary course of business and repayable in accordance with customary trade practices and which are paid in accordance with such terms and not on a deferred basis)	\$ _____
5. all obligations of such Person in respect of Disqualified Equity Interests	\$ _____
6. any obligation of such Person guaranteeing or intended to guarantee (whether directly or indirectly guaranteed, endorsed, co-made, discounted, or sold with recourse) any obligation of any other Person that constitutes Indebtedness under any of clauses (1) through (5) above	\$ _____
Total (1+2+3+4+5+6) =	\$ _____

Schedule I(b): Ratio of Total Indebtedness to Consolidated Equity § 6.13(b)

A. Required Covenant:

<u>Test Date</u>	<u>Ratio</u>
As of each Test Date, commencing with the Fiscal Quarter ending December 31, 2024	Less than or equal to 1.20x

B. Actual Computation: _____ x

1.	Total Indebtedness (see <u>Section C</u> hereof) of Parent and its Subsidiaries	\$ _____
2.	MINUS undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness	\$ (_____)
3.	Subtotal (1-2) =	
	Divided by:	
4.	consolidated equity of Parent and its Subsidiaries	\$ _____
5.	Ratio (3 ÷ 4) =	_____

C. Indebtedness

(i) The amount of any Indebtedness represented by a guaranty or other similar instrument shall be the lesser of the principal amount of the obligations guaranteed and still outstanding and the maximum amount for which the guaranteeing Person may be liable pursuant to the terms of the instrument embodying such Indebtedness, and (ii) the amount of any Indebtedness which is limited or is non-recourse to a Person or for which recourse is limited to an identified asset shall be valued at the lesser of (A) if applicable, the limited amount of such obligations, and (B) if applicable, the fair market value of such assets securing such obligation	
1. All obligations of such Person for borrowed money	\$ _____
2. all obligations of such Person evidenced by bonds, debentures, notes, or other similar instruments and all reimbursement or other obligations in respect of letters of credit, bankers acceptances, or other financial products	\$ _____
3. all obligations or liabilities of others secured by a Lien on any asset of such Person, irrespective of whether such obligation or liability is assumed	\$ _____
4. all obligations of such Person to pay the deferred purchase price of assets (other than trade payables incurred in the ordinary course of business and repayable in accordance with customary trade practices and which are paid in accordance with such terms and not on a deferred basis)	\$ _____

5. all obligations of such Person in respect of Disqualified Equity Interests	\$ _____
6. any obligation of such Person guaranteeing or intended to guarantee (whether directly or indirectly guaranteed, endorsed, co-made, discounted, or sold with recourse) any obligation of any other Person that constitutes Indebtedness under any of clauses (1) through (5) above	\$ _____
7. Subtotal (1+2+3+4+5+6) =	\$ _____
<u>MINUS</u>	
8. Indebtedness under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000	\$ _____
Total (7-8) =	\$ _____

Annex I
(See Attached)

LOAN AGREEMENT

by and between

GENERAL MOTORS HOLDINGS LLC,

(“Lender”)

ASPEN AEROGELS GEORGIA, LLC

(“Borrower”)

and

THE GUARANTORS FROM TIME TO TIME PARTY HERETO

Dated as of November 28, 2022

TABLE OF CONTENTS

	Page
1. DEFINITIONS AND CONSTRUCTION.	1
1.1. Definitions	1
1.2. Code	1
1.3. Schedules and Exhibits	1
1.4. Definition of “Knowledge”	1
2. TERM LOAN, TERMS OF PAYMENT, SECURITY INTEREST, GUARANTY, MATURITY, ASSUMPTION.	1
2.1. Term Loan	1
2.2. Interest; Interest Rate.	2
2.3. Payments; Prepayments.	3
2.4. Grant of Security; Guaranty	4
2.5. Effect of Maturity	5
3. CONDITIONS TO ADVANCES.	5
3.1. Conditions Precedent to the Closing Date	5
3.2. Conditions Precedent to the Funding of each Term Loan	5
4. REPRESENTATIONS AND WARRANTIES.	6
4.1. Due Organization; Authorization; Power and Authority.	6
4.2. Subsidiaries; Investments.	6
4.3. Indebtedness; Liabilities	7
4.4. Collateral.	7
4.5. Binding Obligations; Perfected Liens.	7
4.6. Litigation	7
4.7. Financial Condition	8
4.8. Solvency	8
4.9. Regulatory Compliance	8
4.10. Patriot Act	8
4.11. OFAC	8
4.12. Anti-Bribery and Anti-Corruption Laws.	8
4.13. Tax Returns and Payments; Pension Contributions.	9
4.14. [Reserved].	9
4.15. SEC Reports	9
4.16. Transactions with Affiliates.	10
4.17. Environmental Condition	10
4.18. Material Contracts and Project Documents	10
4.19. Permits, Etc	11
4.20. Insurance	11
4.21. Condemnation	12
4.22. Contractors	12
4.23. Utilities and Access	12
4.24. Zoning	12
4.25. Absence of Material Adverse Effect	12

4.26.	Full Disclosure	13
4.27.	Intellectual Property	13
5.	AFFIRMATIVE COVENANTS.	14
5.1.	Compliance.	14
5.2.	Financial Statements, Reports, Certificates	14
5.3.	Taxes; Pensions	16 17
5.4.	Access to Collateral; Inspections; Books and Records	17
5.5.	Insurance.	18
5.6.	Disbursement Account	20 21
5.7.	Protection of Intellectual Property Rights	20 21
5.8.	Litigation Cooperation	21
5.9.	Formation or Acquisition of Subsidiaries	22
5.10.	Reserved Quarterly Consultation	22
5.11.	Environmental	22
5.12.	Incorporation of Terms	22 23
5.13.	Use of Proceeds	23
5.14.	Delivery of Material Contracts and Project Contracts	23 24
5.15.	Further Assurances	23 24
5.16.	Construction Related Covenants	23 24
5.17.	Collateral	26
5.18.	Full Disclosure	26 27
5.19.	Post-Closing Deliveries	26 27
5.20.	Separate Tax Parcel	27
5.21.	Compliance with Laws	27
6.	NEGATIVE COVENANTS.	27
6.1.	Indebtedness	27
6.2.	Liens	28
6.3.	Dispositions	27 28
6.4.	Mergers or Acquisitions	28
6.5.	Restricted Payments; Investments	28
6.6.	Restricted Debt Payments.	28
6.7.	Transactions with Affiliates	29
6.8.	Amendments to Material Contracts and Other Debt Documents	28 29
6.9.	Limitations on Negative Pledges and other Agreements	29
6.10.	Control Agreement	29
6.11.	Compliance	29
6.12.	OFAC; Patriot Act; Anti-Corruption Laws	30
6.13.	Financial Covenants	30
7.	EVENTS OF DEFAULT.	2930
7.1.	Payment Default	29 30
7.2.	Covenant Default	29 30
7.3.	Attachment; Levy; Restraint on Business	31
7.4.	Insolvency	30 31
7.5.	Other Indebtedness and Agreements	30 31

7.6.	Milestones	31
7.7.	Judgments	32
7.8.	Misrepresentations	31 32
7.9.	Governmental Approvals; Noncompliance; Lapse of Permits	31 32
7.10.	Loan Documents	31 32
7.11.	Change of Control	32 33
7.12.	Proceedings	32 33
7.13.	Bond Defaults	32 33
8.	RIGHTS AND REMEDIES.	32 33
8.1.	Rights and Remedies	32 33
8.2.	Remedies Cumulative	33
<u>8.3.</u>	<u>Equity Cure Rights</u>	<u>34</u>
9.	WAIVERS; INDEMNIFICATION.	33 34
9.1.	Demand; Protest; etc	33 34
9.2.	Lender's Liability for Collateral	33 35
9.3.	Indemnification	33 35
10.	NOTICES.	34 35
11.	CHOICE OF LAW AND VENUE; JURY TRIAL WAIVER; JUDICIAL REFERENCE PROVISION.	35 37
12.	ASSIGNMENTS; SUCCESSORS.	37 39
12.1.	Assignments	37 39
12.2.	Successors	37 39
13.	AMENDMENTS; WAIVERS.	37 39
13.1.	Amendments and Waivers	37 39
13.2.	No Waivers; Cumulative Remedies	37 39
14.	GENERAL PROVISIONS.	38 39
14.1.	Section Headings	38 39
14.2.	Interpretation	38 40
14.3.	Severability of Provisions	38 40
14.4.	Debtor-Creditor Relationship	38 40
14.5.	Counterparts; Electronic Execution	39 41
14.6.	Revival and Reinstatement of Obligations; Certain Waivers	39 41
14.7.	Confidentiality	39 41
14.8.	Survival	40 42
14.9.	Integration	41 43
14.10.	No Setoff	41 43

EXHIBIT AND SCHEDULES

Exhibit A	Form of Security Agreement
Exhibit B	Form of Guaranty Agreement
Exhibit C	Form of Collateral Assignment Agreement
Exhibit D	Form of Promissory Note
Exhibit E	Form of Revolver Intercreditor Agreement
Exhibit F	Form of Borrowing Request
Exhibit G	Form of Title Company “Date Down” Endorsement for Georgia Property
Exhibit H	Form of Compliance Certificate

Schedule 1.1	Definitions
Schedule 1.1(c)	Contracts to be Collaterally Assigned
Schedule 1.1(d)	Excluded Accounts
Schedule 1.1(p)	Project Budget
Schedule 1.1(t)	Term Loan Budget
Schedule 3.1	Conditions Precedent to the Closing Date
Schedule 3.2	Conditions Precedent to the Funding of each Term Loan
Schedule 4.2(b)	Affiliates of Loan Parties
Schedule 4.3	Existing Liabilities
Schedule 4.16(a)	Affiliate Transactions
Schedule 4.16(b)	Assets and Services
Schedule 4.17	Environmental Conditions
Schedule 4.18	Material Contracts and Project Documents
Schedule 4.19	Permits
Schedule 4.22	Contractors
Schedule 5.16(b)	Subcontractors
Schedule 7.6	Milestones

LOAN AGREEMENT

THIS LOAN AGREEMENT (this “Agreement”), is entered into as of November 28, 2022 by and between **GENERAL MOTORS HOLDINGS LLC** (“General Motors” together with its successors and assigns, the “Lender”, as that term is hereinafter further defined), **ASPEN AEROGELS GEORGIA, LLC**, a Georgia limited liability company (“Borrower”), **ASPEN AEROGELS, INC.**, a Delaware corporation (the “Parent”), **ASPEN AEROGELS RHODE ISLAND, LLC**, a Rhode Island limited liability company (“Aspen RI”) and the other Guarantors from time to time party hereto.

Borrower will be constructing and equipping an aerogel manufacturing plant in Bulloch County, Georgia (the “Project”) to produce barrier blankets for use in insulation components that are used in battery modules for electric vehicles. Borrower has requested, and the Lender has agreed to provide, a multi-draw senior secured loan on the terms and conditions set forth in this Agreement, the proceeds of which will be used, together with capital contributions to Borrower from Parent and the proceeds of other financing, for the construction of and equipment for the first phase of the Project.

The parties agree as follows:

1. DEFINITIONS AND CONSTRUCTION.

1.1.**Definitions.** Capitalized terms used in this Agreement shall have the meanings specified therefor in Schedule 1.1.

1.2.**Code.** Any terms used in this Agreement that are defined in the Code shall be construed and defined as set forth in the Code unless otherwise defined herein; provided, that to the extent that the Code is used to define any term herein and such term is defined differently in different Articles of the Code, the definition of such term contained in Article 9 of the Code shall govern.

1.3.**Schedules and Exhibits.** All of the schedules and exhibits attached to this Agreement shall be deemed incorporated herein by reference.

1.4.**Definition of “Knowledge”.** For purposes of the Loan Documents, whenever a representation or warranty is made to any Loan Party’s “knowledge” or “awareness”, to the “best of any Loan Party’s knowledge”, or with a similar qualification, knowledge or awareness means the actual knowledge, after reasonable investigation, of any Responsible Officer.

2. TERM LOAN, TERMS OF PAYMENT, SECURITY INTEREST, GUARANTY, MATURITY, ASSUMPTION.

2.1.**Term Loan.**

(a) **Term Loan Commitment.** Subject to the terms and conditions of this Agreement and the other Loan Documents, including the conditions precedent set forth in Section 3.1 of this Agreement, the Lender hereby agrees to make Term Loans to Borrower in an aggregate amount of up to \$100,000,000 (the “Term Loan Commitment”) after ~~January 1, 2023~~the date that

is twelve (12) months prior to the date of Start of Production at the Georgia Facility, and before the Term Loan Commitment Expiration Date.

(b) **Term Loan Funding.**

(i) Subject to the conditions precedent set forth in Section 3.2 of this Agreement, the Lender agrees to fund Term Loans following not less than five (5) Business Days (or such shorter period as the Lender may agree in writing (including via email)) advance written notice substantially in the form of Exhibit F (such notice, a “Borrowing Request”) from Borrower, together with evidence of the satisfaction of the conditions precedent set forth in Section 3.2 of this Agreement (each, a “Term Loan” and collectively, the “Term Loans”); provided, that: (w) the aggregate amount of all such Term Loans shall not exceed the Term Loan Commitment, (x) the Lender shall not have any obligation to fund any Term Loan more frequently than once in every month and (y) each borrowing shall be in a minimum principal amount of \$~~5~~20,000,000 (or such lesser aggregate amount as is equal to the unused Term Loan Commitment at such time).

(ii) The proceeds of each Term Loan shall be deposited in the Disbursement Account and, to the extent intended to be used for the construction of the Georgia Facility pursuant to the applicable Term Loan Budget, disbursed to Borrower by the Title Company, all in accordance with Section 5.16(g).

(c) **No Re-borrowing.** Any principal amount of the Term Loan that is repaid or prepaid may not be re-borrowed. All principal of, interest on, and other amounts payable in respect of the Term Loan shall constitute Obligations hereunder.

2.2. Interest; Interest Rate.

(a) **Interest Rates.** Except as provided in Section 2.2(b) and Section 2.2(d), interest shall accrue on the Obligations at (i) Term SOFR plus 8.00% (the “Cash Interest Rate”) and shall be payable on a semi-annual basis in arrears on the first day of each January and July following the date of incurrence of the applicable Obligation (the “Interest Payment Date”) or, at the option of Borrower or (ii) at the election of Borrower ~~pursuant to the penultimate sentence of this clause (a)~~ in accordance with the PIK Election process, prior to the delivery of a certificate of occupancy issued by Bulloch County, Georgia (the “Certificate of Occupancy”), Term SOFR plus 9.00% (the “PIK Interest Rate” and together with the Cash Interest Rate, the “Interest Rates”, and each an “Interest Rate”) and shall be payable in kind (“PIK Interest”) on a semi-annual basis in arrears on each Interest Payment Date by increasing the outstanding principal amount of the Term Loan, and thereby the Note, without any further action on the part of Borrower or the Lender. Borrower shall notify the Lender in writing whether it elects to pay PIK Interest at least five (5) Business Days (or such shorter period as the Lender may agree in writing (including via email)) prior to each Interest Payment Date (each such election, a “PIK Election”). All interest hereunder shall be computed on the basis of a year of 360 days. Term SOFR shall be determined by the Lender and such determination shall be conclusive absent manifest error.

(b) **Default Interest.** Following and during the continuance of an Event of Default, upon at least two (2) Business Days’ prior written notice to Borrower and not to be delivered earlier than five (5) Business Days after the Event of Default (which notice once

delivered shall be retroactive), interest on the Obligations shall accrue at the applicable Interest Rate plus 2.00% ("Default Interest") and shall at the option of the Lender, be payable in kind by increasing the outstanding principal amount of the Term Loan, and thereby the Note, on each Interest Payment Date, in arrears.

(c) **Payment of Interest.** Interest hereunder shall be due and payable or capitalized in accordance with the terms hereof before and after any judgment, and before and after the commencement of any Insolvency Proceeding.

(d) **Intent to Limit Charges to Maximum Lawful Rate.** In no event shall any Interest Rate or rates payable under this Agreement, plus any other amounts paid in connection herewith, exceed the highest rate permissible under any law that a court of competent jurisdiction shall, in a final determination, deem applicable. Borrower and the Lender, in executing and delivering this Agreement, intend legally to agree upon the rate or rates of interest and manner of payment stated within it; provided, that, anything contained herein to the contrary notwithstanding, if such rate or rates of interest or manner of payment exceeds the maximum allowable under applicable law, then, *ipso facto*, as of the date of this Agreement, Borrower is and shall be liable only for the payment of such maximum amount as is allowed by law, and payment received from Borrower in excess of such legal maximum, whenever received, shall be applied to reduce the principal balance of the Obligations to the extent of such excess.

2.3. **Payments; Prepayments.**

(a) **Payments by Borrower.** All outstanding principal and interest on the Term Loan and the other Obligations shall be immediately due and payable in immediately available funds, without notice or demand, on the Maturity Date as directed by the Lender in accordance with Section 2.3(b).

(b) **Application.**

All payments remitted to the Lender and all proceeds of Collateral received by the Lender shall be applied as follows:

(i) first, to pay any Lender Expenses (including cost or expense reimbursements) or indemnities then due to the Lender under the Loan Documents, until Paid in Full;

(ii) second, to pay interest accrued in respect of the Term Loan until Paid in Full;

(iii) third, to pay the principal of the Term Loan until Paid in Full;

(iv) fourth, to pay any other Obligations; and

(v) fifth, to Borrower (to be wired to the account designated by Borrower in writing) or such other Person entitled thereto under applicable law.

(c) **Prepayments.**

(i) Optional. Borrower may, upon not less than three (3) Business Days (or such shorter period as the Lender may agree in writing (including via email)) prior notice to the Lender, prepay the outstanding Term Loans and terminate any unused Term Loan Commitment in part or in full. Any prepayment of a Term Loan shall be accompanied by all accrued and unpaid interest (including PIK Interest) on such Term Loan and disbursed in accordance with Section 2.3(b).

(ii) Dispositions: If at any time prior to the Maturity Date, any Loan Party shall receive the Net Proceeds of any voluntary or involuntary sale or disposition of assets of any Loan Party (including Net Proceeds of insurance or arising from casualty losses or condemnations and payments in lieu thereof, but excluding any Transfer permitted pursuant to Section 6.3), in excess of \$1,000,000 in any calendar year, Borrower shall, within two (2) Business Days of the date of receipt (or such longer period as the Lender may agree in writing (including via email)) by any Loan Party of such Net Proceeds, prepay the outstanding principal amount of the Obligations in accordance with Section 2.3(b) in an amount equal to 100% of such excess Net Proceeds received by such Person in connection with such sales or dispositions.

(iii) Indebtedness and Equity. Within two (2) Business Days (or such longer period as the Lender may agree in writing (including via email)) of the date of incurrence by any Loan Party of any Indebtedness (other than under (A) any Additional Financing Facility, (B) Indebtedness permitted under clauses (a), (b), (d), (g) (only to the extent that the terms of Indebtedness incurred under clause (g) do not allow Borrower to use the proceeds of such Indebtedness to prepay the Term Loans), (h), (k), (m) and (n) of the “Permitted Indebtedness” definition or other Permitted Indebtedness (including pursuant to clause (e) of the definition thereof, the proceeds of which are used solely for the Project)) or receipt of proceeds from any equity issuance by Parent (other than proceeds that shall be used solely for the Project) in excess of \$300,000,000 in the aggregate during the term of this Agreement, Borrower shall prepay the outstanding principal amount of the Obligations in accordance with Section 2.3(b) in an amount equal to 100% of the Net Proceeds received by such Person in connection with such incurrence or contribution; provided, that any undrawn commitments under a Revolving Loan Agreement shall be considered “Indebtedness” for purposes of this clause (c)(iii). The provisions of this Section 2.3(c)(iii) shall not be deemed to be implied consent to any such incurrence otherwise prohibited by the terms of this Agreement.

(iv)Equity Cure. The Borrower shall prepay the Obligations with 100% of the proceeds of any Cure Amount within one (1) Business Day (or such longer period as the Lender may agree in writing (including via email)) of issuance thereof.

2.4. Grant of Security; Guaranty. To secure the prompt payment of all the Obligations and to secure the prompt performance by each Loan Party of its covenants and obligations in this Agreement and in the other Loan Documents:

(i) pursuant to that certain Security Agreement attached to this Agreement as Exhibit A (the “Security Agreement”), the Loan Parties shall grant to the Lender a continuing first priority Lien (subject to Permitted Liens) on all presently existing and hereafter acquired or

arising Collateral (including any real property); provided, that in connection with the incurrence of the Revolving Loan Agreement and the execution of the Revolver Intercreditor Agreement, the Lender shall subordinate its Lien on accounts receivable and inventory (inclusive of the proceeds thereof) on the terms set forth therein;

(ii) pursuant to that certain Guaranty Agreement (the “Guaranty”) attached to this Agreement as Exhibit B, the Guarantors shall guarantee Borrower’s Obligations;

(iii) pursuant to that certain Mortgage, Aspen RI shall grant the Lender a continuing first priority Lien (subject to Permitted Liens) in the Rhode Island Property;

(iv) pursuant to that certain Leasehold Deed to Secure Debt, Borrower shall grant to the Lender a continuing first priority Lien (subject to Permitted Liens) in its rights under such deed in connection with the Georgia Property;

(v) pursuant to that certain Revenue Bond Pledge Agreement, Borrower shall grant to the Lender a continuing first priority Lien in its rights under the Revenue Bond;

(vi) pursuant to the Collateral Assignment Agreements, Borrower shall grant to the Lender a continuing first priority Lien (subject to Liens permitted pursuant to clause (c)(ii) of the definition of “Permitted Liens”) in its rights under the agreements and contracts covered by such Collateral Assignment Agreements; and

(vii) pursuant to a Control Agreement, Borrower shall grant to the Lender a continuing first priority Lien (subject to Liens permitted pursuant to the applicable Control Agreement) in the Disbursement Account.

2.5. Effect of Maturity. No termination of the obligations of the Lender shall relieve or discharge Borrower or the other Loan Parties of their respective duties, or covenants hereunder or under any other Loan Document and the Lender’s Liens in the Collateral shall continue to secure the Obligations and shall remain in effect until all Obligations have been Paid in Full. When all of the Obligations have been Paid in Full, the Lender will, at Borrower’s sole expense, execute and deliver any termination statements, lien releases, discharges of security interests, and other similar discharge or release documents (and, if applicable, in recordable form) as are reasonably necessary or requested in writing by Borrower to release, as of record, the Lender’s Liens and all notices of Liens previously filed by the Lender and shall deliver any possessory Collateral.

3. CONDITIONS TO ADVANCES.

3.1. Conditions Precedent to the Closing Date. The Lender’s Term Loan Commitment is subject to the fulfillment, to the satisfaction of the Lender, of each of the conditions precedent set forth on Schedule 3.1.

3.2. Conditions Precedent to the Funding of each Term Loan. The obligation of the Lender to fund any Term Loan shall be subject to the limitations in Section 2.1(b) and to the fulfillment, to the satisfaction of the Lender, of each of the conditions precedent set forth on Schedule 3.2.

4. **REPRESENTATIONS AND WARRANTIES.**

In order to induce the Lender to enter into this Agreement, each of Borrower, Aspen RI and/or Parent, as applicable, makes the following representations and warranties to the Lender which shall be accurate as of the date of this Agreement and the date of the advance of each Term Loan:

4.1. **Due Organization; Authorization; Power and Authority.**

(a) Each Loan Party and each of such Loan Party's respective Subsidiaries is duly existing and in good standing (and with respect to any Loan Party organized in the U.S., as a Registered Organization) in its jurisdiction of organization and each is qualified and licensed to do business and each is in good standing (where such concept is applicable) in any jurisdiction in which the conduct of each of its business or its ownership of property requires that it be qualified, except where the failure to do so could not reasonably be expected to cause a Material Adverse Effect. In connection with this Agreement, Borrower has delivered to the Lender a completed certificate signed by the Loan Parties, entitled "Perfection Certificate", in form reasonably acceptable to the Lender. Borrower and Parent represent and warrant to the Lender that (a) each Loan Party's exact legal name is that indicated on the Perfection Certificate and on the signature page hereof; (b) each Loan Party is an organization of the type and is organized in the jurisdiction set forth in the Perfection Certificate; (c) the Perfection Certificate accurately sets forth each Loan Party's organizational identification number or accurately states that such Loan Party has none; (d) the Perfection Certificate accurately sets forth each Loan Party's place of business, or, if more than one, its chief executive office as well as such Loan Party's mailing address (if different than its chief executive office); (e) except as set forth on the Perfection Certificate, each Loan Party (and each of its predecessors) has not, in the past five (5) years, changed its jurisdiction of organization, organizational structure or type, or any organizational number assigned by its jurisdiction; and (f) all other information set forth on the Perfection Certificate pertaining to each Loan Party and each of their respective Subsidiaries is accurate and complete in all material respects.

(b) The execution, delivery and performance by each Loan Party of the Loan Documents to which it is a party have been duly authorized, and do not (i) conflict with any of such Loan Party's Governing Documents, (ii) contravene, conflict with, constitute a default under or violate any material Requirement of Law, (iii) contravene, conflict or violate any applicable order, writ, judgment, injunction, decree, determination or award of any Governmental Authority by which such Loan Party or any of its Subsidiaries or any of their property or assets may be bound or affected, (iv) require any action by, filing, registration, or qualification with, or Governmental Approval from, any Governmental Authority (except such Governmental Approvals which have already been obtained and are in full force and effect) or (v) constitute an event of default under any Project Contract or Material Contract by which such Loan Party is bound or the Georgia Lease.

4.2. **Subsidiaries; Investments.**

(a) Set forth in the Perfection Certificate is a complete and accurate description of the authorized Equity Interests of each Loan Party as of the Closing Date, by class, and a description of each such class (including economic interests, voting interests, and governance

rights), the number of shares of each such class that are issued and outstanding and the holder of such Equity Interests. Neither Borrower nor Aspen RI is subject to any obligation (contingent or otherwise) to repurchase or otherwise acquire or retire any shares of its Equity Interests or any security convertible into or exchangeable for any of its Equity Interests (other than the Convertible Notes or any other permitted Additional Financing Facility).

(b) As of the Closing Date, (i) neither Borrower nor Aspen RI have direct or indirect Subsidiaries and (ii) set forth on Schedule 4.2(b) is a complete and accurate list of each Loan Party's Affiliates, including direct and indirect parent entities and the ownership of such entity.

4.3. Indebtedness; Liabilities. As of the Closing Date, the Loan Parties have no Indebtedness for borrowed money or other material liabilities (direct or contingent), other than the ~~Convertible Notes~~ or as otherwise disclosed on Schedule 4.3 or in the financial statements filed by Parent with the SEC on November 7, 2022.

4.4. Collateral.

(a) Each Loan Party has good title to, has rights in, and the power to transfer each item of the Collateral upon which it purports to grant a Lien hereunder (including Aspen RI being the sole owner of the fee interest in the Rhode Island Property), free and clear of any and all Liens, except Permitted Liens. No Loan Party has Collateral Accounts at or with any bank or financial institution, other than the Collateral Accounts described in the Perfection Certificate and the Disbursement Account, or of which Borrower has given the Lender notice and taken such actions as are necessary to give the Lender a perfected security interest therein, to the extent required by Section 5.9 and subject to any applicable Intercreditor Agreement.

(b) As of the Closing Date, the Collateral is not in the possession of any third party bailee (such as a warehouse), except as otherwise provided in the Perfection Certificate.

4.5. Binding Obligations; Perfected Liens.

(a) Each Loan Document to which any Loan Party is a party has been duly executed and delivered by such Loan Party and is the legally valid and binding obligation of such Loan Party, enforceable against such Loan Party in accordance with its respective terms, except as enforcement may be limited by equitable principles or by bankruptcy, insolvency, reorganization, moratorium, or similar laws relating to or limiting creditors' rights generally.

(b) The Mortgage and Leasehold Deed to Secure Debt when duly executed, delivered and recorded and the other Security Documents when duly executed, delivered and recorded or filed, as appropriate, and when the financing statements have been filed against each Loan Party in the appropriate filing office, will constitute perfected first priority liens against the Collateral, prior to all other liens and encumbrances, including those which may hereafter accrue, except for Permitted Liens.

4.6. Litigation. There are no actions or proceedings pending or, to the knowledge of any Responsible Officer, threatened in writing by or against any Loan Party or any of their respective Subsidiaries involving more than, individually or in the aggregate, \$750,000, except as otherwise disclosed in Parent's SEC Reports.

4.7.**Financial Condition.** The consolidated financial statements for any Loan Party delivered to the Lender pursuant to Section 5.2 fairly present in all material respects such Loan Party's consolidated financial condition and such Loan Party's consolidated results of operations.

4.8.**Solvency.** The Loan Parties on a consolidated basis are Solvent.

4.9.**Regulatory Compliance.** No Loan Party is an "investment company" or a company "controlled" by an "investment company" under the Investment Company Act of 1940, as amended. No Loan Party is engaged as one of its important activities in extending credit for margin stock (under Regulations X, T and U of the Federal Reserve Board of Governors). Each Loan Party (a) has complied in all material respects with all Requirements of Law which would reasonably be expected to cause a Material Adverse Effect, and (b) has not violated any Requirements of Law the violation of which could reasonably be expected to cause a Material Adverse Effect. Each Loan Party and each of its Subsidiaries have obtained all consents, approvals and authorizations of, made all declarations or filings with, and given all notices to, all Government Authorities that are necessary to continue their respective businesses as currently conducted, the absence of which could reasonably be expected to have a Material Adverse Effect on such Loan Party's business.

4.10.**Patriot Act.** To the extent applicable, each Loan Party is in compliance with the (a) Trading with the Enemy Act, as amended, and each of the foreign assets control regulations of the United States Treasury Department (31 CFR, Subtitle B, Chapter V, as amended) and any other enabling legislation or executive order relating thereto, and (b) Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism (USA Patriot Act of 2001) (the "Patriot Act").

4.11.**OFAC.** Each Loan Party is in compliance with economic and trade sanctions administered and enforced by OFAC, the U.S. Department of State, and any other relevant sanctions authority ("Sanctions"). None of the Loan Parties nor any of their respective directors, officers, employees, or agents (a) is a Sanctioned Person or a Sanctioned Entity, (b) has assets located in Sanctioned Entities, or (c) directly or indirectly engages in dealings with, or derive revenues from investments in or transactions with Sanctioned Persons or Sanctioned Entities.

4.12.**Anti-Bribery and Anti-Corruption Laws.**

(a) Each Loan Party is in compliance with the U.S. Foreign Corrupt Practices Act of 1977, as amended, and the anti-bribery and anti-corruption laws of those jurisdictions in which they do business (collectively, the "Anti-Corruption Laws").

(b) No Loan Party has at any time:

(i) offered, promised, paid, given, or authorized the payment or giving of any money, gift or other thing of value, directly or indirectly, to or for the benefit of any employee, official, representative, or other Person acting on behalf of any foreign (i.e., non-U.S.) Governmental Authority thereof, or of any public international organization, or any foreign political party or official thereof, or candidate for foreign political office (collectively, "Foreign Official"), for the purpose of: (A) influencing any act or decision of such Foreign Official in his, her, or its official capacity; or (B) inducing such Foreign Official to do, or omit to do, an act in

violation of the lawful duty of such Foreign Official; or (C) securing any improper advantage, in order to obtain or retain business for, or with, or to direct business to, any Person; or

(ii) acted or attempted to act in any manner which would subject Borrower to liability under any Anti-Corruption Law.

(c) There are, and have been, no allegations, investigations or inquiries with regard to a potential violation of any Anti-Corruption Law by any Loan Party or any of their current or former directors, officers, employees, stockholders or agents, or other Persons acting or purporting to act on their behalf.

(d) Each Loan Party has adopted, implemented and maintains anti-bribery and anti-corruption policies and procedures that are reasonably designed to ensure compliance with the Anti-Corruption Laws.

4.13. Tax Returns and Payments; Pension Contributions.

(a) Each Loan Party has timely filed all required tax returns and reports, and such Loan Party has timely paid all foreign, federal, state and material local taxes, assessments, deposits and contributions owed by such Loan Party, except to the extent such taxes are being contested in good faith by appropriate proceedings promptly instituted and diligently conducted, so long as such reserve or other appropriate provision, if any, as shall be required in conformity with GAAP shall have been made therefor.

(b) No Loan Party is aware of any claims or adjustments proposed for any of such Loan Party's prior tax years which could result in additional taxes in an amount in excess of \$750,000 becoming due and payable by such Loan Party. Each Loan Party has paid all amounts necessary to fund all present pension, profit sharing and deferred compensation plans in accordance with their terms, and no Loan Party has withdrawn from participation in, and has not permitted partial or complete termination of, or permitted the occurrence of any other event with respect to, any such plan which could reasonably be expected to result in any liability of such Loan Party in an amount in excess of \$750,000, including any liability to the Pension Benefit Guaranty Corporation or its successors or any other governmental agency.

4.14. [Reserved].

4.15. **SEC Reports**. The Parent has filed all reports, schedules, forms, statements and other documents required to be filed by the Parent under the Exchange Act for the two (2) years preceding the date hereof, as and when required by the Exchange Act (the foregoing materials, including the exhibits thereto and documents incorporated by reference therein, being collectively referred to herein as the "**SEC Reports**"). The SEC Reports at the time they were filed complied in all material respects with the Securities Act or the Exchange Act, as applicable, and the applicable rules and regulations of the Securities and Exchange Commission (the "**SEC**") thereunder. There are no contracts, agreements or other documents that are required to be described in the SEC Reports and/or to be filed as exhibits thereto that are not described, in all material respects, and/or filed as required by the Securities Act or Exchange Act, as applicable. There has not been any material change or amendment to, or any waiver of any material right under, any such contract or agreement that has not been described in and/or filed as an exhibit to the SEC

Reports to the extent required by the Securities Act or Exchange Act, as applicable. The SEC Reports taken as a whole do not contain an untrue statement of a material fact or omit to state a material fact required to be stated therein or necessary to make the statements therein, in the light of the circumstances under which they were made, not misleading.

4.16. Transactions with Affiliates.

(a) No Loan Party is a party to a transaction with any of its Affiliates (other than transactions among the Loan Parties, the Convertible Notes, the Equity Investment or any other Additional Financing Facility with Koch Strategic Platforms (and its Affiliates) otherwise constituting “Permitted Indebtedness”), other than transactions (i) entered into prior to the Closing Date (x) in the ordinary course of business each not to exceed \$500,000 individually, or in a series of related transactions, or (y) disclosed on Schedule 4.16(a) to the extent that such transactions were approved by a majority of the Disinterested Board Members of the Parent’s Board of Directors or by Parent’s audit committee in accordance with the requirements for related party transactions under the New York Stock Exchange rules then in effect or (ii) entered into after the Closing Date in accordance with, and subject to, Section 6.7.

(b) Schedule 4.16(b) lists all assets owned by any Affiliate of any Loan Party, other than Affiliates that are Loan Parties, that are or will be used by Borrower connection with the Project or in the Parent’s performance under the GM Purchase Contracts and all services to be provided by any Affiliate, other than Affiliates that are Loan Parties, of such Loan Party (provided, that such assets or services are disclosed on a supplement to Schedule 4.16(b)).

4.17. Environmental Condition. Except for matters as set forth on Schedule 4.17 for the past five (5) years which, either individually or in the aggregate, could not reasonably be expected to result in a material liability, (a) no Property has been used by any Loan Party or, to the knowledge of any Loan Party, after due inquiry, by previous owners or operators for the disposal of, or to produce, store, handle, treat, release, or transport, any Hazardous Materials, where such disposal, production, storage, handling, treatment, release or transport was in violation of any applicable Environmental Law, (b) to the knowledge of the Loan Parties, after due inquiry, no Property is designated or identified pursuant to any environmental protection statute as a Hazardous Materials disposal site, (c) no Loan Party has received or is subject to any outstanding written order, consent decree, or settlement agreement with any Person relating to any Environmental Law or Environmental Liability with respect to any Property and (d) (x) no Property is the subject of any pending, or to any Loan Party’s knowledge threatened (in writing), Environmental Action, and (y) no Loan Party has received any written notice that any Property is subject to a pending investigation or inquiry by a Governmental Authority regarding a potential violation of Environmental Law, in each case, which remains unresolved. No Loan Party has received written notice that a Lien arising under any Environmental Law has attached to any Property.

4.18. Material Contracts and Project Documents.

(a) Set forth on Schedule 4.18 is a complete and accurate list of all Material Contracts and Project Documents to which any Loan Party is a party as of the Closing Date, showing the parties and subject matter thereof and amendments and modifications thereto.

Schedule 4.18 (as may be supplemented by Borrower in writing from time to time) also includes a list of all Material Contracts and Project Documents entered into, or anticipated to be entered into, in connection with the Project and the anticipated timeline for execution of each such Project Document or Material Contract as applicable, and specifically identifies any such Material Contract or Project Document that has been or is anticipated to be entered into by Borrower with an Affiliate. The Georgia Lease, the Revenue Bond, each Material Contract and each Project Document (i) is in full force and effect and is binding upon and enforceable against Borrower and, to the best knowledge of Borrower, all other parties thereto in accordance with its terms, except as enforcement may be limited by equitable principles or by bankruptcy, insolvency, reorganization, moratorium, or similar laws relating to or limiting creditors' rights generally, (ii) has not been otherwise amended or modified in a manner materially adverse to the Lender (in its capacity as such), (unless otherwise consented to by the Lender), and (iii) is not in default due to the action of Borrower or, to the best knowledge of Borrower, any other party thereto which default would result in an Event of Default under Section 7.5(a).

(b) Each Material Contract or Project Document to which any Guarantor is a party in existence as of the Closing Date (and with regard to any Project Document or Material Contract entered into after the Closing Date, following its execution) (i) is in full force and effect and is binding upon and enforceable against such Guarantor and, to the best knowledge of such Guarantor, all other parties thereto in accordance with its terms, except as enforcement may be limited by equitable principles or by bankruptcy, insolvency, reorganization, moratorium, or similar laws relating to or limiting creditors' rights generally, (ii) has not been otherwise amended or modified in a manner materially adverse to the Lender (in its capacity as such), (unless otherwise consented to by the Lender), and (iii) is not in default due to the action of such Guarantor or, to the knowledge of such Guarantor, any other party thereto which default would result in an Event of Default under Section 7.5(a).

4.19. Permits, Etc.

(a) Borrower has, or is in the process of acquiring, and is in compliance with, all Project Permits. Schedule 4.19 (as may be supplemented by Borrower in writing from time to time) sets forth a list of all Project Permits. Borrower has no knowledge of any existing basis that is reasonably likely to delay the issuance of any Project Permit.

(b) Each Loan Party has, or is in the process of acquiring, and is in compliance with, all Permits required for such Loan Party lawfully to own, lease or operate the business to be conducted by such Loan Party.

(c) To the knowledge of the Loan Parties, no condition exists or event has occurred which, in itself or with the giving of notice or lapse of time or both, would reasonably be expected to result in the suspension, revocation, impairment, forfeiture or non-renewal of any Permit.

4.20. Insurance. Each Loan Party maintains the insurance and required services and financial assurance as required by law and by Section 5.5 (including with regard to the commencement date of such insurance requirements).

4.21.**Condemnation.** There are no proceedings pending which have been served upon any Loan Party, or, to the Loan Parties' knowledge, threatened in writing, to acquire any Property or any interest therein through any power of condemnation or eminent domain, or to enjoin or similarly prevent the construction or use of the Project.

4.22.**Contractors.** The names and addresses of all Contractors who were engaged by Borrower or Parent in connection with the Project prior to the date of this Agreement, and/or are engaged or employed in connection with the Project on the date of this Agreement, and the amount or estimated amount of their respective contracts with the General Contractor are set forth on Schedule 4.22 (as may be supplemented by Borrower in writing from time to time), including the amounts owed to them as of the Closing Date.

4.23.**Utilities and Access.** Borrower has received adequate certifications or commitments from all appropriate utility companies and Governmental Authorities to determine that (i) all utility services necessary for (A) the construction of the Improvements are available at the boundaries of the Georgia Property and (B) for the subsequent use of the Project have been included in the Plans and will be completed in sufficient time to allow Borrower to comply with the Milestones; (ii) such utilities are adequate to serve the Improvements for their intended use; and (iii) there are no conditions limiting the use of such utilities, other than normal charges of the utility supplier. All steps appropriate through the applicable stages of the development and construction of the Improvements have been taken by Borrower with utility companies and applicable Governmental Authorities to insure the complete construction, installation and use of such utilities in accordance with the Plans. All streets and easements necessary for access to and the occupancy of the Improvements are available to the boundaries of the Property.

4.24.**Zoning.**

(a) (i) The construction of the Project and the use and operation of the Georgia Property as indicated herein conforms to and complies with all material requirements of such public and private conditions, restrictions, reservations and zoning ordinances, and (ii) the construction of the Project and the use and operation of the Georgia Property complies in all material respects with all rules, regulations and requirements of all relevant Governmental Authorities, and does not result in any material violation of any law, ordinance, regulation or restriction now existing and affecting the Georgia Property.

(b) (i) To Aspen RI's knowledge, the use and operation of the Rhode Island Property as indicated herein conforms to and complies with all material requirements of such public and private conditions, restrictions, reservations and zoning ordinances, and (ii) the use and operation of the Rhode Island Property complies in all material respects with all rules, regulations and requirements of all relevant Governmental Authorities, and does not result in any material violation of any law, ordinance, regulation or restriction now existing and affecting the Rhode Island Property.

4.25.**Absence of Material Adverse Effect.** Since December 31, 2021, there has been no change, event, occurrence or circumstance, either individually or in the aggregate, that has had or would reasonably be expected to have a Material Adverse Effect on any of the Loan Parties.

4.26.Full Disclosure.

(a) All written information with respect to the Loan Parties in any certificate or written statement given to the Lender, including the information contained in each Budget (including, to the Loan Parties knowledge based on information believed by the Loan Parties to be reasonable and accurate at the time furnished, both hard costs and soft costs associated therewith, and all costs for obtaining payment and performance bonds with regard to Major Subcontracts as required by Section 5.16) and in any document furnished to the Lender by any Loan Party in connection with this Agreement or any other Loan Document, when taken as a whole, did not, when furnished, contain any untrue statement of a material fact or omit to state a material fact necessary in order to make the statements contained therein not materially misleading in light of the circumstances under which such statements are made; provided, that this clause (a) shall not apply to projections, forecasts, financial estimates and other forward-looking information and/or other projected information, information of a general economic or industry-specific nature provided in any Budget (collectively, "Projections").

(b) The Projections have been prepared in good faith based upon assumptions believed by Borrower to be reasonable at the time furnished (it being recognized that such Projections are not to be viewed as facts and are subject to significant uncertainties and contingencies many of which are beyond Borrower's control, that no assurance can be given that any particular financial projections will be realized, that actual results may differ from projected results and that such differences may be material).

4.27.Intellectual Property.

(a) Each Loan Party is the sole owner of the Intellectual Property which it owns or purports to solely own, except for (i) non-exclusive licenses granted to its customers in the ordinary course of business, (ii) over-the-counter or off-the-shelf software that is commercially available to the public, (iii) any rights the U.S. Government may have in connection with such Loan Party's contracts with the U.S. Government entities as noted in the Perfection Certificate; (iv) material Intellectual Property licensed to such Loan Party and (v) non-material Intellectual Property licensed to such Loan Party. Except as otherwise disclosed in SEC Reports, (A) each patent which such Loan Party owns or purports to own and which is material to such Loan Party's business is valid and enforceable, and (B) no part of the Intellectual Property such Loan Party owns or purports to own and which is material to such Loan Party's business has been judged invalid or unenforceable, in whole or in part. To the best of any Loan Party's knowledge, no claim has been made that any part of the Intellectual Property owned or purported to be owned by such Loan Party violates the Intellectual Property rights of any third party except to the extent such claim would not have a Material Adverse Effect on such Loan Party's business, taken as a whole.

(b) Except as noted on the Perfection Certificate, no Loan Party is a party to, nor is it bound by, any Restricted License.

5. **AFFIRMATIVE COVENANTS.**

Each of Borrower, Parent and/or Aspen RI, as applicable, covenants and agrees that, until the Obligations are Paid in Full it shall do the following:

5.1. **Compliance.**

(a) Maintain its and all of its Subsidiaries' legal existence and good standing (where such concept is applicable) in their respective jurisdictions of formation and maintain qualification in each jurisdiction.

(b) Maintain all Governmental Approvals necessary for the performance by each Loan Party of its obligations under the Loan Documents to which it is a party and for the grant of a security interest to the Lender in the Collateral. Each Loan Party shall, upon request from the Lender, promptly provide copies of any such obtained Governmental Approvals to the Lender.

5.2. **Financial Statements, Reports, Certificates.** Provide the Lender with the following:

(a) as soon as available, and in any event within forty-five (45) days after the end of each ~~fiscal quarter of Parent~~ (or such longer period as the Lender may agree in writing (including via email)), company prepared consolidated balance sheet ~~and income statement covering~~ as of the last day of such Fiscal Quarter and related statements of income and cash flows for the Parent and its Subsidiaries' on a consolidated operations for such quarter basis for such Fiscal Quarter and the elapsed portion of the fiscal year and comparative figures for the same periods in the immediately preceding fiscal year, prepared in accordance with GAAP consistently applied, certified by a Responsible Officer and in a form reasonably acceptable to the Lender;

(b) as soon as available, and in any event within one hundred fifty (150) days following the end of Parent's fiscal year (or such longer period as the Lender may agree in writing (including via email)), audited consolidated financial statements prepared ~~under~~ in accordance with GAAP, consistently applied, together with an opinion without a "going concern" qualification (except as resulting from the impending maturity of any Indebtedness) on the financial statements from KPMG LLP, or such other independent certified public accounting firm reasonably acceptable to the Lender;

(c) within thirty (30) days after the end of each fiscal year of Borrower (or such longer period as the Lender may agree in writing (including via email)), or more frequently as updated by Parent's board of directors, (A) annual operating budgets (including income statements, balance sheets and cash flow statements, by month) for the upcoming fiscal year of Parent and its Subsidiaries, and (B) annual financial projections for the following fiscal year (on a quarterly basis) as approved by Parent's board of directors, together with any related business forecasts used in the preparation of such annual financial projections;

(d) for so long as Parent is subject to the reporting requirements under the Exchange Act within five (5) days of filing (or such longer period as the Lender may agree in writing (including via email)), copies of all periodic and other reports, proxy statements and other materials filed by Parent with the SEC, any Governmental Authority succeeding to any or all of

the functions of the SEC or with any national securities exchange, or distributed to its shareholders, as the case may be. Documents required to be delivered pursuant to the terms hereof (to the extent any such documents are included in materials otherwise filed with the SEC) may be delivered electronically and if so delivered, shall be deemed to have been delivered on the date on which Parent posts such documents, or provides a link thereto, on Parent's website on the Internet at Parent's website address; provided, however, that Parent shall promptly notify the Lender in writing (which may be by electronic mail) of the posting of any such documents;

(e) within five (5) days of delivery (or such longer period as the Lender may agree in writing (including via email)), copies of (x) all material statements, reports and notices made available to any of the Parent's security holders or to any holders of any Material Indebtedness (including the holders of the Convertible Notes) and (y) notices of conversion of the Convertible Notes delivered thereunder;

(f) Promptly, and in no event no later than three (3) Business Days (or such longer period as the Lender may agree in writing (including via email)) upon any Responsible Officer of Borrower obtaining knowledge thereof, notify the Lender of the following:

(i) the occurrence of any Default or Event of Default;

(ii) any filing or commencement, or written threat to commence, any action, suit or proceeding by any Loan Party (excluding actions, suits or proceedings by any Loan Party (A) in connection with the enforcement of its Intellectual Property Rights and (B) in defense of challenges to its immaterial Intellectual Property Rights; provided, that in the case of challenges to any Loan Party's immaterial Intellectual Property Rights that are (x) subject to any Technology License Agreement and/or (y) necessary for the operation of the Georgia Facility or the Rhode Island Facility, notice of the initial filing only shall be required) or against any Loan Party or by any Governmental Authority, in each case, that would reasonably be expected to result in monetary damages to such Loan Party or any of its Subsidiaries of, individually or in the aggregate, in excess of \$750,000, seeks injunctive relief (including with regard to any Intellectual Property of any Loan Party subject to the preceding parenthetical) or that would reasonably be expected to result in an Event of Default under Section 7.6;

(iii) any matter that has resulted or could reasonably be expected to result in a Material Adverse Effect;

(g) notice of (x) an equity investment in any of the Loan Parties promptly following the consummation of such investment and (y) the incurrence of any Permitted Indebtedness pursuant to clauses (c), (g), (j), (k) (if related to the Project), or (l) of the definition thereof (or any refinancing of the foregoing Indebtedness pursuant to clause (i) of the definition thereof), and subject to any other notice periods set forth thereof, at least five (5) Business Days (or such shorter period as the Lender may agree in writing (including via email)) prior to the incurrence of such Indebtedness (provided, that the foregoing limitation does not apply to any obligations of any Loan Party under any commercial agreement with General Motors);

(h) other financial information reasonably requested by the Lender and reasonably available to Borrower;

(i) if a Loan Party is not now a Registered Organization but later becomes one, such Loan Party shall promptly notify the Lender of such occurrence and provide the Lender with such Loan Party's organizational identification number (if available from the relevant jurisdiction);

(j) ~~(x)~~ concurrently with the delivery of the documents required under Section 5.2(a) and 5.2(b), Borrower shall deliver an updated Perfection Certificate, signed by a Responsible Officer of Borrower, (i) setting forth the information required pursuant to the Perfection Certificate and indicating any changes in such information from the most recent Perfection Certificate delivered pursuant to this clause (j) (or, prior to the first delivery of a Perfection Certificate pursuant to this clause (j), from the Perfection Certificate delivered on the Closing Date) or (ii) certifying that there has been no change in such information from the most recent Perfection Certificate delivered pursuant to this clause (j) (or, prior to the first delivery of a Perfection Certificate pursuant to this clause (j), from the Perfection Certificate delivered on the Closing Date); and (y) concurrently with the delivery of the documents required under Section 5.2(a), Borrower shall deliver a written certificate (a "Compliance Certificate") signed by a Financial Officer of Parent substantially in the form of Exhibit H, setting forth the calculations supporting compliance with the financial covenants set forth in Section 6.13 (provided, that if the audited financial statements would result in any change to the calculations supporting compliance set forth in the Compliance Certificate delivered for the last Fiscal Quarter of any fiscal year, then Borrower shall also be required to deliver a Compliance Certificate in connection with the delivery of the documents required under Section 5.2(b) concurrently therewith).

(k) promptly, and in no event no later than five (5) Business Days (or such longer period as the Lender may agree in writing (including via email)) after receipt of the same by Borrower, Borrower shall deliver to the Lender the monthly Progress Status Report provided to the Loan Parties by Koch Project Solutions, substantially in the same form of the report provided to the Lender by Borrower on September 22, 2022; ~~and~~

(l) on and after the first funding of any Term Loan, within five (5) Business Days (or such longer period as the Lender may agree in writing (including via email)) after the end of each fiscal month, an updated budget tracking template in the form of the "summary budget and tracker" tab of the Excel file provided by Borrower to the Lender on November 22, 2022; and

(m) promptly, and in any event no later than five (5) Business Days (or such longer period as the Lender may agree in writing (including via email)) after any change to the information set forth on Schedule 4.2(b) as of the Closing Date (or on Schedule 4.2(b) as subsequently delivered to the Lender pursuant to this clause 5.2(m)), Borrower shall deliver to the Lender an updated Schedule 4.2(b).

provided, that, for purposes of this Section 5.2, none of Parent or its Subsidiaries shall be required to disclose or provide any information (i) that constitutes non-financial trade secrets or non-financial proprietary information of any Person, (ii) in respect of which disclosure to the Lender (or any of its Party Representatives) is prohibited by applicable law, (iii) that is subject to attorney-client or similar privilege or constitutes attorney work product or (iv) in respect of which Parent or any of its Subsidiaries owes confidentiality obligations to any third party (provided, further that such confidentiality obligations were not entered into in contemplation of the requirements of this Section 5.2); provided, further, that notwithstanding the foregoing, the obligations in Section 5.2(a)

and (b) may be satisfied with respect to information filed by Parent with any securities exchange or with the SEC or any analogous governmental or private regulatory authority with jurisdiction over matters relating to securities (including Form 10-Q Reports and Form 10-K reports) and shall be deemed to have been delivered on the date on which such items have been made available on the SEC website or the website of the relevant analogous governmental or private regulatory authority or securities exchange (including, for the avoidance of doubt, by way of “EDGAR”).

5.3.Taxes; Pensions.

(a) To the extent any Loan Party defers payment of any contested taxes in an amount in excess of \$750,000, such Loan Party shall (i) notify the Lender in writing of the commencement of, and any material development in, the proceedings to the already disclosed in the SEC Reports, and (ii) posts bonds or takes any other steps as such Loan Party may determine (in its commercially reasonable business judgment) are required to prevent the Governmental Authority levying such contested taxes from obtaining a Lien upon any of the Collateral that is other than a “Permitted Lien.”

(b) Timely file, and require each of its Subsidiaries to timely file, all required tax returns and reports and timely pay, and require each of its Subsidiaries to timely pay, all foreign, federal, state and material local taxes, assessments, deposits and contributions owed each Loan Party and each of its Subsidiaries, except for deferred payment of any taxes contested pursuant to the terms of Section 5.3(a), and shall deliver to the Lender, within five (5) days (or such longer period as the Lender may agree in writing (including via email)), appropriate certificates attesting to such payments, and pay all amounts necessary to fund all present pension, profit sharing and deferred compensation plans in accordance with their terms, except as would not reasonably be expected to result in any liability to such Loan Party in an amount in excess of \$750,000.

5.4.Access to Collateral; Inspections; Books and Records.

(a) Provide to the Lender and its agents, at reasonable times, on at least one (1) Business Days’ notice (provided, that no notice is required if an Event of Default has occurred and is continuing), access to inspect the Collateral and the right to audit and copy any Loan Party’s Books. The foregoing inspections and audits shall be conducted no more often than once every twelve (12) months unless an Event of Default has occurred and is continuing in which case such inspections and audits shall occur as often as the Lender shall determine is necessary. The foregoing inspections and audits shall be conducted at Borrower’s expense, including reasonable out-of-pocket expenses. In the event a Loan Party and the Lender schedule an audit more than five (5) days in advance, and such Loan Party cancels or seeks to or reschedules the audit with less than five (5) days written notice to the Lender, then (without limiting any of the Lender’s rights or remedies) Borrower shall reimburse the Lender for any costs incurred by the Lender, plus any out-of-pocket expenses directly relating to the cancellation or rescheduling.

(b) In addition to the foregoing, cooperate (and cause the architect, the engineer and all Contractors to cooperate) with the Lender in arranging for inspections, at Borrower’s expense, of the Project, including of progress of the construction of the Georgia Facility, by any representatives of the Lender at any such times as the Lender may reasonably request with at least one (1) Business Days’ notice and during normal construction hours (provided, that no notice is

required if an Event of Default has occurred and is continuing); provided, further, the Lender and its representatives agree to comply with Borrower's customary safety procedures in connection with such inspections of the Project.

(c) For all purposes of this Section 5.4, (i) any information disclosed to the Lender in connection with the foregoing inspections shall be subject to Section 14.7 and (ii) none of Parent or its Subsidiaries shall be required pursuant to any such inspections to disclose or provide any information (A) that constitutes non-financial trade secrets or non-financial proprietary information of any Person (provided, that the foregoing limitation does not apply to any obligations of any Loan Party under any commercial agreement with General Motors), (B) in respect of which disclosure to the Lender (or any of its Party Representatives) is prohibited by applicable law, (C) that is subject to attorney-client or similar privilege or constitutes attorney work product or (D) in respect of which Parent or any of its Subsidiaries owes confidentiality obligations to any third party (provided, further, that such confidentiality obligations were not entered into in contemplation of the requirements of this Section 5.4).

5.5. Insurance.

(a) Obtain and maintain, or cause to be maintained, in full force and effect, at its sole expense, insurance for the Loan Parties and the Collateral providing at least the following coverages, to the extent applicable:

(i) Insurance with respect to the Collateral insuring against any peril now or hereafter included within the classification "Special Cause of Loss" (sometimes referred to as "All Risk of Physical Loss" or "Special Perils"), together with an "Ordinance and Law" endorsement, in amounts at all times sufficient to prevent the Lender from becoming a co-insurer within the terms of the Policies (as defined herein) and under applicable law, but in any event such insurance shall be maintained in an amount which, after application of deductible, shall be equal to the full insurable value of the Project, the term "full insurable value" to mean the actual replacement cost of the Collateral (without taking into account any depreciation, and exclusive of excavations, footings and foundations, landscaping and paving) determined annually by an insurer, a recognized independent insurance broker or an independent appraiser selected and paid by Borrower and in no event less than the coverage required pursuant to the terms of any Lease (the "Replacement Cost");

(ii) Commercial general liability insurance on the so-called "occurrence" form, including bodily injury, death and property damage liability, insurance against any and all claims, including all legal liability to the extent insurable and imposed upon the Lender and all court costs and legal fees and expenses, arising out of or connected with the possession, use, leasing, operation, maintenance or condition of the Collateral in such amounts as are generally available at commercially reasonable premiums and are generally required by institutional lenders for properties comparable to the Collateral but in any event for a limit per occurrence of at least \$1,000,000 and an annual aggregate of at least \$2,000,000. This requirement may be satisfied by a layering of Commercial General Liability, Umbrella and Excess Liability Policies, but in no event will the Commercial General Liability policy be written for an amount less than \$10,000,000 per occurrence and \$10,000,000 aggregate for bodily injury and property damage liability.

(iii) Statutory workers' compensation insurance with respect to any work on or about the Project.

(iv) Business interruption insurance in an amount sufficient to avoid any co-insurance penalty and to provide proceeds which will cover a period of not less than ninety (90) days from the date of casualty or loss, containing an extended period of indemnity endorsement which provides that after the physical loss to the Collateral has been repaired, the continued loss of income will be insured until such income returns to the same level it was prior to the loss, or the expiration of ninety (90) days from the date of the loss, whichever first occurs, and notwithstanding that the policy may expire prior to the end of such period.

(v) Broad form boiler and machinery insurance (without exclusion for explosion) covering all boilers or other pressure vessels, machinery, and equipment located in, on or about the Rhode Island Property or the Project (including "system breakdown coverage") and insurance against loss of occupancy or use arising from any breakdown in such amounts as are generally required by institutional lenders for properties comparable to the Rhode Island Property or the Project, as applicable;

(vi) If required by the Lender, flood insurance in an amount at least equal to the greater of (A) the Replacement Cost together with business interruption coverage and (B) the maximum limit of coverage available for the Rhode Island Property and the Project under the National Flood Insurance Act of 1968, The Flood Disaster Protection Act of 1973 and the National Flood Insurance Reform Act of 1994, as each may be amended;

(vii) At all times during which any construction, repairs or alterations are being made with respect to the Project which affect the structure of the Project, (A) owner's contingent or protective liability insurance covering claims not covered by or under the terms or provisions of the above mentioned commercial general liability insurance policy; and (B) the insurance provided for in Subsection 5.5(a)(i) written in a so-called builder's risk completed value form (1) on a non-reporting basis, (2) against all risks insured against pursuant to Section 5.5(a)(i) and (3) including permission to occupy the Project;

(viii) Such other insurance with respect to the Collateral or on any replacements or substitutions or additions or increased coverage limits as may from time to time be required by the Lender against other insurable hazards or casualties which at the time are commonly insured against in the case of property similarly situated, including, without limitation, sinkhole, mine subsidence, earthquake and environmental insurance, due regard being given to the height and type of buildings, their construction, location, use and occupancy; and

(ix) Borrower and Aspen RI shall each obtain an ALTA title insurance policy reasonably satisfactory to the Lender (a "Title Policy") within fifteen (15) days of recording the Leasehold Deed to Secure Debt with respect to the Georgia Property, and the Mortgage with respect to the Rhode Island Property, which Title Policy shall be underwritten by an insurer approved by the Lender (the "Title Company") in an amount approved by the Lender and insuring the liens created under the Mortgage and the Leasehold Deed to Secure Debt to be a first-priority liens, subject only to such exceptions and conditions to title as disclosed in policies of title insurance made available to the Lender ("Permitted Exceptions"), and containing such

endorsements as the Lender may reasonably require. In addition, if required by the Lender, one or more other title insurance companies reasonably acceptable to the Lender shall have issued such coinsurance and/or reinsurance as determined by the Lender. No title matter may be insured over by any Title Company without the express written consent of the Lender. Each advance of the Term Loan shall be insured by the Title Company by an endorsement to the Title Policy for the Georgia Property in the form of Exhibit G as set forth in clause (n) to Schedule 3.2.

(b) All insurance provided for in Subsection 5.5(a) shall be for a term of not less than one (1) year and obtained under valid and enforceable policies (the “Policies” or in the singular, the “Policy”), and shall be issued by one or more other domestic primary insurer(s) (except for policies placed outside the U.S. for Aspen exposure outside the U.S.) having a general policy rating of A- or better and a financial class of IX or better by A.M. Best Company, Inc. (or if a rating of A.M. Best Company, Inc. is no longer available, a similar rating from a similar or successor service). All insurers providing insurance required herein shall be authorized and admitted to issue insurance in the state in which the Project, or the Rhode Island Property, as applicable, is located. The Policy referred to in Subsection 5.5(a)(ii) above shall name the Lender as an additional insured and the Policies referred to in Subsection 5.5(a)(i), (iv), (v), (vi) and (vii), and as applicable (viii), above shall provide that all proceeds be payable to the Lender, named as mortgagee and loss payee and shall contain notice of cancellation and notice of material changes and termination provisions, in each case, reasonably acceptable to the Lender. Evidence of insurance with respect to all renewal and replacement Policies shall be delivered to the Lender as promptly as practicable prior to the expiration date of any of the Policies required to be maintained hereunder which evidence shall bear notations evidencing payment of insurance premiums. Originals or evidence of such replacement Policies shall be delivered to the Lender promptly after receipt by the applicable Loan Party thereof but in any case within thirty (30) days (or such longer period as the Lender may agree in writing (including via email)) after the effective date thereof. If any Loan Party fails to maintain and deliver to the Lender the original Policies or evidence of insurance required herein, the Lender may procure such insurance at such Loan Party’s sole cost and expense.

(c) Each Loan Party shall comply with all insurance requirements and shall not bring or keep or permit to be brought or kept any article upon any of the Project or the Rhode Island Property or cause or permit any condition to exist thereon which would be prohibited by an insurance requirement, or would invalidate the insurance coverage required hereunder to be maintained by any Loan Party on or with respect to any part of the Collateral pursuant to this Section 5.5 and shall not purchase any additional amounts of insurance that would cause the Lender to become a co-insurer within the terms of the Policies.

(d) Unless the Lender requires the Loan Parties to obtain a separate Policy or Policies under Section 5.5(a), the insurance coverage required may be effected under a blanket Policy or Policies covering the Collateral; provided, that any such blanket Policy shall specify, except in the case of commercial general liability insurance, the premises address of each building, the portion of the total coverage of such Policy that is allocated to the Collateral, and any sublimit in such blanket Policy applicable to the Collateral, and shall in any case provide the same protection as would a separate policy insuring only the Collateral and otherwise comply with all other respects with the requirements of this Section 5.5.

(e) In the event of a foreclosure of the Mortgage or the Leasehold Deed to Secure Debt or other transfer of title to the Collateral in extinguishment in whole or in part of the Obligations, all right, title and interest of any Loan Party in and to the Policies (other than builder's risk) then in force concerning the Collateral and all proceeds payable thereunder shall thereupon vest in the Lender or the purchaser at such foreclosure or other transferee in the event of such other transfer of title.

(f) At the Lender's request, each Loan Party shall deliver certificates of insurance policies and evidence of all premium payments.

5.6. Disbursement Account. Deposit the proceeds of the Term Loans into a designated Account (the "Disbursement Account") maintained at a bank or financial institution subject to a Control Agreement that perfects the Lender's security interest in such Account.

5.7. Protection of Intellectual Property Rights.

(a) (i) Take such action as, in its reasonable business judgment, it deems necessary to protect, defend and maintain the validity and enforceability of its material Intellectual Property; (ii) reasonably promptly advise the Lender in writing of material infringements of its material Intellectual Property; and (iii) not take any action or omit to take any action which would reasonably be expected to result in any material Intellectual Property being abandoned, forfeited or dedicated to the public without the Lender's written consent; provided, that, for purposes of this Section 5.7(a), "material Intellectual Property" shall be deemed to include, without limitation, Intellectual Property that is (x) subject to any Technology License Agreement and/or (y) necessary for the operation of the Georgia Facility or the Rhode Island Facility.

(b) Provide written notice to the Lender within twenty (20) days (or such longer period as the Lender may agree in writing (including via email)) of entering or becoming bound by any material Restricted License (other than over-the-counter or off-the-shelf software that is commercially available to the public).

5.8. Litigation Cooperation. From the date hereof and continuing through the termination of this Agreement, make available to the Lender, without expense to the Lender, the Loan Parties and their respective officers, employees, agents and Books, to the extent that the Lender may deem them reasonably necessary to prosecute or defend any third-party suit or proceeding instituted by or against the Lender with respect to any Collateral or relating to any Loan Party; provided, that, for purposes of this Section 5.8, none of Parent or its Subsidiaries shall be required to disclose or provide any information (i) that constitutes non-financial trade secrets or non-financial proprietary information of any Person (provided, that the foregoing limitation does not apply to any obligations of any Loan Party under any commercial agreement with General Motors), (ii) in respect of which disclosure to the Lender (or any of its Party Representatives) is prohibited by applicable law, (iii) that is subject to attorney-client or similar privilege or constitutes attorney work product or (iv) in respect of which Parent or any of its Subsidiaries owes confidentiality obligations to any third party (provided, further, that such confidentiality obligations were not entered into in contemplation of the requirements of this Section 5.8).

5.9. **Formation or Acquisition of Subsidiaries.** Notwithstanding and without limiting the negative covenants contained in Sections 6.2 and 6.6 hereof, in the event Borrower or any Guarantor creates or acquires any Subsidiary (including, without limitation, pursuant to a Division), Borrower and such Guarantor shall, prior to the creation or acquisition of such new Domestic Subsidiary, promptly notify the Lender thereof and, at the Lender's request, in its sole discretion, take all such action as may be reasonably required by the Lender to (a) cause each such Domestic Subsidiary to, in the Lender's sole discretion, become a co-borrower or Guarantor hereunder, together with such appropriate financing statements and/or control agreements, all in form and substance reasonably satisfactory to the Lender (including being sufficient to grant the Lender a first priority Lien (subject to Permitted Liens) in and to the assets of such newly formed or acquired Domestic Subsidiary), (b) provide to the Lender appropriate certificates and powers and financing statements, pledging all of the direct or beneficial ownership interest in such new Subsidiary, in form and substance reasonably satisfactory to the Lender; provided, that with respect to any Foreign Subsidiary, Borrower shall only be required to grant and pledge to the Lender a perfected security interest in up to sixty-five percent (65%) of the stock, units or other evidence of ownership of such Foreign Subsidiary; and (c) provide to the Lender all other documentation in form and substance reasonably satisfactory to the Lender, including one or more opinions of counsel reasonably satisfactory to the Lender, which in its opinion is appropriate with respect to the execution and delivery of the applicable documentation referred to above. Any document, agreement, or instrument executed or issued pursuant to this Section 5.9 shall be a Loan Document.

5.10. ~~Reserved~~ **Quarterly Consultation**—

Commencing with the Fiscal Quarter ending December 31, 2023, and during each Fiscal Quarter thereafter (on such Business Day and at such time as agreed by the Loan Parties and the Lender), representatives of the Loan Parties and the Lender shall meet on a quarterly basis, on such Business Day as mutually agreed (not to be unreasonably withheld, conditioned or delayed) by Borrower and the Lender, to review Lender's (or its Affiliate's) production needs at the Georgia Facility. For the avoidance of doubt, (x) if no meeting occurs during any Fiscal Quarter for any reason (unless Lender in its sole discretion agrees to waive the meeting requirement in such Fiscal Quarter in writing (including via email)), the Loan Parties shall be considered in breach of this Section 5.10 and (y) any such meeting may be conducted virtually or via teleconference call as mutually agreed (not to be unreasonably withheld, conditioned or delayed) by Borrower and the Lender.

5.11. **Environmental.**

(a) Keep any property either owned or operated by any Loan Party free of any Environmental Liens or post bonds or other financial assurances sufficient to satisfy the obligations or liability evidenced by such Environmental Liens;

(b) Comply in all material respects with all applicable Environmental Laws (taking into account any applicable cure under any applicable law) and provide to the Lender environmental reports and information prepared by licensed or registered environmental engineers or other qualified parties reasonably acceptable to the Lender, as the Lender may reasonably request in writing from time to time;

(c) Timely notify the Lender of any legally reportable release of which any Loan Party has knowledge of Hazardous Materials from or onto property owned, leased, or operated by such Loan Party and take any Remedial Actions to the extent required by applicable Environmental Law to abate said release or otherwise to come into compliance, in all material respects, with applicable Environmental Law; and

(d) Promptly, but in any event within seven (7) Business Days of its receipt thereof (unless a shorter notice period is appropriate or required under the circumstances), provide the Lender with written notice of any of the following: (i) notice that an Environmental Lien has been filed against any of the real or personal property of any Loan Party, or (ii) commencement of any Environmental Action or written notice that an Environmental Action will be filed against any Loan Party.

(e) Within twenty (20) Business Days of the Closing Date (or such longer period as the Lender may agree in writing (including via email)), Borrower shall deliver to the Lender a certification of the most recent Phase I report with respect to each Property.

5.12. Incorporation of Terms. In the event that any Loan Party enters into any Other Financing Facility that is a debt facility (including the Revolving Loan Agreement) and that includes terms that are more favorable to the lender thereunder (including the inclusion of any financial covenant, securing such debt with assets that are not part of the Collateral or providing guarantees of such debt by entities that are not Guarantors), in the aggregate, in comparison to the terms set forth in this Agreement (excluding (i) any currency, pricing, interest rate provisions, rate floors, fees, premiums (including prepayment premiums), funding discounts and other purely economic terms, (ii) any equity conversion rights applicable to such Other Financing Facility (subject in all respects to the limitations in Sections 6.5 and 6.6 on the Loan Parties ability to make any Restricted Payment or otherwise repay, buyback, repurchase or redeem any such Other Financing Facility that is a debt facility) and (iii) to reflect the priority on the Collateral as agreed in any Revolver Intercreditor Agreement entered into pursuant to the terms hereof) (each such term being herein called a “More Favorable Provision”), then it is agreed that the More Favorable Provision shall be deemed automatically to be incorporated into this Agreement and the Lender shall be entitled to all of the rights and benefits of the More Favorable Provision as if such More Favorable Provision was incorporated into this Agreement as of the date that such provisions became effective in the Other Financing Facility; provided, however, that in the event that the Other Financing Facility is terminated or the More Favorable Provision is amended or waived, and as a result the More Favorable Provision is no longer more restrictive or more beneficial to the Lender than the provisions otherwise set forth in this Agreement, then the Lender shall no longer be entitled to the benefit of the More Favorable Provision. At the request of the Lender, the Loan Parties shall amend this Agreement to specifically incorporate any such More Favorable Provisions (subject to the terms of this Section 5.12).

5.13. Use of Proceeds.

(a) Borrower shall use the proceeds of any Term Loan and any Additional Financing Facility solely for the purposes set forth in the applicable (pre-approved by General Motors) Term Loan Budget and consistent with the Project Budget (subject, in each case, to an aggregate permitted variance not to exceed ~~11.5~~10% of the amounts set forth in the Project Budget collectively (and not on a line by line basis) plus any amounts in excess thereof to the extent funded with equity (the “Permitted Variance”)).

(b) No part of the proceeds of any Term Loan made hereunder will be used by Borrower or any of its Affiliates, directly or indirectly, for any payments to any governmental official or employee, political party, official of a political party, candidate for political office, or anyone else acting in an official capacity, in order to obtain, retain or direct business or obtain any improper advantage, in violation of the United States Foreign Corrupt Practices Act of 1977, as amended.

(c) No proceeds of any Term Loan made hereunder will be used directly or indirectly to fund any operations in, finance any investments or activities in, or make any payments to or for the benefit of, a Sanctioned Person or a Sanctioned Entity, or in any other manner that would result in a violation of Sanctions by any Person.

5.14. Delivery of Material Contracts and Project Contracts. (i) Deliver written notice to the Lender within five (5) Business Days (or such longer period as the Lender may agree (including via email)) of the entry of any Loan Party into a Material Contract or a Project Contract, a material amendment to any of the foregoing or a material amendment to the Governing Documents or the Georgia Lease, (ii) deliver copies of the foregoing, to the extent requested by the Lender, and (iii) enter into a collateral assignment of such Material Contract or Project Contract, to the extent requested by the Lender.

5.15. Further Assurances. Execute any further instruments and take further action as the Lender reasonably requests to perfect or continue the Lender's Lien in the Collateral or to effect the purposes of this Agreement unless specifically waived in writing by the Lender.

5.16. Construction Related Covenants.

(a) **Permits and Licenses.** Borrower shall diligently pursue and maintain (i) all Project Permits and (ii) licenses necessary to construct the Georgia Facility and the construction and operation of the Project.

(b) **Payment and Performance Bond.** Borrower shall obtain or cause to be obtained and deliver to the Lender a payment and performance bond with regard to (i) all Major Subcontracts awarded with work in progress as of the Closing Date, as set forth on Schedule 5.16(b) and (ii) all Major Subcontracts awarded after the Closing Date involved in the construction of the Georgia Facility, in form and substance reasonably acceptable to the Lender, which is (A) issued by a Treasury-listed surety licensed to do business in the state where the Property is located and (B) written in triple-obligee form naming the Lender as co-obligee.

(c) **Contracts.**

(i) Construction Contract. Borrower shall deliver to the Lender copies of each amendment to the General Contractor's Contract that (i) is a guaranteed maximum price amendment, (ii) results in net increase of more than \$5,000,000 or (iii) is a non-monetary amendment thereto not in the ordinary course of business.

(ii) Contractor Information. If requested by the Lender, Borrower shall deliver a list of all contractors, subcontractors, and material suppliers to be employed in connection with the construction of the Georgia Facility, setting forth (i) the nature of the work to be

performed, (ii) the labor and materials to be supplied, and (iii) the dollar amount of (x) the work and materials to be used in connection therewith and (y) the work that has been performed and the materials that have been used, including the outstanding amounts owed in connection therewith; provided, that with respect to any such contractor, subcontractor or material supplier who is not performing work at the Project or party to a contract with any Loan Party with respect to the Project as of the time of such request, such information shall be provided only to the extent then known by such Loan Party.

(d) **Title to Property.** Borrower shall promptly commence to cure and proceed diligently to complete, or cause to be cured, any defect in the title to any Property (other than Permitted Liens) and any real property necessary to the construction (in accordance with the applicable Budget and Milestones, as applicable), operation, use, and/or enjoyment of the Project on such Property or such other real property for the purposes intended under this Agreement.

(e) **Compliance.** Borrower shall comply in all material respects with applicable building codes in construction of the Project, shall obtain all Project Permits and, if requested by the Lender, shall provide the Lender with copies of any such Project Permits.

(f) **Notices by Governmental Authority; Casualty; Condemnation.** Borrower shall timely comply with any written notice or claim by any Governmental Authority pertaining to the Property and shall, if reasonably requested by the Lender, promptly furnish to the Lender a true and complete copy of any such written notice of defect, shall promptly notify the Lender of any fire or other casualty or any written notice of taking or eminent domain action or proceeding affecting the Project, or the written threat of any such action or proceeding of which the applicable Loan Party becomes aware.

(g) **Title Company.** Borrower shall engage the Title Company as disbursement agent, which shall make disbursements of Term Loan proceeds from the Disbursement Account to Borrower in connection with the construction of the Georgia Facility, and shall “down date” the endorsement of the Title Policy for the Georgia Property to reflect each such Term Loan.

(h) **Notices.**

(i) Borrower shall promptly, and in no event no later than five (5) Business Days (or such longer period as the Lender may agree in writing (including via email)) upon any Responsible Officer of Borrower obtaining knowledge thereof, notify the Lender of the following:

(A) any deficiencies asserted or Liens (other than Permitted Liens) filed against or the commencement of similar proceedings with respect to the Project;

(B) the failure to obtain any Project Permit or the loss of any Project Permit;

(C) a claim against any payment or performance bond delivered pursuant to Section 5.16(b) above; and

(D) any changes in the Plans or any construction contract which (a) will impact production capacity of the Project such that it would be inconsistent with Parent’s

obligations under the GM Purchase Contracts, (b) will result in any change in the fundamental design concept of the Project, (c) will result in any increase or decrease in the most recently delivered Term Loan Budget in excess of the Permitted Variance or (d) would be reasonably expected to cause Borrower to fail to meet any Milestone.

(i) **Indemnity.** Notwithstanding anything to the contrary in Section 9.3, to the fullest extent permitted by law, and at Borrower's sole cost and expense, indemnify, defend, satisfy all judgments and hold harmless, the Lender and its officers, agents and employees in their capacities as such, from and against all claims, actions, judgments, costs, liabilities, damages, losses and expenses arising out of or relating to any Property or the Project, excluding, however, liabilities arising from the fraud, gross negligence, bad faith or willful misconduct of the Lender, its officers, agents and employees.

(j) **Survey and Updated Title Policy.** Promptly, and within thirty (30) days (or such longer period as the Lender may agree in writing (including via email))- after the issuance of the Certificate of Occupancy for the Project, an ALTA as-built survey of the Project.

(k) **Zoning.**

(i) (A) The construction of the Project will be in accordance with the Plans in all material respects, (B) the construction of the Project and the use and operation of the Georgia Property as indicated herein will conform to and comply with all material requirements of such public and private conditions, restrictions, reservations and zoning ordinances, and (C) the construction of the Project and the use and operation of the Georgia Property will comply in all material respects with all rules, regulations and requirements of all relevant Governmental Authorities, and will not result in any material violation of any law, ordinance, regulation or restriction now existing and affecting the Georgia Property.

(ii) (A) The use and operation of the Rhode Island Property as indicated herein will conform to and comply with all material requirements of such public and private conditions, restrictions, reservations and zoning ordinances, and (B) the use and operation of the Rhode Island Property will comply in all material respects with all rules, regulations and requirements of all relevant Governmental Authorities, and does not result in any material violation of any law, ordinance, regulation or restriction now existing and affecting the Rhode Island Property.

5.17. Collateral.

(a) The Collateral shall be maintained at the locations provided in the Perfection Certificate or such other location of which Borrower has given the Lender notice pursuant to Section 5.2(j) and delivered a Collateral Access Agreement in favor of the Lender with respect to such other location or as otherwise permitted hereunder or pursuant to the Security Agreement; provided, that, a Collateral Access Agreement shall not be required if the aggregate value of Collateral maintained at any such location is below \$250,000.

(b) Each Loan Party shall maintain good title to, and rights in, and the power to transfer each item of the Collateral upon which it purports to grant a Lien hereunder (including Aspen RI as the sole owner of the fee interest in the Rhode Island Property), free and clear of any

and all Liens, except Permitted Liens. No Loan Party shall have Collateral Accounts at or with any bank or financial institution, other than the Collateral Accounts described in the Perfection Certificate, the Disbursement Account, any other Collateral Account of which Borrower has given the Lender notice pursuant to Section 5.2(j) and delivered a Control Agreement to the Lender and to the extent required by Section 5.9 and subject to any applicable Intercreditor Agreement.

5.18. Full Disclosure. All written information (other than Projections) with respect to the Loan Parties in any certificate or written statement to be provided to the Lender, including the information contained in each Budget and in any document to be furnished to the Lender by any Loan Party in connection with this Agreement or any other Loan Document, when taken as a whole, will not, when furnished, contain any untrue statement of a material fact or omit to state a material fact necessary in order to make the statements contained therein not materially misleading in light of the circumstances under which such statements will be made (after giving effect to all supplements and updates thereto from time to time).

5.19. Post-Closing Deliveries. (x) The items listed in clauses (e), (i), (j), (k), (l), (o), (p), (q), (r) and (s) of Schedule 3.2 shall be delivered on or before December 16, 2022 (or such longer period as the Lender may agree in writing (including via email)) and (y) Borrower shall deliver to the Lender a new Phase I report with respect to the Rhode Island Property which shall be certified to the Lender on or before March 1, 2023 (or such longer period as the Lender may agree in writing (including via email)).

5.20. Separate Tax Parcel. Within ten (10) Business Days following the Closing Date, Borrower shall make a request to the office of the Bulloch County Assessor to have the parcel on which the Georgia Facility will be built to have such land broken out as a separate tax parcel, will cooperate in satisfying all requirements and will keep the Lender informed as to the status of such effort.

5.21. Compliance with Laws. Without duplication of Section 5.11 in respect of Environmental Laws, comply in all respects with the requirements of all laws, orders, writs, injunctions and decrees and duly observe all requirements of any Governmental Authority, in each case, applicable to it or to its business or property, except if the failure to comply therewith would not result in a material liability or have a Material Adverse Effect.

6. NEGATIVE COVENANTS.

Each of Borrower, Parent and Aspen RI, covenants and agrees that until the Obligations are Paid in Full, it will not:

6.1. Indebtedness. Create, incur, assume, or be liable for any Indebtedness, or permit any Subsidiary to do so, other than Permitted Indebtedness; provided, that any Permitted Indebtedness provided by an Affiliate of any Loan Party shall not have an interest payment obligation that requires cash payment prior to the Payment in Full of the Obligations (excluding any cash interest payable pursuant to any mandatory redemption provisions upon a change of control or asset sale event which are subordinated to the Obligations pursuant to a subordination agreement).

6.2.**Liens**. Create, incur, allow, or suffer any Lien on any of its property, except for Permitted Liens.

6.3.**Dispositions**. Convey, sell, lease, license, transfer, assign, or otherwise dispose of (including, without limitation, pursuant to a Division) (collectively, “Transfer”), or permit any of its Subsidiaries to Transfer, all or any part of its business or property (including Intellectual Property), except for Transfers (a) of Inventory or Equipment in the ordinary course of business; (b) of worn out or obsolete Equipment; (c) to the extent constituting a Transfer, Permitted Liens and Permitted Investments and (d) of non-exclusive licenses for the use of property of any Loan Party or its Subsidiaries in the ordinary course of business (including pursuant to clause (h) of the definition of Permitted Liens and any Technology License Agreement); provided, that no permitted Transfer under the foregoing clauses (a) through (c) shall be made to a non-Loan Party Affiliate of any Loan Party.

To the extent that any Collateral is Transferred as permitted by clause (a) or (b) of this Section 6.3, such Collateral shall be Transferred free and clear of the Liens created by the Loan Documents, which Liens shall be automatically released upon the consummation of such Transfer, and the Lender shall any action reasonably requested by Loan Parties in order to effect the foregoing; provided, that in the case of a Transfer made to any Loan Party, the relevant transferred assets shall become part of the Collateral of the transferee Loan Party (except to the extent such assets are Excluded Assets).

6.4.**Mergers or Acquisitions**. Merge or consolidate, or permit any of its Subsidiaries to merge or consolidate, with any other Person, or acquire, or permit any of its Subsidiaries to acquire, all or substantially all of the capital stock or property of another Person (including, without limitation, by the formation of any Subsidiary or pursuant to a Division).

6.5.**Restricted Payments; Investments**. (a) Make any Restricted Payment (other than any distribution from Aspen RI to Parent); or (b) directly or indirectly make any Investment (including, without limitation, any additional Investment in any Subsidiary), other than a Permitted Investment.

6.6.**Restricted Debt Payments**.

(a) Prepay, buyback, repurchase or redeem any outstanding Indebtedness incurred under clauses (b), (c), (g), (h) (to the extent any such obligation under clause (h) is owed by Borrower), (j) and (l) of the “Permitted Indebtedness” definition (in each case except to the extent permitted pursuant to clause (i) of the definition thereof) prior to its stated maturity.

(b) (i) Make or permit any payment on any Subordinated Indebtedness or Disqualified Equity Interests, except to the extent permitted under the terms of the subordination, intercreditor, or other similar agreement to which such Subordinated Indebtedness or Disqualified Equity Interests is subject, or (ii) amend any provision in any document relating to the Subordinated Indebtedness or Disqualified Equity Interests which would increase the amount thereof to the extent not otherwise permitted hereunder, or adversely affect the subordination thereof to Obligations owed to the Lender.

6.7.**Transactions with Affiliates.** Directly or indirectly enter into or permit to exist any material transaction with any Affiliate of any Loan Party (other than Affiliates that are Loan Parties), except for (a) transactions that are in the ordinary course of such Loan Party's business, each not to exceed \$500,000 individually, or in a series of related transactions, (b) the Convertible Notes, the Equity Investment and any Additional Financing Facility provided by Koch Strategic Platforms (and its Affiliates) otherwise constituting "Permitted Indebtedness" and (c) any transaction in excess of \$500,000 individually; provided, that such transaction is upon fair and reasonable terms that are not materially less favorable to such Loan Party than would be obtained in an arm's length transaction with a non-affiliated Person; provided, further, that such transaction under this clause (c) shall be approved by either a majority of the Disinterested Board Members of the Parent's Board of Directors or by Parent's audit committee in accordance with the requirements for related party transactions under the New York Stock Exchange rules then in effect; provided, further, that at any time after the Closing Date, as promptly as practicable following the entrance into any transaction permitted by the foregoing clauses (b) or (c), the appropriate Loan Party shall deliver to the Lender a supplement to Schedule 4.16(a) listing such transaction.

6.8.**Amendments to Material Contracts and Other Debt Documents.** Amend, modify, or change any of the terms or provisions of any term or condition of any Material Contract, Project Document, any Governing Document or the Georgia Lease, that, either individually or in the aggregate, is adverse in any material respect to the Lender.

6.9.**Limitations on Negative Pledges and other Agreements.** Other than as provided in the Loan Documents or the documentation governing any Permitted Indebtedness that is subject to a Permitted Lien (and subject to the applicable Intercreditor Agreement), enter into, incur, or permit to exist, directly or indirectly, any agreement, instrument, deed, lease or other arrangement that prohibits, restricts or imposes any condition upon the ability of any Loan Party to create, incur or permit to exist any Lien upon any of its property or revenues, whether now owned or hereafter acquired, or that requires the grant of any security for an obligation if security is granted for another obligation.

6.10.**Control Agreement.** ~~No Loan Party shall e~~Enter into any control agreement over any Account containing proceeds of an Other Financing Facility.

6.11.**Compliance.** Become an "investment company" or a company controlled by an "investment company", under the Investment Company Act of 1940, as amended, or undertake as one of its important activities extending credit to purchase or carry margin stock (as defined in Regulation U of the Board of Governors of the Federal Reserve System); fail to meet the minimum funding requirements of ERISA, permit a Reportable Event or non-exempt Prohibited Transaction (each as defined in ERISA), to occur; fail to comply with the Federal Fair Labor Standards Act or violate any other law or regulation, or permit any of its Subsidiaries to do so; withdraw or permit any Subsidiary to withdraw from participation in, permit partial or complete termination of, or permit the occurrence of any other event with respect to, any present pension, profit sharing and deferred compensation plan which could reasonably be expected to result in any liability of any Loan Party in an amount in excess of \$750,000, including any liability to the Pension Benefit Guaranty Corporation or its successors or any other governmental agency.

6.12.**OFAC; Patriot Act; Anti-Corruption Laws.** Fail to comply with the laws, regulations and executive orders referred to in Sections 4.10, 4.11 and 4.12.

6.13.**Financial Covenants.** As of each Test Date commencing with the Fiscal Quarter ending December 31, 2024:

(a) the Total Leverage Ratio shall not be greater than 5.00:1.00; and

(b) the ratio of total Indebtedness of Parent and its Subsidiaries (excluding Indebtedness incurred under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000 and undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness) to the consolidated equity of Parent and its Subsidiaries shall not be greater than 1.20x.

The Loan Parties will demonstrate compliance with this Section 6.13 by delivering a Compliance Certificate in accordance with and subject to Section 5.2(j)(y).

7. **EVENTS OF DEFAULT.**

Any one or more of the following events shall constitute an event of default (each, an “Event of Default”) under this Agreement:

7.1.**Payment Default.** Borrower or any other Loan Party fails to (a) make any payment of principal or interest on any Term Loan on its due date, or (b) pay any other Obligations within three (3) Business Days after such Obligations are due and payable (which three (3) Business Day cure period shall not apply to payments due on the Maturity Date). During the cure period, the failure to make or pay any payment specified under clause (b) hereunder is not an Event of Default (but no Term Loans will be funded during the cure period);

7.2.**Covenant Default.**

(a) Any Loan Party fails or neglects to perform any obligation in Sections 5.2 (provided, however, that Borrower shall have ten (10) Business Days from the scheduled due date to cure any default under clauses 5.2(a) and (b)), 5.4, 5.6, 5.12, 5.13, 5.16(b), 5.17, 5.18 or any covenant in Section 6 (subject to the Cure Right with respect to a default under Section 6.13); or

(b) Any Loan Party fails or neglects to perform, keep, or observe any other term, provision, condition, covenant or agreement contained in this Agreement or any Loan Documents, and as to any default (other than those specified in this Section 7) under such other term, provision, condition, covenant or agreement that can be cured, has failed to cure the default within ten (10) days after the occurrence thereof; provided, however, that if the default cannot by its nature be cured within the ten (10) day period or cannot after diligent attempts by such Loan Party be cured within such ten (10) day period, and such default is likely to be cured within a reasonable time, then such Loan Party shall have an additional period (which shall not in any case exceed thirty (30) days) to attempt to cure such default, and within such reasonable time period the failure to cure the default shall not be deemed an Event of Default (but no Term Loans shall be funded during such cure period). Cure periods provided under this section shall not apply, among other things, to any other covenants set forth in clause (a) above;

7.3. Attachment; Levy; Restraint on Business.

(a) (i) The service of process seeking to attach, by trustee or similar process, any funds of any Loan Party or of any entity under the control of such Loan Party (including a Subsidiary), or (ii) a notice of lien or levy (other than a Permitted Lien) is filed against any Property, the Project or any of the assets of any Loan Party by a Governmental Authority or by any other party, and the same under sub-clauses (i) and (ii) hereof are not, within thirty (30) days after the occurrence thereof, discharged or stayed (whether through the posting of a bond or otherwise); provided, however, no Term Loan shall be funded during any thirty (30) day cure period; or

(b) (i) any material portion of any Loan Party's assets is attached, seized, levied on, or comes into possession of a trustee or receiver, or (ii) any court order enjoins, restrains, or prevents such Loan Party from conducting any material part of its business;

7.4. Insolvency. (a) any Loan Party and its Subsidiaries, taken as a whole, is not Solvent, (b) any Loan Party begins an Insolvency Proceeding; or (c) an Insolvency Proceeding is begun against any Loan Party and not dismissed or stayed within forty-five (45) days (but no Term Loan shall be funded while any of the conditions described in clause (a) exist and/or until any Insolvency Proceeding is dismissed);

7.5. Other Indebtedness and Agreements.

(a) a default under any Material Contract to which any Loan Party is a party, or under any GM Purchase Contract, in each case after the passing of the applicable cure period thereunder (if any) and giving effect to any waivers, that results in a right by such third Person, irrespective of whether exercised, to accelerate the maturity of such Loan Party's obligations thereunder or to terminate such Material Contract or any GM Purchase Contract;

(b) the occurrence of an event of default under the Georgia Lease as a result of which, upon the expiration of any applicable grace period provided in such instrument and the exercise of remedies available therein or by applicable laws, the lessor of the Georgia Lease may terminate the Georgia Lease; or

(c) subject to the applicable cure periods of any Project Document, any default or failure of Borrower to comply with any of the terms or conditions of any Project Document that could reasonably be expected to result in a Project MAE.

7.6. Milestones. Any Milestone shall not be satisfied within thirty (30) days (subject to a day for day extension due to any Force Majeure delays (subject to a maximum of sixty (60) days) and as such date may be further extended by the Lender in writing (including via email)) after the date by which such Milestone was to have been achieved or cessation of work on the construction of the Project for any period of thirty (30) consecutive days (except for any work stoppage approved by the Lender or due to Force Majeure (solely on a day to day extension and subject to a maximum of thirty (30) consecutive days)); provided, that notwithstanding the foregoing, the aggregate extension for all Milestones shall not exceed sixty (60) days.

7.7.**Judgments.** One or more final judgments, orders or decrees for the payment of money in an amount, individually or in the aggregate, of at least \$750,000 (not covered by independent third-party insurance as to which liability has been accepted by such insurance carrier) shall be rendered against any Loan Party and the same are not, within thirty (30) days after the entry thereof, discharged or after execution thereof, stayed or bonded pending appeal, or such judgments are not discharged prior to the expiration of any such stay (provided, that no Term Loan will be funded prior to the discharge, stay, or bonding of such judgment, order or decree);

7.8.**Misrepresentations.** Any Loan Party or any Responsible Officer makes any representation, warranty, or other statement now or later in this Agreement, any Loan Document or in any writing delivered to the Lender or to induce the Lender to enter this Agreement or any Loan Document, and such representation, warranty, or other statement is incorrect in any material respect when made;

7.9.**Governmental Approvals; Noncompliance; Lapse of Permits.**

(a) Any Governmental Approval shall have been (i) revoked, rescinded, suspended, modified in an adverse manner or not renewed in the ordinary course for a full term or (ii) subject to any decision by a Governmental Authority that designates a hearing with respect to any applications for renewal of any of such Governmental Approval or that could result in the Governmental Authority taking any of the actions described in clause (i) above, and, in each case, such decision or such revocation, rescission, suspension, modification or non-renewal has, or could reasonably be expected to have, a Material Adverse Effect;

(b) Failure of the Project or any part thereof to comply with any Governmental Requirements, which failure is not cured within thirty (30) days after written notice thereof is delivered by the Lender to Borrower; or

(c) Any Project Permit lapses or ceases to be in full force and effect and is not reinstated or renewed within ten (10) Business Days thereafter.

7.10.**Loan Documents.**

(a) The validity or enforceability of any Loan Document shall at any time for any reason be declared to be null and void, or a proceeding shall be commenced by any Person (other than by the Lender, any Eligible Assignee or any of their respective Affiliates) seeking to establish the invalidity or unenforceability thereof, or any Loan Party or any other party to a Loan Document shall deny that such Person has any liability or obligation purported to be created under any Loan Document (in each case, other than solely as the result of (i) an action or failure to act on the part of the Lender or (ii) upon Payment in Full of the Obligations); or

(b) The liens, mortgages or security interests of the Lender in any material portion of the Collateral for the Project become unenforceable in whole or in part, or cease to be of the priority herein required or by the applicable Intercreditor Agreement, or subordination agreement or the validity or enforceability thereof, in whole or in part, shall be challenged or denied by Borrower or any Guarantor or any Affiliate thereof (other than (i) a release of Collateral in accordance with the terms of the Loan Documents (including in connection with any Transfer permitted under Section 6.3 or upon Payment in Full of the Obligations) or (ii) as a result of any

action or omission by the Lender (including failure of the Lender to maintain possession of any Collateral delivered to it or failure of the Lender to file UCC financing statements, amendments or continuations).

7.11.**Change of Control.** A Change of Control shall occur, whether directly or indirectly that is not a Permitted Change of Control.

7.12.**Proceedings.** The (a) indictment of, or commencement of criminal proceedings against, any Loan Party or any Affiliate under any criminal statute, (b) commencement of civil proceedings against any Loan Party pursuant to which proceedings the penalties or remedies sought or available include forfeiture to any Governmental Authority of any material portion of the property of such Loan Party.

7.13.**Bond Defaults.** The party to any bond for performance of the work or payment of the costs thereof fails to comply with the terms therefore upon which terms the effectiveness of such bond is conditioned, which failure thereof would reasonably be expected to result in a Project MAE.

8. **RIGHTS AND REMEDIES.**

8.1.**Rights and Remedies.** Upon the occurrence and during the continuation of an Event of Default, the Lender may (upon written notice to Borrower), in addition to any other rights or remedies provided for hereunder or under any other Loan Document or by applicable law, declare the principal of, and any and all accrued and unpaid interest and fees in respect of, the Term Loan and all other Obligations, whether evidenced by this Agreement or by any other Loan Documents to be immediately due and payable, whereupon the same shall become and be immediately due and payable and Borrower shall be obligated to repay all of such Obligations in full, without presentment, demand, protest, or further notice or other requirements of any kind, all of which are hereby expressly waived by Borrower, exercise all other rights and remedies available to the Lender under the Loan Documents, under applicable law, or in equity. In addition, upon the occurrence and during the continuation of an Event of Default, any commitment to make Term Loans shall, with any notice to Borrower or any other Person or any act by the Lender, terminate. The foregoing to the contrary notwithstanding, upon the occurrence of any Event of Default described in Section 7.4, in addition to the remedies set forth above, without any notice to Borrower or any other Person or any act by the Lender, the Obligations, inclusive of the principal of, and any and all accrued and unpaid interest and fees in respect of, the Term Loans and all other Obligations, whether evidenced by this Agreement or by any of the other Loan Documents, shall automatically become and be immediately due and payable and Borrower shall automatically be obligated to repay all of such Obligations in full, without presentment, demand, protest, or notice or other requirements of any kind, all of which are expressly waived by Borrower, and any commitment to make Term Loans shall terminate.

8.2.**Remedies Cumulative.** The rights and remedies of the Lender under this Agreement, the other Loan Documents, and all other agreements shall be cumulative. The Lender shall have all other rights and remedies not inconsistent herewith as provided under the Code, by law, or in equity. No exercise by the Lender of one right or remedy shall be deemed an election,

and no waiver by the Lender of any Event of Default shall be deemed a continuing waiver. No delay by the Lender shall constitute a waiver, election, or acquiescence by it.

8.3. **Equity Cure Rights.** In the event the Loan Parties fail to comply with any financial covenant set forth in Section 6.13 (including any financial covenant added after the Closing Date pursuant to Section 5.12), subject to the terms and conditions hereof, Parent and its Subsidiaries shall have the right until the expiration of the 15th Business Day subsequent to the date the applicable financial statements are required to be delivered to receive cash contributions from its direct or indirect shareholders in an aggregate amount equal to, but not greater than, the amount necessary to cure such financial covenant (hereinafter, the "Cure Right"), and within one (1) Business Day (or such longer period as Lender may agree in writing (including via email)) of receipt by Parent or any of its Subsidiaries of such cash pursuant to the exercise by Parent or any of its Subsidiaries of such Cure Right, such amount shall, prior to the expiration of such 15-Business Day period referred to above, be contributed or distributed (to the extent not issued by the Borrower), as the case may be, as cash common equity to the Borrower (the "Cure Amount") and shall be paid to the Lender as a mandatory prepayment of the Term Loans (applied in accordance with Section 2.3(b)), and such financial covenant shall then be recalculated giving effect to the following pro forma adjustments: (a) With regard to clause 6.13(a), (i) Consolidated EBITDA shall be increased for the applicable Fiscal Quarter and for the subsequent three (3) consecutive Fiscal Quarters, solely for the purpose of measuring such financial covenant and not for any other purpose under this Agreement, by an amount equal to the Cure Amount; and (ii) any prepayment of the Term Loans made with respect to such Cure Amount shall not serve as a reduction to Indebtedness for borrowed money for purposes of calculating any financial covenant set forth in Section 6.13 for the applicable Fiscal Quarter (but, notwithstanding anything to the contrary in this Agreement, such Cure Amount shall, if applied to the Term Loans, reduce Indebtedness for Borrowed Money for purposes of calculating any financial covenant set forth in Section 6.13 after the Fiscal Quarter in respect of which the Cure Amount is contributed); (b) with regard to clause 6.13(b), the Cure Amount shall be deemed an increase in the consolidated equity of Parent and its Subsidiaries; and (c) if, after giving effect to the foregoing recalculations, the Loan Parties shall then be in compliance with the requirements of the financial covenant set forth in Section 6.13, the Loan Parties shall be deemed to have been in compliance with such Financial Covenant as of the relevant date of determination with the same effect as though there had been no failure to comply therewith at such date, and the applicable breach or default of the financial covenant that had occurred shall be deemed not to have occurred for this purpose of the Agreement and no Event of Default shall be deemed to have occurred. Notwithstanding anything herein to the contrary, in no event shall Parent and/or the Borrower be permitted to exercise the Cure Right hereunder (x) more than four (4) times in the aggregate during the term of this Agreement or (y) more than two (2) times in any four consecutive Fiscal Quarter period.

9. WAIVERS; INDEMNIFICATION.

9.1. Demand; Protest; etc. Borrower waives demand, protest, notice of protest, notice of default or dishonor, notice of payment and nonpayment, nonpayment at maturity, release, compromise, settlement, extension, or renewal of documents, instruments, chattel paper, and guarantees at any time held by the Lender on which Borrower may in any way be liable.

9.2. Lender's Liability for Collateral. Borrower hereby agrees that: (a) so long as the Lender complies with its obligations, if any, under the Code, the Lender shall not in any way or manner be liable or responsible for: (i) the safekeeping of the Collateral, (ii) any loss or damage thereto occurring or arising in any manner or fashion from any cause, (iii) any diminution in the value thereof, or (iv) any act or default of any carrier, warehouseman, bailee, forwarding agency, or other Person, and (b) all risk of loss, damage, or destruction of the Collateral shall be borne by Borrower.

9.3. Indemnification. Borrower shall pay, indemnify, defend, and hold the Lender-Related Persons (each, an "Indemnified Person") harmless (to the fullest extent permitted by law) from and against any and all claims, demands, suits, actions, investigations, proceedings, liabilities, fines, costs, penalties, and damages, and all reasonable fees and disbursements of attorneys, experts, or consultants and all other costs and expenses actually incurred in connection herewith or in connection with the enforcement of this indemnification (as and when they are incurred and irrespective of whether suit is brought), at any time asserted against, imposed upon, or incurred by any of them (a) in connection with or as a result of or related to the execution and delivery incurred in advising, structuring, drafting, reviewing, or administering the Loan Documents, enforcement, performance, or administration (including any restructuring or workout with respect hereto) of this Agreement, any of the other Loan Documents, or the transactions contemplated hereby or thereby or the monitoring of Borrower's compliance with the terms of the Loan Documents; provided, that the indemnification in this clause (a) shall not extend to disputes solely between or among the Lender and its respective Affiliates that do not involve any acts or omissions of Borrower, and (b) with respect to any actual or prospective investigation, litigation, or proceeding related to this Agreement, any other Loan Document, the making of the Term Loan hereunder, or the use of the proceeds of the Term Loan hereunder (irrespective of whether any Indemnified Person is a party thereto), or any act, omission, event, or circumstance in any manner related thereto (each and all of the foregoing, the "Indemnified Liabilities"). The foregoing to the contrary notwithstanding, Borrower shall have no obligation to any Indemnified Person under this Section 9.3 with respect to any Indemnified Liability that a court of competent jurisdiction finally determines to have resulted from the fraud, gross negligence, bad faith or willful misconduct of such Indemnified Person or its officers, directors, employees, attorneys, or agents. This provision shall survive the termination of this Agreement and the Payment in Full of the Obligations. If any Indemnified Person makes any payment to any other Indemnified Person with respect to an Indemnified Liability as to which Borrower was required to indemnify the Indemnified Person receiving such payment, the Indemnified Person making such payment is entitled to be indemnified and reimbursed by Borrower with respect thereto. Any demand for payment of Indemnified Liabilities shall be made within thirty (30) days of demand therefor. **WITHOUT LIMITATION, THE FOREGOING INDEMNITY SHALL APPLY TO EACH INDEMNIFIED PERSON WITH RESPECT TO INDEMNIFIED LIABILITIES WHICH IN WHOLE OR IN PART ARE CAUSED BY OR ARISE OUT OF ANY ACT OR OMISSION OF SUCH INDEMNIFIED PERSON OR OF ANY OTHER PERSON, OTHER THAN FRAUD, GROSS NEGLIGENCE, BAD FAITH OR WILLFUL MISCONDUCT.**

10. NOTICES.

Unless otherwise provided in this Agreement, all notices or demands relating to this Agreement or any other Loan Document shall be in writing and (except for financial statements

and other informational documents which may be furnished electronically as contemplated in Section 5.2) shall be personally delivered or sent by registered or certified mail (postage prepaid, return receipt requested), overnight courier, electronic mail (at such email addresses as a party may designate in accordance herewith), or facsimile. In the case of notices or demands to Borrower or other Loan Parties or the Lender, as the case may be, they shall be sent to the respective address set forth below:

If to any Loan Party:

ASPEN AEROGELS, INC.

26 Forest Street, Suite 200

Marlborough, MA 01752

Attn: Ricardo C. Rodriguez and
Virginia H. Johnson

Email: rrodriguez@aerogel.com,
gjohnson@aerogel.com, and
legal@aerogel.com

with a copy to:

WEIL, GOTSHAL & MANGES LLP

767 Fifth Avenue

Attn: Andrew Colao, Esq. and
Jacqueline Oveissi, Esq.

Email: andrew.colao@weil.com
jacqueline.oveissi@weil.com

If to the Lender:

GENERAL MOTORS HOLDINGS LLC

Detroit Treasury Office

300 Renaissance Center

Mail code: 482-C26-D41

Detroit, MI 48265

Attention: Yana Shor, Assistant Treasurer

Email: treasurysfrm@gm.com

with copies to:

GENERAL MOTORS HOLDINGS LLC
Global Purchasing and Supply Chain
Cole Engineering Center
29755 Louis Chevrolet Road
Warren, MI 48090-9020
M/C 480-210-8N
Attn: Aaron M. Silver
Email: aaron.silver@gm.com

and to:

HONIGMAN LLP
315 East Eisenhower Parkway
Suite 100
Ann Arbor, Michigan 48108-3330
Attn: Barbara A. Kaye, Esq.
Email: bkaye@honigman.com

Any party hereto may change the address at which they are to receive notices hereunder, by notice in writing in the foregoing manner given to the other party. All notices or demands sent in accordance with this Section 10, shall be deemed received on the earlier of the date of actual receipt or three (3) Business Days after the deposit thereof in the mail; provided, that (a) notices sent by overnight courier service shall be deemed to have been given when received, (b) notices by facsimile shall be deemed to have been given when sent (except that, if not given during normal business hours for the recipient, shall be deemed to have been given at the opening of business on the next Business Day for the recipient) and (c) notices by electronic mail shall be deemed received upon the sender's receipt of an acknowledgment from the intended recipient (such as by the "return receipt requested" function, as available, return email or other written acknowledgment).

11. CHOICE OF LAW AND VENUE; JURY TRIAL WAIVER; JUDICIAL REFERENCE PROVISION.

(a) THE VALIDITY OF THIS AGREEMENT AND THE OTHER LOAN DOCUMENTS (UNLESS EXPRESSLY PROVIDED TO THE CONTRARY IN ANOTHER LOAN DOCUMENT IN RESPECT OF SUCH OTHER LOAN DOCUMENT), THE CONSTRUCTION, INTERPRETATION, AND ENFORCEMENT HEREOF AND THEREOF, THE RIGHTS OF THE PARTIES HERETO AND THERETO WITH RESPECT TO ALL MATTERS ARISING HEREUNDER OR THEREUNDER OR RELATED HERETO OR THERETO, AND ANY CLAIMS, CONTROVERSIES OR DISPUTES ARISING HEREUNDER OR THEREUNDER OR RELATED HERETO OR THERETO SHALL BE DETERMINED UNDER, GOVERNED BY, AND CONSTRUED IN ACCORDANCE WITH THE LAWS OF THE STATE OF MICHIGAN; PROVIDED, THAT WITH RESPECT TO ANY PORTION OF THE COLLATERAL LOCATED OUTSIDE OF THE STATE OF MICHIGAN, THE LAWS OF THE PLACE IN WHICH SUCH COLLATERAL IS OR IS DEEMED TO BE LOCATED IN SHALL APPLY TO THE EXTENT OF PROCEDURAL AND SUBSTANTIVE MATTERS RELATING ONLY

TO THE CREATION, PERFECTION, FORECLOSURE OF LIENS AND ENFORCEMENT OR RIGHTS AND REMEDIES AGAINST THE COLLATERAL OR REAL PROPERTY COLLATERAL.

(b) THE PARTIES AGREE THAT ALL ACTIONS OR PROCEEDINGS ARISING IN CONNECTION WITH THIS AGREEMENT AND THE OTHER LOAN DOCUMENTS SHALL BE TRIED AND LITIGATED ONLY IN THE STATE OF MICHIGAN AND, TO THE EXTENT PERMITTED BY APPLICABLE LAW, THE STATE COURTS LOCATED IN WAYNE COUNTY, MICHIGAN OR THE FEDERAL COURTS LOCATED IN THE EASTERN DISTRICT OF MICHIGAN; PROVIDED, THAT ANY SUIT SEEKING ENFORCEMENT AGAINST ANY COLLATERAL OR OTHER PROPERTY MAY BE BROUGHT, AT THE LENDER'S OPTION, IN THE COURTS OF ANY JURISDICTION WHERE SUCH COLLATERAL OR OTHER PROPERTY MAY BE FOUND. EACH LOAN PARTY AND THE LENDER WAIVES, TO THE EXTENT PERMITTED UNDER APPLICABLE LAW, ANY RIGHT IT MAY HAVE TO ASSERT THE DOCTRINE OF FORUM *NON CONVENIENS* OR TO OBJECT TO VENUE TO THE EXTENT ANY PROCEEDING IS BROUGHT IN ACCORDANCE WITH THIS SECTION 11(b).

(c) TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, EACH LOAN PARTY AND THE LENDER HEREBY WAIVES ITS RIGHTS, IF ANY, TO A JURY TRIAL OF ANY CLAIM, CONTROVERSY, DISPUTE OR CAUSE OF ACTION DIRECTLY OR INDIRECTLY BASED UPON OR ARISING OUT OF ANY OF THE LOAN DOCUMENTS OR ANY OF THE TRANSACTIONS CONTEMPLATED THEREIN, INCLUDING CONTRACT CLAIMS, TORT CLAIMS, BREACH OF DUTY CLAIMS, AND ALL OTHER COMMON LAW OR STATUTORY CLAIMS (EACH A "CLAIM"). EACH LOAN PARTY AND THE LENDER REPRESENTS THAT EACH HAS REVIEWED THIS WAIVER AND EACH KNOWINGLY AND VOLUNTARILY WAIVES ITS JURY TRIAL RIGHTS FOLLOWING CONSULTATION WITH LEGAL COUNSEL. IN THE EVENT OF LITIGATION, A COPY OF THIS AGREEMENT MAY BE FILED AS A WRITTEN CONSENT TO A TRIAL BY THE COURT.

(d) EACH OF THE LENDER AND THE LOAN PARTIES HEREBY IRREVOCABLY AND UNCONDITIONALLY SUBMITS TO THE EXCLUSIVE JURISDICTION OF THE STATE AND FEDERAL COURTS LOCATED IN THE COUNTY OF MICHIGAN AND THE STATE OF MICHIGAN, IN ANY ACTION OR PROCEEDING ARISING OUT OF OR RELATING TO ANY LOAN DOCUMENTS, OR FOR RECOGNITION OR ENFORCEMENT OF ANY JUDGMENT. EACH OF THE PARTIES HERETO AGREES THAT A FINAL JUDGMENT IN ANY SUCH ACTION OR PROCEEDING SHALL BE CONCLUSIVE AND MAY BE ENFORCED IN OTHER JURISDICTIONS BY SUIT ON THE JUDGMENT OR IN ANY OTHER MANNER PROVIDED BY LAW. NOTHING IN THIS AGREEMENT OR ANY OTHER LOAN DOCUMENT SHALL AFFECT ANY RIGHT THAT THE LENDER MAY OTHERWISE HAVE TO BRING ANY ACTION OR PROCEEDING RELATING TO THIS AGREEMENT OR ANY OTHER LOAN DOCUMENT AGAINST ANY LOAN PARTY OR ITS PROPERTIES IN THE COURTS OF ANY JURISDICTION.

(e) NO CLAIM MAY BE MADE BY ANY LOAN PARTY AGAINST THE LENDER OR ANY AFFILIATE, DIRECTOR, OFFICER, EMPLOYEE, COUNSEL, REPRESENTATIVE, OR ATTORNEY-IN-FACT OF THE LENDER FOR ANY SPECIAL, INDIRECT, CONSEQUENTIAL, PUNITIVE OR EXEMPLARY DAMAGES OR LOSSES IN RESPECT OF ANY CLAIM FOR BREACH OF CONTRACT OR ANY OTHER THEORY OF LIABILITY ARISING OUT OF OR RELATED TO THE TRANSACTIONS CONTEMPLATED BY THIS AGREEMENT OR ANY OTHER LOAN DOCUMENT, OR ANY ACT, OMISSION, OR EVENT OCCURRING IN CONNECTION THEREWITH, AND EACH LOAN PARTY HEREBY WAIVES, RELEASES, AND AGREES NOT TO SUE UPON ANY CLAIM FOR SUCH DAMAGES, WHETHER OR NOT ACCRUED AND WHETHER OR NOT KNOWN OR SUSPECTED TO EXIST IN ITS FAVOR.

12. ASSIGNMENTS; SUCCESSORS.

12.1.**Assignments.** The Lender may assign and delegate all or any portion of its rights and duties under the Loan Documents, including the Obligations owed to it, to one or more Eligible Assignees, subject to at least three (3) Business Days prior written notice to Borrower (but without any consent of Borrower). Borrower may not assign all or any portion of its rights and duties under the Loan Documents, including the Obligations, without the Lender's prior written consent.

12.2.**Successors.** This Agreement shall bind and inure to the benefit of the respective successors and assigns of each of the parties.

13. AMENDMENTS; WAIVERS.

13.1.**Amendments and Waivers.** No amendment, waiver or other modification of any provision of this Agreement or any other Loan Document, and no consent with respect to any departure by any Loan Party therefrom, shall be effective, unless the same shall be in writing and signed by the Lender and the Loan Parties party thereto, and then any such waiver or consent shall be effective, but only in the specific instance and for the specific purpose for which given.

13.2.**No Waivers; Cumulative Remedies.** No failure by the Lender to exercise any right, remedy, or option under this Agreement or any other Loan Document, or delay by the Lender in exercising the same, will operate as a waiver thereof. No waiver by the Lender will be effective unless it is in writing, and then only to the extent specifically stated. No waiver by the Lender on any occasion shall affect or diminish the Lender's rights thereafter to require strict performance by Borrower of any provision of this Agreement. The Lender's rights under this Agreement and the other Loan Documents will be cumulative and not exclusive of any other right or remedy that the Lender may have.

14. GENERAL PROVISIONS.

14.1.**Section Headings.** The section headings in this Agreement are for convenience of reference only and are not to affect the construction hereof or to be taken into consideration in the interpretation hereof.

14.2. **Interpretation.**

(a) Neither this Agreement nor any uncertainty or ambiguity herein shall be construed against the Lender or Borrower, whether under any rule of construction or otherwise. On the contrary, this Agreement has been reviewed by all parties and shall be construed and interpreted according to the ordinary meaning of the words used so as to accomplish fairly the purposes and intentions of all parties hereto.

(b) The definitions of terms in this Agreement apply equally to the singular and plural forms of the terms defined. Whenever the context may require, any pronoun includes the corresponding masculine, feminine and gender neutral forms. The words “include,” “includes” and “including” will be deemed to be followed by the phrase “without limitation.” The word “will” is intended to be construed to have the same meaning and effect as the word “shall”, and is intended to mean mandatory and not permissive, and “may not” is intended to mean an absolute prohibition. Unless the context requires otherwise:

(i) any definition of or reference to any agreement, instrument or other document herein will be construed as referring to the agreement, instrument or other document as from time to time amended, restated, supplemented, modified, renewed or extended;

(ii) any reference herein to any Person will be construed to include the Person’s permitted successors and assigns;

(iii) the words “herein,” “hereof” and “hereunder,” and words of similar import, will be construed to refer to this Agreement in its entirety and not to any particular provision hereof;

(iv) all references herein to Sections will be construed to refer to Sections of this Agreement; and

(v) the words “asset” and “property” will be construed to have the same meaning and effect and to refer to any and all tangible and intangible assets and properties, including cash, securities, accounts and general intangibles.

14.3. **Severability of Provisions.** Any provision of this Agreement that is prohibited or unenforceable in any jurisdiction shall, as to such jurisdiction, be ineffective to the extent of such prohibition or unenforceability without invalidating the remaining provisions hereof, and any such prohibition or unenforceability in any jurisdiction shall not invalidate or render unenforceable such provision in any other jurisdiction.

14.4. **Debtor-Creditor Relationship.** The relationship between the Lender, on the one hand, and Borrower, on the other hand, arising out of or in connection with the Loan Documents or the transactions contemplated thereby is solely that of creditor and debtor. The Lender does not have (or shall be deemed to have) any fiduciary relationship or duty to Borrower arising out of or in connection with the Loan Documents or the transactions contemplated thereby, and there is no agency or joint venture relationship between the Lender, on the one hand, and Borrower, on the other hand, by virtue of any Loan Document or any transaction contemplated therein.

14.5. Counterparts; Electronic Execution. This Agreement may be executed in any number of counterparts and by different parties on separate counterparts, each of which, when executed and delivered, shall be deemed to be an original, and all of which, when taken together, shall constitute but one and the same Agreement. Delivery of an executed counterpart of this Agreement by facsimile or other electronic method of transmission shall be equally as effective as delivery of an original executed counterpart of this Agreement. Any party delivering an executed counterpart of this Agreement by facsimile or other electronic method of transmission also shall deliver an original executed counterpart of this Agreement but the failure to deliver an original executed counterpart shall not affect the validity, enforceability, and binding effect of this Agreement. The foregoing shall apply to each other Loan Document *mutatis mutandis*.

14.6. Revival and Reinstatement of Obligations; Certain Waivers. If the Lender repays, refunds, restores, or returns in whole or in part, any payment or property (including any proceeds of Collateral) previously paid or transferred to the Lender in full or partial satisfaction of any Obligation or on account of any other obligation of Borrower or any Guarantor under any Loan Document, because the payment, transfer, or the incurrence of the obligation so satisfied is asserted or declared to be void, voidable, or otherwise recoverable under any law relating to creditors' rights, including provisions of the Bankruptcy Code relating to fraudulent transfers, preferences, or other voidable or recoverable obligations or transfers (each, a "Voidable Transfer"), or because the Lender elects to do so on the reasonable advice of its counsel in connection with a claim that the payment, transfer, or incurrence is or may be a Voidable Transfer, then, as to any such Voidable Transfer, or the amount thereof that the Lender elects to repay, restore, or return (including pursuant to a settlement of any claim in respect thereof), and as to all reasonable costs, expenses, and attorneys' fees of the Lender related thereto, the liability of Borrower with respect to the amount or property paid, refunded, restored, or returned will automatically and immediately be revived, reinstated, and restored and will exist.

14.7. Confidentiality.

(a) The Lender and the Loan Parties agree that all information regarding the Lender and any Loan Party, and either party's operations, assets, and existing and contemplated business plans ("Confidential Information") shall be treated by the other party in a confidential manner, and shall not be disclosed by either party to Persons who are not parties to this Agreement, except (i) to attorneys for and other advisors, accountants, auditors, and consultants to the parties and to employees, directors and officers of the Lender and Borrower (the Persons in this clause (i), "Party Representatives") on a "need to know" basis in connection with this Agreement and the transactions contemplated hereby and on a confidential basis subject to the terms of this Section 14.7; provided, further, that the disclosing party shall be responsible for such Party Representative's compliance with this Section 14.7; (ii) to Subsidiaries and Affiliates of the Lender and Borrower on a "need to know" basis in connection with this Agreement and the transactions contemplated hereby and on a confidential basis; provided, that any such Subsidiary or Affiliate shall have agreed to receive such information hereunder subject to the terms of this Section 14.7; provided, further, that the disclosing party shall be responsible for its Subsidiaries and Affiliates compliance with this Section 14.7; (iii) as may be required by regulatory authorities so long as such authorities are informed of the confidential nature of such information, (iv) as may be required by statute, decision, or judicial or administrative order, rule, or regulation, (v) as may be agreed to in advance in writing by the other party, (vi) as requested or required by any Governmental

Authority pursuant to any subpoena or other legal process, (vii) as to any such information that is or becomes generally available to the public (other than as a result of prohibited disclosure by any Loan Party, the Lender, or their respective Party Representatives), (viii) in connection with any assignment, participation or pledge permitted pursuant to Section 12.1; provided, that prior to receipt of Confidential Information any such assignee, participant, or pledgee shall have agreed in writing to receive such Confidential Information either subject to the terms of this Section 14.7 or pursuant to confidentiality requirements either substantially similar to those contained in this Section 14.7 or otherwise reasonably acceptable to Parent (and such Person may disclose such Confidential Information to Persons employed or engaged by them as described in clause (i) above), (ix) in connection with any litigation or other adversary proceeding involving parties hereto which such litigation or adversary proceeding involves claims related to the rights or duties of the parties under this Agreement or the other Loan Documents and (x) in connection with, and to the extent reasonably necessary for, the exercise of any party hereto of its rights or remedies under this Agreement or under any other Loan Document; provided, that with respect to any disclosure under clauses (iii), (iv), (vi), (ix) or (x), (A) prior to any disclosure thereunder, the disclosing party agrees to provide the other party with prior notice thereof, to the extent that it is practicable to do so and to the extent that the disclosing party is permitted to provide such prior notice to the other party by such regulatory authority, Governmental Authority, statute, decision, judicial or administrative order, rule, regulation or other law or such litigation or adversary proceeding, as applicable, (y) any disclosure thereunder shall be limited to the portion of the Confidential Information as may be required by such regulatory authority, Governmental Authority, statute, decision, judicial or administrative order, rule, regulation or other law or reasonably necessary to defend its rights under such litigation or adversary proceeding, as applicable and (z) the disclosing party shall use commercially reasonable efforts to ensure that any such information so disclosed is accorded confidential treatment.

(b) Without first obtaining the Lender's consent, no Loan Party will in any manner: (i) advertise or publish the fact that the Lender is a party to any Loan Document or performing any of its obligations under any Loan Document nor disclose the terms of this Agreement, the Term Loan or any of the other Loan Documents; (ii) use the Lender's trademarks, trade names or confidential information in its advertising or promotional materials; or (iii) use the Lender's trademarks, trade names or confidential information in any form of electronic communication such as web sites (internal or external), blogs or other types of postings, except as may be required to perform hereunder or as required by law including under the Securities Act or the Exchange Act, as applicable. Notwithstanding the foregoing, any announcement made by the parties hereto in relation to the execution of this Agreement will be shared by Borrower with the Lender reasonably in advance of the Closing Date.

14.8.Survival. All representations and warranties made by any Loan Party in the Loan Documents and in the certificates or other instruments delivered in connection with or pursuant to this Agreement or any other Loan Document shall be considered to have been relied upon by the other parties hereto and shall survive the execution and delivery of the Loan Documents and the making of any Term Loan, regardless of any investigation made by any such other party or on its behalf and notwithstanding that the Lender may have had notice or knowledge of any Default or Event of Default or incorrect representation or warranty at the time any credit is extended hereunder, and shall continue in full force and effect as long as the Obligations have not been Paid in Full.

14.9.**Integration.** This Agreement, together with the other Loan Documents, reflects the entire understanding of the parties with respect to the transactions contemplated hereby and shall not be contradicted or qualified by any other agreement, oral or written, before the date hereof.

14.10.**No Setoff.** All payments made by Borrower hereunder or under any note or other Loan Document will be made without setoff, counterclaim, or other defense and without deduction or withholding for any Taxes.

[SIGNATURE PAGES FOLLOW.]

IN WITNESS WHEREOF, the parties hereto have caused this Agreement to be executed and delivered as of the date first above written.

BORROWER:

ASPEN AEROGELS GEORGIA, LLC

By: _____
Name: _____
Title: _____

GUARANTORS:

ASPEN AEROGELS, INC.

By: _____
Name: _____
Title: _____

ASPEN AEROGELS RHODE ISLAND, LLC

By: _____
Name: _____
Title: _____

Signature Page to Loan Agreement

LENDER:

GENERAL MOTORS HOLDINGS LLC, as the
Lender

By: _____

Name: _____

Title: _____

Signature Page to Loan Agreement

Exhibit A

Form of Security Agreement

(See attached)

Exhibit B

Form of Guaranty Agreement

(See attached)

Exhibit C

Form of Collateral Assignment Agreement

(See attached)

Exhibit D

Form of Promissory Note

(See attached)

Exhibit E

Form of Revolver Intercreditor Agreement

(See attached)

Exhibit F

Form of Borrowing Request

(See attached)

Exhibit G

Form of Title Company “Date Down” Endorsement for Georgia Property

(See attached)

Exhibit H

Form of Compliance Certificate

(See attached).

EXHIBIT H – FORM OF COMPLIANCE CERTIFICATE

Attached to and made a part of that certain Loan Agreement, as it may be amended, restated, amended and restated, supplemented or otherwise modified in accordance with its terms from time to time, including all schedules and exhibits attached thereto (the “Agreement”), dated as of November 28, 2022, by and between General Motors Holdings LLC, a Delaware limited liability company (“General Motors” together with its successors and assigns, “Lender”), Aspen Aerogels Georgia, LLC, a Georgia limited liability company (the “Borrower”), Aspen Aerogels, Inc., a Delaware corporation (the “Parent”), Aspen Aerogels Rhode Island, LLC, a Rhode Island limited liability company (“Aspen RI”) and the other Guarantors (as that term is defined in the Agreement) from time to time party thereto. Capitalized terms used herein but not specifically defined herein shall have the meanings ascribed to them in the Agreement.

This Certificate is submitted pursuant to Section 5.2(j) of the Agreement.

The undersigned, in the undersigned’s capacity as Financial Officer of the Parent (and not in any individual capacity and without assuming any personal liability), hereby certifies to Lender that as of the date of this Certificate:

1. The undersigned is a Financial Officer of the Parent.
2. The financial statements of Parent and its Subsidiaries being concurrently delivered herewith have been prepared in accordance GAAP consistently applied and there have been no material changes in accounting policies or financial reporting practices of each Loan Party since [date of the last Compliance Certificate/date of last financial statements delivered prior to closing] or, if any such change has occurred, such changes are set forth in a writing attached hereto.
3. Attached hereto as Schedule I(a) is a true and correct calculation of the Total Leverage Ratio covenant contained in Section 6.13(a) of the Agreement as of the Test Date ended [insert applicable Fiscal Quarter end].
4. Attached hereto as Schedule I(b) is a true and correct calculation of the ratio of total Indebtedness of Parent and its Subsidiaries (excluding Indebtedness incurred under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000 and undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness) to the consolidated equity of Parent and its Subsidiaries contained in Section 6.13(b) of the Agreement.

[Signature Page follows]

ASPEN AEROGELS, INC.

By:

Name:

Title:

[Chief Financial Officer]

[Signature Page to Compliance Certificate]

Schedule I – Financial Covenant Calculations¹

Schedule I(a): Total Leverage Ratio § 6.13(a)

A. Required Covenant:

<u>Test Date</u>	<u>Ratio</u>
<u>As of each Test Date, commencing with the Fiscal Quarter ending December 31, 2024</u>	<u>Less than or equal to 5.00 to 1.00</u>

B. Actual Computation: _____ : 1.00

<u>1.</u>	<u>Total Indebtedness (see Section D hereof) of Parent and its Subsidiaries</u>	<u>\$ _____</u>
<u>2.</u>	<u>MINUS Indebtedness outstanding under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000 and undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness</u>	<u>\$(_____)</u>
<u>3.</u>	<u>Subtotal (1-2) =</u>	
	<u>Divided by:</u>	
<u>4.</u>	<u>Consolidated EBITDA of Parent and its Subsidiaries for the most recently completed Measurement Period</u>	<u>\$ _____</u>
<u>5</u>	<u>Ratio (3 ÷ 4) =</u>	<u>_____</u>

C. Consolidated EBITDA

<u>1. Consolidated Net Income of Parent and its Subsidiaries for the most recently completed Measurement Period</u>	<u>\$ _____</u>
<u>PLUS (to the extent included in calculating Consolidated Net Income for the most recently completed Measurement Period, without duplication)</u>	
<u>2. the sum of all interest, premium payments, debt discount, fees, charges and related expenses in connection with Indebtedness for borrowed money to the extent treated as interest in accordance with GAAP</u>	<u>\$ _____</u>

¹To be delivered also following the delivery of the audited F/S if such statements would result in a different calculation than as set forth in the certificate delivered concurrently with the unaudited financial statements for the applicable period.

3. <u>provision for federal, state, local and foreign income Taxes, taxes on profit or capital, including, without limitation, state franchise and similar Taxes, and foreign withholding taxes</u>	\$ _____
4. <u>depreciation and amortization expenses</u>	\$ _____
5. <u>noncash (A) charges, expenses, costs or losses solely to the extent constituting extraordinary, unusual or non-recurring items, (B) compensation charges, (C) charges for the excess of GAAP rent expense over actual rent paid during such period due to the use of straight line rent for GAAP purposes or write-off and/or write-downs of inventory or other tangible assets and (D) asset write-offs and/or write-downs or any impairment charges</u>	\$ _____
6. <u>charges or expenses incurred in connection with the issuance or incurrence of permitted Indebtedness or the issuance of Equity Interests (that do not result in a Change of Control) in an aggregate amount during the term of this Agreement not to exceed \$500,000</u>	
7. <u>any other adjustments as shall be mutually agreed</u>	\$ _____
8. <u>Subtotal (1+2+3+4+5+6+7) =</u>	\$ _____
<u>MINUS (to the extent included in calculating such Consolidated Net Income for the most recently completed Measurement Period, without duplication)</u>	
9. <u>federal, state, local and foreign income Tax credits, Tax credits on profit or capital, including, without limitation, state franchise and similar Tax credits, and foreign withholding Tax credits</u>	\$ _____
10 <u>any noncash gain from extraordinary, unusual or non-recurring items</u> ±	\$ _____
11 <u>Subtotal (9+10) =</u> ±	\$ _____
<u>Total (8-11) =</u>	\$ _____

D. Indebtedness

(i) The amount of any Indebtedness represented by a guaranty or other similar instrument shall be the lesser of the principal amount of the obligations guaranteed and still outstanding and the maximum amount for which the guaranteeing Person may be liable pursuant to the terms of the instrument embodying such Indebtedness, and

<u>(ii) the amount of any Indebtedness which is limited or is non-recourse to a Person or for which recourse is limited to an identified asset shall be valued at the lesser of (A) if applicable, the limited amount of such obligations, and (B) if applicable, the fair market value of such assets securing such obligation</u>	
<u>1. All obligations of such Person for borrowed money</u>	<u>\$ _____</u>
<u>2. all obligations of such Person evidenced by bonds, debentures, notes, or other similar instruments and all reimbursement or other obligations in respect of letters of credit, bankers acceptances, or other financial products</u>	<u>\$ _____</u>
<u>3. all obligations or liabilities of others secured by a Lien on any asset of such Person, irrespective of whether such obligation or liability is assumed</u>	<u>\$ _____</u>
<u>4. all obligations of such Person to pay the deferred purchase price of assets (other than trade payables incurred in the ordinary course of business and repayable in accordance with customary trade practices and which are paid in accordance with such terms and not on a deferred basis)</u>	<u>\$ _____</u>
<u>5. all obligations of such Person in respect of Disqualified Equity Interests</u>	<u>\$ _____</u>
<u>6. any obligation of such Person guaranteeing or intended to guarantee (whether directly or indirectly guaranteed, endorsed, co-made, discounted, or sold with recourse) any obligation of any other Person that constitutes Indebtedness under any of clauses (1) through (5) above</u>	<u>\$ _____</u>
<u>Total (1+2+3+4+5+6) =</u>	<u>\$ _____</u>

Schedule I(b): Ratio of Total Indebtedness to Consolidated Equity § 6.13(b)

A. Required Covenant:

<u>Test Date</u>	<u>Ratio</u>
<u>As of each Test Date, commencing with the Fiscal Quarter ending December 31, 2024</u>	<u>Less than or equal to 1.20x</u>

B. Actual Computation: x

1.	<u>Total Indebtedness (see Section C hereof) of Parent and its Subsidiaries</u>	<u>\$ _____</u>
2.	<u>MINUS undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness</u>	<u>\$ (_____)</u>
3.	<u>Subtotal (1-2) =</u>	
	<u>Divided by:</u>	
4.	<u>consolidated equity of Parent and its Subsidiaries</u>	<u>\$ _____</u>
5.	<u>Ratio (3 ÷ 4) =</u>	<u>_____</u>

C. Indebtedness

<u>(i) The amount of any Indebtedness represented by a guaranty or other similar instrument shall be the lesser of the principal amount of the obligations guaranteed and still outstanding and the maximum amount for which the guaranteeing Person may be liable pursuant to the terms of the instrument embodying such Indebtedness, and (ii) the amount of any Indebtedness which is limited or is non-recourse to a Person or for which recourse is limited to an identified asset shall be valued at the lesser of (A) if applicable, the limited amount of such obligations, and (B) if applicable, the fair market value of such assets securing such obligation</u>	
<u>1. All obligations of such Person for borrowed money</u>	<u>\$ _____</u>
<u>2. all obligations of such Person evidenced by bonds, debentures, notes, or other similar instruments and all reimbursement or other obligations in respect of letters of credit, bankers acceptances, or other financial products</u>	<u>\$ _____</u>
<u>3. all obligations or liabilities of others secured by a Lien on any asset of such Person, irrespective of whether such obligation or liability is assumed</u>	<u>\$ _____</u>
<u>4. all obligations of such Person to pay the deferred purchase price of assets (other than trade payables incurred in the ordinary course of business and repayable in accordance with customary trade practices and which are paid in accordance with such terms and not on a deferred basis)</u>	<u>\$ _____</u>
<u>5. all obligations of such Person in respect of Disqualified Equity Interests</u>	<u>\$ _____</u>

6. <u>any obligation of such Person guaranteeing or intended to guarantee (whether directly or indirectly guaranteed, endorsed, co-made, discounted, or sold with recourse) any obligation of any other Person that constitutes Indebtedness under any of clauses (1) through (5) above</u>	<u>\$</u> _____
7. <u>Subtotal (1+2+3+4+5+6) =</u>	<u>\$</u> _____
<u>MINUS</u>	
8. <u>Indebtedness under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000</u>	<u>\$</u> _____
<u>Total (7-8) =</u>	<u>\$</u> _____

Schedule 1.1

Definitions

As used in this Agreement, the following terms shall have the following definitions:

“Account” is any “account” as defined in the Code with such additions to such term as may hereafter be made, and includes, without limitation, all accounts receivable and other sums owing to any Loan Party.

“Account Services” means commercial credit cards, stored value cards, purchasing cards, treasury management services, netting services, overdraft protections, check drawing services, automated payment services (including depository, overdraft, controlled disbursement, ACH transactions, return items and interstate depository network services), employee credit card programs, cash pooling services, and any arrangement and/or service similar to any of the foregoing and/or otherwise in connection with Cash management and any Accounts.

“Additional Financing Facility” means one or more equity or debt financing arrangements with one or more third parties to provide Borrower with cash proceeds, in the aggregate, of not less than ~~(excluding the Equity Investment)~~ \$350,000,000 (inclusive of cash proceeds contributed to the Borrower for use in connection with the construction of the Georgia Facility prior to the First Amendment Effective Date).

“Additional Letters of Credit” has the meaning specified therefor in clause (n) of the definition of “Permitted Indebtedness”.

“Affiliate” means, as applied to any Person, any other Person who controls, is controlled by, or is under common control with, such Person. For purposes of this definition, “control” means the possession, directly or indirectly through one or more intermediaries, of the power to direct the management and policies of a Person, whether through the ownership of Equity Interests, by contract, or otherwise; provided, that with regard to (a) Borrower: (i) any Person which Controls or is Controlled by Borrower shall be deemed an Affiliate of Borrower and (ii) each partnership in which Borrower is a general partner shall be deemed an Affiliate of Borrower and (b) the Guarantor: (i) any Person which Controls or is Controlled by the Guarantor shall be deemed an Affiliate of the Guarantor and (ii) each partnership in which the Guarantor is a general partner shall be deemed an Affiliate of the Guarantor.

“Agreement” means the Loan Agreement to which this Schedule 1.1 is attached, as amended, supplemented or otherwise modified from time to time.

“Anti-Corruption Laws” has the meaning specified therefor in Section 4.12(a) of this Agreement.

“Architect” means Thompson Consultants Inc. who has been retained by Borrower to design and inspect the construction of the Improvements.

“Architect’s Contract” ~~shall~~ means the written agreement between Borrower and the Architect, dated February 16, 2022 (as amended, restated, amended and restated, supplemented or otherwise modified from time to time in a manner not otherwise prohibited by this Agreement), for architectural services relating to the design and construction of the Improvements.

“Bankruptcy Code” means title 11 of the United States Code, as in effect from time to time.

“Board of Directors” means, as to any Person, the board of directors (or comparable managers), sole member or equivalent governing body, as applicable, of such Person, or any committee thereof duly authorized to act on behalf of the board of directors (or comparable managers), sole member or equivalent governing body, as applicable.

“Books” all books and records including ledgers, federal and state tax returns, records regarding assets or liabilities, the Collateral, business operations or financial condition, and all computer programs or storage or any equipment containing such information.

“Borrower” has the meaning specified therefor in the preamble to this Agreement.

“Borrowing Request” has the meaning specified therefor in Section 2.1(b)(i) of this Agreement.

“Budget” means, collectively, (a) the Project Budget and (b) any Term Loan Budget.

“Business Day” means any day that is not a Saturday, Sunday, or other day on which banks are authorized or required to close in the State of Michigan.

“Capital Lease” means, as applied to any Person, any lease of any property (whether real, personal or mixed) by that Person as lessee that, in conformity with GAAP as in effect prior to giving effect to the adoption of ASU No. 2016-02 “Leases (Topic 842)” and ASU No. 2018-11 “Leases (Topic 842)”, is or should be accounted for as a capital lease on the balance sheet of that Person; provided, that for the avoidance of doubt, the amount of obligations attributable to any Capital Lease shall be the amount thereof accounted for as a liability in accordance with GAAP.

“Cash Equivalents” means (a) marketable direct obligations issued or unconditionally guaranteed by the United States or any agency or any State thereof having maturities of not more than one (1) year from the date of acquisition; (b) commercial paper maturing no more than one (1) year after its creation and having the highest rating from either Standard & Poor’s Ratings Group or Moody’s Investors Service, Inc.; (c) certificates of deposit issued maturing no more than one (1) year after issue; and (d) money market funds at least ninety-five percent (95%) of the assets of which constitute Cash Equivalents of the kinds described in clauses (a) through (c) of this definition.

“Certificate of Occupancy” has the meaning specified therefor in Section 2.2(a) of this Agreement.

“Change of Control” means that

- (i) Parent ceases to Control Borrower or Aspen RI; or
-

(ii) the acquisition of ownership or voting control, directly or indirectly, beneficially (within the meaning of Rules 13d-3 and 13d-5 of the Exchange Act) or of record, on or after the Closing Date, by any Person or group (within the meaning of Sections 13d and 14d of the Exchange Act), of shares representing more than fifty percent (50%) of the aggregate ordinary voting power represented by the issued and outstanding capital stock of Parent.

“Claim” has the meaning specified therefor in Section 11(c) of this Agreement.

“Closing Date” means the date on which the conditions set forth in Section 3.1 are satisfied.

“Code” means the Michigan Uniform Commercial Code, as in effect from time to time.

“Collateral” means all property pledged as collateral security for the Obligations under the Security Agreement, the Mortgage, the Leasehold Deed to Secure Debt or any other Security Document, and all other property of the Loan Parties on which the Lender has been granted a Lien by any Loan Party; provided, that, for the avoidance of doubt, (a) any Intellectual Property and Intellectual Property Rights owned or purported to be owned by any Loan Party, (b) any pumps pledged to secure Borrower’s EDGE grant from the Georgia Department of Community Affairs, in each case, does not and shall not constitute Collateral and (c) funds in any Excluded Accounts; provided, that (i) the aggregate balance in such Accounts does not exceed 105% of the amounts permitted pursuant to clause (n) of the definition of “Permitted Indebtedness” at any time and (ii) the sole use of the funds in such accounts is to secure the Parent’s obligations under the Existing Letters of Credit and the Additional Letters of Credit (the foregoing clauses (a), (b) and (c), “Excluded Assets”).

“Collateral Account” is any Deposit Account, Securities Account (as that term is defined in the Code) or Commodity Account (as that term is defined in the Code), in each case, other than any Excluded Accounts.

“Collateral Access Agreement” means an agreement, in form and substance reasonably acceptable to the Lender, between the Lender, the applicable Loan Party and any Person (other than a Loan Party) in possession of any Collateral.

“Collateral Assignment Agreement” means an agreement substantially in the form of Exhibit C hereto, with respect to the contracts listed on Schedule 1.1(c).

“Company Competitor” means any Person that (a) is an entity that designs, develops or manufactures aerogels-based or aerogel-like materials based products or technology, (b) designs, develops or manufactures insulation, fire barrier or other materials for use primarily in energy infrastructure, battery systems and/or sustainable building materials, (c) designs, develops or manufactures battery materials or (d) is an auto part and system supplier or, in each case, any Affiliate of such Person ~~(including the Persons identified via a separate written notice delivered by Borrower to the Lender on the Closing Date).~~

“Compliance Certificate” has the meaning specified therefor in Section 5.2(j) of this Agreement.

“Consolidated EBITDA” means, at any date of determination, an amount equal to Consolidated Net Income of Parent and its Subsidiaries for the most recently completed Measurement Period, plus

the following to the extent included in calculating Consolidated Net Income for the most recently completed Measurement Period, without duplication:

(i) Consolidated Interest Expense;

(ii) the provision for federal, state, local and foreign income Taxes, taxes on profit or capital, including, without limitation, state franchise and similar Taxes, and foreign withholding taxes;

(iii) depreciation and amortization expense;

(iv) noncash (A) charges, expenses, costs or losses solely to the extent constituting extraordinary, unusual or non-recurring items, (B) compensation charges, (C) charges for the excess of GAAP rent expense over actual rent paid during such period due to the use of straight line rent for GAAP purposes or write-off and/or write-downs of inventory or other tangible assets and (D) asset write-offs and/or write-downs or any impairment charges;

(v) charges or expenses incurred in connection with the issuance or incurrence of permitted Indebtedness or the issuance of Equity Interests (that do not result in a Change of Control) in an aggregate amount during the term of this Agreement not to exceed \$500,000; and

(vi) any other adjustments as shall be mutually agreed.

minus

(b) the following to the extent included in calculating such Consolidated Net Income for the most recently completed Measurement Period, without duplication:

(i) federal, state, local and foreign income Tax credits, Tax credits on profit or capital, including, without limitation, state franchise and similar Tax credits, and foreign withholding Tax credits; and

(ii) any noncash gain from extraordinary, unusual or non-recurring items.

“Confidential Information” has the meaning specified therefor in Section 14.7 of this Agreement.

“Consolidated Interest Expense” means, without duplication, for any Measurement Period, the result of the sum of all interest, premium payments, debt discount, fees, charges and related expenses in connection with Indebtedness for borrowed money to the extent treated as interest in accordance with GAAP.

“Consolidated Net Income” means, as of any date of determination, with respect to any Person and its Subsidiaries, for any Measurement Period, the net income (or loss) of such Parent and its Subsidiaries for such Measurement Period, determined on a consolidated basis in accordance with GAAP.

“Construction Contracts” mean (a) the General Contractor’s Contract, (b) all other construction or development contracts executed by Borrower, as opposed to contracts executed by the General Contractor, for the construction of the Improvements, and (c) all contracts executed by Contractors.

“Contractor” ~~shall~~ means each contractor, subcontractor or supplier who will perform work or supply materials for the Georgia Facility pursuant to contracts or work or purchase orders executed or issued by Borrower or the General Contractor.

“Control” means the possession, directly or indirectly, of the power to direct or cause the direction of the management or policies of a Person, even if subject to approval rights with respect to certain decisions, whether through the ability to exercise voting power, by contract or otherwise. A Person shall not be deemed to be “controlled by” another Person solely by means of such other Person’s ownership of Equity Interests in the Person specified unless such other Person holds or beneficially owns, directly or indirectly, 10% or more of the Equity Interest in the Person specified or 10% or more of any class of voting securities of the Person specified; provided, however, the foregoing limitations shall not restrict whether a Person is deemed to be “controlled by” another Person if such other Person has the power to direct or cause the direction of the management or policies of a Person, even if subject to approval rights with respect to certain decisions, and even if through the ability to exercise voting power less than the ownership thresholds set forth in the foregoing clause, by contract or otherwise. “Controlling” and “Controlled” have meanings correlative thereto.

“Control Agreement” means a fully-blocked control agreement, in form and substance reasonably satisfactory to the Lender, executed and delivered by any Loan Party, the Lender and the financial institution in which the applicable account is maintained.

“Convertible Notes” means those certain Aspen Aerogels, Inc. Convertible Senior PIK Toggle Notes due 2027.

“Convertible Notes Subordination Agreement” means that certain Subordination Agreement, dated as of the Closing Date, among the Lender, Parent, Wood River Capital, LLC, Borrower, Aspen RI, subordinating the Holders (as defined in the Convertible Notes) right of payment to the Obligations, as such agreement may be amended, restated, amended and restated, supplemented or otherwise modified from time to time in accordance with its terms.

“Cure Amount” has the meaning assigned to such term in Section 8.3.

“Cure Right” has the meaning assigned to such term in Section 8.3.

“Default” means an event, condition, or default that, with the giving of notice, the passage of time, or both, would be an Event of Default.

“Default Interest” has the meaning specified therefor in Section 2.2(b) of this Agreement.

“Deposit Account” means any deposit account (as that term is defined in the Code).

“Disbursement Account” has the meaning specified therefor in Section 5.6 of this Agreement.

“Disinterested Board Member” means a member of the Board of Directors of any Loan Party who (a) is not a current employee of such Loan Party or any of its Affiliates, (b) does not receive remuneration from such Loan Party or any of its Affiliates, either directly or indirectly, in any capacity other than as a director, except in an amount for which disclosure would not be required pursuant to Item 404(a) of the proxy solicitation rules of the SEC and (c) does not possess an interest in any other transaction, and is not engaged in a business relationship, for which disclosure would be required pursuant to Item 404(a) or (b) of the proxy solicitation rules of the SEC.

“Disqualified Equity Interest” means any Equity Interest that, by its terms (or the terms of any security or other Equity Interests into which it is convertible or for which it is exchangeable), or upon the happening of any event or condition (a) matures or is mandatorily redeemable (other than solely for Equity Interests that are not Disqualified Equity Interests), pursuant to a sinking fund obligation otherwise, (b) contains a mandatory repurchase obligation or any other repurchase obligation at the option of the holder thereof (other than solely for Equity Interests that are not Disqualified Equity Interests), in whole or in part, (c) provides for scheduled payments of dividends in cash, or (d) is or becomes convertible into or exchangeable for Indebtedness or any other Equity Interests that would constitute Disqualified Equity Interests, in each case, prior to the date that is ninety-one (91) days after the Term Loan Commitment Expiration Date; provided, that any Equity Interests that would not constitute Disqualified Equity Interests but for provisions thereof giving holders thereof (or the holders of any security into or for which such Equity Interests are convertible, exchangeable or exercisable) the right to require the issuer thereof to redeem such Equity Interests upon the occurrence of any change of control or asset sale event occurring prior to ninety-one (91) days following the Maturity Date at the time such Equity Interests are issued shall not constitute Disqualified Equity Interests if such Equity Interests provide that the issuer thereof will not redeem any such Equity Interests pursuant to such provisions prior to the Maturity Date).

Notwithstanding the foregoing, (A) if such Equity Interests are issued pursuant to any plan for the benefit of directors, officers, employees, members of management, managers or consultants or by any such plan to such directors, officers, employees, members of management, managers or consultants, in each case, in the ordinary course of business of Parent, Borrower or any of its Subsidiaries, such Equity Interests shall not constitute Disqualified Equity Interests solely because it may be required to be repurchased by the issuer thereof in order to satisfy applicable statutory or regulatory obligations, and (B) no Equity Interests held by any future, present or former employee, director, officer, manager, member of management or consultant (or their respective Affiliates) of Parent, Borrower or its Subsidiaries shall be considered Disqualified Equity Interests because such stock is redeemable or subject to repurchase pursuant to any management equity subscription agreement, stock option, stock appreciation right or other stock award agreement, stock ownership plan, Governing Document, stockholder agreement or similar agreement that may be in effect from time to time.

“Division” means, in reference to any Person which is an entity, the division of such Person into two (2) or more separate Persons, with the dividing Person either continuing or terminating its existence as part of such division, including, without limitation, as contemplated under Section 18-217 of the Delaware Limited Liability Company Act for limited liability companies formed under Delaware law, or any analogous action taken pursuant to any other applicable law with respect to any corporation, limited liability company, partnership or other entity.

“Dollars” or “\$” means United States dollars.

“Domestic Subsidiary” means a Subsidiary organized under the laws of the United States or any state or territory thereof or the District of Columbia.

“Eligible Assignee” means (a) any Affiliate of the Lender or (b) any other Person (other than (i) a natural person and (ii) any Person that is identified by Borrower as a Company Competitor of Parent and/or any of its Subsidiaries (or any Affiliate of such competitor) at the time the Lender provides notice of such assignment pursuant to Section 12.1).

“Environmental Action” means any written complaint, summons, citation, notice of violation, order, Claim, judicial or administrative proceeding or judgment, received by any Loan Party from any Governmental Authority or any third party alleging: (a) violations of or non-compliance with Environmental Law; (b) releases of Hazardous Materials from or at any assets, properties, or businesses of any Loan Party, any Subsidiary of a Loan Party, or any of their predecessors in interest; or (c) releases of Hazardous Materials from or at any off-site facilities which received Hazardous Materials for off-site treatment, storage, disposal, or other management that were generated in connection with the Project or the Rhode Island Property by any Loan Party, any Subsidiary of any Loan Party, or any of their predecessors in interest.

“Environmental Indemnity Agreements” means those two Environmental Indemnity Agreements, dated as of the date hereof, among the Lender and the Loan Parties as indemnitors, governing the Rhode Island Property and the Georgia Property, respectively.

“Environmental Law” means any applicable federal, state, provincial, foreign or local statute, law, rule, regulation, ordinance or code now or hereafter in effect and in each case as amended, including any judicial or administrative order, consent decree or judgment, in each case, to the extent binding on any Loan Party, relating to the protection of the environment or to Hazardous Materials, including, without limitation, those governing the Permits, and also including, without limitation, the Comprehensive Environmental Response, Compensation, and Liability Act, the Resource Conservation and Recovery Act, the Toxic Substances Control Act, the Clean Air Act, the Clean Water Act, and similar state and local requirements.

“Environmental Liabilities” means all liabilities, monetary obligations, losses, damages, costs and expenses (including all reasonable fees, disbursements and expenses of counsel, experts, or consultants, and costs of investigation and feasibility studies), fines, penalties, sanctions, and interest incurred as a result of any Remedial Action required under Environmental Law, and which relate to any Environmental Action.

“Environmental Lien” means any Lien in favor of any Governmental Authority for Environmental Liabilities.

“Equipment” means all “equipment” as defined in the Code with such additions to such term as may hereafter be made, and includes without limitation all machinery, fixtures, goods, vehicles (including motor vehicles and trailers), and any interest in any of the foregoing.

“Equity Interests” means, with respect to a Person, all of the shares, options, warrants, interests, participations, or other equivalents (regardless of how designated) of or in such Person, whether voting or nonvoting, including capital stock (or other ownership or profit interests or units), preferred stock, or any other “equity security” (as such term is defined in Rule 3a11-1 of the General Rules and Regulations promulgated by the SEC under the Exchange Act) and including any of the foregoing which is by its terms or upon the happening of an event convertible into any such other interest.

“Equity Investment” has the meaning specified therefor in Schedule 3.2 of this Agreement.

“Event of Default” has the meaning specified therefor in Section 7 of this Agreement.

“Excluded Accounts” means the Accounts listed on Schedule 1.1(d) (as may be supplemented by Borrower in writing from time to time).

“Exchange Act” means the Securities Exchange Act of 1934, as in effect from time to time.

“Excluded Assets” has the meaning specified therefor in the definition of “Collateral”.

“Existing Letters of Credit” has the meaning specified therefor in clause (n) of the definition of “Permitted Indebtedness”.

“Financial Officer” means the Chief Financial Officer, principal accounting officer, treasurer or corporate controller.

“First Amendment Effective Date” means September 28, 2023.

“Fiscal Quarter” means the fiscal quarter of Parent, ending on the last day of each March, June, September and December.

“Floor” means a rate of interest equal to 1.00%.

“Force Majeure” means any of the following circumstances to the extent they are outside the control of any Loan Party: (a) any act or provision of any present or future law or regulation or governmental authority; (b) acts of God; (c) earthquakes; (d) floods; (e) wars; (f) terrorism; or (g) changes to the design of the Project required by the Lender and which could reasonably be expected to result in a Project MAE.

“Foreign Official” has the meaning specified therefor in Section 4.12(b)(i) of this Agreement.

“Foreign Subsidiary” means any direct or indirect Subsidiary of any Loan Party that is not a Domestic Subsidiary.

“Funding Date” means each date of borrowing on which the conditions set forth in Section 3.2 are satisfied.

“GAAP” means generally accepted accounting principles as in effect from time to time in the United States, consistently applied.

“General Contractor” ~~shall~~ means Turner Construction, who has been retained by Borrower as the general contractor for the construction of the Georgia Facility.

“General Contractor’s Contract” ~~shall~~ means the stipulated sum contract, dated as of February 17, 2022 (as amended, restated, amended and restated, supplemented or otherwise modified from time to time in a manner not otherwise prohibited by this Agreement), between Borrower and the General Contractor for the costs of construction of the Improvements.

“General Motors” has the meaning specified therefor in the preamble to this Agreement.

“Georgia Facility” ~~shall~~ means that certain aerogel manufacturing plant to be built on the Georgia Property.

“Georgia Lease” that certain Lease Agreement dated as of May 1, 2022, between Borrower and the Development Authority of Bulloch County with respect to the Georgia Property.

“Georgia Property” ~~shall~~ means the real property located at Bulloch County, Georgia, upon which the Georgia Facility are to be constructed. The Property is legally described in Exhibit A to the Georgia Lease.

“GM Competitor” is any entity that is:

(a) an original equipment manufacturer of, or any Person that controls or owns substantially all of the Equity Interests in an original equipment manufacturer of, Vehicles (whether gas, electric powered, or hybrid) that are used primarily on public roads, streets, or highways (whether sold direct or through franchised dealers), including, without limitation, any Affiliate, subsidiary, or entity similar to or in competition with an entity that has a trademark, service mark, or brand owned or operated by Ford, Stellantis (or any of its predecessors or subsidiaries including FCA US LLC), Nissan, Toyota, Tesla, Jaguar Land Rover, Rivian, Daimler, Zoox, Nikola Corporation, VW, Kia, or Hyundai; or

(b) a distributor, seller, contract manufacturer, or other entity that manufactures, has manufactured, or otherwise purchases Vehicles that are used to provide (whether directly or through independent contractors) services to, or deliver goods for, third parties including, without limitation, such services that qualify or otherwise constitute transportation as a service, mobility as a service, shared autonomous vehicles, logistics, transportation, or other types of services; or

(c) a manufacturer, distributor, or seller of components and other raw materials that are incorporated into, used in the propulsion system of a Vehicle including EV tractions motors, electric drive units or other electrical architecture.

“GM Purchase Contracts” means all purchase orders and related agreements, as the same may be amended, restated, amended and restated, supplemented or otherwise modified from time to time, entered into by Parent or any of its Affiliates and the Lender (or any of the Lender’s affiliates), in connection with the purchase of the Product.

“Governing Documents” means, with respect to any Person, the certificate or articles of incorporation or formation, by-laws, limited liability company agreement or operating agreement or other equivalent organizational documents of such Person.

“Governmental Approval” is any consent, authorization, approval, order, license, franchise, permit, certificate, accreditation, registration, filing or notice, of, issued by, from or to, or other act by or in respect of, any Governmental Authority.

“Governmental Authority” means the government of any nation or any political subdivision thereof, whether at the national, state, territorial, provincial, municipal or any other level, and any agency, authority, instrumentality, regulatory body, court, central bank or other entity exercising executive, legislative, judicial, taxing, regulatory or administrative powers or functions of, or pertaining to, government.

“Governmental Requirements” means all laws, ordinances, rules, regulations, codes, orders, writs, injunctions or decrees of any Governmental Authority applicable to Borrower, Guarantor, or the Property or any portion of the Project.

“Guarantors” means Parent, Aspen RI and any additional guarantor entering into a Guaranty after the Closing Date in accordance with Section 5.9.

“Guaranty” has the meaning specified therefor in Section 2.4(ii) of this Agreement.

“Hazardous Materials” means: (a) substances that give rise to liability under Environmental Law or are defined or listed in, or otherwise classified pursuant to, any applicable laws or regulations as “hazardous substances,” “hazardous materials,” “hazardous wastes,” “toxic substances,” or substances regulated by reason of deleterious properties such as ignitability, corrosively, reactivity, carcinogenicity, reproductive toxicity, or “EP toxicity”, (b) oil, petroleum, or petroleum derived substances, natural gas, natural gas liquids, synthetic gas, drilling fluids, produced waters, and other wastes associated with the exploration, development, or production of crude oil, natural gas, or geothermal resources, (c) any explosives or any radioactive materials, and (d) asbestos in any form, polychlorinated biphenyls, and per- and polyfluoroalkyl substances.

“Improvements” ~~shall~~ means any physical improvements to the Georgia Property.

“Indebtedness” as to any Person means (a) all obligations of such Person for borrowed money, (b) all obligations of such Person evidenced by bonds, debentures, notes, or other similar instruments and all reimbursement or other obligations in respect of letters of credit, bankers acceptances, or other financial products, (c) all obligations or liabilities of others secured by a Lien on any asset of such Person, irrespective of whether such obligation or liability is assumed, (d) all obligations of such Person to pay the deferred purchase price of assets (other than trade payables incurred in the ordinary course of business and repayable in accordance with customary trade practices and which are paid in accordance with such terms and not on a deferred basis), (e) all

obligations of such Person in respect of Disqualified Equity Interests, and (f) any obligation of such Person guaranteeing or intended to guarantee (whether directly or indirectly guaranteed, endorsed, co-made, discounted, or sold with recourse) any obligation of any other Person that constitutes Indebtedness under any of clauses (a) through (d) above. For purposes of this definition, (i) the amount of any Indebtedness represented by a guaranty or other similar instrument shall be the lesser of the principal amount of the obligations guaranteed and still outstanding and the maximum amount for which the guaranteeing Person may be liable pursuant to the terms of the instrument embodying such Indebtedness, and (ii) the amount of any Indebtedness which is limited or is non-recourse to a Person or for which recourse is limited to an identified asset shall be valued at the lesser of (A) if applicable, the limited amount of such obligations, and (B) if applicable, the fair market value of such assets securing such obligation.

“Indemnified Liabilities” has the meaning specified therefor in Section 9.3 of this Agreement.

“Indemnified Person” has the meaning specified therefor in Section 9.3 of this Agreement.

“Insolvency Proceeding” means any of the following: (a) the filing by any Loan Party of a voluntary petition in bankruptcy under any provision of any bankruptcy law (including the Bankruptcy Code) or a petition to take advantage of any receivership or insolvency laws, including any petition seeking the dissolution, winding up, total or partial liquidation, reorganization, composition, arrangement, adjustment or readjustment or other relief of any Loan Party, any Loan Party’s debts or any Loan Party’s assets or the appointment of a trustee, receiver, liquidator, custodian or similar official for any Loan Party or a material part of any Loan Party’s property; (b) the inability of any Loan Party to pay its debts (including trade debts) generally as they become due; (c) the appointment of a receiver, liquidator, trustee, custodian or other similar official for any Loan Party or all or a material part of any Loan Party’s assets; (d) the filing of any petition against any Loan Party under any bankruptcy law (including the Bankruptcy Code) or other receivership or insolvency law, including any petition seeking the dissolution, winding up, total or partial liquidation, reorganization, composition, arrangement, adjustment or readjustment or other relief of any Loan Party, any Loan Party’s debts or any Loan Party’s assets or the appointment of a trustee, receiver, liquidator, custodian or similar official for any Loan Party or a material part of any Loan Party’s property; (e) the general assignment by any Loan Party for the benefit of creditors or any other marshaling of the assets and liabilities of any Loan Party; or (f) a corporate (or similar) action taken by any Loan Party to authorize any of the foregoing.

“Intellectual Property” ~~shall~~ means all patents, trademarks, service marks, trade names, copyrights, know-how and processes.

“Intellectual Property License Agreements” means all agreements between any Loan Party and any third party pursuant to which (a) such third party grants to any Loan Party a license to any material Intellectual Property Rights, including the Technology License Agreement (other than over-the-counter or off-the-shelf software licenses entered into by a Loan Party that are commercially available to the public) and (b) such Loan Party grants to any third party a license to any material Intellectual Property Rights (other than non-exclusive licenses of material Intellectual Property Rights granted by a Loan Party to its customers in the ordinary course of business) and in each case of clauses (a) and (b), necessary to operate any Loan Party’s business.

“Intellectual Property Rights” means any and all intellectual property rights in any jurisdiction throughout the world, including: (a) patents and patent applications; (b) domain names, trademarks, service marks, trade dress, trade names, logos and corporate names, and registrations and applications for registration thereof together with all translations, transliterations, adaptations, derivations and combinations thereof and including all of the goodwill associated therewith; (c) copyrights and copyrightable works (registered or unregistered); (d) mask works; (e) trade secrets, confidential information, and know-how; (f) rights in software (including source code, object code, and executable code) and (g) registrations and applications for any of the foregoing.

“Intercreditor Agreement” means (a) the Revolver Intercreditor Agreement and (b) any other intercreditor agreement in form and substance reasonably acceptable to the Lender.

“Interest Payment Date” has the meaning specified therefor in Section 2.2(a) of this Agreement.

“Interest Rate” has the meaning specified therefor in Section 2.2(a) of this Agreement.

“Inventory” means all “inventory” as defined in the Code in effect on the date hereof with such additions to such term as may hereafter be made, and includes without limitation all merchandise, raw materials, parts, supplies, packing and shipping materials, work in process and finished products, including without limitation such inventory as is temporarily out of any Loan Party’s custody or possession or in transit and including any returned goods and any documents of title representing any of the above.

“Investment” means, with respect to any Person, any investment by such Person in any other Person (including Affiliates) in the form of loans, guarantees, advances, capital contributions (excluding (a) commission, travel, and similar advances to officers and employees of such Person made in the ordinary course of business, (b) bona fide accounts receivable arising in the ordinary course of business and (c) loans and advances of payroll payments or other compensation (including in connection with deferred compensation plans) to present or former employees, directors, members of management or officers in the ordinary course of business), or acquisitions of Indebtedness, Equity Interests, or all or substantially all of the assets of such other Person (or of any division or business line of such other Person), and any other items that are or would be classified as investments on a balance sheet prepared in accordance with GAAP. The amount of any Investment shall be the original cost of such Investment plus the cost of all additions thereto, without any adjustment for increases or decreases in value, or write-ups, write-downs, or write-offs with respect to such Investment.

“IRC” means the Internal Revenue Code of 1986, as amended, and any successor statutes, and all regulations and guidance promulgated thereunder. Any reference to a specific section of the IRC shall be deemed to be a reference to such section of the IRC and any successor statutes, and all regulations and guidance promulgated thereunder.

“Leasehold Deed to Secure Debt” means that certain Leasehold Deed to Secure Debt, Assignment and Security Agreement, dated as of the date hereof, among Borrower and the Lender.

“Leases” means the Georgia Lease and any other leases or amendment to any leases covering any part of the Properties.

“Lender” has the meaning specified therefor in the preamble to this Agreement, and shall also include any other Person made a party to this Agreement pursuant to the provisions of Section 12.1 of this Agreement and “Lenders” means each of the Lenders or any one or more of them.

“Lender Expenses” means all (a) costs or expenses (including Taxes and insurance premiums) required to be paid by any Loan Party under any of the Loan Documents that are paid, advanced, or incurred by the Lender, (b) reasonable and documented out-of-pocket fees or charges paid or incurred by the Lender in connection with the Lender’s transactions with Loan Party under any of the Loan Documents, including, photocopying, notarization, couriers and messengers, telecommunication, public record searches, filing fees, recording fees, publication, real estate surveys, real estate title policies and endorsements, and environmental audits, (c) the Lender’s reasonable and documented out-of-pocket fees and charges imposed or incurred in connection with any background checks related to Loan Party, (d) reasonable and documented out-of-pocket costs and expenses paid or incurred by the Lender to correct any default or enforce any provision of the Loan Documents, or during the continuance of an Event of Default, in gaining possession of, maintaining, handling, preserving, storing, shipping, selling, preparing for sale, or advertising to sell the Collateral, or any portion thereof, irrespective of whether a sale is consummated, (e) reasonable and documented out-of-pocket costs incurred in connection with Lender’s exercise of its inspection and related collateral rights pursuant to Section 5.4, (f) the Lender’s reasonable costs and expenses (including reasonable and documented out-of-pocket attorneys’ fees and expenses) relative to third party claims or any other lawsuit or adverse proceeding paid or incurred, whether in enforcing or defending the Loan Documents or otherwise in connection with the transactions contemplated by the Loan Documents, the Lender’s Liens in and to the Collateral, or the Lender’s relationship with Borrower, (g) the Lender’s reasonable and documented costs and expenses (including reasonable and documented out-of-pocket attorneys’ fees and due diligence expenses) incurred in advising, structuring, drafting, reviewing, administering, or amending, waiving, or modifying the Loan Documents, and (h) the Lender’s reasonable and documented out-of-pocket costs and expenses (including reasonable and documented out-of-pocket attorneys, accountants, consultants, and other advisors fees and expenses) incurred in terminating, enforcing (including attorneys, accountants, consultants, and other advisors fees and expenses incurred in connection with a “workout,” a “restructuring,” or an Insolvency Proceeding concerning any Loan Party or in exercising rights or remedies under the Loan Documents), or defending the Loan Documents, irrespective of whether a lawsuit or other adverse proceeding is brought, or in taking any enforcement action or any Remedial Action with respect to the Collateral.

“Lender-Related Person” means, with respect to any Lender, such Lender, together with such Lender’s Affiliates, officers, directors, employees, attorneys, and agents.

“Lien” means any mortgage, deed of trust, pledge, hypothecation, assignment, charge, deposit arrangement, encumbrance, easement, lien (statutory or other), security interest, or other security arrangement and any other preference, priority, or preferential arrangement of any kind or nature whatsoever, including any conditional sale contract or other title retention agreement, the interest of a lessor under a Capital Lease and any synthetic or other financing lease having substantially the same economic effect as any of the foregoing.

“Loan Documents” means (a) this Agreement, (b) the Guaranty, (c) the Security Documents, (d) the Perfection Certificate, (e) the Note and any other note or notes executed by Borrower in connection with this Agreement and payable to the Lender, (f) the Convertible Notes Subordination Agreement and any other subordination agreement entered into with Lender and any other holder of Equity Interests or Indebtedness in the future, (g) any Intercreditor Agreement and (h) any other instrument or agreement entered into, now or in the future, by Borrower or any Guarantor and the Lender in connection with this Agreement and designated by Borrower and the Lender as a “Loan Document”.

“Loan Parties” means Borrower and the Guarantors.

“Major Subcontract” ~~shall~~ means any subcontractor contract that is related to the Project with a value that exceeds \$2,000,000 in the aggregate, other than any such subcontractor contract that will be completed in full prior to December 31, 2022 and for which less than \$5,000,000 remains to be paid thereunder as of December 31, 2022.

“Material Adverse Effect” means (a) a material adverse effect on the timing or scope of the Project or a material adverse effect on the ability of the Parent to perform under the GM Purchase Contracts (inclusive of the ability to achieve a Milestone that would be reasonably expected to result in an Event of Default under Section 7.6 or to achieve Start of Production at the Georgia Facility (a Material Adverse Effect pursuant to this clause (a), a “Project MAE”), (b) a material adverse effect on the business, operations, conditions (financial or otherwise) of Borrower or any Guarantor, (c) a material impairment of any Loan Party’s ability to perform its obligations under the Loan Documents to which Borrower is a party or of the Lender’s ability to enforce the Obligations or realize upon the Collateral (other than as a result of an action taken or not taken that is solely in the control of the Lender), or (d) a material impairment of the enforceability or priority of the Lender’s Liens with respect to all or a material portion of the Collateral.

“Material Contracts” ~~shall~~ means, collectively, (i) the agreements evidencing Material Indebtedness; (ii) the documents evidencing the Convertible Notes, (iii) each purchase contract with regard to the purchase of any machinery and equipment funded with the proceeds of the Term Loan, (iv) the agreements to which any Loan Party is subject in connection with the tax exempt financing issued by the Development Authority of Bulloch County in connection with the construction of the Georgia Facility, and (v) all other contracts, leases, instruments, guaranties, licenses, or other arrangements (other than the Loan Documents) to which Borrower is or became a party and as to which the breach, nonperformance, cancellation or failure to renew by any party thereto could reasonably be expected to result in a Material Adverse Effect.

“Material Indebtedness” means any Indebtedness of any Loan Party or its Subsidiaries in a principal amount in excess of \$1,000,000 individually (exclusive of Capital Leases in the aggregate of less than \$2,500,000).

“Maturity Date” means the earliest of: (a) ~~March 31~~ September 30, 2025 and (b) ninety (90) days before the maturity date of any other debt facility any Loan Party may be a party to (other than the Revolving Loan Agreement) and (c) the date that any Term Loan shall become due and payable in full hereunder, whether by acceleration or otherwise.

“Measurement Period” means the four Fiscal Quarters then ended for which Financial Statements have been (or are required to have been) delivered pursuant to Section 5.2(a) or (b), as applicable.

“Milestone” means each of the actions, events or deliverables to be completed by Borrower as set forth on Schedule 7.6.

“More Favorable Provision” has the meaning specified therefor in Section 5.12 of this Agreement.

“Mortgage” ~~shall~~ means that certain Mortgage covering the Rhode Island Property, executed by Aspen RI and delivered to the Lender pursuant to this Agreement, securing the Obligations.

“Net Proceeds” means, with respect to any event, the proceeds actually received (directly or indirectly) from time to time (whether as initial consideration or through the payment of deferred consideration), in cash or Cash Equivalents, by or on behalf of such Loan Party, in connection therewith after deducting therefrom only (i) reasonable fees, commissions, and expenses related thereto and required to be paid by such Loan Party in connection with such event and (ii) taxes paid or payable to any taxing authorities by such Loan Party in connection with such event, in each case to the extent, but only to the extent, that the amounts so deducted are, at the time of receipt of such cash, actually paid or payable to a Person that is not an Affiliate of any Loan Party or any of its Subsidiaries, and are properly attributable to such transaction.

“Note” has the meaning given on Schedule 3.1.

“Obligations” means all loans hereunder, debts, principal, interest (including any interest that accrues after the commencement of an Insolvency Proceeding, regardless of whether allowed or allowable in whole or in part as a claim in any such Insolvency Proceeding), premiums, liabilities, obligations (including indemnification obligations), fees, Lender Expenses (including any fees or expenses that accrue after the commencement of an Insolvency Proceeding, regardless of whether allowed or allowable in whole or in part as a claim in any such Insolvency Proceeding), guaranties, and all covenants and duties for amounts owing by Borrower or Guarantor arising out of, under, pursuant to, in connection with, or evidenced by this Agreement or any of the other Loan Documents, whether direct or indirect, absolute or contingent, due or to become due, now existing or hereafter arising, and including all interest not paid when due and all other expenses or other amounts that Borrower or Guarantor is required to pay or reimburse by the Loan Documents or by law or otherwise in connection with the Loan Documents. Without limiting the generality of the foregoing, the Obligations of Borrower under the Loan Documents include the obligation to pay (i) the principal of the Term Loans (including capitalized interest thereon), (ii) interest accrued on the Term Loans, (iii) Lender Expenses, (iv) fees payable under this Agreement or any of the other Loan Documents, and (v) indemnities and other amounts payable by Borrower under any Loan Document. Any reference in this Agreement or in the Loan Documents to the Obligations shall include all or any portion thereof and any extensions, modifications, renewals, or alterations thereof, both prior and subsequent to any Insolvency Proceeding.

“OFAC” means The Office of Foreign Assets Control of the U.S. Department of the Treasury.

“Other Financing Facility” means any agreement (whether in the form of a loan or credit agreement, stockholder agreement, operating agreement, investment agreement or other documentation), other than the Revolving Loan Agreement and Convertible Notes, entered into by any Loan Party which is binding upon such Loan Party, or to which such Loan Party or its assets is subject, in connection with the provision of equity or debt financing by any third party (including any Additional Financing Facility), and is inclusive of all agreements entered into by such parties in connection with any such equity or debt financing.

“Paid in Full” or “Payment in Full” means all Obligations (other than contingent indemnification obligations, if any, to the extent no unsatisfied claim giving rise thereto has been or may reasonably be expected to be asserted) shall have been paid in full in cash by wire transfer of immediately available funds and any commitment to make Term Loans shall have terminated. For clarity, in the event of a termination of the GM Purchase Contracts (other than by the Lender) prior to satisfaction in full of the Obligations, or an Event of Default under this Agreement at any time as and to the extent provided in Section 7, any unpaid portion of the Obligations must be satisfied in full by indefeasible payment in cash.

“Party Representatives” has the meaning specified therefor in Section 14.7 of this Agreement.

“Patriot Act” has the meaning specified therefor in Section 4.10 of this Agreement.

“Perfection Certificate” has the meaning specified therefor in Section 4.1(a) of this Agreement, as amended, restated, amended and restated, supplemented or otherwise modified from time to time in accordance with, and subject to, Section 5.2(j).

“Periodic Term SOFR Determination Day” has the meaning specified in the definition of “Term SOFR”.

“Permits” means all authorizations, approvals, permits, variances, land use entitlements, consents, licenses, franchises and agreements issued by or entered into with any Governmental Authority now or hereafter required for the applicable Loan Party to conduct its business and complete the Project (including amendments to any of the foregoing).

“Permitted Change of Control” means a Change of Control; provided, that the Person who Controls Borrower or the Parent, as applicable, (i) is not a GM Competitor, (ii) agrees in writing to abide by, and assume all, of the liabilities, obligations and terms of this Agreement, the other Loan Documents, the GM Purchase Contracts and any other related agreement or document and (iii) has the reasonable financial and operational capability to comply with the obligations of Borrower or Parent, as applicable, under the GM Purchase Contracts.

“Permitted Indebtedness” means:

- (a) Borrower’s Indebtedness to the Lender under this Agreement and the other Loan Documents;
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(b) Indebtedness existing on the Closing Date and shown on Schedule 4.3 or in the Perfection Certificate (as delivered as of the date hereof);

(c) Subordinated Indebtedness;

(d) Unsecured obligations to trade creditors so long as they do not remain unpaid for more than 120 days;

(e) Indebtedness secured by Liens permitted under clause (d) of the definition of “Permitted Liens” hereunder;

(f) Indebtedness incurred under the Revolving Loan Agreement, not to exceed (together with Indebtedness incurred under clause (n) below) \$50,000,000 in aggregate principal amount at any time; provided, that no secured party under the Revolving Loan Agreement, as such agreement may be refinanced or modified from time to time, shall be an Affiliate of any Loan Party or Koch Strategic Platforms;

(g) Indebtedness to the U.S. Department of Energy;

(h) Indebtedness (i) of any Loan Party owed to any other Loan Party, (ii) of any non-Loan Party Subsidiary owed to any Loan Party and (iii) of any Loan Party owing to any non-Loan Party in an aggregate amount under this clause (iii), together with Investments permitted in connection with clause (d) of the definition of “Permitted Investments”, not to exceed \$250,000 in any fiscal year plus any amounts necessary to fund payroll in respect of any non-Loan Party not to exceed \$10,000,000 in any fiscal year;

(i) extensions, re-financings, modifications, amendments and restatements of any items of Permitted Indebtedness under clauses (a), (b), (d), (e), (f) (other than Indebtedness incurred under the Convertible Notes or any other Indebtedness owed to any Affiliate of any Loan Parties) above; provided, that (A) the principal amount thereof is not increased or decreased following such extensions, re-financings, modifications, amendments and restatement (without the Lender’s prior written consent), (B) no additional property secures the Indebtedness thereunder nor does any additional party guaranty such Indebtedness beyond the scope of collateral and guaranty provided for the Permitted Indebtedness being refinanced, (C) the terms thereof are not modified to impose more burdensome terms, taken as a whole, upon any Loan Party or its Subsidiary, as the case may be than the terms of the Indebtedness being refinanced, (D) the weighted average life to maturity at the time such refinancing Indebtedness is incurred is not less than the remaining weighted average life to maturity of the Indebtedness being refinanced and (E) with respect to a refinancing of the Permitted Indebtedness in clause (f) above, the Lender shall receive notice at least ten (10) days prior notice (or such notice as the Lender may agree in writing (including via email)) prior to such refinancing;

(j) junior secured Indebtedness, not to exceed (together with any secured Indebtedness incurred pursuant to any Additional Financing Facility pursuant to clause (l) below) \$~~20~~50,000,000 in the aggregate at any time; provided, that the maturity date of such Indebtedness is not less than six months after the maturity date of the Term Loan;

(k) obligations with respect to the agreements entered into by any Loan Party in connection with (i) any performance bonds, surety bonds, bank guaranties or similar instruments in the ordinary course of business (including any such bonds required pursuant to Section 5.16) and (ii) the taxable revenue bond issued by the Development Authority of Bulloch County in connection with the construction of the Georgia Facility as in effect as of the Closing Date;

(l) Indebtedness incurred under any Additional Financing Facility; provided, that solely if such Indebtedness is secured, the aggregate principal amount thereof shall not exceed (together with any junior secured Indebtedness incurred pursuant to clause (j) above) \$2050,000,000 in the aggregate at any time; provided, that the maturity date of such Indebtedness is not less than six months after the maturity date of the Term Loan;

(m) Indebtedness of Borrower in respect of any Account (including the Disbursement Account) and/or any other Account Services in connection therewith; and

(n) Indebtedness in respect of (i) the letters of credit issued under the SVB Revolving Loan Agreement in an amount not to exceed \$906,279.42 plus €256,467.31 (the “Existing Letters of Credit”) and (ii) additional letters of credit to be issued in the ordinary course of business not to exceed \$6,000,000 in the aggregate (and such additional letters of credit in excess thereof that may be required in the ordinary course of business prior to the date Borrower obtains a Revolving Loan Agreement if requested by Borrower and approved by the Lender in writing (including via email) in its sole discretion (the “Additional Letters of Credit”); provided, that Indebtedness incurred under this clause (n), together with any Indebtedness incurred under clause (f) above, shall not exceed \$50,000,000 in the aggregate principal amount at any time;

provided, that with respect to any Indebtedness incurred pursuant to clauses (e) (with regard to up to \$50,000,000 of such Indebtedness), (g) and (l) (to the extent applicable), the proceeds thereof shall be available solely for the Project or as otherwise permitted pursuant to Section 5.13.

“Permitted Investments” are:

(a) Investments shown on the Perfection Certificate (as delivered as of the date hereof) and existing on the Closing Date;

(b) Cash or Cash Equivalents;

(c) Parent’s Investments consisting of the endorsement of negotiable instruments for deposit or collection or similar transactions in the ordinary course of Parent’s business;

(d) Investments (i) by any Loan Party in any other Loan Party, (ii) of any non-Loan Party Subsidiary in any Loan Party and (iii) by any Loan Party in any non-Loan Party Subsidiary, in an aggregate amount, together with any Indebtedness described in clause (h)(iii) of the definition of “Permitted Indebtedness”, not to exceed \$250,000 in any fiscal year plus any amounts necessary to fund payroll in respect of any non-Loan Party not to exceed \$10,000,000 in any fiscal year;

(e) Investments (including debt obligations) received in connection with the bankruptcy or reorganization of customers or suppliers and in settlement of delinquent obligations of, and other disputes with, customers or suppliers arising in the ordinary course of business;

(f) Investments consisting of notes receivable of, or prepaid royalties and other credit extensions, to customers and suppliers who are not Affiliates, in the ordinary course of business; provided, that this clause (f) shall not apply to Investments of any Loan Party in any Subsidiary; and

(g) Investments consisting of (or resulting from) proceeds of (w) Additional Financing Facilities in Borrower, (x) Permitted Indebtedness incurred by Parent pursuant to clauses (c), (f) or (j) of the definition thereof and invested in Aspen RI or Borrower or (y) additional equity issuances in Aspen RI to the extent not constituting the proceeds of any Additional Financing Facility.

“Permitted Liens” means:

(a) Liens existing on the Closing Date and shown on the Perfection Certificate (as delivered as of the date hereof) or arising under this Agreement and the other Loan Documents;

(b) Liens granted pursuant to (i) the Revolving Loan Agreement, subject to the Revolver Intercreditor Agreement and (ii) any Additional Financing Facility; provided, that prior to or concurrently with the granting of any Liens over an Additional Financing Facility, the Lender, the Loan Parties and the secured party under any Additional Financing Facility shall enter into a customary first-lien second-lien Intercreditor Agreement of the tenor of a silent second lien, in form and substance reasonably satisfactory to the Lender.

(c) (i) Liens for taxes, fees, assessments or other government charges or levies, either (A) not due and payable or (B) being contested in good faith and for which the applicable Loan Party maintains adequate reserves on its Books, provided, that no notice of any such Lien has been filed or recorded under the Internal Revenue Code of 1986, as amended, and the Treasury Regulations adopted thereunder, (ii) statutory Liens, carriers, warehousemen, mechanics, repairmen, workmen and materialmen, and other Liens imposed by applicable law, in each case, incurred in the ordinary course of business that are either (A) not overdue for a period of more than 30 days or (B) being contested in good faith and by appropriate proceedings diligently conducted, if adequate reserves with respect thereto are maintained on the books of the applicable Person, (iii) landlords’ and banks’ Liens incurred in the ordinary course of business, (iv) Liens in favor of customs and revenue authorities arising as a matter of law to secure payment of customs duties in connection with the importation of goods and (v) Liens in connection with any zoning, building or similar laws or right reserved to or vested in any Governmental Authority to control or regulate the use of any or dimensions of real property or the structures thereon;

(d) Capital Leases and purchase money Liens on Equipment acquired or held by any Loan Party incurred for financing the acquisition of the Equipment securing no more than \$100,000,000 in the aggregate amount outstanding;

(e) Liens to secure payment of workers' compensation, employment insurance, old-age pensions, social security and other like obligations incurred in the ordinary course of business (other than Liens imposed by ERISA);

(f) Liens incurred in the extension, renewal or refinancing of the Indebtedness secured by Liens described in clauses (a) through (d), but any extension, renewal or replacement Lien must be limited to the property encumbered by the existing Lien (and the secured party thereof must be subject to the same Intercreditor Agreement as the existing secured party, if any, and the principal amount of the Indebtedness may not increase;

(g) leases or subleases of real property granted in the applicable Loan Party's ordinary course of business (including, without limitation, the Georgia Lease), and leases, subleases, non-exclusive licenses or sublicenses of property (other than real property, Intellectual Property and Intellectual Property Rights) granted in the ordinary course of such Loan Party's business, if the leases, subleases, licenses and sublicenses do not prohibit granting the Lender a security interest;

(h) non-exclusive licenses of Intellectual Property or Intellectual Property Rights granted to third parties in the ordinary course of business;

(i) Liens securing Indebtedness incurred pursuant to clause (j) of the "Permitted Indebtedness" definition (and any refinancing thereof pursuant to clause (i) of the definition of "Permitted Indebtedness"); provided, that prior to or concurrently with the granting of any Liens over such Indebtedness, the Lender, the Loan Parties and the secured party under such Indebtedness shall enter into a customary first-lien second-lien Intercreditor Agreement of the tenor of a silent second lien, in form and substance reasonably satisfactory to the Lender;

(j) Liens arising from judgments, decrees or attachments in circumstances not constituting an Event of Default under Section 7.3 or 7.7 of this Agreement;

(k) (i) Permitted Exceptions and (ii) non-monetary Liens that, individually and in the aggregate, do not and would not reasonably be expected to impair or adversely affect the use, occupancy or value of any Property;

(l) Banks Liens and rights and remedies as to the Accounts and related Account Services referred to in clause (m) of the definition of "Permitted Indebtedness" definition and/or arising as a matter of law, under the Code or under customary general terms and conditions encumbering such Accounts that are within the general parameters customary in the banking industry or arising pursuant to such banking institution's general terms and conditions; and

(m) Liens securing obligations in respect letters of credit, bank guaranties, surety bonds, performance bonds or similar instruments permitted pursuant to clauses (n) and (k) of the definition of "Permitted Indebtedness" (including pledges and deposits of Cash or Cash Equivalents with respect to the foregoing).

“Person” means natural Persons, corporations, limited liability companies, limited partnerships, general partnerships, limited liability partnerships, joint ventures, trusts, land trusts, business trusts, or other organizations, irrespective of whether they are legal entities, and governments and agencies and political subdivisions thereof.

“PIK Election” has the meaning specified therefor in Section 2.2(a) of this Agreement.

“PIK Interest” has the meaning specified therefor in Section 2.2(a) of this Agreement.

“Plans” ~~shall~~ means the final working drawings and specifications for the construction of the Georgia Facility and shall include all architectural and engineering studies, plans and specifications.

“Policy” and “Policies” have the meaning specified therefor in Section 5.5(b) of this Agreement.

“Product” refers to the “Goods” as such term is defined in the GM Purchase Contracts.

“Project” has the meaning specified therefor in the preamble to this Agreement.

“Project Budget” ~~shall~~ means the project budget delivered to the Lender by Borrower on the Closing Date substantially in the form of budget set forth in Schedule 1.1(p) (which shall be subject to the Permitted Variance), as amended, restated, amended and restated, supplemented or otherwise modified by Borrower from time to time (it being understood and agreed that any amendments, restatements, amendments and restatements, supplements or other modifications in excess of the Permitted Variance subject to the reasonable approval of the Lender); provided, that, for the avoidance of doubt, the Project Budget includes up to \$2,500,000 to finance the cost of the payment and performance bonds to be delivered pursuant to Section 5.16(b).

“Project Contracts” ~~shall~~ means the Construction Contracts, the Architect’s Contract and any and all other contracts between Borrower and any third party relating to the development, construction, management and maintenance of the Project (including amendments to any of the foregoing).

“Project Documents” ~~shall~~ means the Plans, the Project Contracts and all Permits.

“Project MAE” ~~shall have~~ has the meaning specified therefor in the “Material Adverse Effect” definition.

“Project Permits” means all permits necessary for any work, the nature and extent of which requires a permit, in connection with the construction or equipping of the Project.

“Projections” ~~shall have~~ has the meaning specified therefor in Section 4.26(a).

“Property” ~~shall~~ means the Rhode Island Property or the Georgia Property, and “Properties” ~~shall~~ means both collectively.

“Real Property” means any estates or interests in real property now owned or hereafter acquired by Borrower and the improvements thereto.

“Registered Organization” is any “registered organization” as defined in the Code with such additions to such term as may hereafter be made.

“Remedial Action” means all actions taken to (a) clean up, remove, remediate, contain, treat, monitor, assess, evaluate, or in any way address Hazardous Materials in the indoor or outdoor environment, (b) prevent or minimize a release of Hazardous Materials, (c) restore or reclaim natural resources or the environment, (d) perform any pre-remedial studies, investigations, or post-remedial operation and maintenance activities, (e) conduct any other actions in response to a release of Hazardous Materials required by Environmental Law, or (f) correct or resolve any violation of or non-compliance with Environmental Law.

“Requirement of Law” is, as to any Person, any law (statutory or common), treaty, rule or regulation or determination of an arbitrator or a court or other Governmental Authority, in each case, applicable to or binding upon such Person or any of its property or to which such Person or any of its property is subject.

“Responsible Officer” is any of the Chief Executive Officer, President, Chief Financial Officer, Vice President (Finance), Controller or equivalent officer of any Loan Party, as applicable.

“Restricted License” is any license or other agreement, the absence of which would be reasonably expected to cause a Material Adverse Effect on any Loan Party’s business, with respect to which such Loan Party is the licensee (a) that prohibits or otherwise restricts such Loan Party from granting a security interest in such Loan Party’s interest in such license or agreement or any other property, or (b) for which a default under or termination of interferes with the Lender’s right to sell any Collateral.

“Restricted Payment” means to (a) declare or pay any dividend or make any other payment or distribution, directly or indirectly, on account of Equity Interests issued by any Loan Party (including any payment in connection with any merger or consolidation involving any Loan Party) or to the direct or indirect holders of Equity Interests issued by any Loan Party in its capacity as such, or (b) purchase, redeem, make any sinking fund or similar payment, or otherwise acquire or retire for value (including in connection with any merger or consolidation involving any Loan Party) any Equity Interests issued by any Loan Party, (c) make any payment to retire, or to obtain the surrender of, any outstanding warrants, options, or other rights to acquire Equity Interests of any Loan Party now or hereafter outstanding, or (d) pay any fees or other remuneration to any Affiliate of any Loan Party (except pursuant to the terms of the Convertible Notes, ~~the documentation for the Equity Investment~~ and any other Permitted Indebtedness entered into with an Affiliate to the extent permitted pursuant to Section 6.7).

“Revenue Bond” means that certain Development Authority of Bulloch County Taxable Revenue Bond (Aspen Aerogels, LLC Project), Series 2022 in the principal amount of up to \$650,000,000 issued by the Development Authority of Bulloch County in connection with the construction of the Georgia Facility.

“Revenue Bond Pledge Agreement” means that certain pledge agreement entered into in connection with that certain Revenue Bond.

“Revolver Intercreditor Agreement” means an intercreditor agreement between the Lender and the Revolving Lender substantially in the form of Exhibit E (with such changes to such form as may be reasonably acceptable to the Lender).

“Revolving Lender” means a lender providing working capital financing to any Loan Party with such advances being based on working capital assets.

“Revolving Loan Agreement” means a revolving credit agreement between any Loan Party and a Revolving Lender in form reasonably acceptable to the Lender.

“Rhode Island Facility” ~~shall~~ means that certain aerogel manufacturing plant built on the Rhode Island Property.

“Rhode Island Property” ~~shall~~ means the real property located at 3 Dexter Road, East Providence, Rhode Island 02914. The Property is legally described in the Mortgage.

“Sanctioned Entity” means (a) a country or a government of a country, (b) an agency of the government of a country, (c) an organization directly or indirectly controlled by a country or its government, (d) a Person resident in or determined to be resident in a country, in each case, that is subject to a country sanctions program administered and enforced by OFAC.

“Sanctioned Person” means a Person named on the list of Specially Designated Nationals maintained by OFAC.

“Sanctions” has the meaning specified therefor in Section 4.11 of this Agreement.

“SEC” has the meaning specified in Section 4.15 of this Agreement.

“SEC Reports” has the meaning specified in Section 4.15 of this Agreement.

“Securities Act” means the Securities Act of 1933, as amended.

“Security Agreement” has the meaning specified therefor in Section 2.4(i) of this Agreement.

“Security Documents” means, collectively, any Intercreditor Agreement, the Convertible Notes Subordination Agreement, the Security Agreement, the Guaranty, the Leasehold Deed to Secure Debt, the Mortgage, the Revenue Bond Pledge Agreement, the Collateral Assignment Agreements, the Control Agreement, the Collateral Access Agreements and each other agreement, document or instrument entered into by a Loan Party that creates or purports to create a Lien over all or any part of its assets in respect of the Obligations in favor of the Lender.

“SOFR” means a rate equal to the secured overnight financing rate as administered by the SOFR Administrator.

“SOFRA Administrator” means the Federal Reserve Bank of New York (or a successor administrator of the secured overnight financing rate).

“Solvent” means, with respect to any Person as of any date of determination, that (a) at fair valuations, the sum of such Person’s debts (including contingent liabilities) is less than all of such Person’s assets, (b) such Person is not engaged or about to engage in a business or transaction for which the remaining assets of such Person are unreasonably small in relation to the business or transaction or for which the property remaining with such Person is an unreasonably small capital, and (c) such Person has not incurred and does not intend to incur, or reasonably believe that it will incur, debts beyond its ability to pay such debts as they become due (whether at maturity or otherwise), and (d) such Person is “solvent” or not “insolvent”, as applicable within the meaning given those terms and similar terms under applicable laws relating to fraudulent transfers and conveyances. For purposes of this definition, the amount of any contingent liability at any time shall be computed as the amount that, in light of all of the facts and circumstances existing at such time, represents the amount that can reasonably be expected to become an actual or matured liability (irrespective of whether such contingent liabilities meet the criteria for accrual under Statement of Financial Accounting Standard No. 5).

“Start of Production” means the date agreed to by Borrower and the Lender for the start of production at the Georgia Facility under the applicable GM Purchase Contracts.

“Subordinated Indebtedness” means any unsecured Indebtedness (including Indebtedness under the Convertible Notes) incurred from time to time that is subordinated in right of payment to the Obligations pursuant to a subordination agreement in form and substance reasonably acceptable to the Lender and (a) that is not guaranteed by any other party, (b) that is not subject to scheduled amortization, redemption, sinking fund or similar payment and does not have a final maturity, in each case, on or before the date that is six months after the Maturity Date, (c) that does not include any financial covenants or any covenant or agreement that is more restrictive or onerous on any Loan Party in any material respect than any comparable covenant in this Agreement and is otherwise on terms and conditions reasonably acceptable to the Lender, (d) shall be limited to cross-payment default and cross-acceleration to designated “senior debt” (including the Obligations), and (e) the terms and conditions of the subordination are reasonably acceptable to the Lender in its sole discretion.

“Subsidiary” of a Person means a corporation, partnership, limited liability company, or other entity in which that Person directly or indirectly owns or controls the Equity Interests having ordinary voting power to elect a majority of the Board of Directors of such corporation, partnership, limited liability company, or other entity.

“SVB Revolving Loan Agreement” has the meaning specified therefor in clause (l) of Schedule 3.1.

“Taxes” means any taxes, levies, imposts, duties, fees, assessments or other charges of whatever nature now or hereafter imposed by any jurisdiction or by any political subdivision or taxing authority thereof or therein, and all interest, penalties or similar liabilities with respect thereto.

“Technology License Agreement” means an Intellectual Property royalty free license agreement between Borrower and/or Aspen RI and Parent providing that Borrower and Aspen RI may carry out their respective operations using any of the Intellectual Property owned by the Parent.

“Title Company” has the meaning specified therefor in Section 5.5(a)(ix) of this Agreement.

“Title Policy” has the meaning specified therefor in Section 5.5(a)(ix) of this Agreement.

“Term Loan” has the meaning specified therefor in Section 2.1(b) of this Agreement.

“Term Loan Budget” ~~shall~~ means the budget delivered prior to each Funding Date, substantially in the form of the budget consistent with AIA Form G703 and set forth in Schedule 1.1(t) (subject to the Permitted Variance), or such other form that provides substantially the same scope of information, as of the time of such Funding Date, with respect to use of proceeds, schedule of the Project and remaining sources and uses, including setting forth the expected use of proceeds of the Term Loans being funded on the applicable Funding Date (and the Equity Investment and/or any Additional Financing Facility funded concurrently therewith), as amended, restated, amended and restated, supplemented or otherwise modified by Borrower from time to time (it being understood and agreed that any amendments, restatements, amendments and restatements, supplements or other modifications in excess of the Permitted Variance are subject to the reasonable approval of the Lender).

“Term Loan Commitment” has the meaning specified therefor in Section 2.1(a) of this Agreement.

“Term Loan Commitment Expiration Date” means the earlier of (i) with respect to Term Loan Commitments that are funded, the date of borrowing of the applicable Term Loan and (ii) with respect to Term Loan Commitments that are not funded, ~~September 30~~ March 31, 2024 (or such later date as approved in writing by the Lender in its sole discretion).

“Term SOFR” means, the Term SOFR Reference Rate for a six-month interest period on the day (such day, the “Periodic Term SOFR Determination Day”) that is two (2) U.S. Government Securities Business Days prior to the first day of such interest period, as such rate is published by the Term SOFR Administrator; provided, however, that if as of 5:00 p.m. (New York City time) on any Periodic Term SOFR Determination Day the Term SOFR Reference Rate for the applicable tenor has not been published by the Term SOFR Administrator, then Term SOFR will be the Term SOFR Reference Rate for such tenor as published by the Term SOFR Administrator on the first preceding U.S. Government Securities Business Day for which such Term SOFR Reference Rate for such tenor was published by the Term SOFR Administrator so long as such first preceding U.S. Government Securities Business Day is not more than three (3) U.S. Government Securities Business Days prior to such Periodic Term SOFR Determination Day, and; provided, further, that if Term SOFR determined as provided above shall ever be less than the Floor, then Term SOFR shall be deemed to be the Floor.

“Term SOFR Administrator” means CME Group Benchmark Administration Limited (CBA) (or a successor administrator of the Term SOFR Reference Rate selected by the Lender in its reasonable discretion).

“Term SOFR Reference Rate” means the forward-looking term rate based on SOFR.

“Test Date” means the last day of each Fiscal Quarter for which a Compliance Certificate has been (or is required to have been) delivered pursuant to Section 5.2(j)(y).

“Total Leverage Ratio” means, as of any Test Date, the ratio of (i) total Indebtedness of Parent and its Subsidiaries (excluding Indebtedness incurred under a Revolving Loan Agreement that benefits the Borrower in a principal amount not to exceed \$50,000,000 and undrawn letters of credit permitted under clause (n) of the definition of Permitted Indebtedness) as of such Test Date to (ii) Consolidated EBITDA of Parent and its Subsidiaries for the most recently ended Measurement Period.

“United States” means the United States of America.

“U.S. Government Securities Business Day” means any day except for (a) a Saturday, (b) a Sunday or (c) a day on which the Securities Industry and Financial Markets Association recommends that the fixed income departments of its members be closed for the entire day for purposes of trading in United States government securities.

“Vehicle” means any motor vehicle (including any car, van, or truck), regardless of whether used for household, commercial (including fleets, first to last mile delivery, or other transportation and logistics services), or other purposes.

“Voidable Transfer” has the meaning specified therefor in Section 14.6 of this Agreement.

ASPEN AEROGELS, INC.
BONUS PLAN
(Amended and Restated Effective as of January 1, 2024)

ARTICLE 1

Purpose

1.1 Purpose. The purpose of the Bonus Plan (the “Plan”) is to promote the creation of value for shareholders of Aspen Aerogels, Inc. (the “Company”) by closely aligning the interests of Participants with those goals and expectations established for the Company by the Board of Directors of the Company (the “Board”).

ARTICLE 2

Eligibility

2.1 Eligibility. Each individual who has been selected to participate in the Plan with respect to a Performance Period (the “Participants”) shall be issued a Participation Statement setting forth the terms and conditions of such Participant’s participation in the Plan as determined by the Administrator in its sole discretion. For purposes of the Plan, “Participation Statement” means one or more writings in such form as approved by the Administrator to set forth terms and conditions of a Participant’s bonus opportunity under the Plan for the applicable Performance Period. A Participation Statement may be in an electronic format and may consist of one or more of the following (including a combination thereof): (a) an agreement to be executed by the Participant, (b) a notice or statement issued by the Company which need not be executed by the Participant, and/or (c) an appendix to, or subplan established under, the Plan. An individual who is selected as a Participant for a particular Performance Period is not guaranteed or assured of being selected as a Participant with respect to any subsequent Performance Period.

ARTICLE 3

Bonus Opportunity

3.1 Bonus Opportunity. Each Participant’s Participation Statement shall set forth a target bonus opportunity level for the Participant (whether as a percentage of the Participant’s then current base salary or as a dollar amount or otherwise as determined by the Administrator). The incentive bonus opportunity represents a target cash award (“Target Award”) with payment contingent upon achievement of specified performance metrics set forth in the Participation Statement over the applicable performance period set forth in the Participation Statement (the “Performance Period”).

3.2 Performance Metrics.

3.2.1 Participant Designations. The performance metrics and other terms and conditions of the bonus opportunities provided to Participants under the Plan need not be uniform with respect to any or all Participants. A Participation Statement may, but shall not be required to,

classify a Participant as a member of a sub-group of Participants, such as, for example, sales representatives, corporate personnel, etc.

3.2.2 Performance Metrics. Each Participant's Participation Statement shall set forth and establish the performance metrics or goals to be applicable for the Participant and the relative weightings thereof. Such performance metrics shall consist of measurable or subjective performance goals based upon any financial measures, business measures, strategic objectives or other criteria as determined by the Administrator in its sole discretion, and may be determined with regard to the performance of the Company (absolute or relative to peers or to an index) or any affiliate, business unit, segment, division, region or function, the performance of any individual Participant, or upon any other basis determined by the Administrator in its sole discretion. Performance metrics established by the Administrator under this Plan may include, but shall not be limited to, attaining a "segment revenue" goal, attaining a Company revenue goal, attaining individual performance ratings or individual "motivation by objective" goals, attaining a Company EBITDA, and/or such other performance metrics that may be established from time to time, (collectively, "Goals"). "Segment revenue" means that portion of Company revenue derived from the geographical region and/or business group assigned to a Participant or group of Participants as detailed in a Participation Statement. "Company revenue" and "EBITDA" shall be based upon and conclusively determined by reference to the Company's unaudited financials.

3.2.3 Allocation of Performance Metrics. A Participation Statement may set forth and establish a percentage allocation of a Target Award that will correlate to the Goals set forth in the Participation Statement. The designated percentage of the Target Award will be achieved based on achievement of the applicable Goals.

3.2.4 Levels of Achievement. A Participation Statement may set forth and establish minimum threshold achievement levels, maximum achievement levels, and one or more intermediate achievement levels applicable to one or more Goals, with corresponding payout levels for a portion, all, or a multiple of a Participant's Target Award.

3.3 Additional Goals/Special Awards. The Administrator may in its sole discretion include such other goals and awards in a Participation Statement.

3.4 Payment. Following the completion of the Performance Period, the Administrator shall determine the level of performance achieved in respect of the applicable Goals for the Performance Period and the amount (if any) of the bonus payable to each Participant with respect to that Performance Period. Upon approval and certification by the Administrator of the achieved performance, payment in cash, less any applicable tax withholdings, shall be made to each eligible Participant as soon as practicable but no later than March 15 of the calendar year immediately following the Performance Period (the "Payment Date") in a single lump sum.

3.4.1. Adjustment of Goals. The Administrator may, in its sole discretion and at any time, modify the Goals and/or or the related levels of achievement, in whole or in part, as the Administrator deems appropriate and equitable to reflect a change in the business, operations, corporate structure or capital structure of the Company, the manner in which the Company conducts its business, or any other events or circumstances.

3.4.2 Adjustment of Payout Amounts. The Administrator may, in its sole discretion, increase, decrease or eliminate the amount of any Award otherwise earned by any Participant to reflect such Participant's individual performance or such other factors as the Administrator deems relevant.

3.5 Continued Service. Except as otherwise may be provided in an applicable Participation Statement or a written agreement between the Participant and the Company, or as otherwise determined by the Administrator in its sole discretion, a Participant must remain employed in good standing (and not on notice of termination or resignation) through the applicable Payment Date in order to be eligible to receive any bonus payment under the Plan.

ARTICLE 4 Administration

4.1 Administration. The Compensation Committee of the Board shall be the administrator of the Plan until such time as the Board shall otherwise determine. The Administrator may delegate such of its duties to officers of the Company as it shall determine.

4.2 Authority of Administrator. The Administrator shall have authority, duty and power to interpret and construe the provisions of the Plan as it deems appropriate, to adopt, establish and revise rules, procedures and regulations relating to the Plan, to determine the conditions subject to which any benefits may be payable, to resolve all factual and legal questions concerning the status of the Participants and others under the Plan, including but not limited to, eligibility for benefits and to make any other determinations which it believes necessary or advisable for the administration of the Plan. Benefits under this Plan will be payable only if the Administrator decides in its sole discretion that the applicant is entitled to them under the Plan. The determinations, interpretations, and regulations of the Administrator and the calculations of the Administrator shall be final and binding on all persons and parties concerned.

ARTICLE 5 Amendment and Termination of the Plan

5.1 Amendment and Termination of the Plan. The Administrator reserves the power to alter, amend, wholly revise or terminate the Plan at any time and from time to time and the interest of each Participant is subject to the powers so reserved; provided, however, that no amendment or termination shall be effective without the consent of a Participant to the extent that it would have a materially adverse impact on a Participant's economic benefit under any Participation Statement.

ARTICLE 6 Miscellaneous

6.1 Unsecured General Creditor. Participants and their beneficiaries, heirs, successors and assigns shall have no legal or equitable rights, interests or claims in any property or assets of the Company. For purposes of the payment of benefits under this Plan, any and all of the Company's assets shall be, and remain, the general, unpledged unrestricted assets of the Company.

The Company's obligation under the Plan shall be merely that of an unfunded and unsecured promise to pay money in the future.

6.2 Nonassignability. Neither a Participant nor any other person shall have any right to commute, sell, assign, transfer, pledge, anticipate, mortgage or otherwise encumber, transfer, hypothecate, alienate or convey in advance of actual receipt, the amounts, if any, payable hereunder, or any part thereof, which are, and all rights to which are expressly declared to be, unassignable and non-transferable. No part of the amounts payable shall, prior to actual payment, be subject to seizure, attachment, garnishment or sequestration for the payment of any debts, judgments, alimony or separate maintenance owed by a Participant or any other person, be transferable by operation of law in the event of a Participant's or any other person's bankruptcy or insolvency or be transferable to a spouse as a result of a property settlement or otherwise.

6.3 Not a Contract of Employment. The terms and conditions of this Plan shall not be deemed to constitute a contract of employment between the Company and the Participant. Nothing in this Plan shall be deemed to give a Participant the right to be retained in the service of the Company.

6.4 Compensation Recoupment Policy. Notwithstanding any other provision of this Plan or a Participation Statement, each bonus opportunity granted to a Participant under this Plan shall be subject to forfeiture, and each bonus paid to a Participant under this Plan shall be subject to recoupment, pursuant to the terms of the Company's Compensation Recoupment Policy, as in effect from time to time.

6.5 Governing Law. Except as otherwise may be provided pursuant to the Compensation Recoupment Policy, all issues concerning this Plan shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts, without giving effect to any choice of law or conflict of law provision or rule (whether of the Commonwealth of Massachusetts or any other jurisdiction) that would cause the application of the law of any jurisdiction other than the Commonwealth of Massachusetts.

6.6 Waiver of Jury Trial. By executing a Participation Statement or accepting any benefit under this Plan, each Participant shall waive, to the fullest extent permitted by law, any right to trial by jury of any claim, demand, action, or cause of action (i) arising under this Plan or (ii) in any way connected with or related or incidental to the dealings of the parties hereto in respect of this Plan or any of the transactions related hereto, in each case whether now existing or hereafter arising, and whether in contract, tort, equity, or otherwise. By executing a Participation Statement or accepting any benefit under this Plan, each Participant shall agree and consent that any such claim, demand, action, or cause of action shall be decided by court trial without a jury and that an original counterpart or a copy of the Plan and any applicable Participation Statement with any court as written evidence of the Participant's consent to the waiver of the Participant's right to trial by jury.

[END OF DOCUMENT]

EXECUTIVE EMPLOYMENT AGREEMENT

This Executive Employment Agreement (this “**Agreement**”) is dated and shall be effective as of September 5, 2023 (the “**Effective Date**”), by and between Aspen Aerogels, Inc., a Delaware corporation (the “**Company**”), and Stephanie Pittman (the “**Executive**”).

Recitals:

WHEREAS, the Company wishes to employ the Executive and the Executive wishes to accept such employment on the terms and conditions set forth in this Agreement;

NOW, THEREFORE, in consideration of the mutual covenants and agreements herein contained and other good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, the parties hereto agree as follows:

1. **Definitions.** As used herein, the following terms shall have the following meanings.

“**Board**” means the Company’s board of directors.

“**Cause**” means: (i) willful misconduct, dishonesty, fraud or breach of fiduciary duty to the Company; (ii) deliberate disregard of the lawful rules or policies of the Company, failure to perform assigned duties, or material breach of an employment or other agreement with the Company, which results in material harm to the Company; (iii) the unauthorized disclosure of any trade secret or confidential information of the Company; (iv) the commission of an act which constitutes unfair competition with the Company or which induces any customer or supplier to breach a contract with the Company; or (v) the indictment of the Executive for any felony involving deceit, dishonesty or fraud, or any criminal conduct by the Executive that would reasonably be expected to result in material injury or reputational harm to the Company. For purposes hereof, whether or not the Executive has committed an act or omission of the type referred to in subparagraphs (i) through (vi) above shall be determined by the Company in its reasonable, good faith discretion, based upon the facts known to the Company at the relevant time.

“**Change of Control**” shall mean the first to occur of any of the following events: (i) the consummation of a reorganization, merger, consolidation or other similar transaction of the Company with or into any other Person or Group (within the meaning of Section 13(d)(3) of the Securities Act of 1934, as amended) in which holders of the Company’s voting securities immediately prior to such reorganization, merger, consolidation or other similar transaction shall not, directly or indirectly, continue to hold at least a majority of the outstanding voting securities of the Company; (ii) a sale, lease, exchange or other transfer (in one transaction or a related series of transactions) of all or substantially all of the Company’s assets; (iii) the acquisition by any Person or any Group of such quantity of the Company’s voting securities as causes such Person or Group (other than a Person or Group who is a shareholder of the Company on the Effective Date) to own beneficially, directly or indirectly, as of the time immediately after such transaction or series of transactions, more than fifty percent (50%) of the combined voting power of the voting securities of the Company other than as a result of (a) an acquisition of securities directly from the Company or (b) an acquisition of securities by the Company which by reducing the voting securities outstanding increases the proportionate voting power represented by the voting securities

owned by any such Person or Group to more than fifty percent (50%) of the combined voting power of such voting securities; or (iv) a change in the composition of the Board within a two (2) year period such that a majority of the members of the Board are not Continuing Directors. As used herein, the term **“Continuing Directors”** shall mean as of any date of determination, any member of the Board who (a) was a member of the Board immediately after the Effective Date, or (b) was nominated for election or elected to the Board with the approval of, or whose election to the Board was ratified by, at least a majority of the Continuing Directors who were members of the Board at the time of that nomination or election; provided, however, that in no case shall (1) the public offering and sale of the Company’s common stock by its shareholders pursuant to a registered secondary offering, (2) the voluntary or involuntary bankruptcy of the Company, or (3) any transaction or series of transactions that would not qualify as a change in control within the meaning of Section 409A of the Internal Revenue Code of 1986, as amended (the **“Code”**) constitute a Change of Control.

“Good Reason” means: (i) any material breach by the Company of this Agreement; (ii) a change in the Executive’s reporting relationships such that the Executive no longer directly reports to the President and Chief Executive Officer; (iii) a material reduction or material adverse change in the Executive’s current duties, responsibilities and authority, without the Executive’s consent; (iv) the demand by the Company for the Executive to relocate Executive’s primary office location from Northborough, Massachusetts, provided (1) such relocation increases the Executive’s regular vehicular one-way in-office commute to such new location by more than forty (40) miles, and (2) the Company does not offer the Executive a remote working arrangement as part of the office relocation, without the Executive’s consent; or (v) any reduction by the Company in the Executive’s Base Salary or the Executive’s Performance Bonus Target without the Executive’s consent, except for across-the-board compensation reductions based on the Company’s financial performance similarly affecting all or substantially all senior management employees of the Company. For purposes hereof, whether or not the Executive has Good Reason to terminate the Executive’s employment by the Company pursuant to subparagraphs (i) through (v) above shall be determined by the Company in its reasonable, good faith discretion, based upon the facts known to the Company at the relevant time.

“Permanent Disability” means the Executive is unable to perform, by reason of physical or mental incapacity, the Executive’s then duties or obligations to the Company, for a total period of one hundred eighty (180) days in any three hundred sixty (360) day period.

“Person” means an individual, a partnership, a corporation, a limited liability company, an association, a joint stock company, a trust, a joint venture, an unincorporated organization or any other entity, including a governmental entity or any department, agency or political subdivision thereof.

“Qualifying Termination” means the date on which the Executive’s employment is terminated by the Company without Cause as provided in Section 3(e), or the Executive terminates employment for Good Reason as provided in Section 3(f).

2. Employment.

(a) **Term.** The term of this Agreement shall commence on the Effective Date and continue until the date that is one (1) year after the Effective Date (the “**Expiration Date**”), unless terminated earlier by the Company or by the Executive in accordance with the provisions of Section 3 of this Agreement (the “**Initial Employment Period**”). The Initial Employment Period shall automatically renew for additional one (1) year terms (each such term being a “**Renewal Employment Period**”), and the Expiration Date shall be the last day of such Renewal Employment Period, unless either the Company or the Executive provides written notice of non-renewal of the Employment Period to the other party at least sixty (60) calendar days before the applicable Expiration Date, and if extended, any Renewal Employment Period may be terminated earlier by the Company or by the Executive in accordance with the provision of Section 3 of this Agreement. As used in this Agreement, the “**Employment Period**” shall refer to the Initial Employment Period and any Renewal Employment period, as applicable.

(b) **Position and Duties.**

(i) During the Employment Period, the Executive shall serve as Chief Human Resources Officer of the Company and shall have the duties, responsibilities and authority consistent with such position that are designated by the Company’s President and Chief Executive Officer, subject to the direction and supervision of the Board.

(ii) The Executive shall devote the Executive’s best efforts and full business time and attention (except for permitted vacation periods and reasonable periods of illness or other incapacity) to the business and affairs of the Company. The Executive shall perform the Executive’s duties and responsibilities to the best of the Executive’s ability in a diligent, trustworthy businesslike and efficient manner. Notwithstanding the foregoing, the Executive may, to the extent not otherwise prohibited by this Agreement, devote such amount of time as does not interfere or compete with the performance of the Executive’s duties under this Agreement to any one or more of the following activities: (A) engaging in charitable activities, including serving on the boards of directors of charitable organizations, consistent with such policies, rules and regulations as the Company may adopt from time to time, or (B) serving on the board of directors of any other company with the prior written approval of the Company.

(iii) The Executive agrees to abide by the Company’s Code of Business Conduct and Ethics, Anti- Corruption Policy, Insider Trading Policy, Disclosure Controls & Procedures Policy, Form 8-K Disclosure Compliance Policy, Regulation FD Disclosure Policy, Compensation Recoupment Policy, Environmental & Sustainability Policy, Health & Safety Policy, and the Delegation of Authority Policy, each as in effect from time to time and such other policies, rules and regulations as the Company may adopt from time to time.

(c) **Salary and Benefits.** During the Employment Period, Executive shall be entitled to the following compensation and benefits:

(i) **Base Salary.** During the Employment Period, the Executive’s base salary shall be \$400,000 (such annual salary, as it may be adjusted upward by the Board in its discretion, being referred to as the “**Base Salary**”). The Base Salary shall be payable in regular installments in accordance

with the Company's general payroll practices, shall be subject to customary withholding. Base Salary may be increased at the discretion of the Board, and may be decreased in the discretion of the Board only in connection with across-the-board compensation reductions based on the Company's financial performance similarly affecting all or substantially all senior management employees of the Company.

(ii) **Annual Performance Bonus.** The Executive shall be eligible to earn a discretionary annual cash incentive bonus (the "**Performance Bonus**") of not less than 50% of the Executive's then effective Base Salary (each, a "**Performance Bonus Target**"), subject in all respects to the terms and conditions established by the Board, and provided that, notwithstanding the foregoing, the Executive's Performance Bonus Target for the 2023 fiscal year shall be pro-rated based on the portion of such fiscal year from the Effective Date through the end of such fiscal year.

(iii) **Expense Reimbursement.** The Company shall reimburse the Executive for all reasonable travel and other expenses (including periodic dues and membership fees pertaining to pertinent licensing, registrations, professional associations and industry groups with the prior approval of the Company) incurred by the Executive in connection with the performance of the Executive's duties and obligations under this Agreement. The Executive shall comply with such reasonable limitations and reporting requirements with respect to expenses as may be established by the Company from time to time.

(iv) **Benefit Plans and Programs.** The Executive shall be entitled to participate in all compensation or employee benefit plans or programs and receive all benefits and perquisites for which salaried employees of the Company generally are eligible under any plan or program now or established later by the Company on the same basis as similarly situated senior executives of the Company. The Executive may participate to the extent permissible under the terms and provisions of such plans or programs, in accordance with program provisions. Nothing in this Agreement shall preclude the Company from amending or terminating any of the plans or programs applicable to salaried employees or senior executives of the Company as long as such amendment or termination is applicable to all salaried employees or senior executives, as the case may be, so long as such plans or programs are replaced with plans no less favorable, in the aggregate, than existing plans.

(v) **Long-Term Compensation.** The Executive shall also be eligible for grants of long-term incentive compensation, at the discretion of and subject to such terms and conditions as established by the Compensation and Leadership Development Committee of the Board (the "**Compensation Committee**"). Within thirty (30) days after the Effective Date, the Company will request that the Compensation Committee approve a new hire equity grant to the Executive under the Aspen Aerogels 2023 Equity Incentive Plan (the "**Equity Incentive Plan**"), with an aggregate fair market value as of the date of grant of approximately \$400,000, allocated approximately 75% to non-qualified stock options and 25% to restricted share units, with vesting in substantially equal annual installments over a three-year period from the Effective Date (or if later, from the date of grant), and with such other terms as established by the Compensation Committee and evidenced by an award agreement substantially in the form approved by the Compensation Committee under the Equity Incentive Plan.

(vi) **Clawback Policy.** All compensation shall be subject to any forfeiture or recoupment pursuant to the Company's Compensation Recoupment Policy, as then in effect, or any successor compensation recoupment or "clawback" policy established by the Company or the Board generally for senior executives from time to time, and any other such policy required by applicable law.

(d) **Change of Control: Equity Awards.** In the event of a Change of Control, notwithstanding anything to the contrary in any then outstanding stock option agreement or other equity award agreement (unless otherwise specifically stated in any equity award agreement that provides for vesting based wholly or in part upon the achievement of performance objectives), to the extent any outstanding stock options and other equity awards are not assumed by the Company's successor in a Change of Control, the vesting of all stock options and other equity awards (unless otherwise specifically stated in any equity award agreement that provides for vesting based wholly or in part upon the achievement of performance objectives) outstanding and held by the Executive as of the Change of Control shall immediately accelerate and become fully vested and exercisable, subject to any permitted action by the Board upon a Change of Control under the Company's applicable equity plan to terminate the stock options or other equity awards upon a Change of Control, provided, however, that the foregoing shall not apply to any outstanding equity award to the extent such acceleration of vesting would result in a violation of Section 409A of the Code.

3. **Termination.** The Executive's employment may be terminated under the following circumstances:

(a) **Expiration.** The Executive's employment hereunder shall terminate upon the Expiration Date as a result of a non-renewal by either party, following notice of non-renewal in accordance with Section 2(a).

(b) **Death.** The Executive's employment hereunder shall terminate upon Executive's death.

(c) **Disability.** The Company may terminate the Executive's employment upon the Executive's Permanent Disability. If any question shall arise as to whether the Executive has a Permanent Disability so as to be unable to perform the essential functions of the Executive's then existing position or positions with or without reasonable accommodation, the Executive may, and at the request of the Company shall, submit to the Company a certification in reasonable detail by a physician selected by the Company to whom the Executive or the Executive's guardian has no reasonable objection as to whether the Executive is so disabled or how long such disability is expected to continue, and such certification shall for the purposes of this Agreement be conclusive of the issue. The Executive shall cooperate with any reasonable request of the physician in connection with such certification. If such question shall arise and the Executive shall fail to submit such certification, the Company's determination of such issue shall be binding on the Executive. Nothing in this Section 3(c) shall be construed to waive the Executive's rights, if any, under existing law including, without limitation, the Family and Medical Leave Act of 1993, 29 U.S.C. §2601 et seq. and the Americans with Disabilities Act, 42 U.S.C. §12101 et seq.

(d) **Termination by Company for Cause.** The Company may terminate the Executive's employment hereunder for Cause.

(e) **Termination by Company Without Cause.** The Company may terminate the Executive's employment hereunder at any time without Cause.

(f) **Termination by the Executive.** The Executive may terminate the Executive's employment hereunder at any time for any reason, including but not limited to Good Reason consistent with the Good Reason Process (hereinafter defined). "**Good Reason Process**" shall mean that (i) the Executive reasonably determines in good faith that a Good Reason condition has occurred; (ii) the Executive notifies the Company in writing of the Good Reason condition within sixty (60) days of the first occurrence of such condition; (iii) the Executive cooperates in good faith with the Company's efforts, for a period not less than thirty (30) days following receipt of such notice to remedy the condition (the "**Cure Period**"); (iv) notwithstanding such efforts, the Good Reason condition continues to exist; and (v) the Executive terminates the Executive's employment within sixty (60) days after the end of the Cure Period by notifying the Company in writing of such termination for Good Reason. If, during the Cure Period, the Company cures the Good Reason condition or demonstrates to the Executive's reasonable satisfaction, to be exercised in good faith, that a Good Reason condition has not occurred, Good Reason shall be deemed not to have occurred. For the avoidance of doubt, a notice of non-renewal of the Employment Period by Executive shall not constitute a termination for Good Reason.

(g) **Notice of Termination.** Except for a termination as specified in Section 3(a) and 3(b), any termination of the Executive's employment by the Company or any such termination by the Executive shall be communicated by written Notice of Termination to the other party hereto. For purposes of this Agreement, a "**Notice of Termination**" shall mean a notice which shall indicate the specific termination provision in this Agreement relied upon. Failure to provide a Notice of Termination by either Party pursuant to this Section 3(g) shall be a material breach of the Agreement.

(h) **Date of Termination.** "**Date of Termination**" shall mean: (i) if the Executive's employment is terminated because the Agreement's Employment Period expires under Section 3(a), the date of the Expiration Date; (ii) if the Executive's employment is terminated by Executive's death, the date of Executive's death; (iii) if the Executive's employment is terminated on account of Permanent Disability under Section 3(c) or by the Company for Cause under Section 3(d), the date on which Notice of Termination is given; (iv) if the Executive's employment is terminated by the Company under Section 3(e), the date on which a Notice of Termination is given; (v) if the Executive's employment is terminated by the Executive under Section 3(f) without Good Reason, thirty (30) days after the date on which a Notice of Termination is given, and (vi) if the Executive's employment is terminated by the Executive under Section 3(f) with Good Reason, the date which is specified in the Notice of Termination, provided that such date must occur within the sixty (60) day period after the end of the Cure Period. Notwithstanding the foregoing, in the event that either party gives a Notice of Termination, the Company may unilaterally accelerate the Date of Termination, without any additional compensation in lieu of notice being due to the Executive as a result of such acceleration.

4. **Compensation on, and Effect of, Termination.**

(a) **Termination Generally.** If the Executive's employment with the Company is terminated for any reason, the Company shall pay or provide to the Executive (or to the Executive's authorized

representative or estate) (i) any Base Salary earned through the Date of Termination, unpaid expense reimbursements (subject to, and in accordance with, Section 2(c) of this Agreement) and, to the extent required by law, unused vacation that accrued through the Date of Termination, such amounts to be paid no more than thirty (30) days after the Executive's Date of Termination (unless required to be paid at an earlier time by law); and (ii) any vested benefits the Executive may have under any employee benefit plan of the Company through the Date of Termination, which vested benefits shall be paid and/or provided in accordance with the terms of such employee benefit plans (collectively, the "**Accrued Benefits**").

(b) **Qualifying Termination Prior to A Change of Control.** If the Executive incurs a Qualifying Termination during the Employment Period and prior to a Change of Control, then, in addition to the Accrued Benefits, and subject to the Executive signing a separation agreement containing, among other provisions, a general release of claims in favor of the Company and related persons and entities and a non-competition provision (as applicable), in a form and manner reasonably satisfactory to the Company (the "**Separation Agreement and Release**") and the Separation Agreement and Release becoming fully effective, all within sixty (60) days of the Date of Termination (the "**Release Period**"):

(i) the Company shall pay the Executive an amount equal to one hundred percent (100%) of the sum of (A) the Executive's then effective Base Salary and (B) an amount equal to the Executive's then effective Performance Bonus Target (the "**Severance Amount**").

(ii) the Company shall pay the Executive any accrued but unpaid Performance Bonus for the prior fiscal year then owed or fully earned by the Executive in accordance with Section 2(c)(ii) above (the "**Earned Performance Bonus**").

(iii) the Company shall pay the Executive a pro-rata portion of the Performance Bonus based upon actual achievement of the performance metrics for the fiscal year in which the Date of Termination occurs (calculated by dividing the number of full months of the applicable fiscal year through the Date of Termination by twelve (12), and multiplying this fraction by the Executive's then effective Performance Bonus Target) (the "**Pro-Rata Bonus**").

(iv) the COBRA eligible health care insurance benefits (e.g., health, dental) being provided by the Company to the Executive on the Date of Termination shall continue in place at the same cost to the Executive as applied to "active" participants on the Date of Termination for a period equal to the lesser of (i) the COBRA Benefit Period or (ii) twelve (12) months ("**Health Care Continuation Benefit**"). The "COBRA Benefit Period" means the period of time after such termination during which COBRA benefits are available to the Executive as of the Date of Termination as set forth in the Company's health care plan. The Executive shall be responsible for applying for the COBRA eligible health care insurance benefit, paying for the same and submitting evidence of such premium costs to the Company for reimbursement during the COBRA Benefit Period. The Company shall reimburse the Executive for the employer's portion of such premiums (as applicable to the active rate) within fifteen (15) days of receipt of evidence of the payment of the premium costs to the Company ("**Premium Reimbursement Payments**"). Notwithstanding the foregoing, if the Company determines, in its sole discretion, that such reimbursement of the premiums would result in a violation of the nondiscrimination rules of Section 105(h)(2) of the Code or any statute or regulation of similar effect (including but not

limited to the 2010 Patient Protection and Affordable Care Act, as amended by the 2010 Health Care and Education Reconciliation Act), then in lieu of reimbursing the premiums, the Company, in its sole discretion, may elect to instead pay the Executive on the first day of each month of such period, a fully taxable cash payment equal to the premiums for that month, subject to applicable tax withholdings (such amount, the “**Special Severance Payment**”), for the remainder of such period. The Executive may, but is not obligated to, use such Special Severance Payment toward the cost of premiums.

(v) if the Executive requests, the Company shall pay (“**Outplacement Payments**”) for an outplacement service (to be selected by the Company) for services rendered in assisting the Executive in locating another job, for a period of six (6) months following the Date of Termination or until the Executive begins working for another employer, whichever occurs first

(“**Outplacement Services**”). These Outplacement Payments, which the Company shall make directly to the vendor providing Outplacement Services, are contingent upon the Executive’s cooperation with the outplacement service and upon active efforts by the Executive to locate another position.

(vi) Notwithstanding anything to the contrary in any then outstanding stock option agreement or other equity award agreement (unless otherwise specifically stated in any equity award agreement that provides for vesting based wholly or in part upon the achievement of performance objectives), (a) the vesting of such number of stock options and other equity awards outstanding and held by the Executive as would have vested in the twelve (12) months immediately following the Date of Termination had the Executive continued the Executive’s employment for such twelve (12) month period shall immediately accelerate and become vested and exercisable as of the Date of Termination, and (b) subject to any permitted action by the Board upon a Change of Control or other merger, sale, dissolution or liquidation of the Company under Company’s applicable equity plan to terminate the stock options or other equity awards, all vested stock options held by the Executive shall be exercisable for one (1) year from the Date of Termination.

(c) **Timing and Form of Severance Payments**. The benefits provided to Executive under Sections 4(b)(i), (ii), (iii), (iv), and (v) shall be paid in the form and at the time specified below:

(i) The Severance Amount shall be paid in substantially equal installments in accordance with the Company’s payroll practice over twelve (12) months commencing within sixty (60) days after the Date of Termination; provided, however, that if the sixty (60) day period begins in one calendar year and ends in a second calendar year, the Severance Amount shall begin to be paid in the second calendar year by the last day of such sixty (60) day period; provided, further, that the initial payment shall include a catch-up payment to cover amounts retroactive to the day immediately following the Date of Termination. Each payment pursuant to this Agreement is intended to constitute a separate payment for purposes of Treasury Regulation Section 1.409A-2(b)(2).

(ii) Earned Performance Bonus shall be paid in a lump sum within sixty (60) days after the Date of Termination; provided, however, that if the sixty (60) day period begins in one calendar year and ends in a second calendar year, the Earned Performance Bonus shall be paid in the second calendar year by the last day of such sixty (60) day period.

(iii) The Pro-Rata Bonus shall be paid when the annual performance bonus would have been otherwise paid if Executive had continued the Executive's employment through the applicable performance period.

(iv) Health Care Continuation Benefit shall commence immediately upon the Executive's Date of Termination and the Executive shall immediately become eligible for Reimbursement Payments in accordance with Section 4(b)(iii), provided however if an executed Separation Agreement and Release has not become fully effective within Release Period, the Company shall immediately cease making Premium Reimbursement Payments (or, if applicable, Special Severance Payments) and the Executive shall be obligated to promptly repay to the Company any previously received Premium Reimbursement Payments (or, if applicable, any Special Severance Payments).

(v) Outplacement Services shall commence immediately upon the Executive's Date of Termination, provided however if an executed Separation Agreement and Release has not become fully effective within the Release Period, Outplacement Services shall immediately cease and the Executive shall be obligated to promptly repay to the Company any previously made Outplacement Payments.

(d) **Rights to Severance.** The receipt of any severance payments or benefits pursuant to Section 4(b) shall be subject to the Executive's submission to the Company of an executed Separation Agreement and Release that becomes fully effective within the Release Period and the Executive's continued compliance with the covenants in Sections 7, 8, and 9 herein, so long as the Company provides the form of Severance Agreement to the Executive no less than twenty-one (21) days prior to the end of the Release Period. In the event the Executive materially breaches any of the provisions set forth in Sections 7, 8 or 9 herein or in the Separation Agreement and Release, and does not cure said breach(es) within thirty (30) days of receiving detailed written notice describing the alleged breach(es), in addition to all other legal and equitable remedies, the Company shall have the right to terminate or suspend all continuing payments and benefits to which the Executive may otherwise be entitled pursuant to Section 4 without affecting the effectiveness of the Executive's release or the Executive's obligations under the Separation Agreement and Release.

(e) **Other Termination Events.** The Executive hereby agrees that no severance payments or benefits under Section 4(b) shall be payable or provided upon termination of the Executive's employment with the Company (i) by the Company with for Cause; (ii) by the Executive without Good Reason; (iii) as a result of the Executive's death or Permanent Disability; or (iv) as a result of non-renewal of the Employment Period by either the Executive or the Company pursuant to Section 2(a), and the Executive hereby waives any claim for such severance payment or benefits except for the Accrued Benefits.

5. **Compensation on Termination after a Change of Control.** The provisions of this Section 5 shall apply in lieu of, and expressly supersede, other than with respect to the requirement for the Executive's submission to the Company of an executed Separation Agreement and Release that becomes fully effective within the Release Period as described in Section 4(d), the provisions of Section 4(b) regarding severance payments and benefits upon a Qualifying Termination, if a Qualifying Termination occurs within twenty-four (24) months after the occurrence of a Change of Control ("**COC Qualifying Termination**"). This Section 5 shall terminate and be of no force or effect beginning twenty-four (24) months after the occurrence of a Change of Control.

(a) **Qualifying Termination after a Change of Control.** During the Employment Period, if the Executive incurs a COC Qualifying Termination, then in addition to the Accrued Benefits, and subject to the signing of the Separation Agreement and Release by the Executive and the Separation Agreement and Release becoming irrevocable within the Release Period:

(i) the Company shall pay the Executive an amount equal to two hundred percent (200%) of the sum of (A) the Executive's then effective Base Salary and (B) the Executive's then effective Performance Bonus Target ("**COC Severance Amount**").

(ii) the Company shall pay the Executive any accrued but unpaid Performance Bonus for the prior fiscal year then owed or fully earned by the Executive in accordance with Section 2(c)(ii) above ("**COC Earned Performance Bonus**").

(iii) the Company shall pay the Executive the Pro-Rata Bonus.

(iv) the COBRA eligible health care insurance benefits (e.g., health, dental) being provided by the Company to the Executive on the Date of Termination shall continue in place at the same cost to the Executive as applied to "active" participants on the Date of Termination for a period equal to the lesser of (i) the COBRA Benefit Period or (ii) twenty-four (24) months ("**COC Health Care Continuation Benefits**"). The Executive shall be responsible for applying for the COBRA eligible health care insurance benefit, paying for the same and submitting evidence of such premium costs to the Company for reimbursement during the COBRA Benefit Period. The Company shall reimburse the Executive for the employer's portion of such premiums (as applicable to the active rate) within fifteen (15) days of receipt of evidence of the payment of the premium costs to the Company ("**COC Premium Reimbursement Payments**"). Notwithstanding the foregoing, if the Company determines, in its sole discretion, that such reimbursement of the premiums would result in a violation of the nondiscrimination rules of Section 105(h)(2) of the Code or any statute or regulation of similar effect (including but not limited to the 2010 Patient Protection and Affordable Care Act, as amended by the 2010 Health Care and Education Reconciliation Act), then in lieu of reimbursing the premiums, the Company, in its sole discretion, may elect to instead pay the Executive on the first day of each month of such period a fully taxable cash payment equal to the premiums for that month, subject to applicable tax withholdings (such amount, the "**COC Special Severance Payment**"), for the remainder of such period. The Executive may, but is not obligated to, use such Special Severance Payment toward the cost of premiums.

(v) if the Executive wishes, the Company shall pay for an outplacement service ("**COC Outplacement Payments**") (to be selected by the Company) for services rendered in assisting the Executive in locating another job, for a period of six (6) months following the Date of Termination or until the Executive begins working for another employer, whichever occurs first ("**COC Outplacement Services**"). These COC Outplacement Payments, which the Company shall make directly to the vendor providing the COC Outplacement Services, are contingent upon the Executive's cooperation with the outplacement service and upon active efforts by the Executive to locate another position.

(vi) COC Severance Amount shall be payable in a lump sum within sixty (60) days after the Date of Termination; provided, however, that if the sixty (60) day period begins in one calendar year

and ends in a second calendar year, such payment shall be paid or commence to be paid in the second calendar year by the last day of such sixty (60) day period. Each payment pursuant to this Agreement is intended to constitute a separate payment for purposes of Treasury Regulation Section 1.409A-2(b)(2).

(vii) COC Earned Performance Bonus shall be paid in a lump sum within sixty (60) days after the Date of Termination; provided, however, that if the sixty (60) day period begins in one calendar year and ends in a second calendar year, the Earned Performance Bonus shall be paid in the second calendar year by the last day of such sixty (60) day period.

(viii) The Pro-Rata Bonus shall be paid when the annual performance would have been paid if Executive had continued the Executive's employment through the payment date.

(ix) COC Health Care Continuation Benefit shall commence immediately upon the Executive's Date of Termination and the Executive shall immediately become eligible for Reimbursement Payments in accordance with Section 5(a) (iv), provided however if an executed Separation Agreement and Release has not become fully effective within the COC Release Period, the Company shall immediately cease making COC Premium Reimbursement Payments (or, if applicable, any COC Special Severance Payments) and the Executive shall be obligated to promptly repay to the Company any previously received COC Premium Reimbursement Payments (or, if applicable, any COC Special Severance Payments).

(x) COC Outplacement Services shall commence immediately upon the Executive's Date of Termination, provided however if an executed Separation Agreement and Release has not become fully effective within the COC Release Period, COC Outplacement Services shall immediately cease and the Executive shall be obligated to promptly repay to the Company any previously made COC Outplacement Payments.

(xi) Notwithstanding anything to the contrary in any then outstanding stock option agreement or other equity award agreement (unless otherwise specifically stated in any equity award agreement that provides for vesting based wholly or in part upon the achievement of performance objectives), the vesting of all stock options and other equity awards outstanding and held by the Executive shall immediately accelerate and become fully vested and exercisable as of the Date of Termination, and subject to any permitted action by the Board upon a Change of Control pursuant to the Company's applicable equity plan to terminate the stock options or other equity awards upon a Change of Control, all vested stock options shall be exercisable for one (1) year from the Date of Termination.

(b) **Rights to Severance.** The receipt of any severance payments or benefits in Section 5 is subject to the Executive's compliance with the provisions of Section 4(d) above.

(c) **Other Termination Events.** The Executive hereby agrees that no severance payments or benefits under Section 5(a) shall be payable or provided upon termination of the Executive's employment with the Company (i) by the Company with Cause; (ii) by the Executive without Good Reason; (iii) as a result of the Executive's death or Permanent Disability; or (iv) as a result of non-renewal of the Employment Period by either the Executive or the Company pursuant to

Section 2(a), and the Executive hereby waives any claim for such severance payments or benefits except for the Accrued Benefits.

(d) **Parachute Payments**. If Independent Tax Counsel (as that term is defined below) determines that the aggregate payments and benefits provided or to be provided to the Executive pursuant to this Agreement, and any other payments and benefits provided or to be provided to the Executive from the Company or any of its subsidiaries or other affiliates or any successors thereto constitute “parachute payments” as defined in Section 280G of the Code (“**Parachute Payments**”) that would be subject to the excise tax imposed by Section 4999 of the Code (the “**Excise Tax**”), then, except as otherwise provided in the next sentence, such Parachute Payments shall be reduced to the extent the Independent Tax Counsel shall determine is necessary (but not below zero) so that no portion thereof shall be subject to the Excise Tax. If Independent Tax Counsel determines that the Executive would receive in the aggregate greater payments and benefits on an after tax basis if the Parachute Payments were not reduced pursuant to this Section 5(f), then no such reduction shall be made. The determination of which payments or benefits shall be reduced to avoid the Excise Tax shall be made by the Independent Tax Counsel, provided that the Independent Tax Counsel shall reduce or eliminate, as the case may be, payments or benefits in the following order (1) cash payments not subject to Section 409A of the Code; (2) cash payments subject to Section 409A of the Code; (3) equity-based payments and acceleration; and (4) non-cash forms of benefits. To the extent any payment is to be made over time (e.g., in installments, etc.), then the payments shall be reduced in reverse chronological order. The determination of the Independent Tax Counsel under this Section 5(f) shall be final and binding on all parties hereto. For purposes of this Section 5(f), “**Independent Tax Counsel**” shall mean a lawyer, a certified public accountant with a nationally recognized accounting firm, or a compensation consultant with a nationally recognized actuarial and benefits consulting firm with expertise in the area of executive compensation tax law, who shall be selected by the Board, and whose fees and disbursements shall be paid by the Company. The Executive shall have the right, at the Executive’s own expense, to retain Independent Tax Counsel to rebut any decision made by the Company’s Independent Tax Counsel, who may consider such rebuttal before making its final and binding determination.

6. **Tax, Insurance and Indemnity.**

(a) **Insurance and Indemnity**. In no event shall the termination of the Executive’s employment by the Company or any such termination by the Executive pursuant to this Agreement release any claim by the Executive for indemnification that he or she is otherwise entitled to under any director or officer’s insurance policy or any articles, bylaws or other foundation documents of the Company. The Executive shall be entitled to the protections set forth the Company’s Bylaws (as may be amended and restated from time to time) with respect to Indemnification of Directors. Without limiting the foregoing, the Company shall provide Executive with reasonable director’s and officer’s insurance coverage that is at least as favorable as the coverage in existence on the date of this Agreement (the “**Existing D&O Coverage**”); provided, however, that in no event shall the Company be obligated to maintain director’s and officer’s insurance coverage to the extent that premiums thereunder exceed two hundred percent (200%) of the premiums payable by the Company under the Existing D&O Coverage on the date hereof (the “**Threshold**”); provided, further, that to the extent such premiums exceed the foregoing Threshold, the Company shall obtain director’s and officer’s insurance coverage on terms as similar as reasonably practicable to the terms of the Existing D&O Coverage without exceeding the Threshold. Such insurance

coverage shall continue in effect during the Employment Period and after the Employment Period ends for a period of six (6) years thereafter. The cost of such coverage shall be paid by the Company. Notwithstanding anything to the contrary in this Agreement, upon the occurrence of a Change of Control, the obligations set forth in this section shall terminate, provided that the Company shall (i) secure “tail insurance” with respect to the Existing D&O Coverage on reasonable terms and conditions of coverage, and (ii) require any successor (whether direct or indirect, by purchase, merger, consolidation or otherwise) to all or substantially all of the business or assets of the Company to honor any indemnification obligations that the Executive is otherwise entitled to under any articles, bylaws or other foundation documents of the Company in the same manner as the Company’s directors and officers immediately prior to such Change of Control.

(b) **409A.**

(i) Anything in this Agreement to the contrary notwithstanding, if at the time of the Executive’s separation from service within the meaning of Section 409A of the Code, the Company determines that the Executive is a “specified employee” within the meaning of Section 409A(a)(2)(B)(i) of the Code, then to the extent any payment or benefit that the Executive becomes entitled to under this Agreement on account of the Executive’s separation from service would be considered deferred compensation otherwise subject to the twenty percent (20%) additional tax imposed pursuant to Section 409A(a) of the Code as a result of the application of Section 409A(a)(2)(B)(i) of the Code, such payment shall not be payable and such benefit shall not be provided until the date that is the earlier of (A) six (6) months and one (1) day after the Executive’s separation from service, or (B) the Executive’s death. If any such delayed cash payment is otherwise payable on an installment basis, the first payment shall include a catch-up payment covering amounts that would otherwise have been paid during the six-month period but for the application of this provision, and the balance of the installments shall be payable in accordance with its original schedule.

(ii) All in-kind benefits provided and expenses eligible for reimbursement under this Agreement shall be provided by the Company or incurred by the Executive during the time periods set forth in this Agreement. All reimbursements shall be paid as soon as administratively practicable, but in no event shall any reimbursement be paid after the last day of the taxable year following the taxable year in which the expense was incurred. The amount of in-kind benefits provided or reimbursable expenses incurred in one (1) taxable year shall not affect the in-kind benefits to be provided or the expenses eligible for reimbursement in any other taxable year (except for any lifetime or other aggregate limitation applicable to medical expenses). Such right to reimbursement or in-kind benefits is not subject to liquidation or exchange for another benefit.

(iii) To the extent that any payment or benefit described in this Agreement constitutes “non-qualified deferred compensation” under Section 409A of the Code, and to the extent that such payment or benefit is payable upon the Executive’s termination of employment, then such payments or benefits shall be payable only upon the Executive’s “separation from service.” The determination of whether and when a separation from service has occurred shall be made in accordance with the presumptions set forth in Treasury Regulation Section 1.409A-1(h).

(iv) The parties intend that this Agreement shall be administered in accordance with Section 409A of the Code. To the extent that any provision of this Agreement is ambiguous as to its compliance

with Section 409A of the Code, the provision shall be read in such a manner so that all payments hereunder comply with Section 409A of the Code. Each payment pursuant to this Agreement is intended to constitute a separate payment for purposes of Treasury Regulation Section 1.409A-

2(b)(2). The parties agree that this Agreement may be amended, as reasonably requested by either party, and as may be necessary to fully comply with Section 409A of the Code and all related rules and regulations in order to preserve the payments and benefits provided hereunder without additional cost to either party.

(v) The Company makes no representation or warranty and shall have no liability to the Executive or any other person if any provisions of this Agreement are determined to constitute deferred compensation subject to Section 409A of the Code but do not satisfy an exemption from, or the conditions of, such Section.

7. Confidentiality.

(a) **Definition of Confidential Information.** “**Confidential Information**” shall mean: trade secrets and confidential and proprietary information of the Company, or any information provided to the Executive or the Company under an obligation of confidentiality to a third party, or any confidential, trade secret, or proprietary information acquired by the Company from others with whom the Company or any affiliate has a business relationship, whether in written, oral, electronic or other form, including, but not limited to: (i) trade secrets, inventions, mask works, ideas, processes, algorithms, formulae, software in source or object code, data programs, other works of authorship, know-how, improvements, technology direction, product or technology development methodology, technology assessment, experimental procedures, results, process development, product plans, development plans, testing procedures, quality control and testing processes, discoveries, developments, designs and techniques, any other proprietary technology and all Inventions (as defined in Section 9); (ii) information regarding research, development, new products, marketing and selling, business plans, budgets and unpublished financial statements, licenses, prices and costs, margins, discounts, credit terms, pricing and billing policies, quoting procedures, methods of obtaining business, forecasts, future plans and potential strategies, financial projections and business strategies, operational plans, financing and capital-raising plans, activities and agreements, internal services and operational manuals, methods of conducting Company business, suppliers and supplier information, sales techniques and strategies, training methods and materials, and purchasing; (iii) information regarding customers and potential customers of the Company, including customer lists, names, representatives, their needs or desires with respect to the types of products or services offered by the Company, proposals, bids, contracts and their contents and parties, the type and quantity of products and services provided or sought to be provided to customers and potential customers of Company and other non-public information relating to customers and potential customers; (iv) information regarding any of the Company’s business partners and its services, including names, representatives, proposals, bids, contracts and their contents and parties, the type and quantity of products and services received by the Company, and other non-public information relating to business partners; (v) information regarding personnel, employee lists, compensation, and employee skills; (vi) any other non-public information which a competitor of the Company could use to the competitive disadvantage of the Company; and (vii) any other scientific, technical or trade secrets of the Company or of any third party provided to the Executive or the Company under a condition of confidentiality, provided that Confidential Information shall not include information that was known to the Executive prior to

joining the Company, general industry knowledge, or is in the public domain other than through any fault or act by the Executive.

(b) **Protection and Non-Disclosure of Confidential Information.** The Executive expressly acknowledges and agrees that all Confidential Information is and shall remain the sole property of the Company or the third party to whom the Company owes an obligation of confidentiality and that the Executive shall hold it in strictest confidence. The Executive shall at all times, both during the Executive's employment with the Company and after the termination of the Executive's employment for any reason or for no reason, maintain in confidence and shall not, without the prior written consent of the Company, use (except in the course of performance of the Executive's duties for the Company or by court order), disclose, or give to others any Confidential Information. The terms of this Section 7 are in addition to, and not in lieu of, any statutory or other contractual or legal obligation that the Executive may have relating to the protection of the Company's Confidential Information. The terms of this Section 7 shall survive indefinitely any termination of the Executive's employment with the Company for any reason or for no reason.

(c) **Notification to Company.** In the event the Executive is questioned by anyone not employed by the Company or by an employee of or a consultant to the Company not authorized to receive Confidential Information, in regard to any Confidential Information or concerning any fact or circumstance relating thereto, the Executive agrees to promptly notify the Company's Chief Legal Officer and General Counsel.

(d) **Return of Confidential Information.** Upon the termination of the Executive's employment with the Company for any reason or for no reason, or if the Company otherwise requests at any other time before such termination, the Executive shall: (i) return to the Company all tangible Confidential Information and copies thereof (regardless how such Confidential Information or copies are maintained); and (ii) deliver to the Company any property of the Company which may be in the Executive's possession, including, but not limited to, products, materials, memoranda, notes, records, reports, or other documents or photocopies of the same. The Executive agrees that, when the Executive leaves the employ of the Company, or at any time the Company may request, the Executive shall deliver to the Company any and all Company equipment provided to the Executive including laptops and other electronic devices, as well as drawings, notes, memoranda, specifications, devices, formulas, and any other documents pertaining to the Company and/or the Company's business, including, but not limited to, computer files, together with all copies thereof, and any other material containing or disclosing any Confidential Information as defined in Section 7(a) above (collectively, "**such Documents**"). The above shall include any and all such Documents contained on, for example, a home computer system, tablet or smart phone. The Executive further agrees not to retrieve or retain in any way any such Documents, and the Executive shall, for example, first return such Documents to the Chief Legal Officer and General Counsel and then consult with the Company's Chief Legal Officer and General Counsel regarding the removal and deletion of such Documents from any home computer system, personal electronic device, or other personal property and act in accordance with their instructions. The Executive expressly authorizes the Company's designated representatives to access such equipment or devices for this limited purpose and shall provide any passwords or access codes necessary to accomplish this task. The Executive further agrees that any property situated on the Company's premises and/or owned by the Company, including disks and other storage media, filing cabinets

or other work areas, is subject to inspection by Company personnel at any time with or without notice.

(e) **Notice Pursuant to Defend Trade Secrets Act.** Notwithstanding any provision of this Agreement prohibiting the disclosure of Inventions or other Confidential Information, the Executive understands that they may not be held criminally or civilly liable under any federal or state trade secret law for the disclosure of a Company trade secret that: (i) is made (A) in confidence to a federal, state or local government official, either directly or indirectly, or to an attorney; and (B) solely for the purpose of reporting or investigating a suspected violation of law; or (ii) is made in a complaint or other document filed in a lawsuit or other proceeding, if such filing is made under seal. In addition, if the Executive files a lawsuit or other court proceeding against the Company for retaliating against them for reporting a suspected violation of law, the Executive may disclose the Company trade secret to the attorney representing them and use the Company trade secret in the court proceeding, if the Executive files any document containing the Company trade secret under seal and do not disclose the trade secret, except pursuant to court order.

8. **Non-Competition and Non-Solicitation.**

(a) **Interests to be Protected.** The Company and the Executive acknowledge that Executive shall perform essential services for the Company, its employees, and/or its stockholders during the Employment Period. The Executive shall be exposed to, have access to, and work with, a considerable amount of Confidential Information. The Company and the Executive also expressly recognize and acknowledge that the personnel of the Company have been trained by, and are valuable to, the Company and that the Company shall incur substantial recruiting and training expenses if the Company must hire new personnel or retrain existing personnel to fill vacancies. The Company and the Executive expressly recognize that it could seriously impair internal and external goodwill and diminish the value of Company's business should the Executive unfairly compete with the Company in any manner whatsoever. The Company and the Executive acknowledge this covenant is reasonable and it is necessary for the protection of the Company, its stockholders, and employees. For these and other reasons, and the fact that there are many other employment opportunities available to the Executive if the Executive's employment is terminated, the Company and the Executive are in full and complete agreement that the covenants in this Section 8 are fair and reasonable and are entered into freely, voluntarily, and knowingly.

(b) **Non-Competition and Non-Solicitation Definitions.**

(i) **Non-Competition Period.** "Non-Competition Period" shall mean the period commencing on the Effective Date and continuing until one (1) year from the date on which the Executive's employment with the Company terminates for Cause, or the Executive resigns for any reason or no reason, including Good Reason, or because of a non-renewal of the Employment Period by the Executive pursuant to Section 2(a). If the Executive breaches a fiduciary duty to the Company or takes any property belonging to or in the custody of the Company, physically or electronically, the Non-Competition Period shall be extended to two (2) years from the date on which the Executive's employment with the Company terminates. In the case of such an extension, the Company shall not increase the Special Consideration (as defined below).

(ii) Non-Solicitation Period. “**Non-Solicitation Period**” shall mean the period commencing on the Effective Date and continuing until one (1) year from the date on which the Executive’s employment with the Company terminates for any reason or no reason.

(iii) Geographic Reach. “**Geographic Reach**” shall mean the geographic area in which the Executive provided services or had a material presence or influence within the last two (2) years of the Executive’s employment by the Company.

(iv) Competitive Business. “**Competitive Business**” shall mean the research, development, manufacturing, commercialization, marketing, sales, distribution of products, services and technologies involving aerogels.

(v) Competitive Services. “**Competitive Services**” shall mean those activities or services which the Executive provided regardless of job title, at any time within the last two (2) years of the Executive’s employment by the Company.

(vi) Customer or Potential Customer. “**Customer or Potential Customer**” shall mean any person or entity who or which, at any time during the one year period prior to the Executive’s contact with such person or entity, if such contact occurs during the Executive’s employment or, if such contact occurs following the termination of the Executive’s employment, during the one year period prior to the date the Executive’s employment with the Company ends: (A) contracted for, was billed for, or received from the Company any product, service, or process with which the Executive worked directly or indirectly during the Executive’s employment by the Company or about which the Executive acquired Confidential Information; or (B) was in contact with the Executive or in contact with any other employee, owner, or agent of the Company, of which contact the Executive was or should have been aware, concerning the sale or purchase of, or contract for, any product, service, or process with which the Executive worked directly or indirectly during the Executive’s employment with the Company or about which the Executive acquired Confidential Information; or (C) was solicited by the Company in an effort in which the Executive was involved or of which the Executive was aware.

(c) Covenant Not-To-Compete. During the Non-Competition Period, within the Geographic Reach, and without the prior written consent of the Company (which consent the Company may grant or withhold in its sole discretion) the Executive shall not engage in any Competitive Services through or on behalf of the Executive, a third party or another person or entity engaged in a Competitive Business (the “**Covenant Not-to-Compete**”), unless the Company, at its sole discretion and option, chooses not to enforce the Covenant Not-to-Compete by informing the Executive in writing of its election at the time of or prior to the Date of Termination. For the avoidance of doubt, the Covenant Not-to-Compete shall not apply if the Executive has been terminated without Cause or in the event of non-renewal of the Employment Period by the Company pursuant to Section 2(a), provided that, in order for the Executive to be eligible for severance payments and benefits as described in Section 4 or 5 of this Agreement (as applicable), the Executive shall agree to a non-competition covenant as a term of the Separation Agreement and Release in a form substantially similar to the Covenant Not-to-Compete contained in this Section 8.

(d) **Fair and Reasonable Consideration.** The Executive acknowledges and agrees that the Covenant Not-to-Compete is supported by fair and reasonable consideration independent of, and in addition to, the Executive's continued employment by the Company and the Special Consideration (as defined below). Without limiting the foregoing, the Executive acknowledges and agrees that the Executive's receipt of a lump sum payment in the amount of \$500.00 (the "**Consideration Payment**"), constitutes fair and reasonable consideration to support the Executive's Covenant Not-to-Compete. The Consideration Payment shall be paid to the Executive on the Company's next regularly scheduled payroll date following the Effective Date, subject to customary withholding.

(e) **Special Consideration.** As consideration for the Covenant Not-To-Compete, if the Company wishes to enforce this Covenant Not-To-Compete, then Company shall pay the Executive, and the Executive agrees to accept, a payment of fifty percent (50%) of the Executive's highest annualized base salary over the two years prior to the Date of Termination in substantially equal installments in accordance with the Company's general payroll practices and subject to customary withholding during the Non-Competition Period (the "**Special Consideration**"), provided that in the event the Executive is eligible for severance payments or benefits as described in Section 4 or 5 herein (as applicable), such severance payments and benefits shall instead be considered the Special Consideration defined herein. The Special Consideration shall not include any other form of compensation, including but not limited to, commissions, bonuses, equity, reimbursement of expenses, travel discounts or other fringe benefits. In the event that the Company waives the Covenant Not-to-Compete pursuant to Section 8(c), the Executive shall not be entitled to the Special Consideration. Notwithstanding the foregoing, if the Company waives the Covenant Not-to-Compete, the Executive may be eligible to receive severance payments and benefits pursuant to the terms of Section 4 or 5 of this Agreement, as applicable, provided that the Separation Agreement and Release shall not contain a non-competition covenant.

(f) **Covenant Not-to-Solicit.** During the Non-Solicitation Period, the Executive agrees that they shall not, directly or indirectly, for the Executive's own account or on behalf of any other person or entity, (i) solicit, call upon or accept business from, any Customer or Potential Customer; or (ii) interfere with the business relationship between any Customer or Potential Customer and the Company; or (iii) solicit, induce, persuade or hire, or attempt to solicit, induce, persuade or hire, or assist any third party in the solicitation, inducement, persuasion or hiring of, any employee of the Company who worked for the Company during the Executive's tenure with the Company, to leave the employ of the Company (the "**Covenant Not-to-Solicit**"). Executive acknowledges and agrees that the Non-Solicitation Period shall be tolled and shall not run, during any period in which the Executive is in violation of the terms herein.

(g) **Acknowledgement; Opportunity to Review.** The Executive acknowledges and agrees that (i) the Executive has read and understands this Section 8 and agrees to it knowingly and voluntarily; (ii) the Executive has been given a reasonable opportunity to consider the terms of this Section 8, and to consult with an attorney of the Executive's choosing, prior to executing this Agreement; and (iii) with respect to the Covenant Not-to-Compete only, the Executive has been provided with notice of the Covenant Not-to-Compete for the Executive's consideration ten (10) business days before the Covenant Not-to-Compete becomes effective.

9. **Inventions.**

10. **Non-Disparagement.** Executive agrees that during the Executive's employment by the Company and at any time thereafter, the Executive shall not take any action or make any statement, verbally or in writing, or via social media which has the purpose or effect of disparaging the Company, or its directors, officers, employees or its products, provided that nothing in this Section shall restrict the Executive from making any disclosures mandated by state or federal law, from providing information or documents to a state or federal agency if requested by the agency to do so, or from participating in an investigation with a state or federal agency if requested by the agency to do so.

11. **Provisions Necessary and Reasonable; Change and Reaffirmation.**

(a) **Reasonableness of Restrictions.** The Executive acknowledges and agrees that the provisions of Sections 7, 8 and 9 of this Agreement are necessary and reasonable to protect the Company's Confidential Information, property rights, trade secrets, goodwill and business interests. The Executive further acknowledges and agrees that the types of employment which are prohibited by Section 8 are narrow and reasonable in relation to the skills which represent the Executive's principal salable asset both to the Company and to the Executive's other prospective employers, and that the specific but broad temporal and geographical scope of Section 8 is reasonable and fair in light of the Company's need to market its services and develop and sell its products in a large geographic area in order to maintain a sufficient customer base, and in light of the Executive's material presence or influence in the Geographic Reach during the last two years of the Executive's employment with the Company.

(b) **Change and Reaffirmation.** The Executive understands, acknowledges and agrees that: (i) the Executive's obligations under Sections 7, 8, 9 and 10 shall continue in accordance with their express terms regardless of any material changes in the Executive's title, position, duties, salary, compensation or benefits or other terms and conditions of employment, including but not limited to any renewal of the Employment Period, and the Executive agrees to comply with such obligations; and (ii) if the Executive should transfer between or among any affiliates of the Company, wherever situated, or be promoted or reassigned to functions other than the Executive's present functions, the terms of Section 7, 8, 9 and 10 shall continue to apply with full force, and the Executive agrees to comply with such terms. The Executive expressly consents to be bound by the provisions of Sections 7, 8, 9 and 10 for the benefit of the Company or any parent, subsidiary or affiliate to whose employ the Executive may be transferred, without the necessity that this Agreement be resigned at the time of such transfer.

12. **Representations and Warranties of the Executive.** Executive hereby represents and warrants to the Company that:

(a) the Company has advised the Executive that at no time should the Executive divulge to or use for the benefit of the Company any trade secret or confidential or proprietary information of any previous employer or other third party, and that the Executive has not divulged or used and shall not divulge or use any such information for the benefit of the Company. The Executive further agrees that if they have signed a confidentiality, non-competition or similar type of agreement with

any former employer or other entity, the Executive shall comply with the terms of such agreement to the extent that its terms are lawful under applicable law, or otherwise disclose it to the Company.

(b) after undertaking a careful search (including searches of the Executive's computers, cell phones, electronic devices, and documents), the Executive has returned all property and confidential information belonging to all prior employers (and/or third parties the Executive performed services for in accordance with the terms of those applicable agreements);

(c) the Executive shall indemnify and hold the Company harmless against loss, damage, liability or expense arising from any claim based upon circumstances alleged to be inconsistent with the representations and warranties in Sections 11(a)-(b) above;

(d) the Executive has not been convicted within the last five (5) years of any felony or misdemeanor in connection with the offer, purchase, or sale of any security or any felony involving fraud or deceit, including, but not limited to, forgery, embezzlement, obtaining money under false pretenses, larceny, or conspiracy to defraud;

(e) the Executive has not been and is not currently (i) the target of an internal complaint of discrimination or harassment at a prior employer that resulted in any adverse remedial action against the Executive; or (ii) subject to any actual or threatened judicial, administrative or arbitral actions, suits or proceedings (public or private) by or before a court, government agency or arbiter, or before any arbitrator, mediator or other alternative dispute resolution provider, relating to discrimination or harassment in employment;

(f) the Executive is not currently subject to any state administrative enforcement order or judgment entered by a state securities administrator within the last five (5) years and is not subject to any state's administrative enforcement order or judgment in which fraud or deceit (including, but not limited to, making untrue statements of material facts and omitting to state material facts) was found in which the order or judgement was entered within the last five (5) years;

(g) the Executive is legally authorized to work in the United States of America; and

(h) This Agreement constitutes the legal, valid and binding obligations of the Executive, enforceable in accordance with its terms, and execution, delivery and performance of this Agreement by the Executive does not and shall not conflict with, violate or cause a breach of any agreement, contract or instrument to which the Executive is a party or any judgment, order or decree to which the Executive is subject.

13. Representations and Warranties of the Company. The Company hereby represents and warrants to the Executive that:

(a) The Company is a corporation duly organized, validly existing and in good standing under the laws of the State of Delaware. The Company has all requisite corporate power and authority to carry out the transactions contemplated by this Agreement.

(b) The execution, delivery and performance of this Agreement has been duly authorized by the Company. This Agreement constitutes a valid and binding obligation of the Company, enforceable in accordance with its terms. The execution and delivery by the Company of this Agreement, and

the fulfillment of and compliance with the respective terms hereof by the Company, do not and shall not (i) conflict with or result in a breach of the terms, (ii) constitute a default under, (iii) result in the creation of any lien, security interest, charge or encumbrance upon the Company's capital stock or assets pursuant to, (iv) give any third party the right to modify, terminate or accelerate any obligation under, (v) result in a violation of, or (vi) require any authorization, consent, approval, exemption or other action by or notice to any court or administrative or governmental body pursuant to, the charter or bylaws of the Company, or any law, statute, rule or regulation to which the Company is subject, or any agreement, instrument, order judgement or decree to which the Company is subject.

14. **Notices.** All notices, demands or other communications to be given or delivered under or by reason of the provisions of this Agreement shall be in writing and shall be deemed to have been given when delivered personally, mailed by certified or registered mail, return receipt requested and postage prepaid, or sent via a nationally recognized overnight courier, or sent via facsimile to the recipient with a confirmation of receipt and accompanied by a certified or registered mailing. Such notices, demands and other communications shall be sent to the address indicated below:

To the Company:

Aspen Aerogels, Inc.
26 Forest Street, Suite 200
Marlborough, MA 01752
Telephone: (508) 691-1111
Facsimile: (508) 691-1200
Attention: Chief Legal Officer & General Counsel

To the Executive:

The address on file in the Company's records.

15. **Miscellaneous.**

(a) **Severability; Blue Pencil.** Whenever possible, each provision of this Agreement shall be interpreted in such manner as to be effective and valid under applicable law, but if any provision of this Agreement is held to be invalid, illegal or unenforceable in any respect under any applicable law or rule in any jurisdiction, such invalidity, illegality or unenforceability shall not affect any other provision or any other jurisdiction, but this Agreement shall be reformed, construed and enforced in such jurisdiction as if such invalid, illegal or unenforceable provision had never been contained herein. If the provisions contained in Sections 7, 8 or 9 of this Agreement are for any reason held to be excessively broad as to duration, geographical scope, activity or subject, it shall be construed by limiting and reducing it, so as to be enforceable to the extent compatible with the applicable law at the time. The deletion of specific words and phrases ("**blue penciling**") and giving effect to the reduced or blue-penciled form is allowed where permitted by applicable law.

(b) **Complete Agreement.** This Agreement and the agreements referred to herein embody the complete agreement and understanding among the parties and supersede and preempt any prior

understandings, agreements or representations by or among the parties, written or oral, which may have related to the subject matter hereof.

(c) **Waiver of Jury trial.** The parties to this Agreement each hereby waives, to the fullest extent permitted by law, any right to trial by jury of any claim, demand, action, or cause of action (i) arising under this Agreement or (ii) in any way connected with or related or incidental to the dealings of the parties hereto in respect of this Agreement or any of the transactions related hereto, in each case whether now existing or hereafter arising, and whether in contract, tort, equity, or otherwise. The parties to this Agreement each hereby agrees and consents that any such claim, demand, action, or cause of action shall be decided by court trial without a jury and that the parties to this Agreement may file an original counterpart or a copy of this Agreement with any court as written evidence of the consent of the parties hereto to the waiver of their right to trial by jury.

(d) **Counterparts; Facsimile Transmission.** This Agreement may be executed simultaneously in two or more counterparts, any one of which need not contain the signatures of more than one party, but all such counterparts taken together shall constitute one and the same Agreement. This Agreement may also be executed and delivered by facsimile or other electronic transmission.

(e) **Successors and Assigns.** The provisions hereof shall inure to the benefit of, and be binding upon and assignable to, successors of the Company by way of merger, consolidation or sale. The Executive may not assign or delegate to any third person the Executive's obligations under this Agreement. The rights and benefits of the Executive under this Agreement are personal to him and no such right or benefit shall be subject to voluntary or involuntary alienation, assignment or transfer. The Company shall require any successor (whether direct or indirect, by purchase, merger, consolidation or otherwise) to all or substantially all of the business and/or assets of the Company expressly to assume and agree to perform this Agreement in the same manner and to the same extent that the Company would have been required to perform it if no such succession had taken place. As used in this Agreement, "the Company" shall mean both the Company as defined above and any such successor that assumes and agrees to perform this Agreement, by operation of law or otherwise.

(f) **Governing Law.** All issues concerning this Agreement shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts, without giving effect to any choice of law or conflict of law provision or rule (whether of the Commonwealth of Massachusetts or any other jurisdiction) that would cause the application of the law of any jurisdiction other than the Commonwealth of Massachusetts.

(g) **Remedies; Injunctive Relief.** The parties hereto agree and acknowledge that money damages may not be an adequate remedy for any breach of the provisions of the Agreement (including but not limited to Sections 7, 8 or 9), and that any party may in its sole discretion apply to any court of law or equity of competent jurisdiction (without posting any bond or deposit) for specific performance and/or injunctive or other relief in order to enforce or prevent any violations of the provisions of this Agreement.

(h) **Amendment and Waiver.** The provisions of this Agreement may be amended and waived only with the prior written consent of the Company and the Executive.

(i) **Certain Expenses.** Subject to any restrictions that may be imposed under the Company's Compensation Recoupment Policy, the Company agrees to pay, as incurred, to the fullest extent permitted by law, or indemnify the Executive if such payment is not legally permitted, for all legal fees and expenses that the Executive may in good faith incur as a result of any contest by the Company, the Executive or others of the validity or enforceability of or liability under, or otherwise involving, any provision of this Agreement; provided, however, that the Executive shall reimburse the Company for all such payments made by the Company in connection with a contest by the Company if a court of competent jurisdiction or an arbitrator shall find that the Executive did not act in good faith in connection with such contest.

(j) **Withholding.** All payments made by the Company to the Executive under this Agreement shall be net of any tax or other amounts required to be withheld by the Company under applicable law.

(k) **Litigation and Regulatory Cooperation.** During and for a reasonable period of time after the Executive's employment, the Executive shall cooperate fully with the Company in the defense or prosecution of any claims or actions now in existence or which may be brought in the future against or on behalf of the Company which relate to events or occurrences that transpired while the Executive was employed by the Company. The Executive's full cooperation in connection with such claims or actions shall include, but not be limited to, being available to meet with counsel to prepare for discovery or trial and to act as a witness on behalf of the Company at mutually convenient times. During and after the Executive's employment, the Executive also shall reasonably cooperate with the Company in connection with any investigation or review of any federal, state or local regulatory authority as any such investigation or review relates to events or occurrences that transpired while the Executive was employed by the Company. The Company shall reimburse the Executive for any reasonable out-of-pocket expenses incurred in connection with the Executive's performance of obligations pursuant to this Section 11(k), to include reasonable attorneys' fees and costs incurred by the Executive.

16. **Survival.** The provisions of Sections 1, 2, 4, 5, 6, 7, 8, 9, 10, 11 and 12 of this Agreement shall survive any termination of this Agreement in accordance with the terms of such sections.

[REMAINDER OF THIS PAGE INTENTIONALLY LEFT BLANK]

IN WITNESS WHEREOF, the parties hereto have executed this Agreement as of the date first above written.

THE COMPANY:

ASPEN AEROGELS, INC.

By: /s/ Donald R. Young
Name: Donald R. Young
Title: President and Chief Executive Officer

THE EXECUTIVE:

By: /s/ Stephanie Pittman
Name: Stephanie Pittman

EXHIBIT A

PRIOR INVENTIONS

TO: Aspen Aerogels, Inc. (the Company) FROM:
Stephanie Pittman
DATE: _

The following is a complete list and description of all Inventions (as defined in Section 9 of my Employment Agreement (the "Agreement")) that (i) I have, alone or jointly with others, conceived, developed, or reduced to practice or caused to be conceived, developed, or reduced to practice prior to the commencement of my employment with the Company, (ii) I consider to be my property or the property of third parties, and (iii) I wish to have excluded from the scope of this Agreement:

Check appropriate lines below:

_____ No Inventions to disclose
_____ See description of Inventions below:

See additional sheets attached with description of Inventions.

Due to a confidentiality agreement, I cannot complete the disclosure above for the inventions listed below, identified only by a cursory name, the party(ies) to whom it belongs, and my relationship to such party(ies):

Cursory Name of Invention	Party(ies)	Relationship
1. _____	_____	_____
2. _____	_____	_____
3. _____	_____	_____

_____ See additional sheets attached.

If this Exhibit is left blank, I represent that there are no Prior Inventions.

Executive Signature

SEPARATION AND RELEASE AGREEMENT

This Separation and Release Agreement ("Agreement") is made and entered into by Aspen Aerogels, Inc. ("Aspen" or the "Company") and Kelley Conte, on behalf of herself, her heirs, executors, administrators, successors and assigns (collectively referred to throughout this Agreement as "Employee," and, together with Aspen, the "Parties"). The Parties agree that:

1. **Separation of Employment.**

(a) Employee's employment with Aspen shall terminate on September 5, 2023 ("Effective Termination Date").

(b) Employee acknowledges that from and after the Effective Termination Date, Employee shall be deemed to have resigned from all boards of, and any officer or other positions with, the Company Group (as that term is defined below). Notwithstanding the foregoing, such resignation(s) shall not be construed to eliminate or reduce Employee's entitlements under this Agreement. At the request of the Company, Employee shall execute such other documents as may be necessary to evidence the resignations covered by this Paragraph 1(b). For purposes of this Agreement, "Company Group" means, individually and collectively, Aspen Aerogels, Inc., a company organized under the laws of Delaware, and its direct and indirect subsidiaries and affiliates.

(c) Regardless of whether Employee signs this Agreement, Employee will be entitled to the following "Accrued Obligations":

(i) Employee's final earned but unpaid wages accrued through the Effective Termination Date, less applicable withholdings, to be paid within thirty (30) days after the Employee's Effective Termination Date;

(ii) the monetary equivalent of Employee's accrued but unused vacation time as of the Effective Termination Date, if any, to be paid within thirty (30) days after the Employee's Effective Termination Date;

(iii) any expenses that Employee has properly incurred but that has not been reimbursed as of the Effective Termination Date, provided Employee submits documentation within thirty (30) days after the Effective Termination Date and as otherwise consistent with the Company's expense reimbursement policy; and

(iv) in the event that Employee participates in the Company's health insurance program, the right to continue in such program under the provisions of the Consolidated Omnibus Budget Reconciliation Act of 1985 ("COBRA"). Employee will receive Employee's COBRA notice under separate cover. If Employee does not elect COBRA, Employee's health insurance coverage will cease on the last day of the month of the Effective Termination Date.

(v) Director's and officer's insurance coverage for a period six years after the Effective Termination Date, in accordance, and subject to the terms, conditions and limitations of Section 6(a) of that certain Amended and Restated Executive Employment Agreement dated as of January 1, 2023 (the "Executive Agreement").

2. **Separation Benefit.**

(a) In consideration for Employee (i) signing and returning this Agreement within twenty-one (21) days of receipt and not revoking this Agreement during the seven (7) day revocation period after it is signed and returned; (ii) complying with the terms of this Agreement; (iii) waiving all of Employee's claims (except as provided in this Agreement) and releasing Aspen and certain other parties, all as further described below in this Agreement; and (iv) waiving and releasing any rights or entitlements to severance or similar post-termination payments or benefits, other than those provided in this Agreement, Aspen will provide Employee with the following "Separation Benefit":

(i) Pro-Rated Discretionary Annual Bonus. Employee acknowledges that by virtue of Employee's Effective Termination Date, Employee would not have been eligible for a discretionary bonus under the Aspen Aerogels, Inc. Bonus Plan ("Bonus Plan") for the 2023 fiscal year. However, as additional consideration, the Company will continue to make Employee eligible for such discretionary bonus (the "Bonus"), determined based on actual achievement of the applicable performance goals for the 2023 fiscal year and *pro-rated based upon the Effective Termination Date*, and consistent with the other applicable terms and conditions of the Bonus Plan. The Company will pay to Employee the Bonus, if any, in the year following the year to which such Bonus relates when annual bonuses for that year are paid to other executives of the Company generally, but in no case later than March 15, 2024.

3. **Consulting Agreement.** Aspen and Employee have executed and delivered simultaneously with this Agreement the Consulting Agreement, a copy of which is attached as Exhibit A (the "Consulting Agreement"), pursuant to which Employee will provide certain services to Aspen beginning immediately following the Effective Termination Date, without a break in Employee's continuous service.

4. **Additional Employee Covenants.** As a condition of entering into this Agreement, Employee hereby expressly represents, acknowledges and agrees that:

(a) neither this Agreement nor the furnishing of the consideration for this Agreement shall be deemed or construed at any time for any purpose as an admission by Releasees (as that term is defined in Paragraph 7 below) of wrongdoing or evidence of any liability or unlawful conduct of any kind;

(b) Employee would not receive the Separation Benefit specified in Paragraph 2 above, except for Employee's execution of this Agreement and the fulfillment of the promises contained in this Agreement;

(c) as of the date Employee executes this Agreement (the "Execution Date"), Employee has not filed, caused to be filed, or presently is a party to any claim, complaint, charge

or lawsuit against Aspen and/or any other Releasees; and Employee has not been retaliated against for reporting any allegations of wrongdoing by Aspen or any of its officers, including any allegations of corporate fraud;

(d) the Separation Benefit provided to Employee in Paragraph 2(a) above exceed anything of value to which Employee would otherwise be entitled absent this Agreement, and apart from (i) the Accrued Obligations, and (ii) any entitlement to vested and unpaid benefits under any benefit plans, policies, programs or arrangements in which Employee participated as of the Effective Termination Date, Employee is not entitled to any further compensation, including, without limitation, any other wages, commissions, bonuses, severance, paid time off, holiday pay, or any other form of compensation or benefit;

(e) Employee has been granted any leave to which Employee was entitled under the Family and Medical Leave Act or related state or local leave or disability accommodation laws, and Employee has no known workplace injuries or occupational diseases;

(f) during the course of Employee's employment with the Company, Employee had access to Confidential Information (as that term is defined below) relating to the Company Group's business affairs that is not generally known by persons not employed by the Company and that could not easily be determined or learned by someone outside of the Company, and Employee agrees that Employee shall not disclose or use such Confidential Information at any time in the future, except as may be required by law, court order or subpoena, and Employee will abide by any and all common law and/or statutory obligations relating to protection and non-disclosure of the Company's Confidential Information. "Confidential Information" means information disclosed to Employee or to which Employee had access that includes, but is not limited to, trade secrets and confidential and proprietary information of the Company Group, or any information provided to the Employee or the Company under an obligation of confidentiality to a third party, or any confidential, trade secret, or proprietary information acquired by the Company from others with whom the Company or any affiliate has a business relationship, whether in written, oral, electronic or other form, including, but not limited to (i) trade secrets, inventions, mask works, ideas, processes, algorithms, formulae, software in source or object code, data programs, other works of authorship, know-how, improvements, technology direction, product or technology development methodology, technology assessment, experimental procedures, results, process development, product plans, development plans, testing procedures, quality control and testing processes, discoveries, developments, designs and techniques, any other proprietary technology and all Inventions (as defined in Section 3); (ii) information regarding research, development, new products, marketing and selling, business plans, budgets and unpublished financial statements, licenses, prices and costs, margins, discounts, credit terms, pricing and billing policies, quoting procedures, methods of obtaining business, forecasts, future plans and potential strategies, financial projections and business strategies, operational plans, financing and capital-raising plans, activities and agreements, internal services and operational manuals, methods of conducting Company business, suppliers and supplier information, sales techniques and strategies, training methods and materials, and purchasing; (iii) information regarding customers and potential customers of the Company, including customer lists, names, representatives, their needs or desires with respect to the types of products or services offered by the Company, proposals, bids, contracts and their contents and parties, the type and quantity of products and services provided or sought to be provided to customers and potential customers of

Company and other non-public information relating to customers and potential customers; (iv)

information regarding any of the Company's business partners and its services, including names, representatives, proposals, bids, contracts and their contents and parties, the type and quantity of products and services received by the Company, and other non-public information relating to business partners; (v) information regarding personnel, employee lists, compensation, and employee skills; (vi) any other non-public information which a competitor of the Company could use to the competitive disadvantage of the Company; and (vii) any other scientific, technical or trade secrets of the Company or of any third party provided to the Employee or the Company under a condition of confidentiality, provided that Confidential Information shall not include information that was (a) known to the Employee prior to joining the Company, (b) general industry knowledge, (c) required to be disclosed by Employee in connection with a judicial, special or arbitral proceeding or pursuant to court order or subpoena, (d) is in the public domain other than through any fault or act by the Employee, or (e) is approved for release with the Company's written authorization;

(g) the term "trade secrets" as used in this Agreement shall be given its broadest possible interpretation under applicable law and shall mean all forms and types of financial, business, scientific, technical, economic, or engineering information, including patterns, plans, compilations, program devices, formulas, designs, prototypes, methods, techniques, processes, procedures, programs, or codes, whether tangible or intangible, and whether or how stored, compiled, or memorialized physically, electronically, graphically, photographically, or in writing that (i) the Company has taken reasonable measures to keep secret, and that (ii) derives independent economic value, actual or potential, from not being generally known to, and not being readily ascertainable through proper means by, another person who can obtain economic value from the disclosure or use of the information;

(h) by executing this Agreement, Employee has been notified by this writing, in accordance with the Defend Trade Secrets Act of 2016, that (i) an individual shall not be held criminally or civilly liable under any federal or state trade secret law for the disclosure of a trade secret that is made in confidence to a federal, state, or local government official, or to an attorney, solely for the purpose of reporting or investigating a suspected violation of law, (ii) an individual shall not be held criminally or civilly liable under any federal or state trade secret law for the disclosure of a trade secret that is made in a complaint or other document filed in a lawsuit or other proceeding, if such filing is made under seal; and (iii) an individual who files a lawsuit for retaliation by an employer for reporting a suspected violation of law may disclose the trade secret to the attorney of the individual and use the trade secret information in the court proceeding, if the individual files any document containing the trade secret under seal and does not disclose the trade secret except pursuant to court order;

(i) Employee (i) has, as of the Execution Date, not used or disclosed any Confidential Information of the Company Group except as required in connection with the performance of Employee's job duties and as otherwise authorized by the Company; (ii) has, as of the Execution Date, returned all of the Company Group's property, equipment, documents, and/or any Confidential Information in Employee's possession or control, including, but not limited to, products, materials, memoranda, notes, records, reports, drawings, specifications, devices, formulas, or other documents or copies thereof; and (iii) hereby consents and authorizes the Company to deduct as an offset from the above-referenced severance payments the value of any

Company Group property not returned or returned in a damaged condition as well as any monies paid by the Company on Employee's behalf;

(j) in consideration for the severance pay and other benefits offered under this Agreement, and except as otherwise prohibited by applicable law, Employee will not disparage or criticize the Company Group, or cause or assist any third party to disparage or criticize the Company Group, or issue any communication, written or otherwise, or cause or assist any third party to issue any communication, written or otherwise, that reflects adversely on or that encourages any adverse action against the Company Group, including but not limited to any statement to or response to an inquiry by any member of the press or media, whether written, verbal, electronic, or otherwise;

(k) Employee will not disclose any information regarding the underlying facts leading up to or the existence or substance of this Agreement, except to Employee's spouse/domestic partner, tax advisor, and/or an attorney with whom Employee chooses to consult regarding Employee's consideration of this Agreement, such facts as are publicly disclosed by the Company in a filing with the Securities and Exchange Commission;

(l) Employee acknowledges that all original works of authorship that were made by Employee (solely or jointly with others) within the scope of the Employee's employment or which relate to the business of the Company or a Company affiliate and which are protectable by copyright are "works made for hire" pursuant to the United States Copyright Act (17 U.S.C. § 101). Employee hereby assigns to the Company or its designee all of the Employee's right, title and interest in and to all of the foregoing. Following the Effective Termination Date, Employee agrees to promptly inform and disclose all Inventions (as defined below) to the Company in writing, to assign all Inventions to the Company, to provide all assistance reasonably requested by the Company to preserve its interests in the Inventions (such as by executing assignments and other documents, testifying, etc.), such assistance to be provided at the Company's expense but without additional compensation to Employee. All ideas, discoveries, creations, manuscripts and properties, innovations, improvements, know-how, inventions, designs, developments, apparatus, techniques, methods, laboratory notebooks, formulae, data, protocols, writings, specifications, sound recordings, and pictorial and graphical representations; the exclusive legal right to reproduce, perform, display, distribute and make derivative works of a work of authorship ("Copyright"); and all paternity, integrity, disclosure, modification, withdrawal, special and any other similar rights recognized by the laws of any jurisdiction or country as "moral rights," "artist's rights," "droit moral," or the like (collectively, "Moral Rights") shall be collectively referred to herein as "Inventions";

(m) At any time after Employee's employment with the Company, Employee shall fully cooperate with the Company and its attorneys and agents in securing and protecting the Company's rights to Inventions, including but not limited to the preparation and filing of all papers and other documents as may be required to perfect the Company's rights in and to any of such Inventions, and joining in any proceeding to obtain letters patent, copyrights, trademarks or other legal rights with respect to any such Inventions in the United States and in any and all other countries, provided that the Company shall bear the expense of such proceedings, and that any patent or other legal right so issued to Employee personally shall be assigned by Employee to the Company or its designee without charge by Employee. If the Company is unable, after reasonable

effort, to secure Employee's signature on any such papers and/or other documents, Employee hereby irrevocably designates and appoints each officer of the Company as Employee's agent and attorney-in-fact to execute any such papers on Employee's behalf, and to take any and all actions as the Company may deem necessary or desirable in order to protect its rights and interests in any Invention;

(n) Employee will voluntarily cooperate with the Company in connection with any pending or future litigation, administrative proceeding, intellectual property proceeding, or other proceeding or other matter which may be filed against or by the Company with any agency, court, or other tribunal and concerning or relating to any matter falling within Employee's knowledge or former area of responsibility. Employee will provide reasonable assistance and completely truthful testimony in such matters including, without limitation, facilitating and assisting in the preparation of any underlying defense, responding to discovery requests, preparing for and attending deposition(s) as well as appearing in court to provide truthful testimony. Such cooperation shall include appearing from time to time at the offices of the Company or its affiliates or their counsel for conferences and interviews and in general providing the officers of the Company and its affiliates and their counsel with the full benefit of Employee's knowledge with respect to any such matter. Employee shall render such cooperation in a timely fashion and at such times as the Company may determine is reasonably necessary with respect to the applicable matter. The Company agrees to reimburse Employee for all reasonable out of pocket expenses incurred at the request of the Company associated with such assistance and testimony; and

(o) material to the inducement for the Company to enter this Agreement are the covenants set forth in this Paragraph 4, and that if Employee breaches or threatens to breach any of covenants set forth in this Paragraph 4, it will cause the Company substantial and irreparable injury to the Company, and therefore, in the event of a breach or threatened breach by Employee of any of these covenants, the Company shall be entitled to seek and obtain equitable relief, in the form of specific performance, and/or temporary, preliminary or permanent injunctive relief, or any other equitable remedy which then may be available restraining Employee from such breach or threatened breach; provided, however, that the right to apply for such relief shall not be construed as prohibiting the Company from pursuing any other available remedies for such breach or threatened breach, and further, without limiting the foregoing, the Company shall be entitled to recover the net amounts of any payments to Employee under Paragraph 2 of this Agreement.

(p) Employee acknowledges and agrees that she shall remain subject to the terms and conditions of the Company's Compensation Recoupment Policy following the termination of her employment and during and following the Term of the Consulting Agreement to the extent provided in the Compensation Recoupment Policy.

5. Waiver of Covenant Not-To-Compete and Continuation of Other Existing Restrictive Covenants. Sections 7 through 11 of the Executive Agreement contain certain restrictive covenants, generally applicable during and after the term of Employee's employment with Aspen, including covenants regarding non-disclosure of confidential information, non-competition (the "Covenant Not-To-Compete", as defined in Section 8(c) of the Executive Agreement), non-solicitation and non-disparagement (collectively, the "Restrictive Covenants").

Notwithstanding the foregoing, and in its discretion as provided pursuant to Section 8(c) of the Executive Agreement, the Company has chosen not to enforce the Covenant Not-To-Compete and

hereby waives any right to enforce the Covenant Not-To-Compete. Accordingly, Employee acknowledges and agrees that Employee shall have no right to receive any “Special Consideration” as defined in Section 8(e) of the Executive Agreement. Further, Employee acknowledges and agrees that Employee is bound by and will comply with the terms of the Restrictive Covenants as set out in the Executive Agreement, other than the Covenant Not-To-Compete, which Restrictive Covenants by their terms survive the termination of Employee’s employment with Aspen.

6. Limitation; Protected Rights. Notwithstanding the foregoing, nothing in this Agreement prohibits or otherwise restricts Employee from (1) initiating, testifying, assisting, complying with a subpoena from, or participating in any manner with an investigation conducted by a Government Agency, including without limitation, with respect to any unfair labor practice charge; (2) filing or disclosing any facts necessary to receive unemployment insurance, Medicaid, or other public benefits to which Employee may be entitled; (3) discussing or disclosing information about unlawful acts in or related to the workplace, including, but not limited to discrimination, harassment, sexual assault, retaliation, wage and hour violations, conduct that is against a clear mandate of public policy, or any other conduct Employee has reason to believe is unlawful; (4) engaging in protected activities under Section 7 of the National Labor Relations Act (“NLRA”), including filing unfair labor practice charges, assisting Company employees in filing unfair labor practice charges, discussing the improvement of terms and conditions of employment (including regarding the terms of this Agreement) with former and current Company employees or union representatives or other third parties for the purpose of engaging in concerted activity under Section 7 of the NLRA; or (5) making any necessary disclosures as otherwise required by law.

7. Release of Claims.

(a) **General Release of Claims.** Employee knowingly and voluntarily releases and forever discharges Aspen, its parent corporation, affiliates, subsidiaries, divisions, predecessors, insurers, successors and assigns, and its current and former employees, attorneys, officers, directors and agents thereof, both individually and in their business capacities, and its employee benefit plans and programs and their administrators and fiduciaries (each a “Releasee” and collectively the “Releasees”), of and from any and all claims, known and unknown, asserted or unasserted, which the Employee has or may have against Releasees as of the Execution Date, including, but not limited to, any claims related to the Employee’s employment or termination of employment. This release includes, but is not limited to, the following: Title VII of the Civil Rights Act of 1964; Sections 1981 through 1988 of Title 42 of the United States Code; The Americans with Disabilities Act of 1990; The Americans with Disabilities Act Amendments Act of 2008; Genetic Information Nondiscrimination Act of 2008; The Age Discrimination in Employment Act of 1967; The Older Workers Benefit Protection Act of 1990; The Rehabilitation Act of 1973; The Equal Pay Act of 1963; The Worker Adjustment and Retraining Notification Act of 1988; The Family and Medical Leave Act of 1993; The Employee Retirement Income Security Act of 1974 (except for any vested benefits under any tax qualified benefit plan); The Immigration Reform and Control Act of 1986; The Fair Credit Reporting Act of 1970; The Massachusetts Fair Employment Practices Statute; The Massachusetts Equal Rights Act; The Massachusetts Civil Rights Act; The Massachusetts Privacy Statute; The Massachusetts Sexual Harassment Statute; The Massachusetts Wage Act; The Massachusetts Minimum Fair Wages Act; The Massachusetts Equal Pay Act; The Massachusetts Paid Family and Medical Leave Law; any and all state or local laws addressing

matters similar to those addressed by the foregoing federal and state laws; any other federal, state or local law, rule, regulation, or ordinance; any public policy, contract claim, tort, or common law; and any basis for recovering costs, fees, or other expenses including attorneys' fees incurred in these matters. This release covers both claims Employee knows about and those Employee may not know about, and is binding upon Employee and Employee's successors, assigns and heirs. Employee also represents that Employee has not pledged, given or sold any portion of any claim discussed in this Agreement to anyone else. If any claim is not subject to release, to the extent permitted by law, Employee waives any right or ability to be a class or collective action representative or to otherwise participate in any putative or certified class, collective or multi-party action or proceeding based on such a claim in which Aspen or any other Releasee identified in this Agreement is a party.

(b) Covenant Not to Sue. Except as otherwise provided below, Employee further agrees and covenants not to sue any Releasee with respect to any released claims, demands, liabilities or obligations released by this Agreement. If Employee breaks Employee's promise in this Paragraph 7 and files a complaint or lawsuit based on legal claims Employee has released, Employee will pay for all reasonable costs incurred by the Company, any related or affiliated entities or the employees, agents, representatives, shareholders or directors of any of them, including reasonable attorneys' fees, in defending against Employee's claim. Notwithstanding the foregoing, this Paragraph 7 does not:

(i) release the Company from (1) any obligations expressly set forth in this Agreement or (2) any other claim that Employee cannot waive as a matter of law;

(ii) prohibit or restrict Employee from filing or limit Employee's ability to file a charge or complaint with the Equal Employment Opportunity Commission or a state or local equivalent, the National Labor Relations Board, the Occupational Safety and Health Administration, the U.S. Department of Justice, the U.S. Securities and Exchange Commission or any other U.S. federal, state or local governmental agency or commission that has applicable jurisdiction to regulate the Company (each a "Government Agency"); or

(iii) prohibit or restrict Employee from communicating with, providing documents or other relevant information to, or otherwise cooperating with any Government Agency, the U.S. Congress, and any agency inspector general, law enforcement or any attorney Employee retains, including, but not limited to, responding to any inquiry, including an inquiry about the existence of this Agreement, its release or its underlying facts, or making other disclosures that are protected under the whistleblower provisions of federal law or regulation. The Company acknowledges and agrees that Employee does not need the prior authorization of the Company to make any such reports or disclosures and Employee is not required to notify the Company that Employee may make or has made such reports or disclosures.

(c) Waiver of Right to Recover in Administrative Action. Notwithstanding the foregoing Paragraph 7(b), Employee acknowledges and agrees that with Employee's release of claims in this Agreement, Employee has waived any right Employee may have to recover monetary or other personal relief in any proceeding based in whole or in part on claims released by Employee in this Agreement. For example, Employee waives any right to monetary damages or reinstatement if an administrative charge is brought against the Company whether by Employee, a Government Agency, or any other person or entity, including but not limited to any federal, state, or local

agency. Further, with Employee's release of claims in this Agreement, Employee specifically assigns to the Company Employee's right to any recovery arising from any such proceeding, provided however, that nothing in this Paragraph 7(c) limits Employee's right to receive an award for information provided to any Government Agency.

(d) Waiver is Knowing and Voluntary. Because Employee is over the age of 40 and consistent with the provisions of the Older Workers Benefits Protection Act, Employee understands and agrees that: (i) this is the full and final release of all claims against the Releasees through the date Employee signs this Agreement; (ii) Employee knowingly and voluntarily releases claims hereunder for valuable consideration; (iii) Employee hereby is and has been advised of Employee's right to have an attorney review this Agreement (at Employee's cost) before signing it; (iv) Employee has twenty-one (21) days to consider whether to sign this Agreement; (v) Employee may, at Employee's sole option, revoke Employee's assent to this Agreement within seven (7) days after signing it, upon return of the written notice included as Exhibit B to this Agreement by delivering it, via first class mail and e-mail, to:

Virginia H. Johnson
Chief Legal Officer, General Counsel & Corporate Secretary Aspen Aerogels, Inc.
26 Forest Street, Suite 200
Marlborough, MA 01752
E-mail: gjohnson@aerogel.com

(vi) this Agreement will not become effective until this 7-day period has expired and will be void if Employee revokes it within such period; and (vii) the eighth (8th) day following Employee's signing of this Agreement is the "Effective Date" of this Agreement.

(e) Employee acknowledges that any obligation of the Company to provide Employee with the Separation Benefit under this Agreement and any other consideration set forth in this Agreement is expressly conditioned on Employee's execution of this Agreement within twenty-one (21) days of the date the Company provides Employee with this Agreement without revocation.

8. Section 409A. It is intended that payments and benefits made or provided to Employee under this Agreement shall comply with Section 409A of the Internal Revenue Code of 1986 (as amended) (the "Code") or an exemption to Section 409A of the Code. Employee acknowledges and agrees, however, that the Company does not guarantee the tax treatment or tax consequences associated with any payment or benefit arising under this Agreement, including, without limitation, to consequences related to Section 409A of the Code. Any payments that qualify for the "short-term deferral" exception, the separation pay exception or another exception under Section 409A of the Code shall be paid under the applicable exception. For purposes of the limitations on nonqualified deferred compensation under Section 409A of the Code, each payment of compensation under this Agreement shall be treated as a separate payment of compensation for purposes of applying the exclusion under Section 409A of the Code for short-term deferral amounts, the separation pay exception or any other exception or exclusion under Section 409A of the Code. Notwithstanding anything in this Agreement to the contrary, if the period during which Employee may review and revoke this Agreement begins in one calendar year and ends in a second

calendar year, the Separation Benefit will begin to be paid in the second calendar year, with the first such payment including a catch-up payment (if any) to cover any amounts that would otherwise have been paid during the prior calendar year.

9. **Miscellaneous.**

(a) **Governing Law and Interpretation.** This Agreement shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts, without regard to conflicts of law principles.

(b) **Waiver.** The failure of the Company to seek enforcement of any provision of this Agreement in any instance or for any period of time shall not be construed as a waiver of such provision or of the Company's right to seek enforcement of such provision in the future.

(c) **Severability.** Should any provision of this Agreement be declared illegal or unenforceable by any court of competent jurisdiction and cannot be modified to be enforceable, such provision shall immediately become null and void, leaving the remainder of this Agreement in full force and effect, provided however, that if the Release in Paragraph 7(a) or any portion of such Release in Paragraph 7(a) is found unenforceable, this Agreement shall be null and void.

(d) **Amendment.** This Agreement may not be modified, altered or changed except in writing and signed by both Parties wherein specific reference is made to this Agreement.

(e) **Costs and Attorneys' Fees.** If Employee breaches any aspect of this Agreement, Aspen shall be entitled to recover from Employee its reasonable costs and attorneys' fees in pursuing the enforcement of this Agreement.

(f) **Execution of the Agreement.** This Agreement may be executed in any number of counterparts, each of which when so executed and delivered shall be deemed an original, and such counterparts together shall constitute one and the same instrument. No party to this Agreement shall be bound until such time as each party hereto has executed this Agreement. A facsimile or PDF copy of this Agreement (or DocuSign or similar electronic signature application), including the signature pages hereto, shall be deemed to be an original.

(g) **Integration.** This Agreement sets forth the entire agreement between the Parties and fully supersedes any prior agreements or understandings between the Parties in relation to the subject matter hereof, (i) except as otherwise set forth in this Agreement; and (ii) except it expressly does not supersede the Restrictive Covenants contained in the Executive Agreement other than the Covenant Not-To-Compete, which Restrictive Covenants shall remain in full force and effect in accordance with their terms. Employee acknowledges that Employee has not relied on any representations, promises, or agreements of any kind made to Employee in connection with Employee's decision to accept this Agreement, except for those set forth in this Agreement.

EMPLOYEE ACKNOWLEDGES THAT THE COMPANY PROVIDED EMPLOYEE WITH A COPY OF THIS AGREEMENT ON SEPTEMBER 5, 2023.

EMPLOYEE IS ADVISED THAT EMPLOYEE HAS UP TO TWENTY- ONE (21) CALENDAR DAYS TO CONSIDER THIS AGREEMENT, WHICH MEANS EMPLOYEE MAY EXECUTE THIS AGREEMENT ANY TIME UNTIL AND THROUGH SEPTEMBER 26, 2023 AND ABSENT SUCH EXECUTION, THIS AGREEMENT WILL BECOME NULL AND VOID.

EMPLOYEE ACKNOWLEDGES AND AGREES THAT ANY MODIFICATIONS, MATERIAL OR OTHERWISE, MADE TO PRIOR VERSIONS OF THIS AGREEMENT DID NOT RESTART OR AFFECT IN ANY MANNER THE ORIGINAL TWENTY-ONE (21) CALENDAR DAY CONSIDERATION PERIOD.

EMPLOYEE FREELY AND KNOWINGLY, AND AFTER DUE CONSIDERATION, ENTERS INTO THIS AGREEMENT INTENDING TO WAIVE, SETTLE AND RELEASE ALL CLAIMS EMPLOYEE HAS OR MIGHT HAVE AGAINST RELEASEES. NEITHER ASPEN NOR ITS AGENTS, REPRESENTATIVES OR ATTORNEYS, MADE ANY REPRESENTATIONS CONCERNING THE TERMS OF THIS AGREEMENT OTHER THAN THOSE CONTAINED HEREIN.

(Signature Page Follows)

The Parties knowingly and voluntarily sign this Agreement as of the date(s) set forth below:

EMPLOYEE
Kelley Conte

COMPANY
Aspen Aerogels, Inc.

By: /s/ Kelley Conte
Name: Kelley Conte
Date: August 21, 2023

By: /s/ Donald Young
Name: Donald Young
Date: August 20, 2023

Return Address for Signed Agreement

Virginia H. Johnson
Chief Legal Officer, General Counsel & Corporate Secretary Aspen Aerogels, Inc.
26 Forest Street, Suite 200
Marlborough, MA 01752
E-mail: gjohnson@aerogel.com

EXHIBIT A
CONSULTING AGREEMENT

EXHIBIT B

NOTICE TO REVOKE CONFIDENTIAL SEPARATION AND RELEASE AGREEMENT DATE:

Virginia H. Johnson
Chief Legal Officer, General Counsel & Corporate Secretary Aspen Aerogels, Inc.
26 Forest Street, Suite 200
Marlborough, MA 01752
E-mail: gjohnson@aerogel.com

Re: Seven Day Notice To Revoke

Dear Ms. Johnson:

I hereby exercise my right to revoke the CONFIDENTIAL SEPARATION AND RELEASE AGREEMENT executed and delivered to you. I understand this revocation must be signed and delivered to Virginia Johnson within seven (7) calendar days from the date the CONFIDENTIAL SEPARATION AND RELEASE AGREEMENT was signed by me.

Very truly yours,

By: _____
Name: _____
Date: _____

Consulting Agreement

This Agreement (hereinafter "Agreement") is made and entered into as of September 5, 2023 (the "Effective Date") by and between **Aspen Aerogels, Inc.**, a Delaware corporation, its affiliates, successors or assigns (together the "Company"), and the undersigned, **Kelley Conte** ("Consultant"), with the principal place of residence at 19 Causeway Street, Hudson, MA 01749, in consideration of the compensation, now and hereafter paid to Consultant by Company under this Agreement. The Company and Consultant have also entered into a Separation and Release Agreement and General Release dated September 5, 2023 (the "Separation Agreement") to which this Agreement is attached as Exhibit A.

For the purposes of this Agreement, the term "Business of the Company" means, any and all of: (i) the development, manufacture, commercial exploitation, marketing, licensing and sales of aerogels and related products and services; or (ii) any primary or substantial work responsibilities carried out by Consultant on behalf of the Company; or (iii) actual or demonstrably anticipated research or development of the Company. The parties agree that the "Business of the Company" is national and international in scope.

1. Statement of Work. Consultant shall provide the services ("Services") described in the Statement of Work attached as Schedule A ("Statement of Work"). The Services shall be provided for the time period provided in the Statement of Work ("Term"), and Consultant shall receive as compensation for the Services the compensation provided for in the Statement of Work ("Compensation").

2. Confidential Information.

(a) Company Information. Consultant hereby agrees at all times during the term of this Agreement and for five years following the termination of the Agreement, to hold in strictest confidence and not to use, except for the benefit of the Company, or to disclose to any person, firm or corporation without written authorization of the Company, any Confidential Information of the Company. This obligation is in addition to Consultant's obligation not to use or disclose Confidential Information obtained by Consultant during her employment by the Company. "Confidential Information" means any Company proprietary information, technical data, including, without limitation, research, product plans, products, services, customer lists and customers (including, without limitation, customers of the Company on whom Consultant called or with whom Consultant became acquainted during the term of this Agreement), markets, software, developments, inventions, processes, formulas, technology, designs, drawings, engineering, hardware configuration information, customer lists, employee lists, marketing, finances or other business information disclosed to Consultant by the Company either directly or indirectly in writing, orally or by drawings or observation of parts or equipment. "Confidential Information" does not include any of the foregoing items which has become publicly known and made generally available through no wrongful act of the Consultant or of others who were under confidentiality obligations with respect to the item or items involved. If Consultant is required by law to disclose Confidential Information,

such as in response to a lawfully issued subpoena or court order, Consultant shall first provide the Company advance notice of his intention to make such disclosure so that the Company can take appropriate steps to protect its Confidential Information from disclosure. Consultant further understands that the trade secrets or know-how of the Company also constitute Confidential Information and must be kept in the strictest confidence and may never be used or disclosed by Consultant other than as provided in this Section 2(a).

(b) Third Party Information. Consultant recognizes that the Company has received, and in the future will receive, from third parties their confidential or proprietary information subject to a duty of the Company to maintain the confidentiality of such information and to use it only for certain limited purposes. Consultant hereby agrees to hold all such confidential or proprietary information of a third party in the strictest confidence and not to disclose it to any person, firm, or corporation or to use it except as necessary in carrying out Consultant's work for the Company consistent with the Company's agreements or other arrangements with any such third parties.

3. Inventions.

(a) Inventions Retained and Licensed. Consultant has attached hereto, as Schedule B, a list describing all inventions, original works of authorship, developments, improvements, and trade secrets which were made by Consultant prior to this Agreement including those conceived, developed or reduced to practice prior to execution of this agreement (collectively referred to herein as "Prior Inventions"), which belong to Consultant, which relate to the Business of the Company, and which are not assigned to the Company hereunder; or, if no such list is attached, Consultant represents that there are no such Prior Inventions. If in the course of this Agreement with Company, Consultant incorporates into a Company product, process or machine, including product literature, website or reports a Prior Invention owned by Consultant or in which Consultant has an interest, the Company is hereby granted and shall have a nonexclusive, transferable royalty-free, irrevocable, perpetual, world-wide license to make, have made, modify, use, sell reproduce, prepare derivative works, and distribute copies to public of such Prior Invention as part of or in connection with Company's business including such product, process machine, product literature, website and reports.

(b) Assignment of Inventions. Consultant hereby agrees that Consultant has made or will promptly make full written disclosure to the Company, will hold in trust for the sole right and benefit of the Company, and hereby assigns to the Company, or its designee, all of Consultant's right, title, and interest in and to any and all inventions, original works of authorship, developments, business plans, concepts, improvements, designs, discoveries, ideas, trademarks or trade secrets, whether or not patentable or registrable under copyright or similar laws, which Consultant solely or jointly conceives or develops or reduces to practice, or authors or cause to be conceived or developed or reduced to practice, during the term of this Agreement with the Company (collectively referred to herein as "Inventions"), except as otherwise provided by applicable law. For

the purposes of this paragraph, Inventions shall not include inventions that Consultant develops entirely on Consultant's own time without using the Company's equipment, supplies, facilities, or trade secret information except for those inventions that either (i) relate at the time of conception or reduction to practice of the invention to the Business of the Company, or (ii) result from any work performed by Consultant for the Company. Consultant further acknowledges that all original works of authorship which are made by Consultant (solely or jointly with others) within the scope of and during the term of this Agreement with the Company and which are protectable by copyright are "works made for hire," as that term is defined in the United States Copyright Act. To the extent applicable law treats any such original works of authorship as not a work made for hire, Consultant hereby assigns as described above, to the Company, or its designee, all of Consultant's right, title, and interest in such works to the Company. Consultant understands and agrees that the decision whether or not to commercialize or market any invention developed by Consultant solely or jointly with others is within the Company's sole discretion and for the Company's sole benefit and that no royalty will be due to Consultant as a result of the Company's efforts to commercialize or market any such invention.

(c) Inventions Assigned to the United States. Consultant hereby agrees to assign to the United States government all of Consultant's right, title, and interest in and to any and all Inventions whenever such full title is required to be in the United States by a contract between the Company and the United States or any of its agencies.

(d) Maintenance of Records. Consultant hereby agrees to keep and maintain adequate and current written records of all Inventions made by Consultant (solely or jointly with others) during the term of this Agreement with the Company. The records will be in the form of notes, sketches, drawings, whether in electronic or hardcopy form, and any other format that may be specified by the Company. The records will be available to and remain the sole property of the Company at all times.

(e) Patent and Copyright Registrations. Consultant hereby agrees to assist the Company, or its designee, at the Company's expense, in every proper way to secure the Company's rights in the Inventions and any copyrights, patents, mask work rights or other intellectual property rights relating thereto in any and all countries, including the disclosure to the Company of all pertinent information and data with respect thereto, the execution of all applications, specifications, oaths, assignments and all other instruments which the Company shall deem necessary in order to apply for and obtain such rights and in order to assign and convey to the Company, its successors, assigns, and nominees the sole and exclusive rights, title and interest in and to such Inventions, and any copyrights, patents, mask work rights or other intellectual property rights relating thereto. Consultant further agrees that Consultant's obligation to execute or cause to be executed, when it is in Consultant's power to do so, any such instrument or papers shall continue after the termination of this Agreement. If the Company is unable because of Consultant's mental or physical incapacity or for any other reason to secure Consultant's signature to apply for or to pursue any application for any United States or foreign patents or copyright

registrations covering Inventions or original works of authorship assigned to the Company as above, then Consultant hereby irrevocably designates and appoints the Company and its duly authorized officers and agents as its agent and attorney in fact, to act for and in its behalf and stead to execute and file any such applications and to do all other lawfully permitted acts to further the prosecution and issuance of letters patent or copyright registrations thereon with the same legal force and effect as if executed by Consultant.

4.Returning Company Documents. Consultant hereby agrees that, upon termination of this Agreement, Consultant will promptly deliver to the Company (and will not keep in her possession, recreate or deliver to anyone else) any and all devices, records, data, notes, reports, proposals, lists, correspondence, specifications, drawings blueprints, sketches, materials, equipment, other documents or property, or reproductions in any medium of any aforementioned items developed during the term of this Agreement by, or otherwise belonging to, the Company, or Consultant's successors or assigns.

5.Restrictive Covenants. Nothing in this Agreement is intended to modify the post-employment obligations applicable to Consultant under the terms of the Amended and Restated Executive Employment Agreement she entered into with the Company dated as of January 1, 2023 (the "Executive Agreement") and the Separation Agreement.

6.Export Control Compliance. Consultant shall not export any information furnished by the Company without first complying with all requirements of US Export Administration Regulations including the International Traffic in Arms Regulations and the Export Administration Regulations, including the requirement for obtaining any export license, if applicable. Consultant shall first obtain the written consent of the Company prior to submitting any request for authority to export any such information. Consultant shall defend, indemnify and hold Aspen Aerogels, Inc., its officers and Directors harmless from all claims, demands, damages, costs, fines, penalties, attorney's fees and all other expenses and costs arising from Consultant's failure to comply with this paragraph, the Arms Export Control Act, the Export Administration Act and applicable regulations, or other applicable regulations.

7.Compliance with Laws and Code of Conduct. Consultant agrees (i) to comply with all applicable laws, statutes and regulations, including without limitation the U.S. Foreign Corrupt Practices Act and any similar foreign laws; and (ii) that Consultant, and every other person working for Consultant, or on her behalf, has not and will not, in connection with any transactions related to this Agreement or any other work in connection with sale of Products, make, offer or promise to make, or transfer any payment or anything of value, directly or indirectly, to any Government Official, or to any third party for payment to any Government Official, to improperly obtain, retain, or direct business or secure an improper advantage or take any other action, directly or indirectly, to violate any applicable laws and regulations prohibiting public or commercial bribery, extortion, kickbacks, or other unlawful or improper means of conducting business. For purposes of this Agreement, "Government Official" means the following: officer or employee of any government; officer or employee of any public international organization (e.g., the United Nations, World Bank, or International Monetary Fund); officer or employee of any department, agency, or instrumentality of any government or of any public international organization; officer or employee

of any government-owned or government-controlled company; political party; political party official; or anyone, whether a private person or otherwise, acting in an official capacity on behalf of any of the above or of any government entity. Consultant acknowledges receipt of the current copy of Aspen's Code of Business Conduct and Ethics ("Code of Conduct") available at <http://ir.aerogel.com/investors/governance/default.aspx>; understands how the Code of Conduct applies to the Consultant and agrees to strictly comply with the Code of Conduct.

8. Representations. Consultant hereby agrees to execute any proper oath or verify any proper document required to carry out the terms of this Agreement. Consultant hereby represents Consultant's performance of all the terms of this Agreement will not result in a breach of any agreement with a third party, including the breach of any agreement to keep in confidence proprietary information acquired by Consultant in confidence or in trust prior to this Agreement. Consultant has not entered into, and Consultant agrees Consultant will not enter into, any oral or written agreement in material conflict herewith.

9. Assignment. Neither this Agreement nor any obligations hereunder may be assigned or delegated by either party without the prior written consent of the other party.

10. ARBITRATION/JURY TRIAL WAIVER

(a) EXCEPT AS PROVIDED IN SECTION 11 HEREOF, EACH OF THE CONSULTANT AND THE COMPANY AGREES THAT ANY DISPUTE OR CONTROVERSY ARISING OUT OF, RELATING TO, OR CONCERNING CONSULTANT'S CONSULTING RELATIONSHIP WITH THE COMPANY OR ANY INTERPRETATION, CONSTRUCTION, PERFORMANCE OR BREACH OF THIS AGREEMENT SHALL BE RESOLVED BY ARBITRATION TO BE HELD IN WORCESTER COUNTY, MASSACHUSETTS, IN ACCORDANCE WITH THE COMMERCIAL ARBITRATION RULES THEN IN EFFECT OF THE AMERICAN ARBITRATION ASSOCIATION. THE DECISION OF THE ARBITRATOR SHALL BE FINAL, CONCLUSIVE, AND BINDING ON THE PARTIES TO THE ARBITRATION. JUDGMENT MAY BE ENTERED ON THE ARBITRATOR'S DECISION IN ANY COURT HAVING JURISDICTION. THE COMPANY AND CONSULTANT SHALL EACH PAY ONE-HALF OF THE COSTS AND EXPENSES OF SUCH ARBITRATION, INCLUDING THE ARBITRATOR'S FEES, AND THE COMPANY AND CONSULTANT SHALL EACH PAY THEIR RESPECTIVE COUNSEL FEES AND EXPENSES.

(b) THE PARTIES HEREBY WAIVE ANY RIGHT EITHER OF THEM MAY HAVE TO A JURY TRIAL AND AGREE THAT ALL DISPUTES BETWEEN THEM SHALL BE RESOLVED SOLELY BY ARBITRATION, EXCEPT AS PROVIDED IN SECTION 11 HEREOF.

(c) CONSULTANT UNDERSTANDS THAT EACH PARTY'S PROMISE TO RESOLVE CLAIMS BY ARBITRATION IN ACCORDANCE WITH THE PROVISIONS OF THIS AGREEMENT, RATHER THAN THROUGH THE COURTS, IS CONSIDERATION FOR THE OTHER PARTY'S LIKE PROMISE. CONSULTANT

FURTHER UNDERSTANDS THAT CONSULTANT HAS BEEN OFFERED TO PROVIDE THE SERVICES BY THE COMPANY IN CONSIDERATION OF ITS PROMISE TO ARBITRATE CLAIMS.

11. EQUITABLE REMEDIES. CONSULTANT AGREES THAT IT WOULD BE IMPOSSIBLE OR INADEQUATE TO MEASURE AND CALCULATE THE COMPANY'S DAMAGES FROM ANY BREACH OF THE COVENANTS SET FORTH IN SECTIONS 2, 3, 4, 5, 6, 7 AND 8 OF THIS AGREEMENT AND THAT SUCH A BREACH WOULD BE IRREPARABLE. ACCORDINGLY, CONSULTANT AGREES THAT IF CONSULTANT BREACHES ANY OF SUCH SECTIONS, THE COMPANY WILL HAVE THE RIGHT, IN ADDITION TO ANY OTHER RIGHT OR REMEDY AVAILABLE TO IT, TO OBTAIN AN INJUNCTION FROM A COURT OF COMPETENT JURISDICTION RESTRAINING SUCH BREACH OR THREATENED BREACH AND TO SPECIFIC PERFORMANCE OF ANY SUCH PROVISION OF THIS AGREEMENT. CONSULTANT FURTHER AGREES THAT NO BOND OR OTHER SECURITY SHALL BE REQUIRED IN OBTAINING SUCH EQUITABLE RELIEF AND CONSULTANT HEREBY CONSENTS TO THE ISSUANCE OF SUCH INJUNCTION AND TO THE ORDERING OF SPECIFIC PERFORMANCE.

12. Governing Law; Consent to Personal Jurisdiction. This Agreement will be governed by the laws of the Commonwealth of Massachusetts. Consultant hereby expressly consents to the exclusive personal jurisdiction of the state and federal courts located in Massachusetts for any lawsuit filed against Consultant by the Company as provided in Section 11 of this Agreement.

13. Termination. This Agreement shall continue to apply to any Statement of Work until such Statement of Work is terminated or otherwise expires on its terms. Notwithstanding the foregoing, Company may terminate this Agreement, including the Statement of Work, for Cause (as hereinafter defined) at any time immediately upon written notice to Consultant. Consultant may terminate this Agreement, including any Statement of Work, for any or no reason upon ten (10) days' written notice to Company. All the terms of this Agreement that by their nature are intended to survive the termination of this Agreement shall survive such termination. For the purposes of this Agreement, "Cause" means: (i) willful misconduct, dishonesty, fraud or breach of fiduciary duty to the Company; (ii) deliberate disregard of the lawful rules or policies of the Company, failure to perform the Services, or material breach of an agreement with the Company, which results in direct harm to the Company; (iii) the unauthorized disclosure of any trade secret or confidential information of the Company; (iv) the commission of an act which constitutes unfair competition with the Company or which induces any customer or supplier to breach a contract with the Company; (v) the indictment of Consultant for any felony involving deceit, dishonesty or fraud, or any criminal conduct by the Consultant that would reasonably be expected to result in material injury or reputational harm to the Company; or (vi) the Consultant's revocation of the Separation Agreement. For purposes hereof, whether or not the Consultant has committed an act or omission of the type referred to in subparagraphs (i) through (vi) above will be determined by the Company in its reasonable, good faith discretion, based upon the facts known to the Company at the relevant time.

14.Entire Agreement. This Agreement sets forth the entire agreement and understanding between the Company and Consultant relating to the subject matter herein and merges all prior discussions between us, but does not affect or supersede the Separation Agreement or Consultant's post-employment obligations under the Executive Agreement, which shall survive any termination of this Agreement and the Statement of Work in accordance with their terms. Notwithstanding the foregoing, Consultant acknowledges and agrees that she shall remain subject to the terms and conditions of the Company's Compensation Recoupment Policy following the termination of her employment and during and following the Term of this Agreement to the extent provided in the Compensation Recoupment Policy. No modification of or amendment to the Agreement, nor any waiver of any rights under this agreement, will be effective unless in writing signed by the party to be charged. Any subsequent change or changes in Consultant's duties, or compensation will not affect the validity or scope of this Agreement.

15.Severability. If one or more of the provisions in this Agreement are deemed void or voidable under applicable law, then the remaining provisions will continue in full force and effect without the inclusion of any such provisions.

16.Survival. The following provisions shall survive any termination of this Agreement: Sections 2 through 8, along with such other provisions of this Agreement, the Separation Agreement and the Executive Agreement as may be necessary to interpret the same.

17.Independent Contractor. Consultant is an independent contractor and shall be solely responsible for (and indemnifies and holds harmless the Company for) any unemployment or disability insurance payments, and any social security, income tax or other withholdings, deductions or payments which may be required by federal, state or local law with respect to any sums paid to Consultant hereunder or for any and all Services or materials provided or rendered by Consultant.

CONSULTANT
Kelley Conte

COMPANY
Aspen Aerogels, Inc.

By: /s/ Kelley Conte
Name: Kelley Conte
Date: August 21, 2023

By: /s/ Donald Young
Name: Donald Young
Date: August 20, 2023

Schedule A – Statement of Work

This Statement of Work dated September 5, 2023 describes Services to be performed by Consultant for Aspen Aerogels, Inc. (the “Company”) and is issued pursuant to the Consulting Agreement dated September 5, 2023 (the “Agreement”). This Statement of Work may be extended or modified only by a written agreement of both Company and Consultant.

A. SERVICES

Consultant to provide Services to the Company in connection with Company’s operations as reasonably requested by the Company from time to time during the term of the Statement of Work, including:

(i) Consultant will provide Company personnel with her historical perspective on, and ideas to improve the Company’s human resources management and operations.

(ii) Consultant will assist with the transition of Chief Human Resources Officer role and related responsibilities to Consultant’s successor.

Consultant will perform the Services free of the direction and control of the Company, but consistent with the objectives and deadlines set by the Company.

Consultant agrees to promptly advise the Company in case of any potential conflict between Consultant’s Services to be provided under this Statement of Work and Consultant’s other engagement or employment with third parties.

B. TERM

Subject to the right of Consultant to terminate this Statement of Work earlier for no or any reason under Section 13 of the Agreement and the Company’s right to terminate this Statement of Work for Cause, the Services pursuant to the Statement of Work shall begin September 5, 2023 and end on March 31, 2024 (the “Initial Term”). Thereafter, the Initial Term may be extended upon the mutual written consent of the Company and Consultant (the Initial Term and any such extensions, the “Term”).

C. COMPENSATION

In connection with Consultant’s Services pursuant to this Statement of Work, as Compensation for her Services, Company agrees that, subject to and conditioned upon the Consultant not revoking the Separation Agreement:

(i) The Company shall pay consultant cash compensation for her services at an hourly rate of \$200.00 per hour, which shall be paid monthly in arrears within 15 calendar days following the last day of each calendar month during the Term, subject to Consultant’s provision of documentation acceptable to the Company, within 5 days after the last day of each such calendar month, of the number of hours of Services performed by Consultant during such calendar month;

(ii) any unvested restricted stock units (“RSUs”) that were previously granted to the Consultant in her capacity as an executive of the Company shall remain outstanding in accordance

with their terms and continue to vest in accordance with their terms as per the applicable grant documents during the Term;

(iii)The unvested stock options (“Options”) that were previously granted to the Consultant in her capacity as an executive of the Company shall not lapse upon the termination of her employment with the Company, and shall remain outstanding in accordance with their terms and continue to vest during the Term, provided that the Consulting Agreement and this Statement of Work have not been terminated prior thereto; and

(iv)Consultant may exercise any then-vested Options until the earlier of (A) two years from the expiration the Term, and (B) the tenth anniversary of the date of grant of the applicable Option.

For the avoidance of doubt, Consultant’s right to receive any and all of the benefits set forth in clauses (i), (ii), (iii) and (iv) of the preceding sentence shall be forfeited and shall not be provided by the Company if the Consultant revokes the Separation Agreement. For the further avoidance of doubt, all then-unvested RSUs and Options shall be forfeited and not vest in the event that either party terminates this Statement of Work and the Consulting Agreement, in accordance with the terms hereof.

* * * * *

CERTIFICATIONS UNDER SECTION 302

I, Donald R. Young, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 2, 2023

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

CERTIFICATIONS UNDER SECTION 302

I, Ricardo C. Rodriguez, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 2, 2023

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez

Chief Financial Officer and Treasurer (principal financial officer)

CERTIFICATIONS UNDER SECTION 906

Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code), each of the undersigned officers of Aspen Aerogels, Inc., a Delaware corporation (the “Company”), does hereby certify, to such officer’s knowledge, that:

The Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 (the “Form 10-Q”) of the Company fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, and the information contained in the Form 10-Q fairly presents, in all material respects, the financial condition and results of operations of the Company.

Dated: November 2, 2023

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Dated: November 2, 2023

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer)

A signed original of this written statement required by Section 906 has been provided to the Company and will be retained by the Company and furnished to the Securities and Exchange Commission or its staff upon request.
